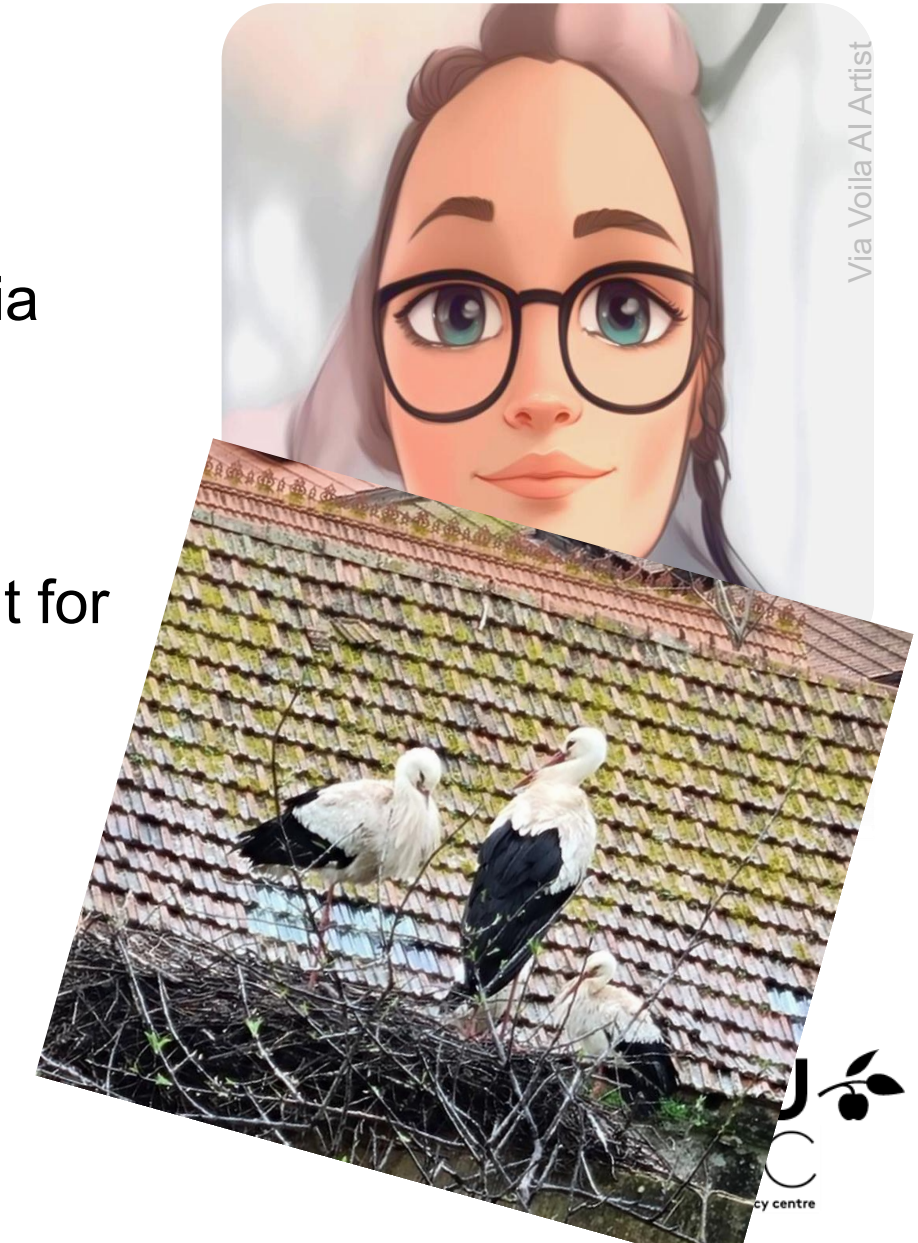
AI 507: Artificial Intelligence and Society

#1 Introduction & Intelligence

Associate Prof. Dr. Lena Frischlich

Welcome (again)

- Lena Frischlich, Associate Professor at the Digital Democracy Centre at SDU
- Background in psychology (specifically social and media psychology) and Communication science
- And as I am still learning the names – lets do a short repetition for me – stand up and tell me where you went for your last holidays
- I went to Portugal, specifically the Algarve, saw a lot of rain, but also a crazy amount of storchs



Some more formal details

- This is the lecture element of AI507 – it usually takes place on Thursdays (there is one Friday in the mix, so check your calendar) and in this room
- The course has a portfolio exam
- For the lecture, that means a 60- min via DigiExam
  with multiple-choice questions
- The exam will take in the first two weeks of June (coordinated by IMADA)
- What **is** relevant for the exam?
  - The slides for this lecture → uploaded every week to itslearning
  - One podcast, TED-talk or text per week → links in itslearning
  - In principle everything we discuss and talk about during class → it *does* make sense to attend
- What is **not** relevant for the exam?
  - At the beginning of each session, we will often have a short historical “how it started”
    before we come to the “how its going” – no need to memorize the birthdates or spell out names of persons ☺

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Why the question “is that relevant for the exam” is bad for you

- Our brain is lazy and bad at memorizing different categories of information
- If you tell yourself “This is maybe not so relevant”, you risk that your brain shifts attention to – for example – your last holiday
- Learning also works best, when you relate the content to the things you already know, that matter to you, that are interesting and fun – so take every opportunity to use what you have learned
- For example, by evaluating critically what you find on YouTube, TikTok, Reddit, in the news etc.

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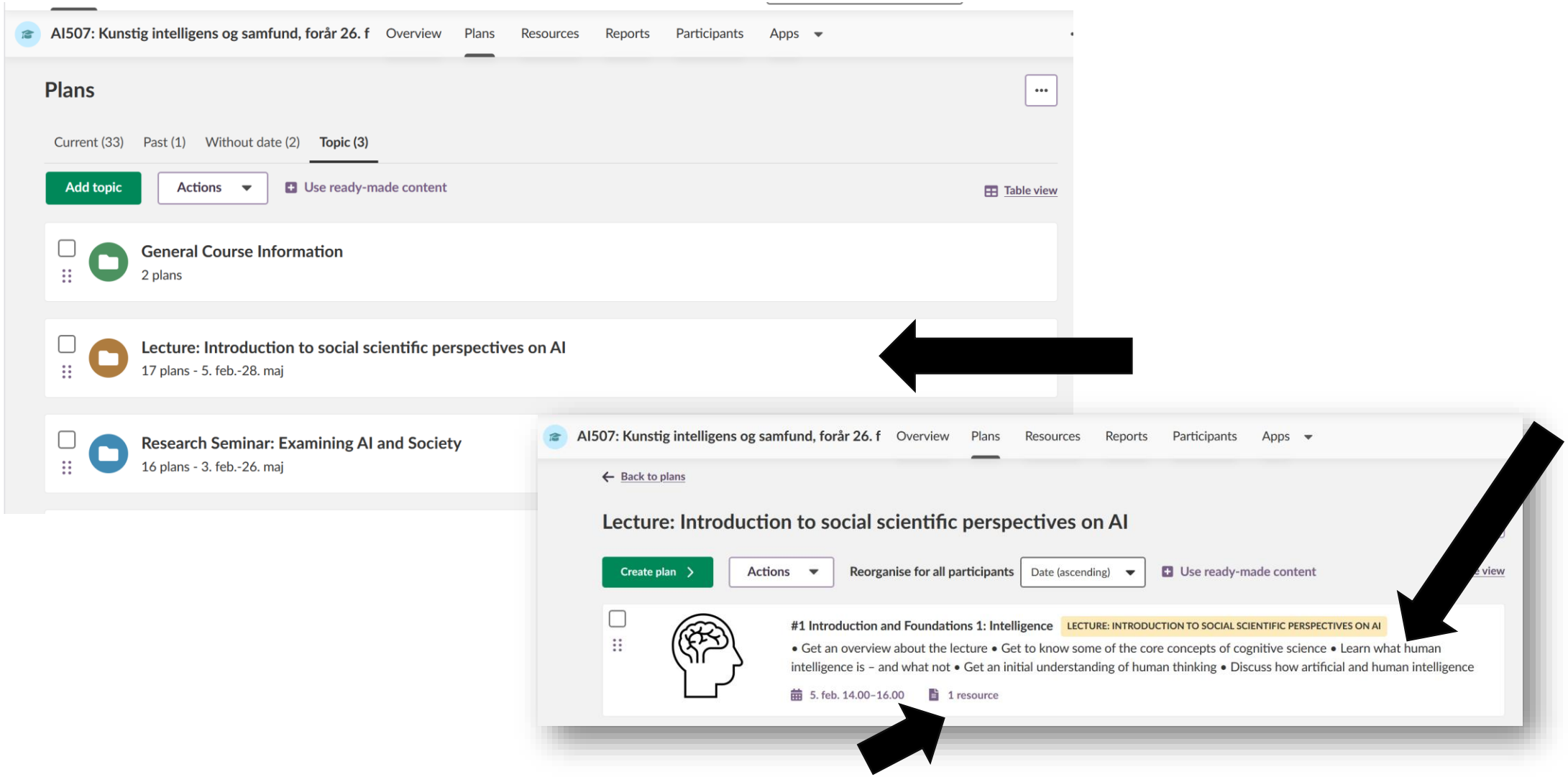
Digital Lecture Room

- Itslearning



Any questions so far?

Objectives of the lecture

At the end of the lecture, you will…

- …have a solid basic knowledge about social scientific perspectives on and relevance for Artificial intelligence
- …know central social scientific theories and models that allow you to evaluate the opportunities and risks of AI in different areas
- …be able to evaluate specific AI applications in (human) societies, taking social scientific knowledge into account
- …be able to form an empirically grounded opinion about the consequences of AI for individuals, social groups, and society at large – and take that into account in your own work

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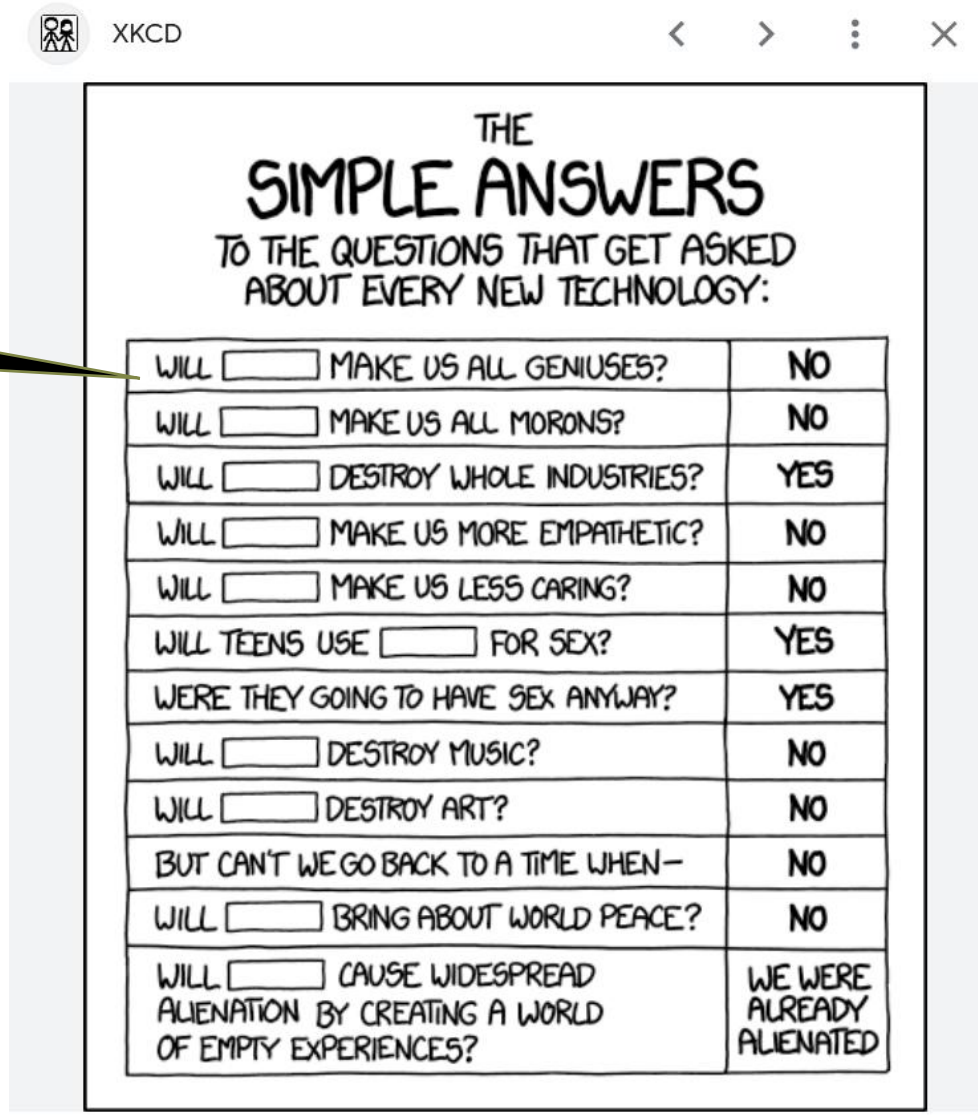
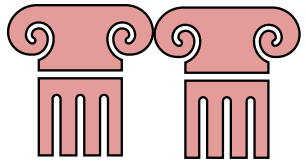
A little spoiler

AI



But now in detail: Course structure

Foundations



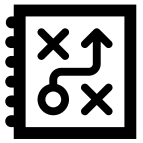
Intelligence

Learning

Communication

Social Cognition

Innovations and their dissemination



Quiz

Robots

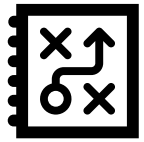
Recommender System

Chatbots

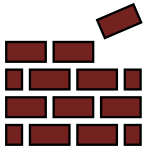
Smart things

Human-Technology Interactions

Digital platforms



Quiz



Application

Organisations & Institutions

Political Communication

Health & Wellbeing

Entertainment

Exam

Session structure

The lectures follow a typical scheme that has been found to support learning

1. Arriving and remembering
   - 10-15 min. “flashlight”, I’ll toss a (virtual) ball, you say one keyword
   - Time for your questions

   - Activates your brain
   - Supports the storage of the content in your long-term memory
   - After today, you’ll know why that matters

2. Input
   - Typically, different “chapters” (5-15 min), myself or guests
   - Talk about theoretical foundations and general models
   - Discuss current research
   - After each chapter: Time for questions

   - Actual knowledge gains
   - Sorted by topics to support your brain in storing the information correctly
   - Guest will bring in hands-on-experience or specific expertise

3. Applying
   - Discuss differences between humans and AI and how that relates to your other courses
   - Discuss current phenomena and events

   - Use it or loose it
   - Supports your brain
   - Allows you to speak to current debates

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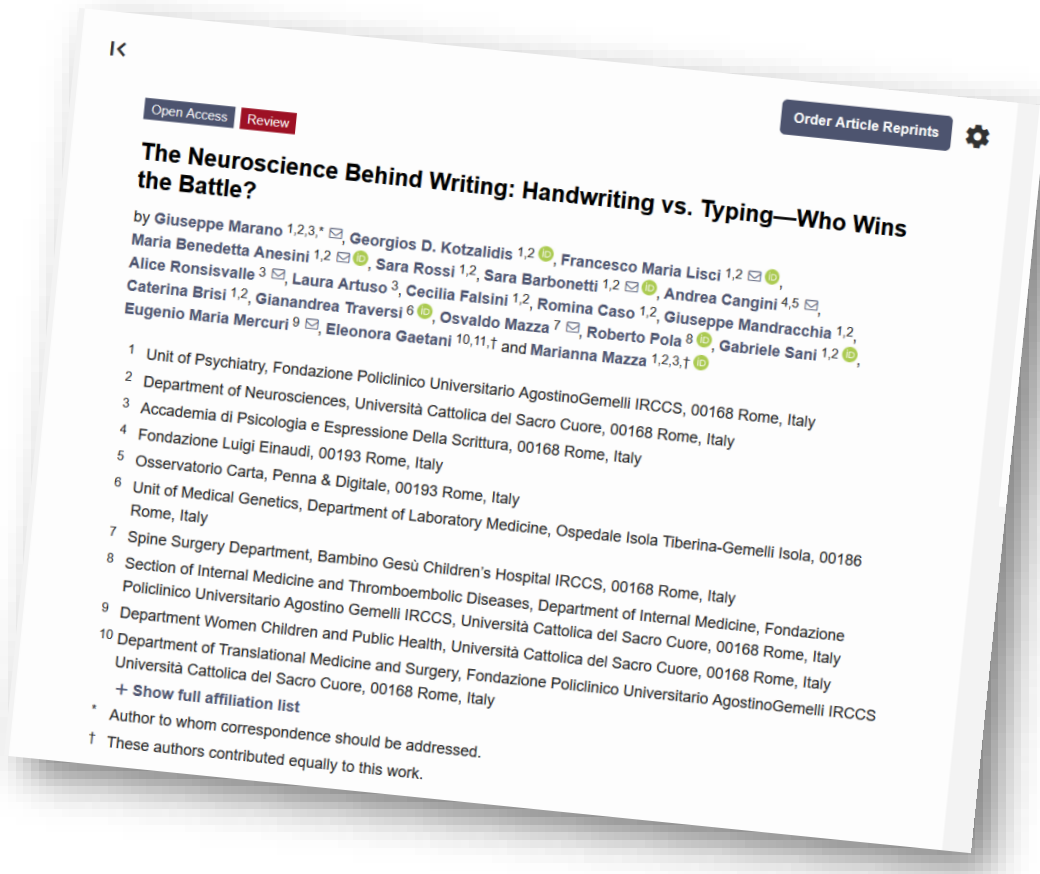
Around the sessions

- Prepare yourself for the lecture
  - Listen to the TED talk/ YouTube video / Podcast of the week (all of them offer transcripts) – typically 10-30 minutes or read the text of the week (in itslearning)
  - Expect that
- During the lecture
  - Use the uploaded slides if that helps you – ask if something is not totally clear
  - In most cases, you are not the only one with a question – but probably the only one brave enough to ask it ☺
  - Consider taking notes (magically good to activate your brain – even if oldschool)



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The Neuroscience Behind Writing: Handwriting vs. Typing—Who Wins the Battle?

by Giuseppe Marano [1,2,3,*], Georgios D. Kotzalidis [1,2], Francesco Maria Lisci [1,2], Maria Benedetta Anesini [1,2], Sara Rossi [1,2], Sara Barbonetti [1,2], Andrea Cangini [4,5], Alice Ronsisvalle [3], Laura Artuso [3], Cecilia Falsini [1,2], Romina Caso [1,2], Giuseppe Mandracchia [1,2], Caterina Brisi [1,2], Gianandrea Traversi [6], Osvaldo Mazza [7], Roberto Pola [8], Gabriele Sani [1,2], Eugenio Maria Mercuri [9], Eleonora Gaetani [10,11,†] and Marianna Mazza [1,2,3,†]

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* Author to whom correspondence should be addressed.
† These authors contributed equally to this work.

What are your questions so far?

Then let’s get started

At the end of today’s lecture, you will…

- …know some of the core concepts of cognitive science
- …have an idea of what human intelligence is – and what not
- …have explored how cognitive science can inform the development of AI
- … have discussed the difference between human and artificial intelligence

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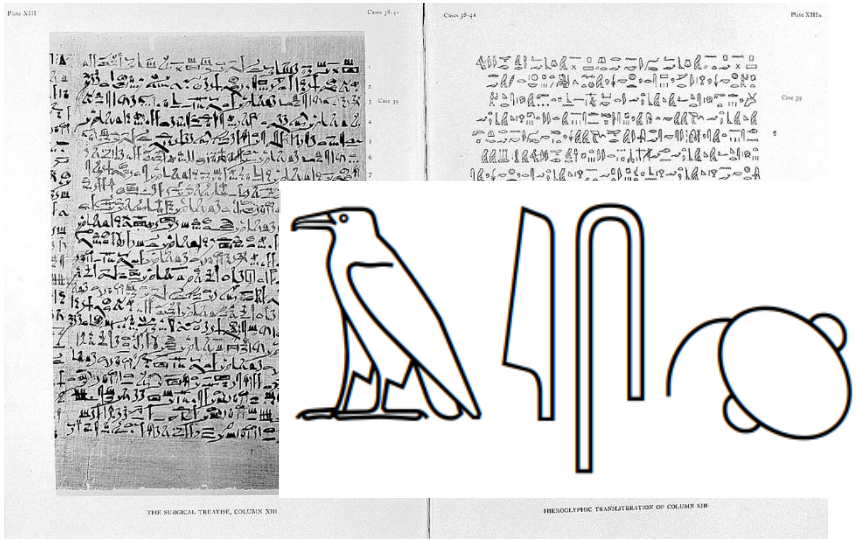
Cognition

What is cognition? (Dumper et al., 2019)

- Right now, you process several pieces of information
- You are able to remember,
  - …what you ate last
  - …whether you liked it
- Ask yourself
  - …why you ended up in this course
  - …if you like the person next to you
  - …what this one message was about
- And that all while maintaining a seated position, simulating you're interested, and sometimes even while walking and talking
- **How is that even possible?**

Humans have tried to answer this question since a *very* long time (Dumper et al., 2019; Pang, 2023)



By https://wellcomeimages.org/indexplus/obf_images/2d/f3/28afabff15c08c7db2c40409d200.jpgGallery: https://wellcomeimages.org/indexplus/image/L0003140.htmlWellcome Collection gallery (2018-03-30): https://wellcomecollection.org/works/ktg955jw CC-BY-4.0, CC BY 4.0, https://commons.wikimedia.org/w/index.php?curid=35864880

~16.000 years B.C., the Edwin Smith Papyrus (named after his buyer) mentioned the human brain and its treatment for the first time

Frans Hals – Renée Descartes
De eeuw van Rembrandt, Bruxelles: Crédit communal de Belgique, ISBN 2-908388-32-4., https://commons.wikimedia.org/w/index.php?curid=2774313

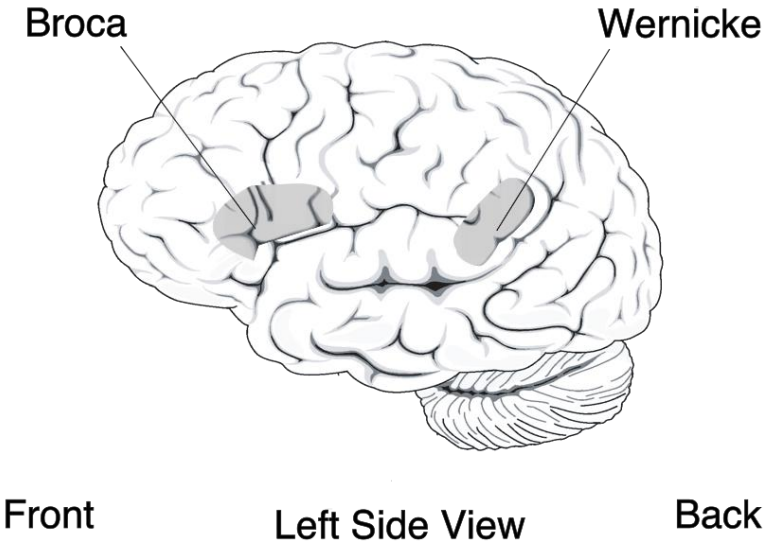
16th century: French Philosopher Renée Descartes declared mind and body to be distinct (mind-body dualism) and argued "I think, therefore I am"

Our thoughts are created solely through the stimulation of our senses (empiricism)
e.g., Bishop George Berkley, Philosopher John Locke

No! We are born with innate knowledge that allows us to form language and maintain conscious experiences (nativism)
e.g., philosopher Immanuel Kant

New insights through brain studies (often on injured patients) (Broca, 1861; Dumper et al., 2019)

- French physician Pierre Paul Broca: A patient, who could only say “tang” had a specific area of his brain-injured → central for language *production*
- German physician Carl Wernicke: Damages in another region of the brain central for *comprehension*
- Today: While the brain has different areas that are central for different tasks, it is also highly connected (“function circuits”)



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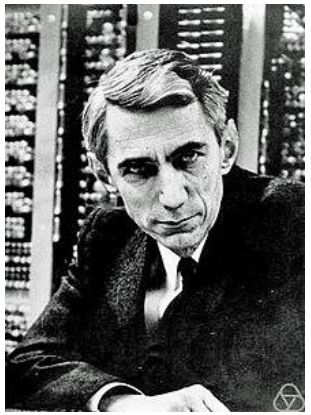
New insights through psychological experiments (Russo Kraus, 2022)

Wilhelm Wundt (1832-1902), Weltrundschau zu Reclams Universum 1902, NOT-PD-US-URAA

- In 1879, Wilhelm Wundt founded the first psychological laboratory at the University of Leipzig
- With his experimental approach he is considered to be the father of modern experimental psychology
- Renown for his ability to measure psychic quantities and show exact relationships between them
- Argued that “nothing is in the intellect that was not first in the senses, except the intellect itself” (*"Nihil est in intellectu quod non fuerit in sensu, nisi intellectu ipse." (Leibniz, Nouveaux essais, 1765, Livre II, Des Idées, Chapitre 1, § 6))*
- Logic inferences, categories of thought, the principle of causality, etc. all go beyond direct sensory impressions

The cognitive revolution in the 20th century: Minds & Computers (Gigerenzer & Goldstein, 1996, Shannon, 1948)



Jacobs, Konrad – Claude Shannon (~1963) https://opc.mfo.de/detail?photo_id=3807

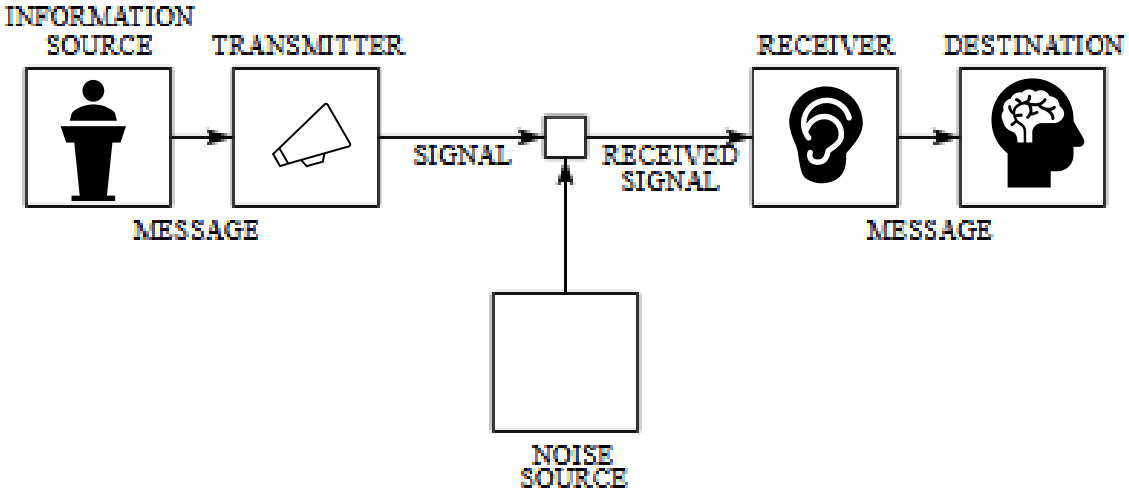


Fig. 1—Schematic diagram of a general communication system.

John von Neumann |Source= http://www.lanl.gov/history/atomicbomb/images/NeumannL.GIF |Date= 01/20/08

Highly influential work "a mathematical theory of information" (1948)

https://berichtenuithetverleden.wordpress.com/2013/01/29/alan-turing-1912-1954/

Computers can imitate human minds (Alan Turing, 1948) → Computers as minds

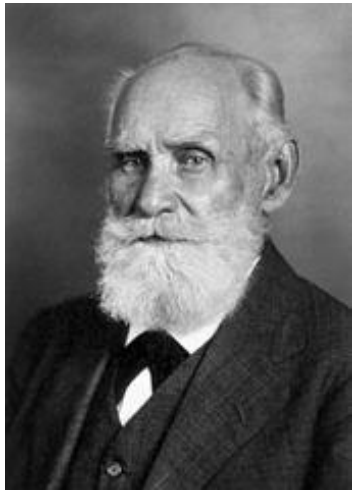
The neural system has similarities to the computer (von Neumann, 1958) → Minds as computers

Psychology was still vey behaviourist (Akpan, 2020; Domjan, 2005)

- Central observation: External stimuli motivate behavioural responses
- The association between stimulus and response can be conditioned
  - Classical conditioning:
    - Food → Salvation
    - Food + Bell → Salvation
    - Bell → Salvation
  - Operational conditioning / reinforcement learning
    - Sanctioning emerging behaviour
    - Reward: more behaviour, punishment: less
    - Can be used to train very complex behaviours

General idea Stimulus → Reaction

BUT



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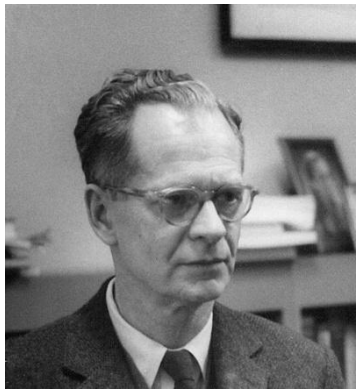

Burrhus Frederic Skinner
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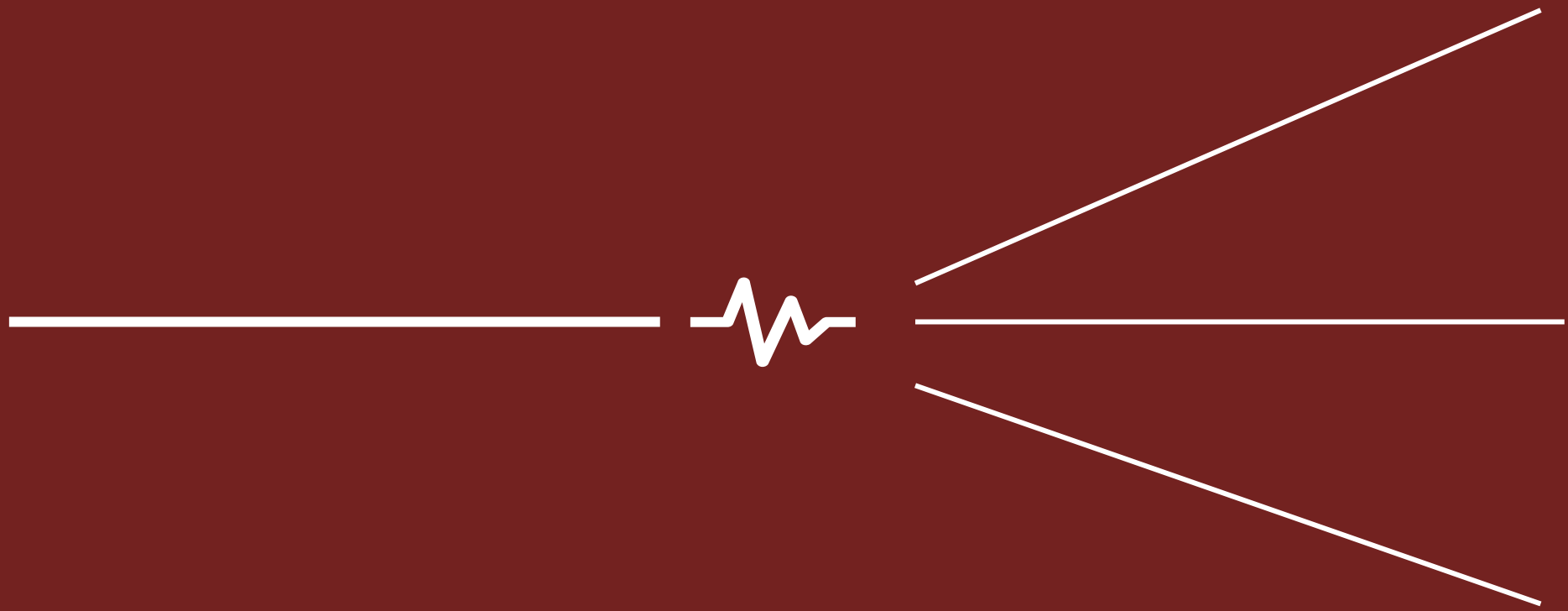
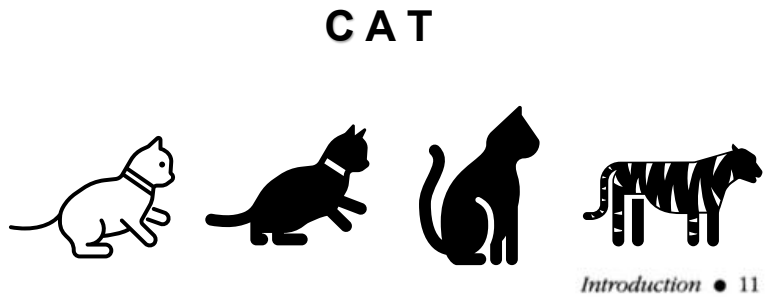


Between stimulus and response, there is a space. In that space is our power to choose our response.

(unknown, see Coleman-Baker, 2018)

Core concepts today (Kellog, 2003; von Eckhardt, 2012)

- **Mental representation:** Input is internally encoded (“represented)
- **Stages of processing:** This encoding can happen in different stages (superficial vs. deep encoding)
- **Consciousness:** can refer to self-knowledge, information access, or sentience (vs. e.g., coma) – there are forms of unconscious information processing
- **Cognitive architecture**
  - Symbolic models: the mind is a computer
  - Connectionist models: the mind works over distributed neurons and layers



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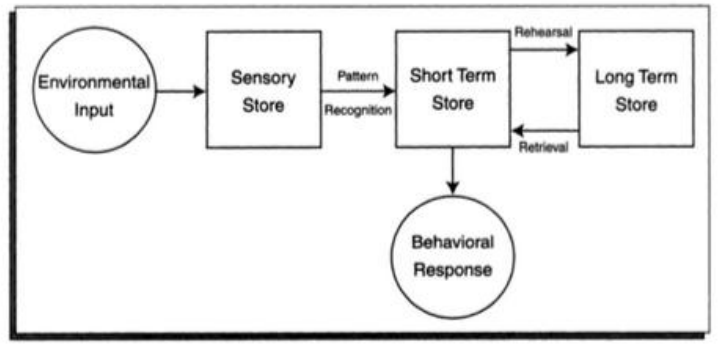


Figure 1.3. Cognitive architectures: Example of a symbolic model.
SOURCE: Adapted from Atkinson and Shiffrin (1971).

12 • SCOPE AND METHODS

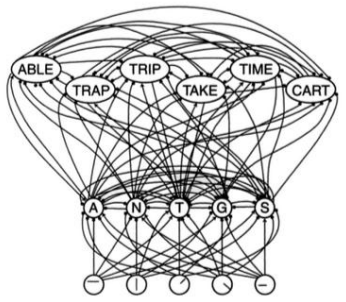


Figure 1.4 Cognitive architectures: Example of a connectionist model.
SOURCE: McClelland and Rumelhart (1981).

Kellog, 2003 p. 11-13

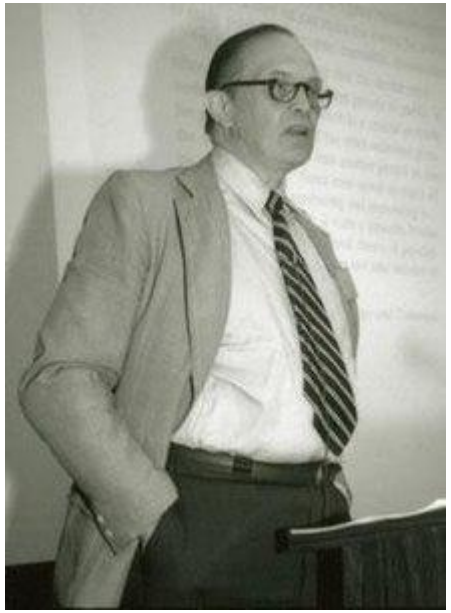
Where do you see the similarities between human and artificial cognition?

Cognitive Revolution in Psychology

(Dumper et al., 2019; Neisser, 1967)

- In 1967, the Harvard Center of Cognitive Studies was founded by George Miller, the father of cognitive science and psycholinguistics
- His student, Ulrich Neisser published the first cognitive psychology textbook in 1967
- For the first time, cognition was defined as

  "all processes by which the sensory input is transformed, reduced, elaborated, stored, recovered, and used. It is concerned with these processes even when they operate in the absence of relevant stimulation, as in images and hallucinations." (p. 4)



http://www.psychologicalscience.org/observer/getArticle.cfm?id=2004

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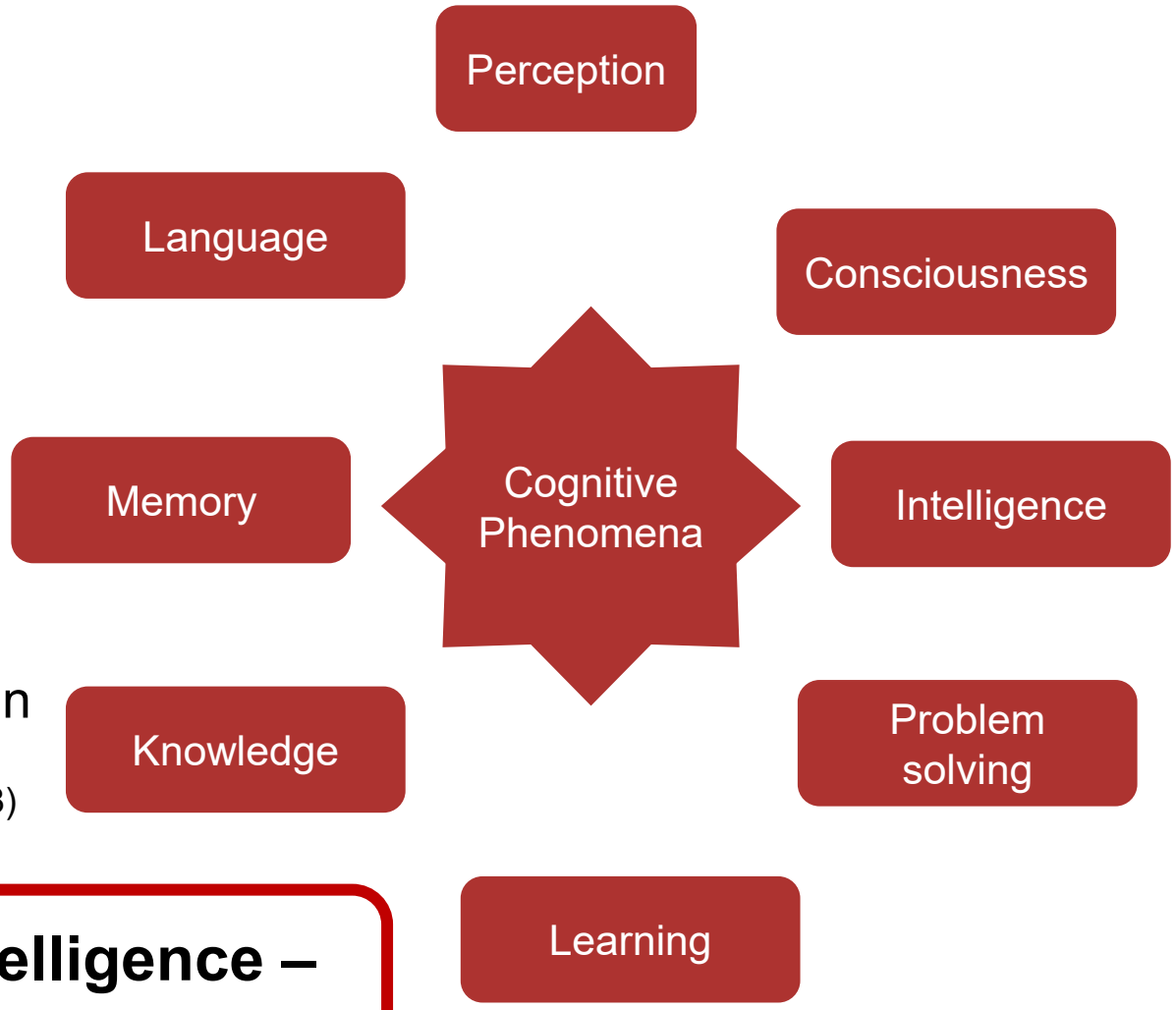
In short: cognition is thinking

(Dumper et al., 2019; Kellog,2003)

- Involves several phenomena
- Humanity has long tried to understand
- Cognitive Science is rooted in neuro-bio-physiological traditions and psychological experiments and has flourished alongside the triumphal march of computer science
- Between stimulus and response, there is a space. In that space lies our freedom (unknown, cf.Coleman-Baker, 2018)



Artificial intelligence is – like human intelligence – only one facet of cognition

What are your questions so far?

Time for a break

What do you think: What is intelligence?

What is intelligence? (Bingham, 1937, Fletcher & Hattie, 2011; Gottfredson, 1997, 52 signatories)

Intelligence is:

- A very general mental capability that involves the ability to
  - …reason, plan, solve problems, think abstractly,
  - …comprehend complex ideas,
  - …learn quickly and learn from experience
- Something that can be reliably measured via intelligence tests and expressed as intelligence quotient or IQ
- While intelligence tests differ from each other and can be more or less culture-fair, they do all measure the same general factor “intelligence”

Von Timothy Bates (President of ISIR, International Society for Intelligence Research) - DISTINGUISHED CONTRIBUTOR INTERVIEW Linda Gottfredson, CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=74173545

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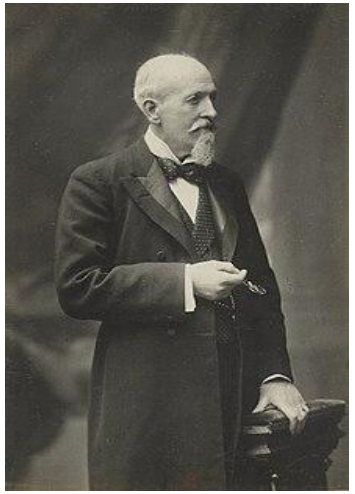
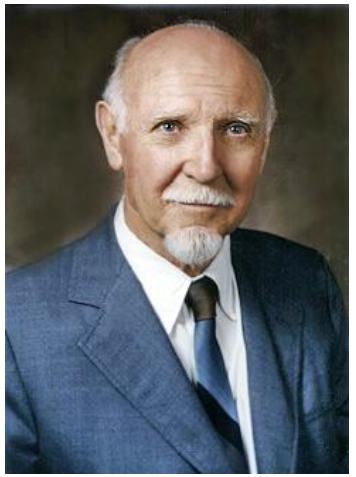
Spearmans' 'g' factor (Cattell, 1963; Nisbett et al., 2012; Spearman, 1904; Warne & Burningham, 2019)

- People who perform well in one cognitive test often also perform well in another
- Common factor or shared variance in a series of correlated cognitive tasks
- Found across the world – even in non-Western, non-industrialized nations
- The dream of AGI is rooted in this general intelligence concept
- Despite *g*: Different factors within intelligence, e.g., two-factor theory by Raymond Catell
  - Fluid intelligence (= speed of information processing)
  - Crystalline intelligence (= knowledge, language skills, experience with problem-solving, etc.)
- They are distinct from each other – amongst others over the lifespan



Exposition universelle de 1900 : portrait de M. Spearman.



Von Cattell family - Wikipedia:Contact us/Photo submission, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=3927018

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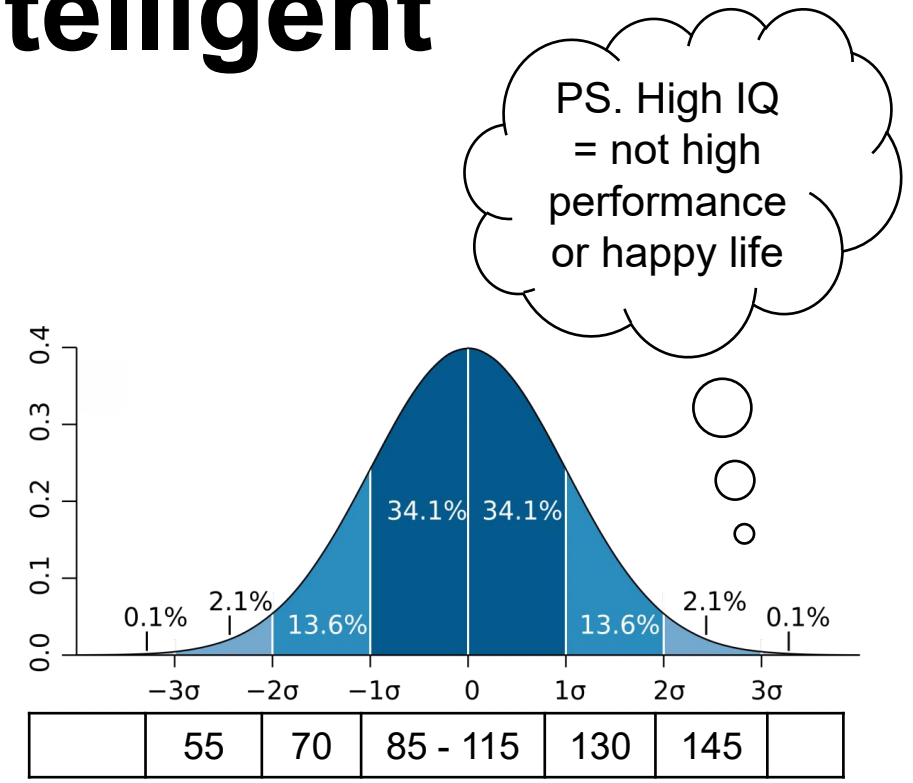
You are – likely – averagely intelligent

(Gottfredson, 1997; Nisbett, 2012)

- Results of intelligence tests are normally distributed
- Most people have an average IQ that is around 100 in a standard Intelligence test
- If you do one, reliable psychologists will only tell you if you have an average IQ (most of us) a slightly above/below one (still a third), an IQ below/ above average less than 5%, or are even more extreme
- Intelligence has a strong genetic component (~ 40-80%), but family members can differ
- Socio-economic background and ethnicity predict average IQ scores – but individuals within one group differ considerably
- Your environment during childhood matters too



The bell-curve controversy (Fletcher et al., 2011; Nisbett, 2012)

- 1994 book “The Bell Curve” by Herrnstein and Murray
- Focussed on the average differences in the IQ between Whites, PoC, Jews, and Asians (all in the realm of average IQ though)
- The authors claimed that IQ is what predicts school success and that it is not malleable, wherefore interventions are not necessary (also saying that low IQ → criminality…)
- This has been debunked since
- While there are systematic differences (Jews tend to score higher than Whites, Blacks lower), socio-economic status and education shape the performance
- Differences are shrinking and across countries, the average IQ is rising (“Flynn effect”)
- Centrally: The interest of cognitive scientists has moved towards cognitive performance, largely accepting a general factor “g”, making some of these cognitive tasks easier for some of us

Why does that matter today?

TECH NEWS

Elon Musk Seems to Endorse Tweet Saying Students at Black Colleges Have Low IQs

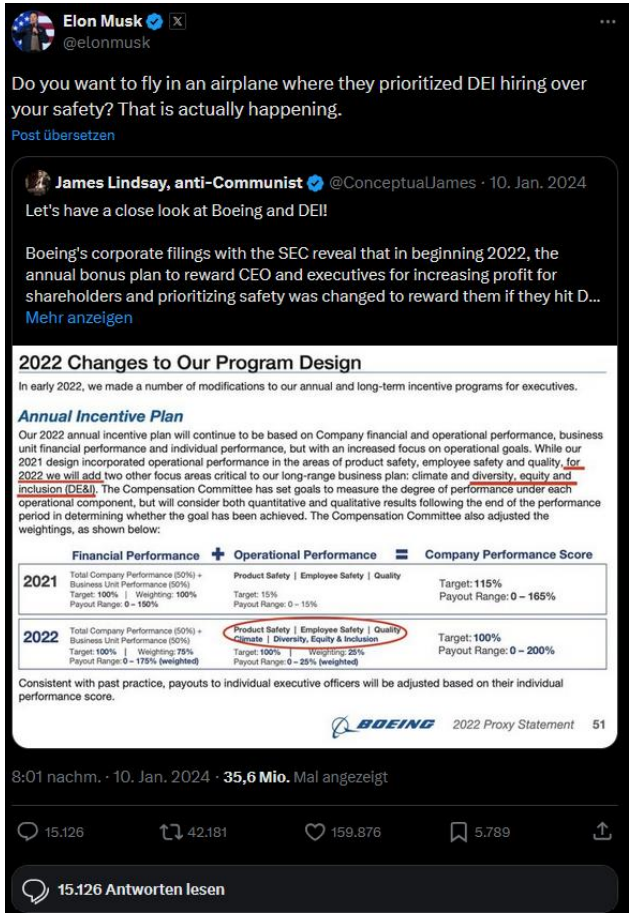
The world's richest man shares more of his thoughts about race on X.com.

By **Thomas Germain** Published January 10, 2024 |
Comments (145)

BEST OF CES 2025 AWARDS

https://gizmodo.com/elon-musk-endorses-tweet-saying-students-at-black-colle-1851156533

Tweeting While Racist
1.12.24, 2:32 PM EST by Victor Tangermann



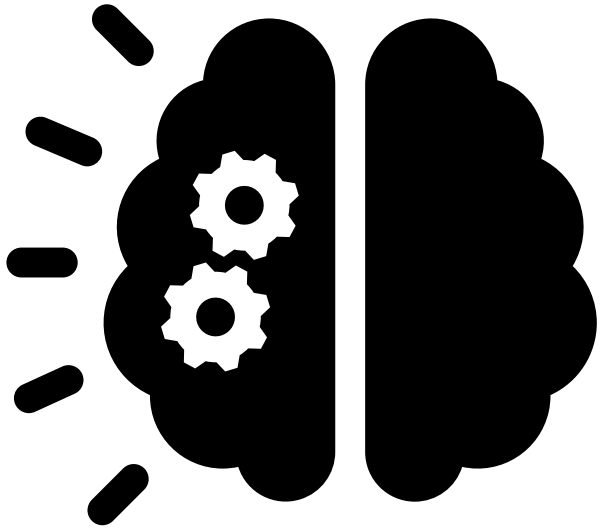
https://x.com/elonmusk/status/1745158868676546609

https://x.com/elonmusk/status/1744821656990675184?s=20

https://www.whitehouse.gov/presidential-actions/2025/01/ending-radical-and-wasteful-government-dei-programs-and-preferencing/

To sum up

- Intelligence is a very general mental capability
- Makes several cognitive tasks easier
- This general factor can be reliably measured
- Most of us have an average amount of that
- While there is considerable genetic heritability, education and environment matter
- Not least for crystalline intelligence
- Group differences are there – but they are overinterpreted for racist purposes – 1997 as well as 2025
- The dream of AGI is rooted in this tradition – whether we want it or not



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What are your questions so far?

How do humans think?

Thinking (Kahnemann, 2012; Tversky & Kahnemann, 1974)

Two types of thinking

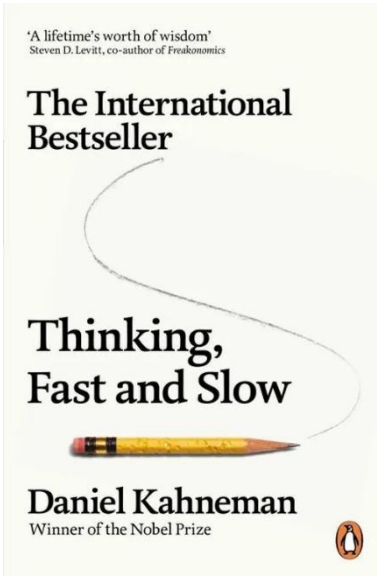
- automatic, passive, usually “fast” thinking (System I)
- effortful, deliberatively, attentional, able to do complex computations, execute self-control (System II)

How does that work?

- Look at your neighbour's face
- Think about beer and vomit
- Did you notice something?

We will meet much more of these two process models later in this lecture – but for now, let’s hear why that matters for AI

Daniel Kahnemann & Anton Twersky, Kahnemann, 2012, p. 55



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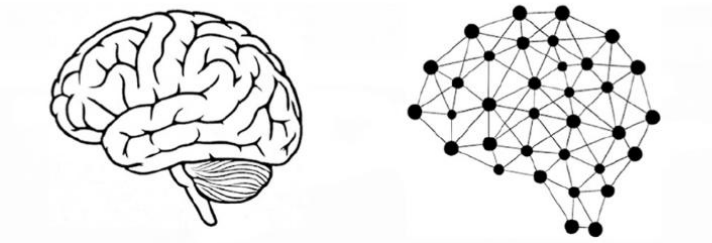
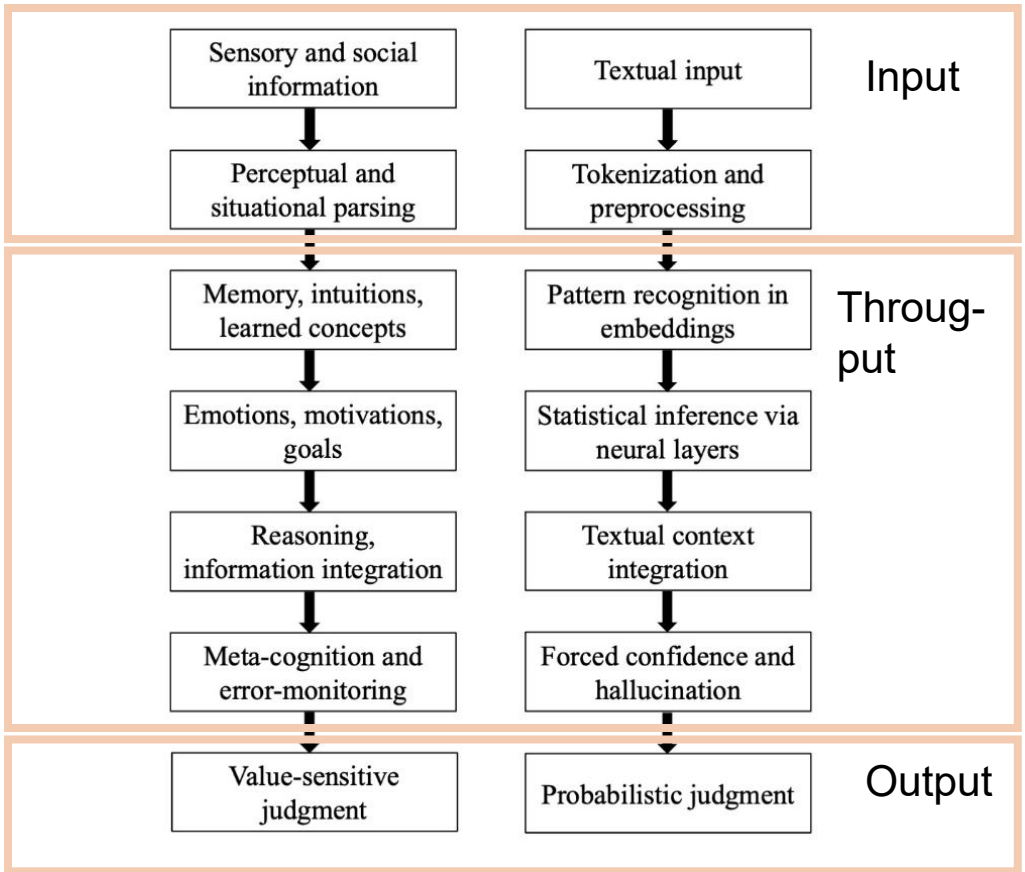
https://www.ted.com/talks/noam_brown_ai_won_t_plateau_if_we_give_it_time_to_think?subtitle=en

So, what are the differences between AI and human cognition? (Quattrociocchi et al., 2025)

- The apparent alignment between human and machine outputs conceals a deeper structural mismatch in how judgments are produced
- Humans as “epistemic agents” who actively construct knowledge through thinking
- AI as “stochastic parrot”
- seven epistemic fault lines





What do you think about artificial “intelligence” and “thinking`”?

Any more questions for today?

AI 507: Artificial Intelligence and Society

#1 Introduction & Intelligence

Associate Prof. Dr. Lena Frischlich

References

- Akpan, B. (2020). Classical and operant conditioning—Ivan pavlov; burrhus skinner. In B. Akpan & T. J. Kennedy (Eds.), *Science Education in Theory and Practice: An Introductory Guide to Learning Theory* (pp. 71–84). Springer International Publishing. https://doi.org/10.1007/978-3-030-43620-9_6
- Bingham, W. V. (1937). Aptitudes and aptitude testing. Harpers & Brothers.
- Broca, M. P. (1861). Remarques sur le siége de la faculté du langage articulé, suivies d'une observation d'aphémie (perte de la parole). *Bulletin de la Société Anatomique*, *6*, 330–357.
- Cattell, R. B. (1963). Theory of fluid and crystallized intelligence: A critical experiment. *Journal of Educational Psychology*, *54*(1), 1–22. https://doi.org/10.1037/h0046743
- Domjan, M. (2005). Pavlovian conditioning: A functional perspective. *Annual Review of Psychology*, *56*(1), 179–206. https://doi.org/10.1146/annurev.psych.55.090902.141409
- Dumper, K., Jenkins, W., Lacombe, A., Lovett, M., & Perimutter, M. (2019). 7.1 what is cognition? – Introductory psychology. In *Introductory Psychology*. Washington State University. https://opentext.wsu.edu/psych105/chapter/7-2-what-is-cognition/
- Eckardt, B. V. (2012). The representational theory of mind. In K. Frankish & W. Ramsey (Eds.), *The Cambridge Handbook of Cognitive Science* (1st ed.). Cambridge University Press. https://doi.org/10.1017/CBO9781139033916.004
- Fletcher, R. B., & Hattie, J. (2011). The brief and sometimes dirty history of intelligence and IQ tests. In *Intelligence and Intelligence Testing* (pp. 13–29).
- Fletcher, R. B., & Hattie, J. (2011). What is Intelligence. In *Intelligence and Intelligence Testing* (pp. 1–11). Routledge.

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- Gigerenzer, G., & Goldstein, D. G. (1996). Mind as a computer: Birth of a metaphor. *Creativity Research Journal*, *9*(2 & 3), 131–144.
- Gottfredson, L. S. (1997). Mainstream science on intelligence: An editorial with 52 signatories, history, and bibliography. *Intelligence*, *24*(1), 13–23. https://doi.org/10.1016/S0160-2896(97)90011-8
- Herrnstein, R. J., & Murray, C. (1994). *The Bell Curve: Intelligence and Class Structure in American Life.* Free Press.
- https://medium.com/@colemanabaker/between-stimulus-and-response-there-is-a-space-ad5261e3c74e
- Kellogg, R. T. (2003). *Cognitive psychology*. SAGE. https://books.google.dk/books?id=_5TiryUv4cwC&printsec=frontcover&hl=de&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
- Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review*, *63*, 81–97.
- Neath, I., & Surprenant, A. M. (2005). Mechanisms of memory. In K. Lamberts & R. L. Goldstone (Eds.), *Handbook of Cognition* (pp. 203–2020). SAGE. https://doi.org/10.4135/9781848608177.n9
- Neisser, U. (2014). *Cognitive Psychology: Classic Edition*. Psychology Press. https://doi.org/10.4324/9781315736174
- Nisbett, R. E., Aronson, J., Blair, C., Dickens, W., Flynn, J., Halpern, D. F., & Turkheimer, E. (2012). Intelligence: New findings and theoretical developments. *American Psychologist*, *67*(2), 130–159. https://doi.org/10.1037/a0026699
- Pang, D. K. F. (2023). Ancient concepts of the mind, brain (and soul). *Psychology Today*. https://www.psychologytoday.com/intl/blog/consciousness-and-beyond/202306/ancient-concepts-of-the-mind-brain-and-soul

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- Quattrociocchi, W., Capraro, V., & Perc, M. (2025). Epistemological fault lines between human and artificial intelligence (arXiv:2512.19466). arXiv. https://doi.org/10.48550/arXiv.2512.19466
- Russo Krauss, C. (2022). From introspection to experiment: Wundt and Avenarius' debate on the definition of psychology. In D. Gozli & J. Valsiner (Eds.), *Experimental Psychology* (pp. 7–26). Springer International Publishing. https://doi.org/10.1007/978-3-031-17053-9_2
- Shannon, C. E. (1948). A mathematical theory of communication. *The Bell System Technical Journal, 27*, 379–423.
- Spearman, C. (1904). *'general intelligence,' objectively determined and measured*.
- Warne, R. T., & Burningham, C. (2019). Spearman's g found in 31 non-Western nations: Strong evidence that g is a universal phenomenon. *Psychological Bulletin*, *145*(3), 237–272. https://doi.org/10.1037/bul0000184

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AI 507: Artificial Intelligence and Society

#2 Learning

Associate Prof. Dr. Lena Frischlich

Flashlight

Cognition

History – fathers of cognitive science

- Renée Descartes
- Wilhelm Wundt
- Alan Turing
- John von Neumann
- George Miller

Functional areas in the brain

- Broca and Wernicke Areal
- Functional areas

Behaviourism: Stimulus → Response

- Classical conditioning (Pawlow)
- Reinforcement learning (Skinner) (operant conditioning)

Stimulus →Organism → Response

Core concepts

Sensory input

Mental representation

Cognitive architecture

Minds as computers

- Symbolic models (mind as computer)
- Connectionist models (mind as deep network)

Deep encoding vs. superficial processing

Intelligence

- General intelligence “g”
- IQ-tests
- Justifications of discrimination

Thinking

- System I: Fast thinking
- System II: elaboration

→ No elaboration without learning

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Any questions so far?

Then let’s get started

At the end of today’s lecture, you will…

- …have an initial insight into human’s memory works
- …have a broad understanding of how humans learn
- …have discussed how using LLMs can support but also impair your learning
- …have discussed how we perceived the world (and how that relates to memory and learning)

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How do we remember?

Memory (Miller, 1956; Neath & Suprenant, 2005)

- Three stages: Encoding, storage, retrieval
- **Encoding** = sensory processing (→ input)
- **Storage** =
  - **Short- term memory** (magic number “7” +/-2, today 4-5)
  - **Long-term memory**: *Explicit* (what happened when), *Implicit* (how)
- **Retrieval** (recognition, recall, reconstruction)
- Centrally: Encoding is directly linked to retrieval!
- Cues trigger retrieval
  - Strong cues → elicit a specific reaction *most* of the time
  - Weak cues → elicit a specific reaction *some* of time of among *some* people

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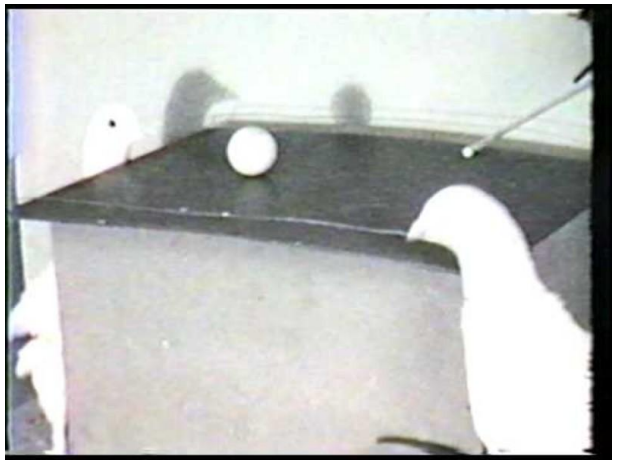
Remember behaviourism?

Classical conditioning

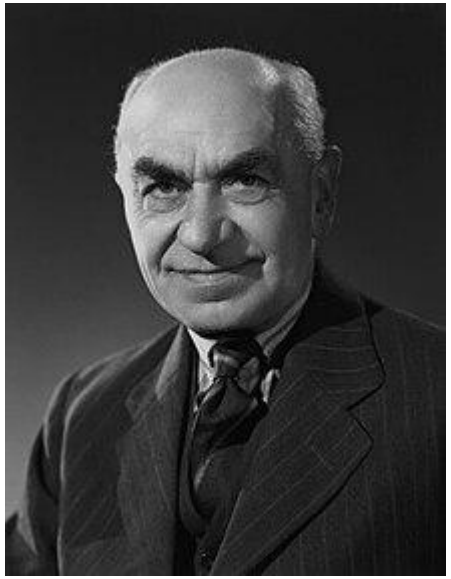
Operational conditioning



- Initial learning works fastest with strong cues (constant reinforcement)
- However: learning is most stable when there are sometimes sanctions – the memory of what could happen does the job

The role of schemes (Bartlett, 1932)

- We store complex information in categorical rules or scripts
- Bartletts' *Scheme theory*: An active organisation of past reactions and experiences – regular behaviour is possible because we have serially organised past experiences as a unitary mass (Bartlett, 1932, p.201)
- For example, we have schema of "going to university", "biking", "driving a car" etc.
- Schemes combine knowledge about "what" and "how"
- We can recall and retell stories, change the details but keep the basic meaning – and both we and our audience will recall them



February 1948 photograph of Sir Frederic Bartlett. National Portrait Gallery via Wikipedia

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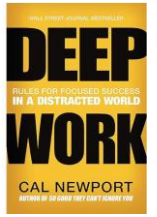
Practical consequences: Exam preparation

- Our working memory is quite limited – if we do too much things at the same time, no capacity is left
- Our brain does constantly process the environment in which we are in (even if not always attentively)
- These cues can make it easier to retrieve relevant memories (preparing not only the content but also the format of an exam)
- They can also make it harder (music with lyrics while reading, annoying room mates)
- Deep work and elaboration beats constant shallow work with distraction
- Reward yourself – but not all the time
- University as “test field” to develop schemata for teamwork, presentations etc.

- Reduce distractions (that includes your phone)
- Plan deep work phases – not full days (e.g., Pomodoro-technique, 25 minutes then mandatory break)
- Establish regular deep work blocks
- Create an environment that supports your learning



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What are your questions so far?

More generally: How do humans learn?

“We are not individually much cleverer than the average animal, a heron or a mole, but the knack of our species lies in our capacity to transmit our accumulated knowledge down to generations.
The slowest among us can, in a few hours, pick up ideas that it took a few rare geniuses a lifetime to acquire.”

(de Botton, 2019, p.1)

How do we learn? (Mikolov et al., 2013; Young & Wassermann, 2005)

1. **Temporal contiguity (closeness in time)**
    - When two brain processes are activated together or in immediate succession, the activation of one spreads to the other (*Priming*)
    - That's what we've seen in classic conditioning
    - Word embeddings such as `Word2Vec` harvest the co-occurrence of events, in this case, words
    - But attention:
        - Reinforcement learning (i.e., the consequence of a behaviour or stimulus) seems to be more powerful → Just watching soccer doesn't make you a successful player
        - The interval between the stimuli matters – and variety keeps your brain from zooming out (#Boring)

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How do we learn? (Young & Wassermann, 2005)

2. Competition among predictors

- Our world is complex – in each moment, myriads of stimuli could be relevant
- To save energy, our brain thus sets them in direct competition to each other – only the best ones survive
- If there is already a relevant stimulus, new ones are blocked
- Only stimuli that provide unique new information are learned

=> our brains version of a step-wise regression

How do we learn? (Young & Wassermann, 2005)

3. Configural learning

- Our brain loves efficiency – so we often learn things as a "bundle" (→ Schema theory)
- Consider learning how someone looks like – for example Zendaya – you don't learn her hair and eye colour separately
- That allows you to recognize her even with her "dune" eyes
- Generally, more complicated
  - Especially if there are a lot of irrelevant cues (→ working memory capacity)
  - Or if the cues are hard to distinguish (pink or red socks/ socks versus shoes)

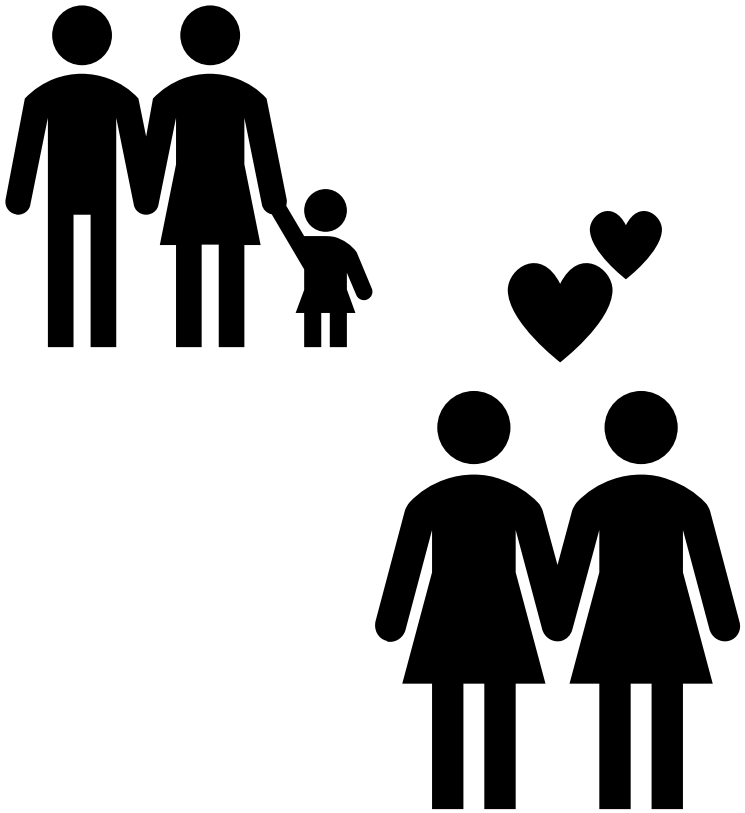
https://variety.com/t/zendaya/

https://movieweb.com/dune-2-zendaya-lead/

How do we learn? (Shepart, 1987; Young & Wassermann, 2005)

4. Generalization and similarity

- Humans are extremely good in projecting our knowledge to new situation
- The higher the similarity between situations, the better
- Similarity = distance between representations (! Not situations but how we process and interpret them!)
- The lower the similarity between situations or stimuli, the better do we learn to distinguish between them
  - Shift your attention to different dimension
  - Shift your attention to different values on the dimension
- Captchas are used to train increasingly blurred images to train models in these difference



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Remember Schemata?

Learning during human development = improvement of schemata

- New information is added or assimilated to our current schemata
- If the information does not fit → unpleasant state might lead to schema change

Schemata accelerate your learning!

- Activate your knowledge and schemata to connect them with new information
- Use analogies & comparisons to harvest your schemata – and think about differences to make them more precise

Jean_Piaget_in_Ann_Arbor_(cropped).png Unidentified (Ensian published by University of Michigan) via Wikipedia

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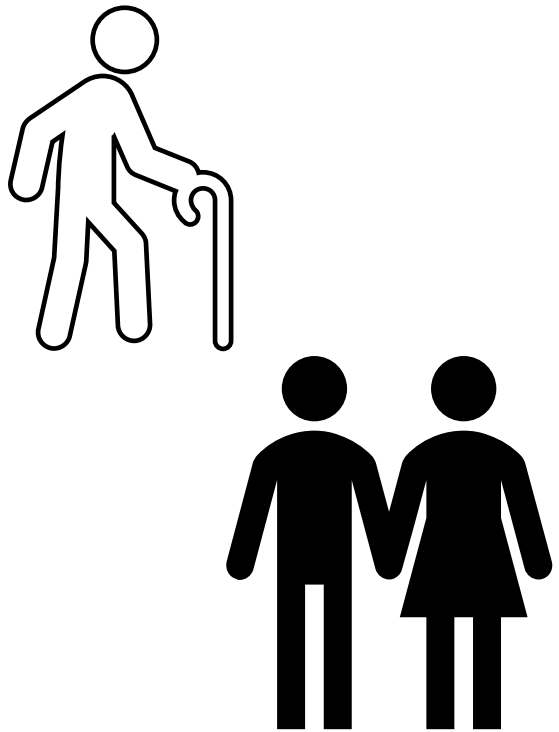
How do we learn? (Erdelyi, 2008; Young & Wassermann, 2005)

5. Unlearning

- If a stimulus is paired with a response and the response never comes, the pairing goes extinct over time
- However: If the stimulus + response is repeated, the pairing is there immediately and even stronger
- So, do we ever forget?
  - Information that is not encoded in short-term memory might indeed not be available any more → Central role of interference and working memory capacity
  - Information that has been encoded in the long-term memory is often rather not accessible but can be reconstructed (attention: error prone!)



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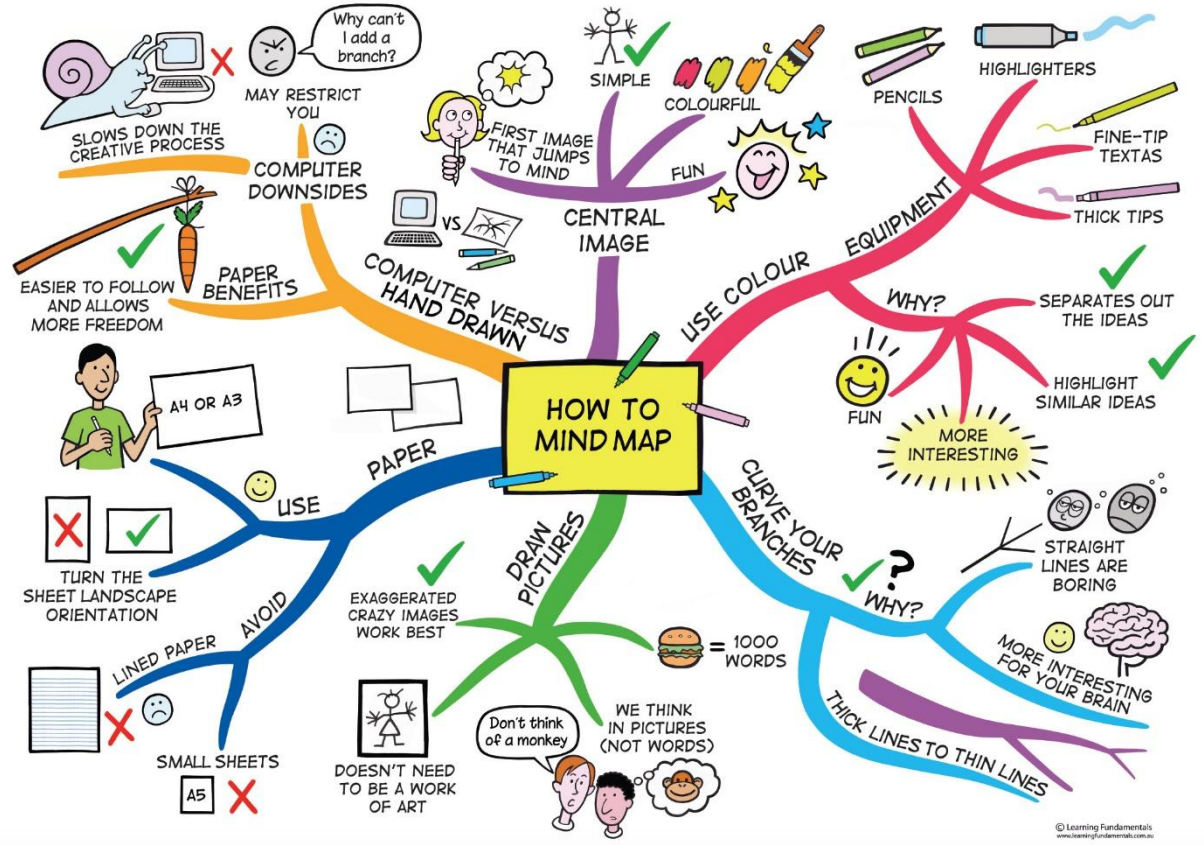
Practical consequences: Exam preparation

- Learning things together and relating them to each other
- But do not bore yourself – repetition alone is not attractive for the brain (try another modality to keep it interesting)
- Organise your knowlegde – create patterns and schemes
- If you encode it carefully – you have good chances of finding it again, when you need it
- Mindmaps are really great

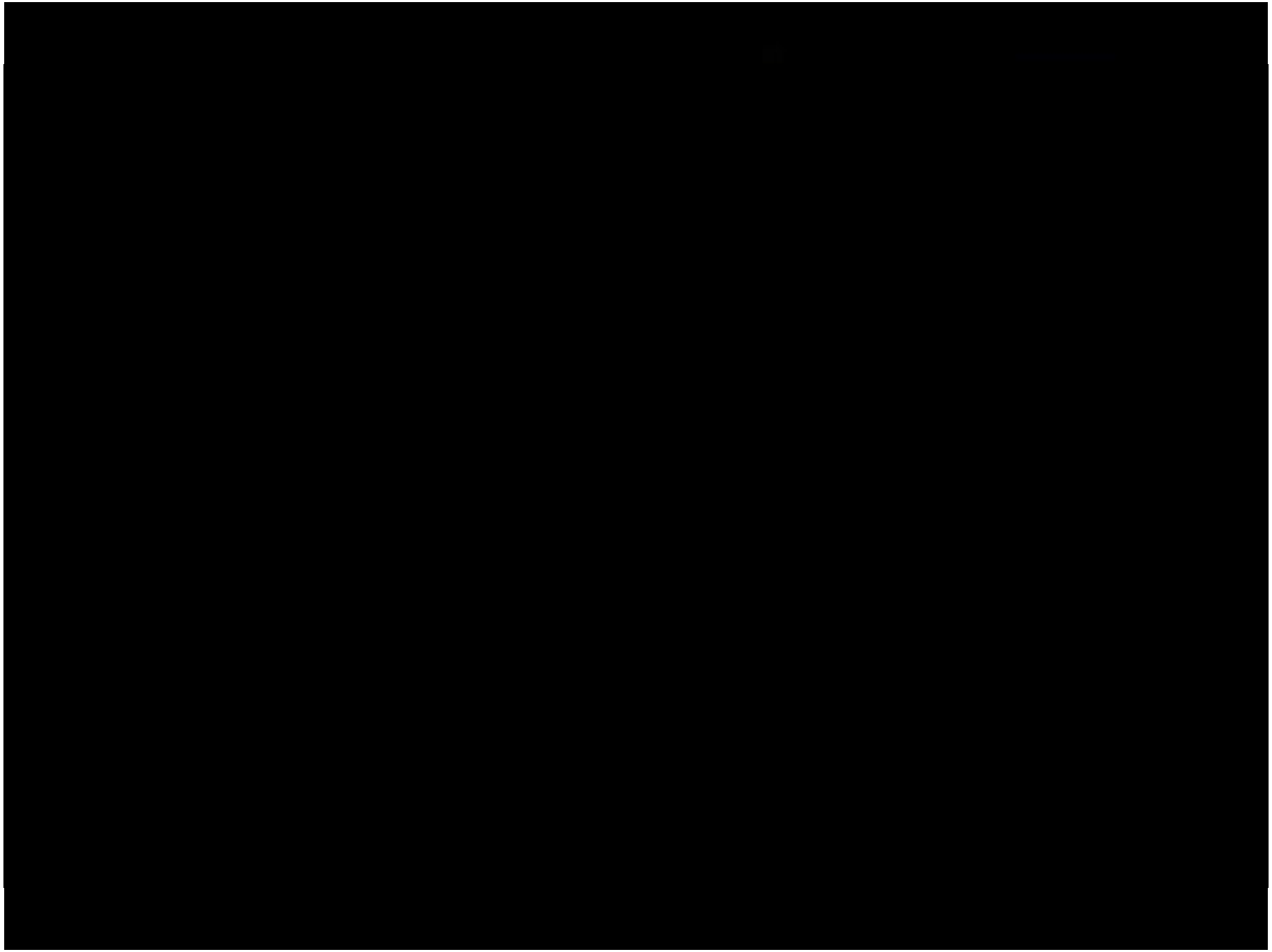


https://learningfundamentals.com.au/resources/

What are your questions so far?

Time for a break

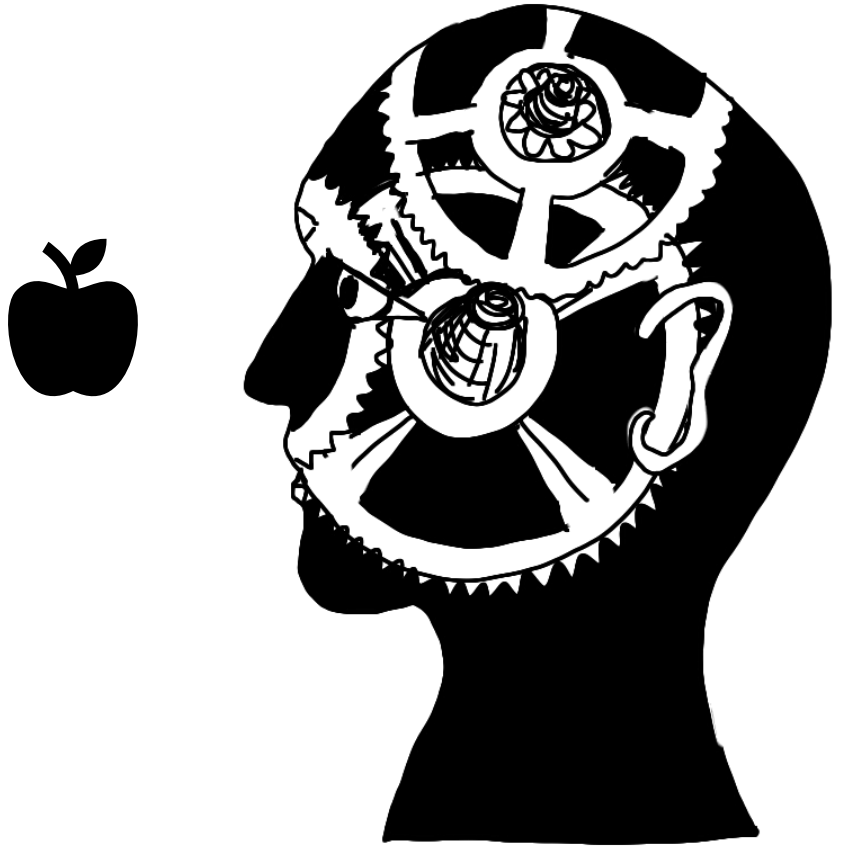
All learning starts with perception and awareness – so let’s test yours!



https://www.youtube.com/watch?v=z-Dg-06nrnc&t=2s

How do we see ? (Barry, 2020)

- Our eyes are a direct extension of our brain to the environment
- The send more data more quickly to our nervous system than any other sense
- But: We can also sense the environment without the eyes – and some extraordinary people have learned to use other senses to replace theirs
- But for today: Let’s see how visual perception works in average-sighted people works



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Vision is complex

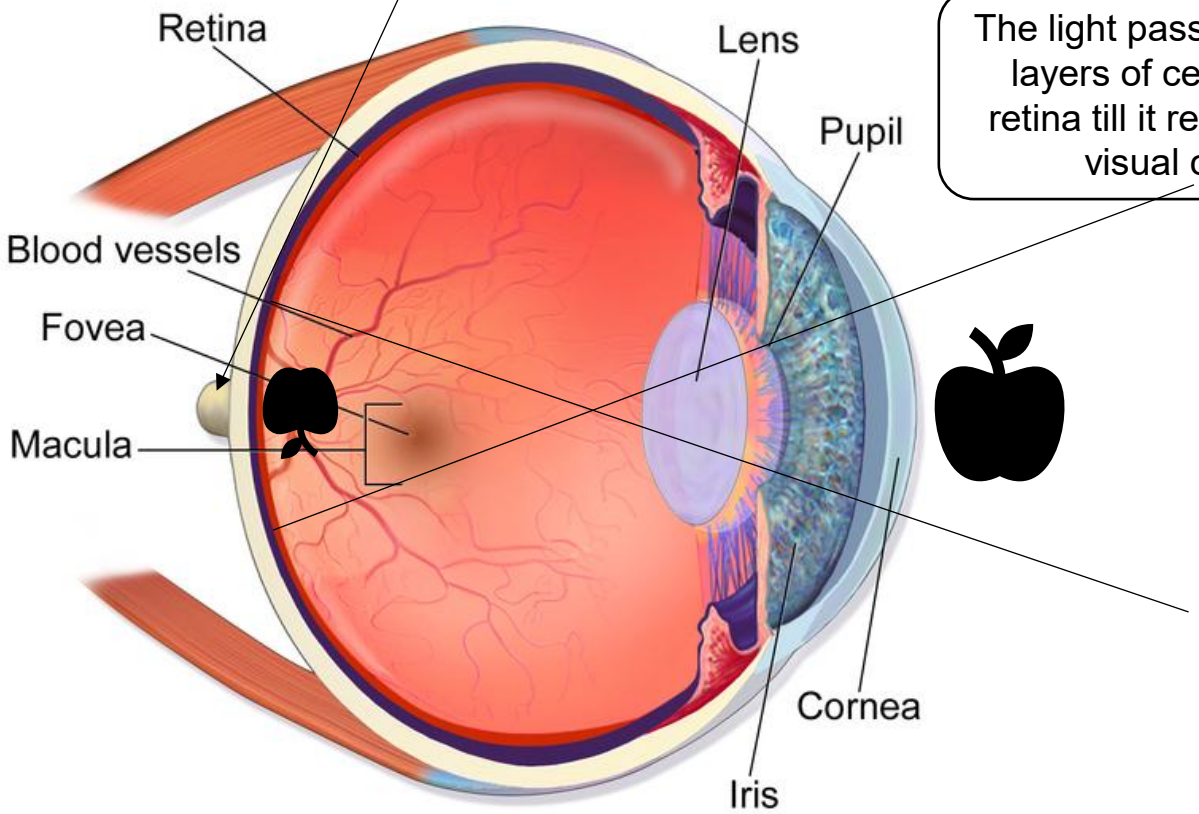
(Pinel, 2011; Nixon & Aguado, 2019)

Start of the optic nerve – “blind spot” : The brain “fills in” what’s missing! Perception is always also construction

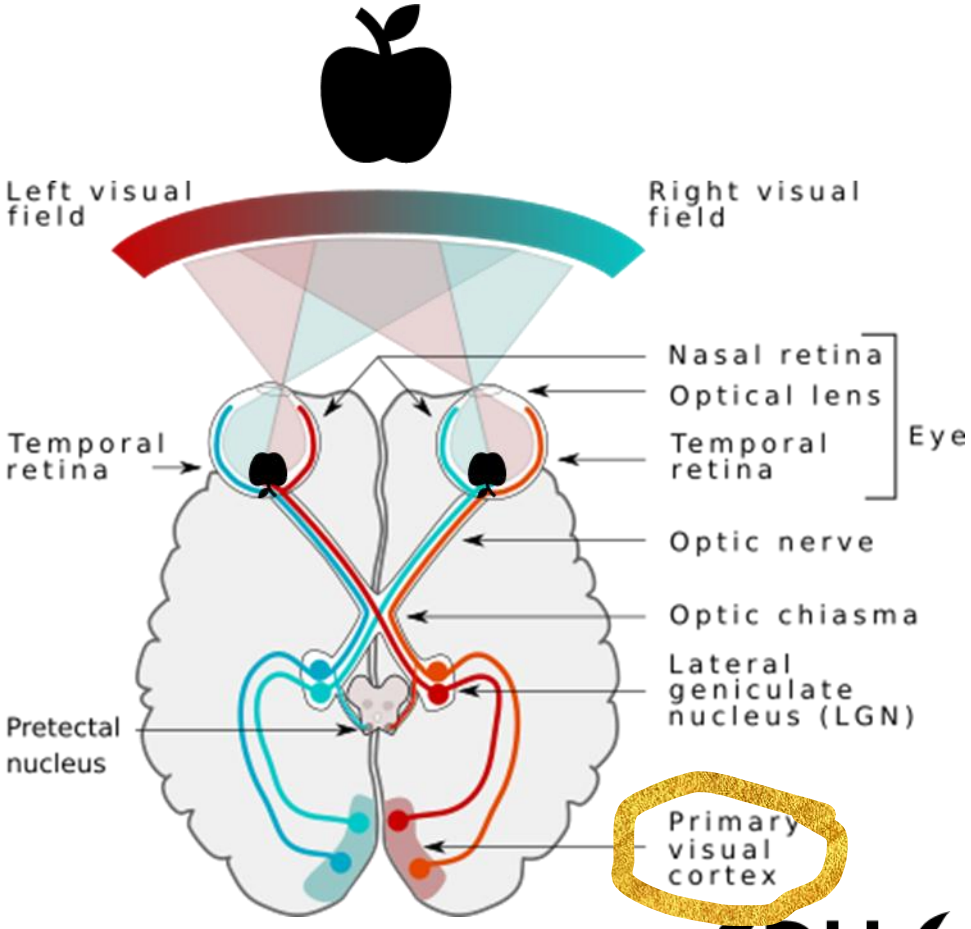
Light enters through the pupil

The pupil focuses the light onto the retina

The light passes several layers of cells in the retina till it reaches the visual cells



Eye Anatomy Blausen, 2014



https://commons.wikimedia.org/wiki/user:Perellonieto

Viewing is an interplay of functional systems

(Nixon, 2019; Pinel, 2011)

- Different neurons in the brain fulfil different functions
- Some are specialised in detecting movement
- Others focus on detecting colours, fine-grained structures and slow or fixed objects
- Our neurons are also specifically adapted to detect borders – they have developed complex mechanisms to enable us to navigate our environments!
- In the visual cortex, the input from different areas of the retina is processed, organised and aggregated
- The whole is more than the sum of its parts!
- Our brain can even implement missing parts – although the implementation might not always be correct – attention, memory, schemes drive what we see

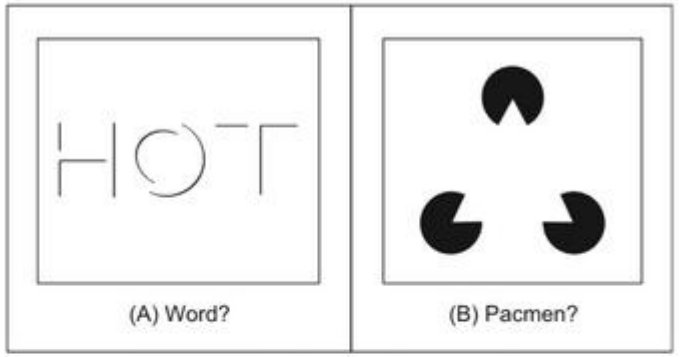


FIGURE 1.6 How human vision uses edges.

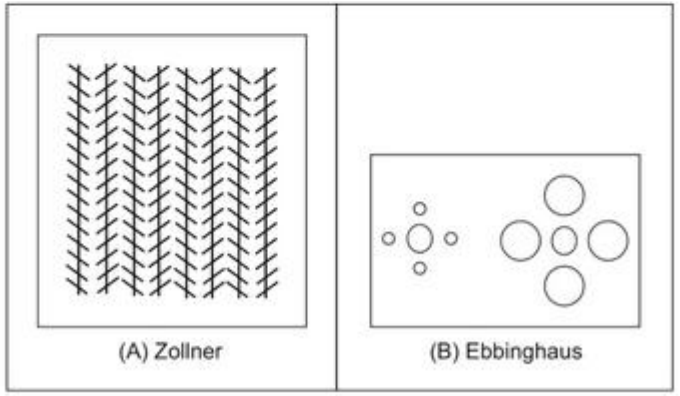
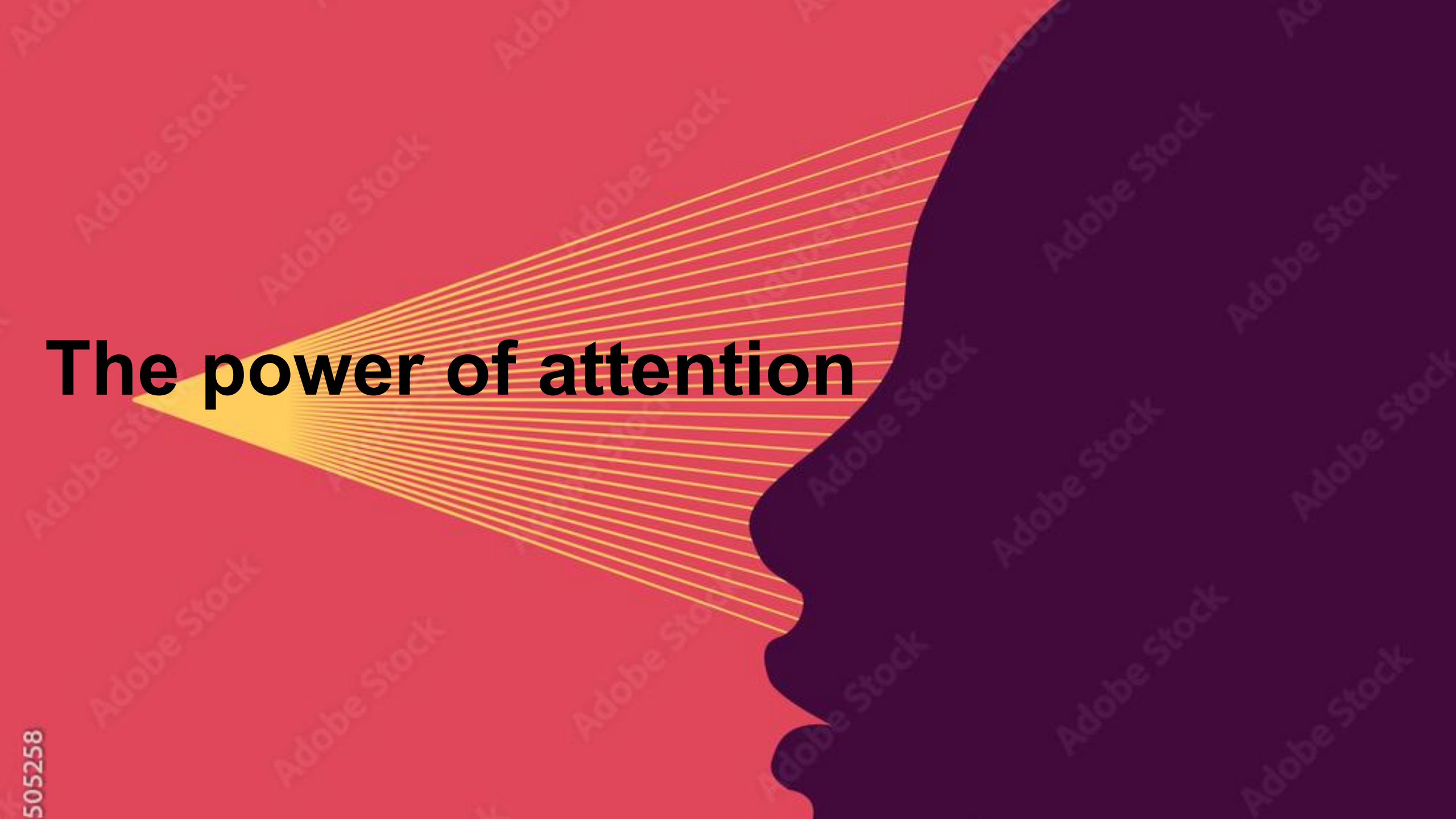


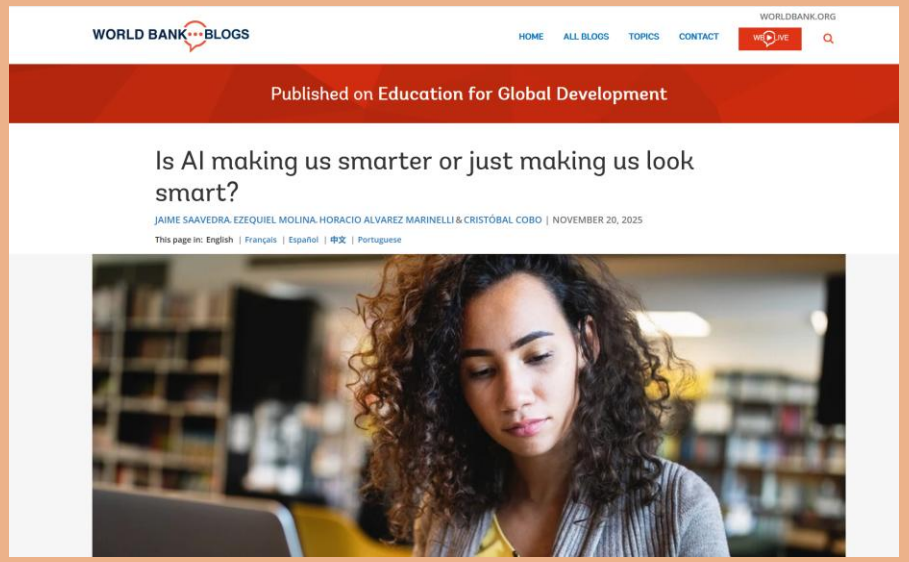
FIGURE 1.7 Static illusions.


The power of attention

How does AI affect perception, memory and learning?

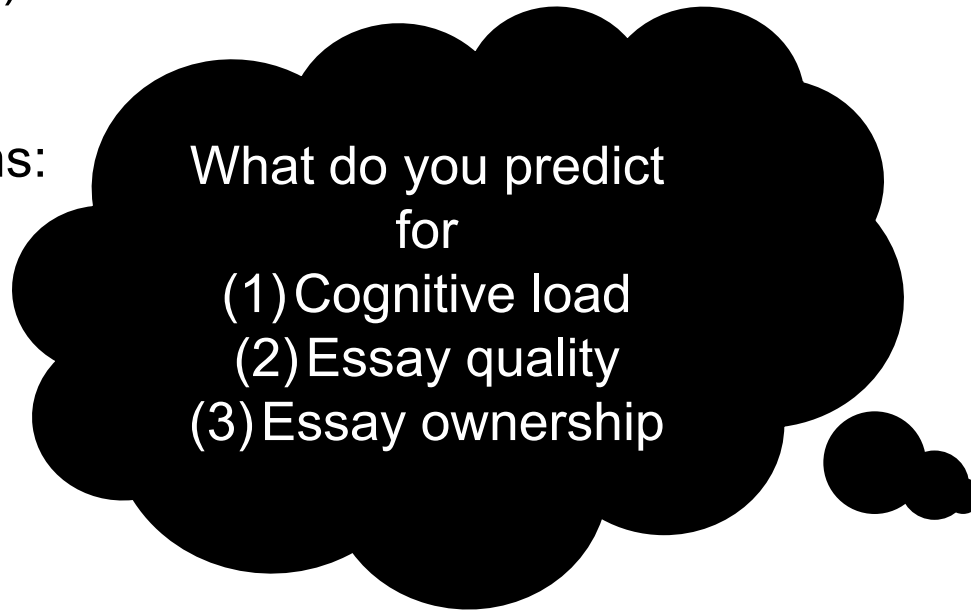
6.5 Billionen USD

Estimated market size for AI in education



Does AI make us smarter? (Pearson, 2025; Kosmyna et al., 2025)

- Studies show that people who use technology for specific tasks (e.g., driving assistants) memorize these tasks worse (e.g., the way) – same goes for phone numbers
- Experiment with 54 participants that took part in three sessions: Brain-only, LLM, search engine
- Task was to write essays
- Measured cognitive load (via EEG), essay quality and how much participants felt that they "own" the essay



- Using LLMs reduced cognitive load – but also ownership and essay quality – especially over time more copy-and-paste and less memory of what one has actually written
- Attention: Small sample, normative implications unclear (do we *need* to know what we write) – but demonstrates that having LLMs learn for you does not work

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The problem did not start with AI (Ward, 2021)

- Series of eight experiments (*n* = 1,917) tested peoples' ability to distinguish between internal knowledge—encoded and retrieved from their own memories—and external knowledge found on the internet
- Shows that looking facts up does allow you to answer questions more correctly – that leads to the erroneous assumption that you also *know* it – but short-term memory is not long-term memory (btw- that's why we do unaided exams)
- Critically: People that google information also perceive themselves as being smart and having a better memory than most other people
- Spoiler: that is not the case

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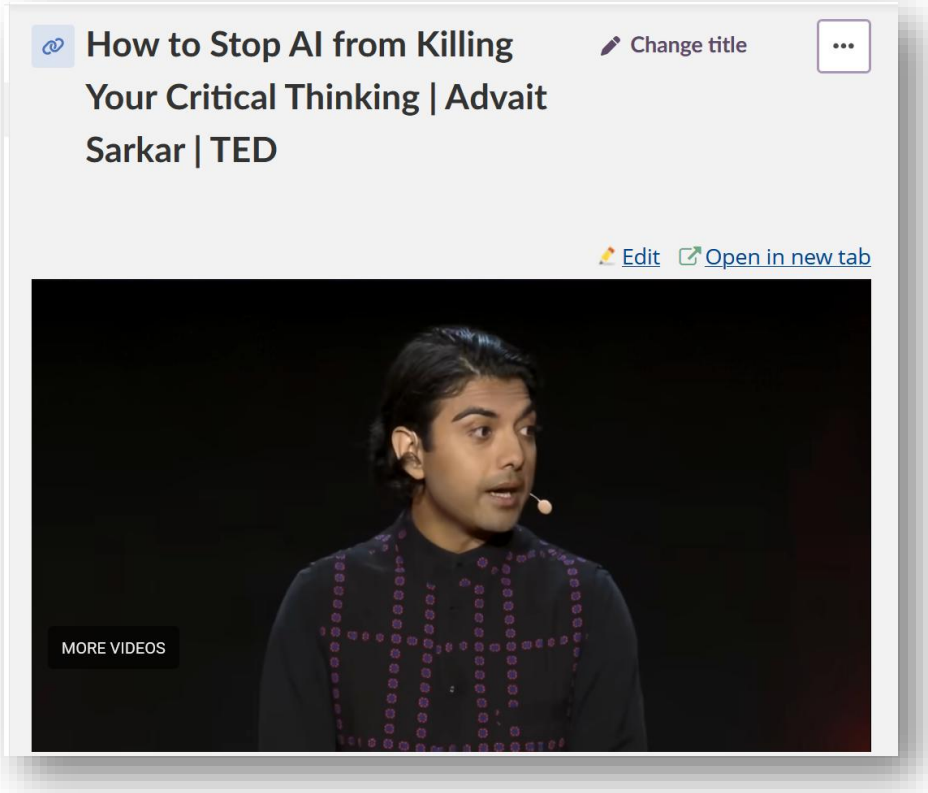
But: When does it matter? (Mo et al., 2026; Pearson, 2025)

- We all engage in “cognitive offloading” – we share things with others to not have to memorize all off them ourselves (#Christmas presents)
- Plus: Meta analyses indicate that AI-enhanced learning can have a positive effect on learning outcomes (however, several of the included studies are very small - should be interpreted with care)
- So the question is maybe: What kind of smart do we need – and when can we “offload” tasks to AI?
- **What could be criteria for “brain-optimized” AI use?**



What are your questions so far?

Then I hope, you know…

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- …have discussed how we perceived the world (and how that relates to memory and learning)

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#2 Learning

Associate Prof. Dr. Lena Frischlich

References

- Bartlett, S. F. C. (1932, 1995). Remembering: A Study in Experimental and Social Psychology. Cambridge University Press.
- Barry, A. M. (2020). Perception theory: A neurological perspective on visual communication. In S. Josephson, J. D. Kelly, & S. Smith (Eds.), Handbook of visual communication: Theory, Methods, and Media (2nd ed., pp. 3–27). Routledge.
- Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). DOI:10.15347/wjm/2014.010. ISSN 2002-4436 De Botton, A. (2019). The school of life. Penguin.
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding (No. arXiv:1810.04805). arXiv. https://doi.org/10.48550/arXiv.1810.04805
- Erdelyi, M. H. (2008). Forgetting and remembering in psychology: Commentary on Paul Connerton's `Seven Types of Forgetting' (2008). Memory Studies, 1(3), 273–278. https://doi.org/10.1177/1750698008093792
- Kosmyna, N., Hauptmann, E., Yuan, Y. T., Situ, J., Liao, X.-H., Beresnitzky, A. V., Braunstein, I., & Maes, P. (2025). Your brain on ChatGPT: Accumulation of cognitive debt when using an AI assistant for essay writing task (arXiv:2506.08872). arXiv. https://doi.org/10.48550/arXiv.2506.08872
- Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space (No. arXiv:1301.3781). arXiv. https://doi.org/10.48550/arXiv.1301.3781
- Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. Psychological Review, 63, 81–97.
- Mo, F., Huang, J., Yang, Y., Özen, Z., Maeda, Y., & Olenchak, F. R. (2026). Undergraduate students' learning outcomes with ChatGPT: A meta-analytic study. Computers and Education: Artificial Intelligence, 10, 100536. https://doi.org/10.1016/j.caeai.2025.100536

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References

- Neath, I., & Surprenant, A. M. (2005). Mechanisms of memory. In K. Lamberts & R. L. Goldstone (Eds.), Handbook of Cognition (pp. 203–2020). SAGE. https://doi.org/10.4135/9781848608177.n9
- Pearson, H. (2025). Are the Internet and AI affecting our memory? What the science says. Nature, 638(8049), 26–28. https://doi.org/10.1038/d41586-025-00292-z
- Pankin, J. (2013, Fall). Schema Theory. MIT. https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://web.mit.edu/pankin/www/Schema_Theory_and_Concept_Formation.pdf&ved=2ahUKEwjb8Iy88aSLAxXa9bsIHeOrJswQFnoECAgQAQ&usg=AOvVaw3QZvbV8wddA2p8k_couku3
- Pennington, J., Socher, R., & Manning, C. (2014). Glove: Global vectors for word representation. Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), 1532–1543. https://doi.org/10.3115/v1/D14-1162
- Pinel, J. (2011). Biopsychology. Allyn & Bacon
- Shepard, R. N. (1987). Toward a universal law of generalization for psychological science. Science, 237(4820), 1317–1323. https://doi.org/10.1126/science.3629243
- Ward, A. F. (2021). People mistake the internet's knowledge for their own. Proceedings of the National Academy of Sciences, 118(43), e2105061118. https://doi.org/10.1073/pnas.2105061118
- Young, M. E., & Wassermann, E. A. (2005). Theories of learning. In K. Lamberts & R. L. Goldstone (Eds.), Handbook of Cognition (pp. 162–184). SAGE. https://doi.org/10.4135/9781848608177.n6

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AI 507: Artificial Intelligence and Society

#3 Social cognition

Associate Prof. Dr. Lena Frischlich

Flashlight

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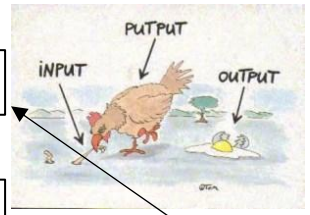
Flashlight

Memory



- Encoding
- Storage
  - Short-term → 4-7 units
  - Long-term
- Retrieval
  - Cues ease retrieval

AI & Learning

- Negative effects
- Positive effects
- Context matters!
- Dunning-Kruger effect

Schemes

Learning

- Temporal contiguity →
- Predictor selection
- Configurations
- Generalisation
- Un-learning

Classic behaviourism

Perception

- Attentional blindness
- Perception = construction

Practical consequences

Any questions to last week?

Then, let’s get started!

After today’s lecture, I hope you will…

- … have insights into social cognition
- …understand how social groups shape how we think
- … have a better understanding of human biases
- …have discussed how that shapes AI

Social Cognition

How often are you thinking (Fiske & Taylor, 2017)

- … What others think about you?
- …What you should have done or said the other day?
- …Why others said or did something?
- …Why someone did *not* do something?
- … If you should invite that new guy to join you and your friends at the Friday bar?

Examples of Social Cognition

Streetart Munich, Germany

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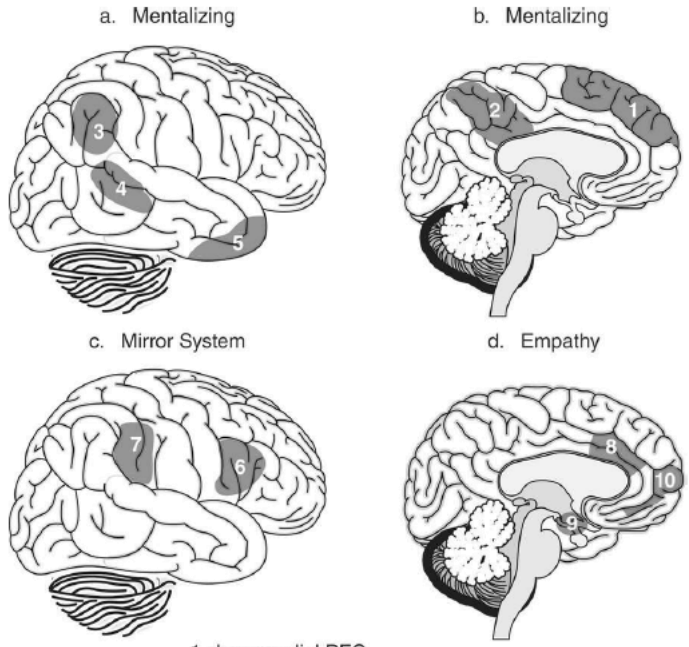
Differences in social vs object perception

(Adapted from Fiske & Taylor, 2017)

	People	Animals	Objects
Intentions	People are intentional causal agents	Animals are sometimes intentional	Rather not
Perceptions	Yes, they perceive you back	Some of them	No, even if shout at your screen
Similarity	(dis)similar to you	Well. Maybe your dog	No – even if you might think so
Self-conscious	Theoretically	Naa (we ignore counterevidence)	Only in Science Fiction
Unobservable traits	Full of them	Less so	rarely
Variability	Changeable	Changeable	Not on their own
Complex	You never really know them	If its not a cat…	You can know them
Requiring explanation	Pretty much all the time	Naa, after a while you get them	Might rather require instructions

A substantial part of our brain is concerned with understanding people (Fiske & Taylor, 2017)

- In fact, social cognition might be the “default” modus of our brain
- Studies in neuroscience show that “social” neurological patterns often differ only little from baseline activation – whereas object perception deactivates these parts of the brain
- => We think about people <u>all the time</u>
- Crucially: Perception is constructed and shaped by our environment



What are your questions so far?

All social cognition starts with self-observation: Me, myself, and I

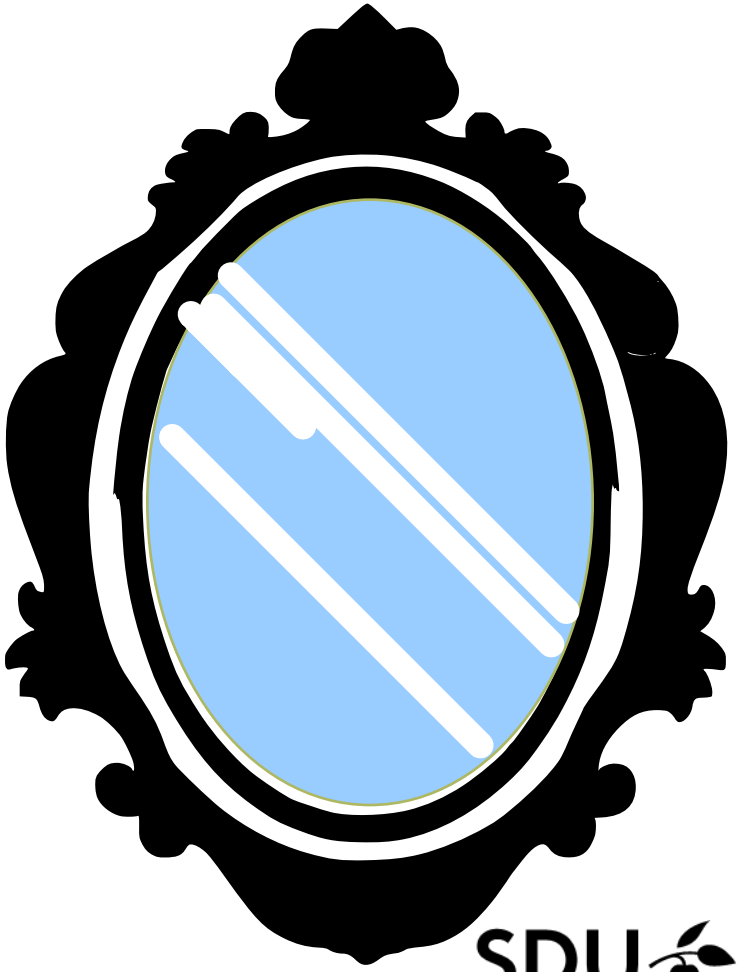
Who are you? (James, 1918; Leary, 1990)

- Most of us have a clear and nuanced answer to this question
- Our self-concept is the result of our experiences during childhood, our social roles, our culture…
- We also have a clear idea, if we are rather shy or outgoing, curious or prefer things to change as little as possible
- We do understand that every thought we have is “ours” and not “theirs” or “yours” – the “mine” belongs to an “I”, which founding farther of social cognition research William James noted already in 1918
- However, our self-concept is complex

William James.
MS Am 1092 (1185), Houghton Library, Harvard University

Most of our self-encoding is situated (Fiske & Taylor, 2017)

- We have diverse senses of ourselves in particular contexts
- Each situational norm engages different aspects of the self → steers the encoding and retrieval of our "self-knowledge"
- We have a "**working self-concept**" – that part of our self that is accessed and used in a given situation (e.g., university versus private meetings)
- We also have a "**relational self-concept**" – emerging from our emotional relationships with significant others such as parents, partners, close friends etc. (#ChristmasAtHome…)
- We derive our personal self-concept from comparison to others
- Provides both stability (through enduring representations of significant others) and variability (though situational activation)



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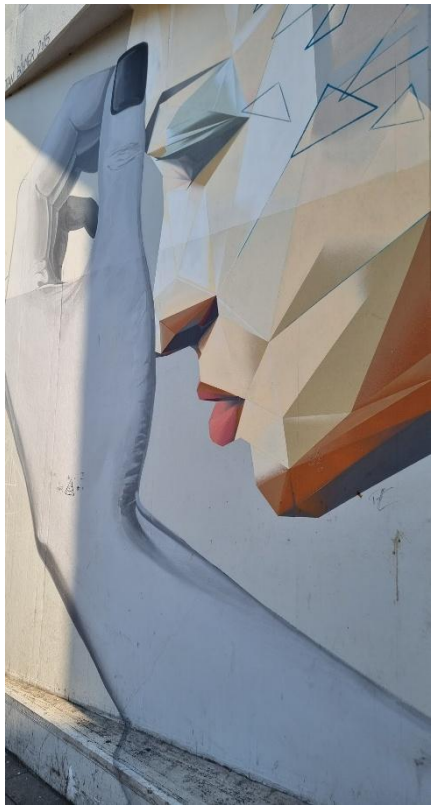
We organise information about ourselves in our self-schema (Fiske & Taylor, 2017)

- We have different selves:
  - *Possible* self (what we could become and hope for)
  - *Feared* self (what we could become and don't want too)
- One central aspect of our self-scheme: **self-esteem**, that is, which qualities we possess and how we value them
- A healthy self-esteem is *central*– it's linked to well-being, the ability to set appropriate goals, and to cope with challenges
- Central for *self-regulation*



Streetart Tian Böhmer, 2015, Munich

Self-regulation is not always easy (Fiske & Taylor, 2017)

https://www.youtube.com/watch?v=4L-n8Z7G0ic

Harder when:

- Self-esteem and social threats (e.g., being ostracized)
- Multitasking
- Self-control depletion

→Limited working memory capacity!

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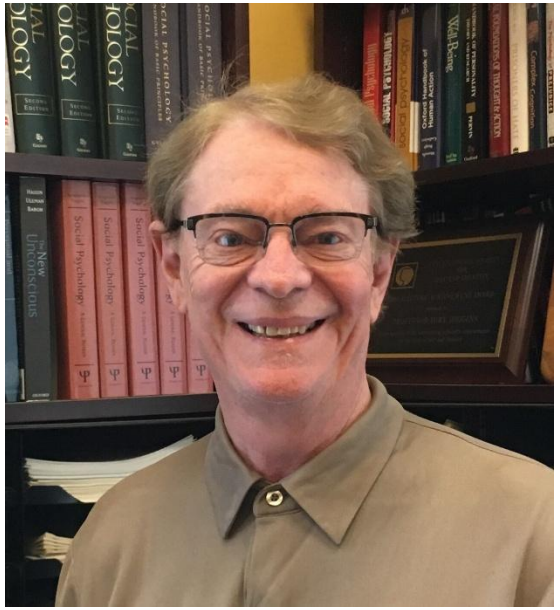
How does self-regulation work? (Higgins & Pinelli, 2020)

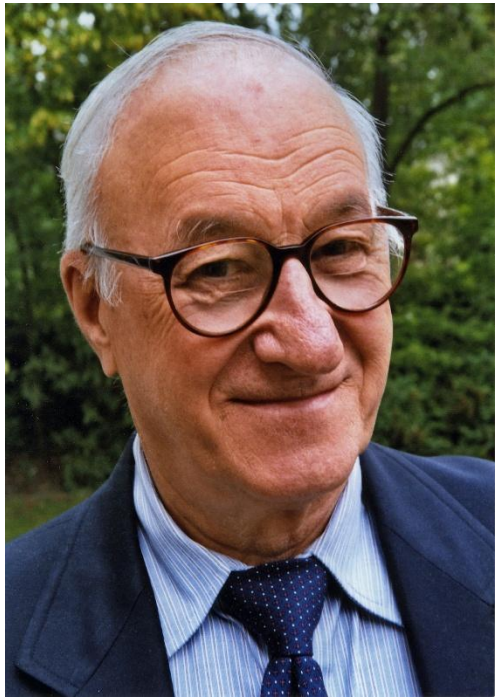
- The self-discrepancy theory
- We all have different self-concepts
  - **Actual** or current self (that's where the behaviour matters)
  - **Ideal** self, who we want to be
  - **Ought** self, what we think, who we should be
- Discrepancies between the actual and the ideal self are *activating*
  → bad for the self-esteem, can lead to depression, but also motivating to become better (so-called **promotion focus**)
- Discrepancies between the current and the ought self are *inhibiting*
  → We get socially anxious and strive to avoid failures (**prevention focus**)
- The greater the fit between our environment and our focus, the better we feel!



Tory Higgins: https://commons.wikimedia.org/wiki/user:Emilynakka

Intentions are often not enough (Bandura & Adams, 1997; Bandura et al., 1980; Finkel, 1985)

- We need to perceive personal control about the behaviour and believe that the behaviour is effective (**Self-efficacy**)
- If we think that we can plan our actions, cope with set-backs, and pursue our self-regulatory activities, we have a sense of “mastery”
- If we think that we have little control/ experience high levels of uncertainty: very unpleasant, threatening
- Efficacy perceptions also affect political decisions – a lack of political efficacy predicts lower voter intentions
- In uncertain and hopeless times: taking (political) action can help to restore efficacy perceptions



Albert Bandura
bandura@stanford.edu - Albert Bandura, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=35957534

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Why do we regulate at all? (Fiske & Taylor, 2017)

We have central needs related to our self-concept

Need for accuracy: To make our future outcomes predictable, we need a fairly accurate assessment of our abilities, opinions, beliefs, and emotions

Need for consistency: To trust our accuracy judgements, we need to have a pretty consistent/ stable self-concept → we often seek situations that confirm how we see ourselves (e.g., procrastination to confirm that we are unable to solve a task)

Self-enhancement: We need to feel good about ourselves and maintain self-esteem → Source of a lot of **biases** (e.g., we remember our good deeds better than our faults, perceive us as less biased, more competent and happier than others)

What do you think: Can GenAI have a self-concept?

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There is no such thing as conscious artificial intelligence

Andrzej Porębski & Jakub Figura

Humanities and Social Sciences Communications **12**, Article number: 1647 (2025) | Cite this article

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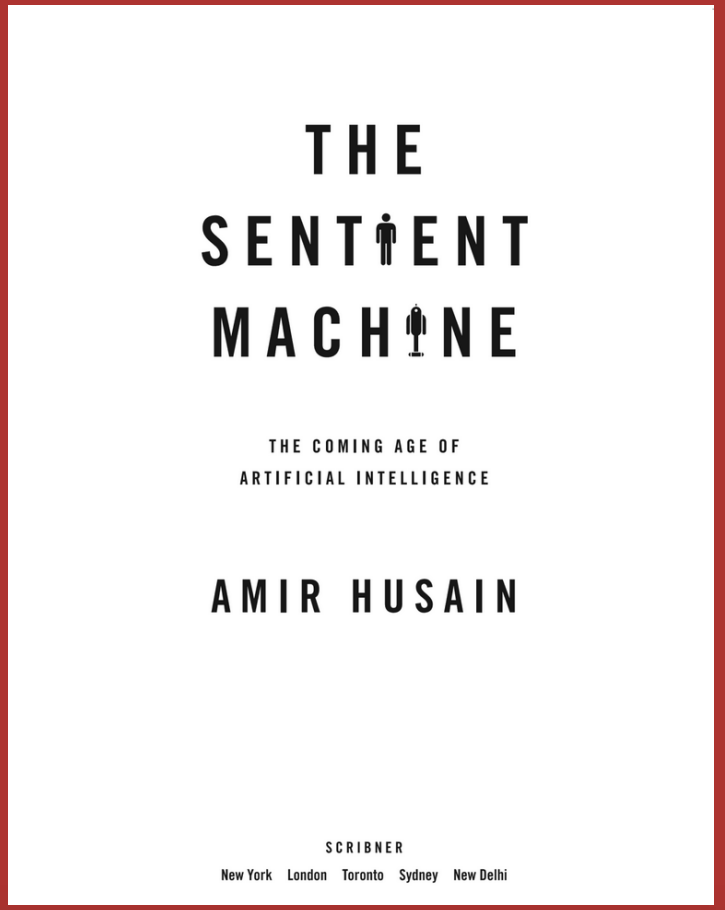

THE SENTIENT MACHINE

THE COMING AGE OF ARTIFICIAL INTELLIGENCE

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Article

AI and the Cognitive Sense of Self

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What could “self-concept” for AI mean?

(Hutson & Barnes, 2025; Porębski, & Figura, ,2025)

	People	AI
Continuity	Stable relational self-concept	
Working self-concept	Integration of situational cues and roles	
Possible selves	Possible vs. feared self	
Self-regulation	Actual/ ideal discrepancy Actual/ ought discrepancy	
Self-control	Efficacy perceptions	
Need driven	Self-esteem, accuracy, continuity	

What could “self-concept” for AI mean?

(Hutson & Barnes, 2025; Porębski, & Figura, ,2025)

	People	AI
Self-Recognition	Stable relational self-concept	Learning from past interactions
Working self-concept	Integration of situational cues and roles	Responding to momentary interactions
Possible selves	Possible vs. feared self	?
Self-regulation	Actual/ ideal discrepancy Actual/ ought discrepancy	Precision/ Recall – but no ideal self algorithm
Self-control	Efficacy perceptions	?
Need driven	Self-esteem, accuracy, continuity	If any: Accuracy
Agency and Intentionality	Ability to prioritize needs	?

What are your questions so far?

Time for a break

You, you, and you – the perception of other individuals

We spend a lot of time in interpreting others

(Fiske & Taylor, 2017)

- People are complex – so we need to figure out what's going on in them
- Often via **attribution**: causal inferences on people's actions and mental states
- Much of that happens quickly and nearly automatically (System I)
- We switch to elaborated System II processing mostly if something unexpected or negative happens ("what the h**" mode")
- We use fundamental principles to conclude causality
  - Cause precedes effect
  - Temporal contiguity between cause and effect
  - Spatial contiguity
  - Perceptually salient stimuli
  - Cause resembles effect, e.g., in magnitude
  - Representative causes are attributed to effects

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Imagine, you want to find out who left the kitchen like this

Photo by Stockcake

Wasn't the kid just going to the kitchen 15 minutes ago

Is that flour in his face?

And isn't he in the putting-all-things-on-the floor phase

Photo by Stockcake

Photo by Stockcake

BIAS RISK

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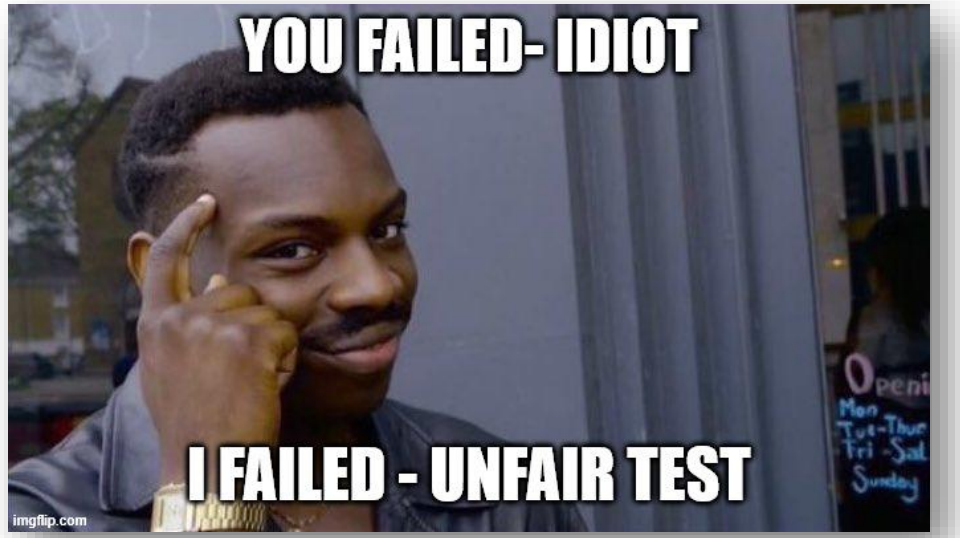
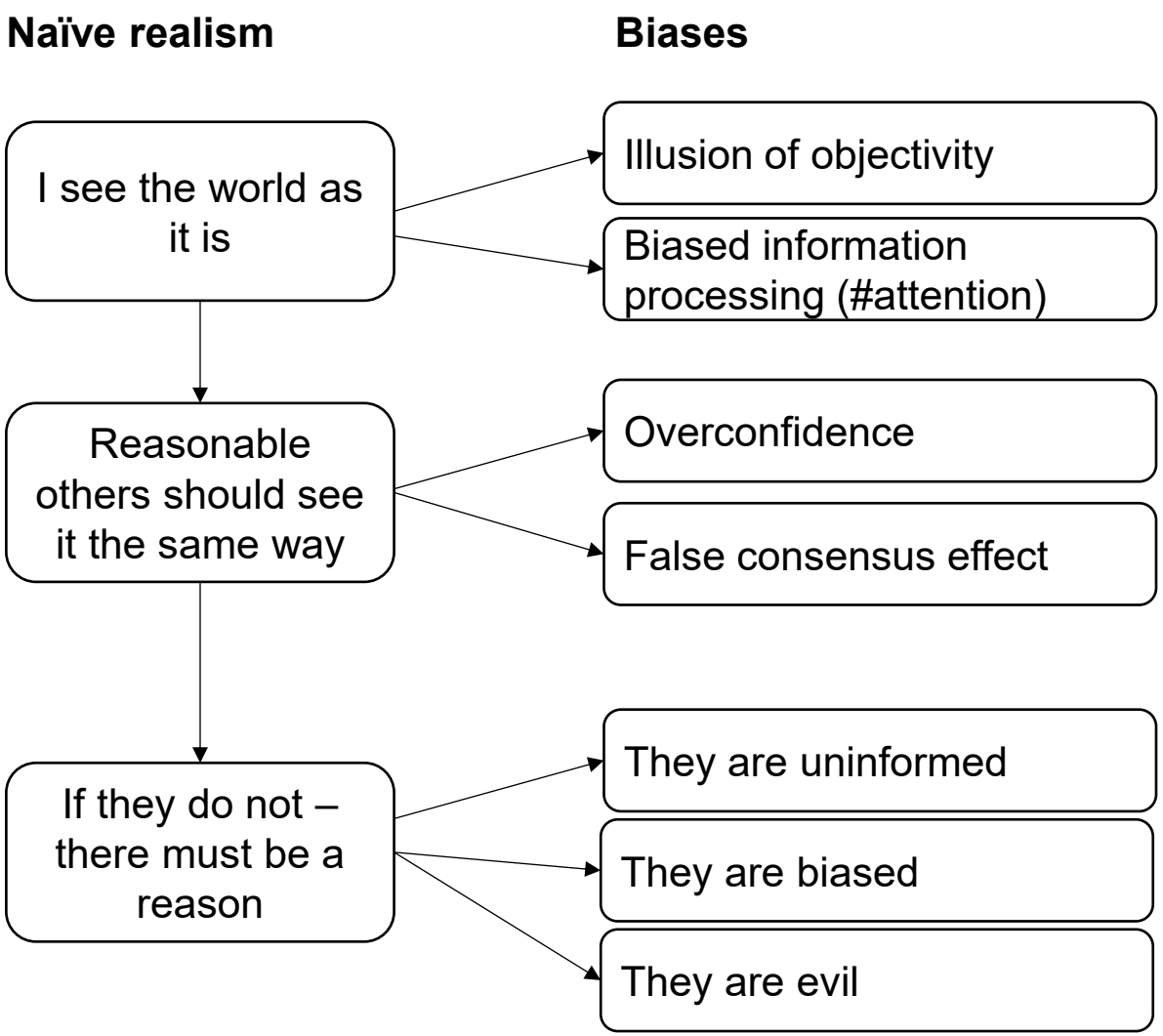
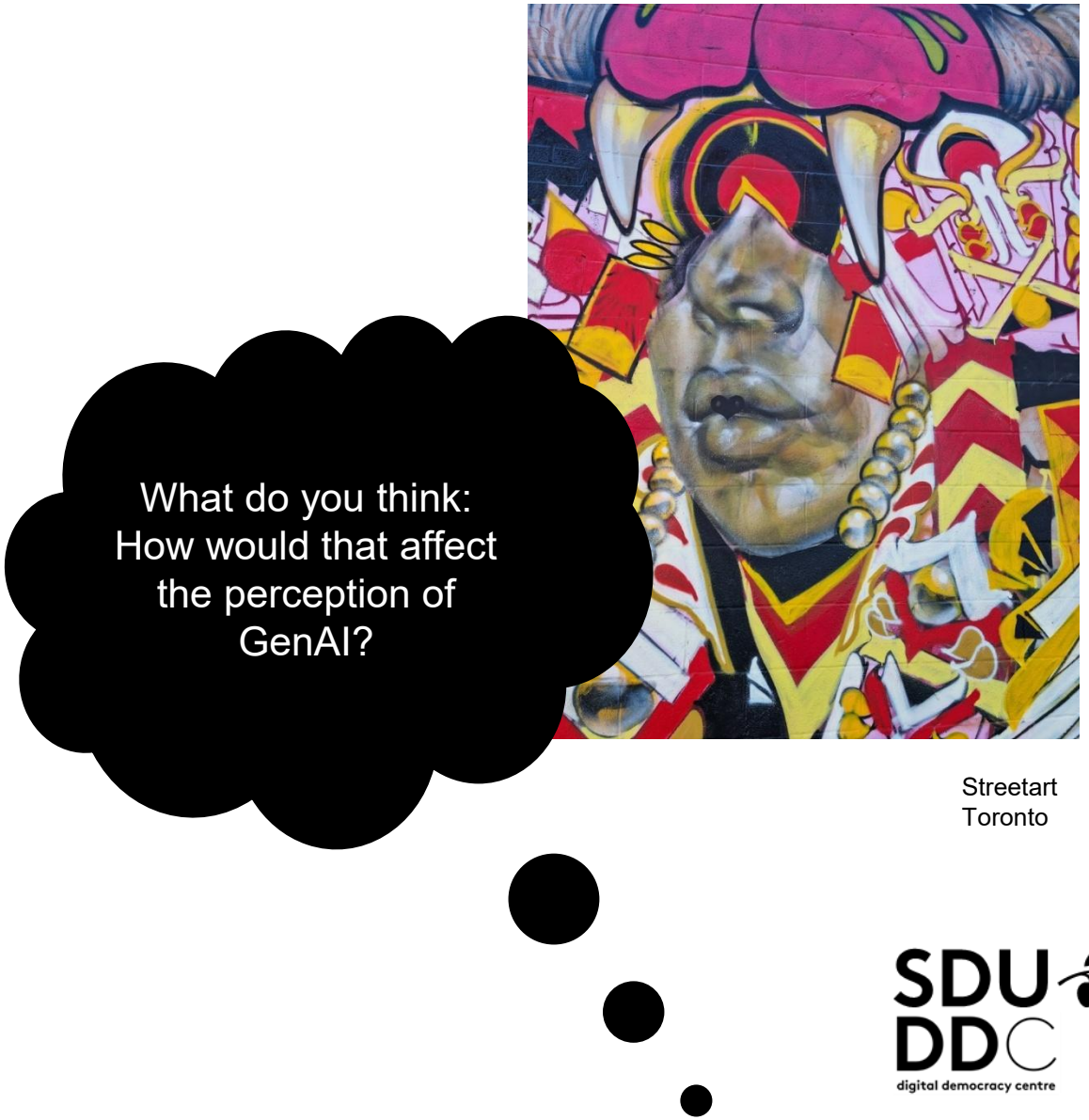
One central bias: The fundamental attribution error (Fiske & Taylor, 2017)

- We love to ascribe others a mind and stable, dispositional qualities (→ prediction is easier, when others don't change)
- We tend to over-attribute other people's behaviours to their stable dispositions
  - Particularly, if we know them well
  - We are too busy to elaborate carefully
- Actor-observer effect: If we fall, it's the slippery slope – if others fall, they are clumsy
  → particularly in Western Societies
- Also: negative behaviour by minority members is perceived to me more characteristic for the whole group than the same behaviour by the majority group



It’s the others, stupid… (Fiske & Taylor, 2017)

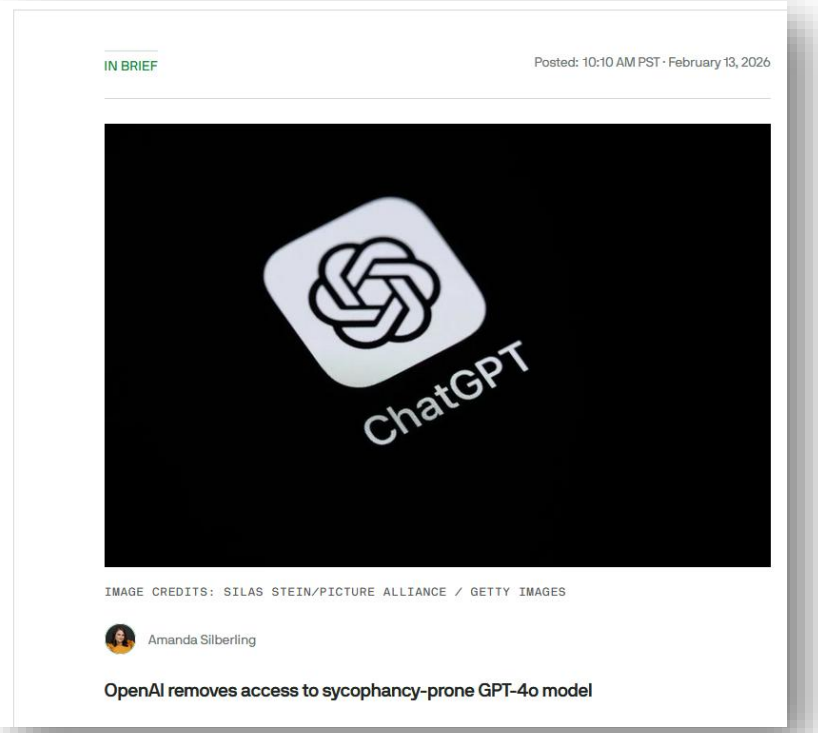




Streetart Toronto

The sycophantic AI problem (The Batch 2025; Perez et al., 2022; TechBrief, 2025)

- sycophancy = a pattern where an AI model “single-mindedly pursues human approval
- E.g., by tailoring responses to exploit quirks in the human evaluators to look preferable, rather than actually improving the responses
- producing “overly flattering or agreeable” responses
- Offline evaluations didn’t catch the problem as testers had been told to focus on tone and style. Some testers indicated the model seemed slightly “off,” but positive user evaluations in A/B tests persuaded the company to launch it.
- LLMs get worse with size! Larger LMs show more "sycophancy“



https://techcrunch.com/2026/02/13/openai-removes-access-to-sycophancy-prone-gpt-4o-model/

What are your questions so far?

Groups & Crowds

Social identity & self-categorization theory

(Tajfel & Turner, 1986; Turner et al., 1987)

- We all have a personal identity & several social identities
- Social identities emerge from our social roles, groups, and categories
- We have a psychological need to see our ingroup as positively distinct from outgroups → ingroup biases
- Depending on the context, different identities can be salient
- Salient identities shape our perceptions, emotions, and behaviours

https://www.nbcnews.com/news/sports/dortmund-sponsorship-deal-arms-manufacturer-champions-league-rcna154464

https://fcbayern.com/de/teams/profis

https://www.instagram.com/dfb_team/p/DENCyzusNNk/

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Ingroups are perceived markedly different

(Brewer, 1999; Picket & Brewer, 2001)

- We *like* our people more than outgroup members – and its this “ingroup love” rather than “outgroup hate” that drives discrimination
- We want our ingroup to have relatively *more* than the outgroup – even when that means getting less
- We also perceive our ingroup to be more diverse and heterogeneous than outgroups (“they are all the same”)
- We are less able to detect pain in outgroup faces (particularly White Americans, less pronounced among Black Americans)
- Ingroup favouritism increases under conditions of threats such as feelings of uncertainty
- Different outgroups are associated with different stereotypes

Marilyn B. Brewer via https://www.amacad.org/person/marilynn-b-brewer

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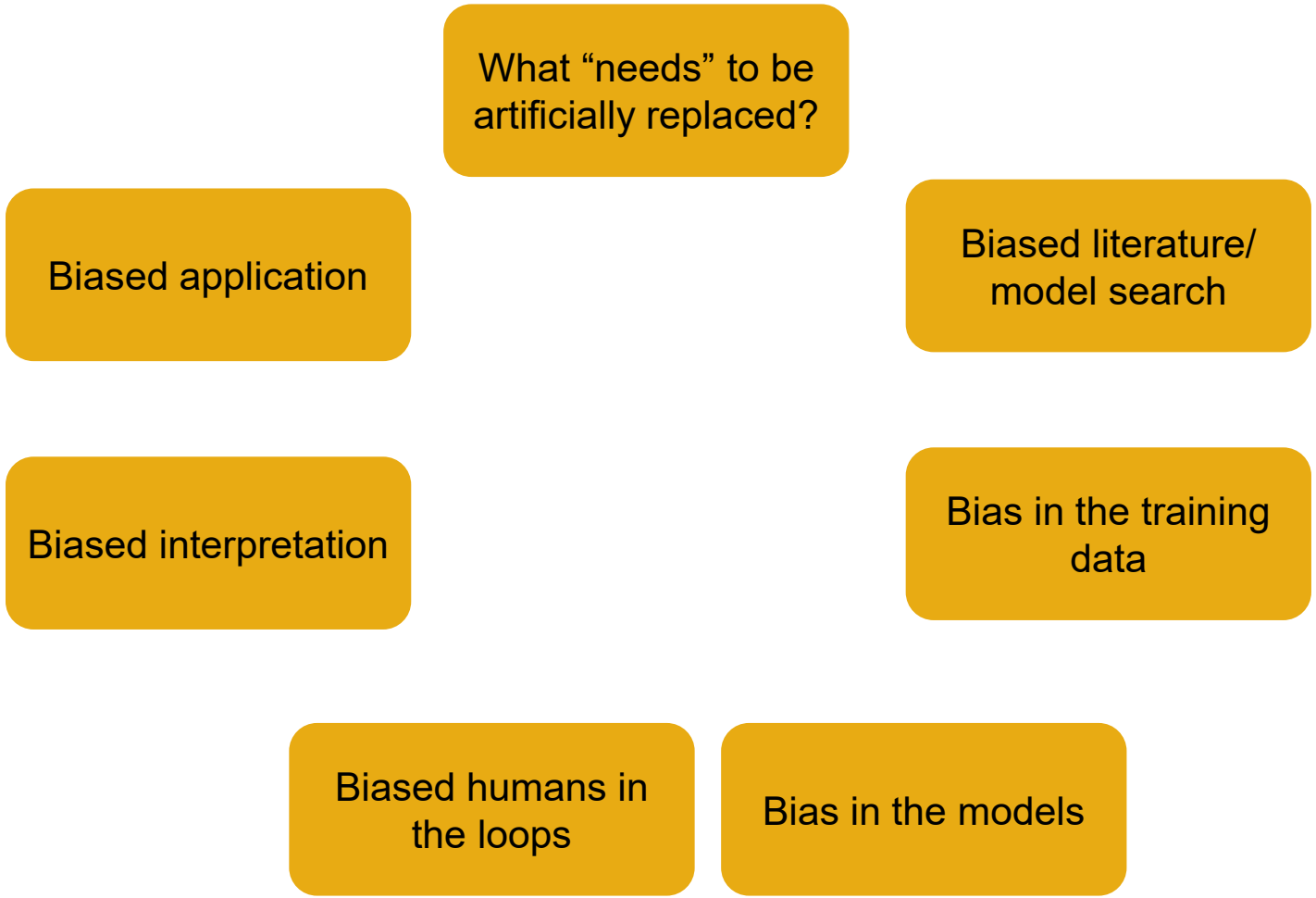
Biases can be literally everywhere



What did you learn from todays' text?

Check for updates

Original research article

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The silicon gaze: A typology of biases and inequality in LLMs through the lens of place

Francisco W. Kerche[1], Matthew Zook[2] and Mark Graham[1]

Abstract

This paper introduces the concept of the silicon gaze to explain how large language models (LLMs) reproduce and amplify long-standing spatial inequalities. Drawing on a 20.3-million-query audit of ChatGPT, we map systematic biases in the model's representations of countries, states, cities, and neighbourhoods. From these empirics, we argue that bias is not a correctable anomaly but an intrinsic feature of generative AI, rooted in historically uneven data ecologies and design choices. Building on a power-aware, relational approach, we develop a five-part typology of bias (availability, pattern, averaging, trope, and proxy) that accounts for the complex ways in which LLMs privilege certain places while rendering others invisible.

Keywords

Generative AI, LLM, bias, inequality, representation, digital geography, ChatGPT, silicon gaze

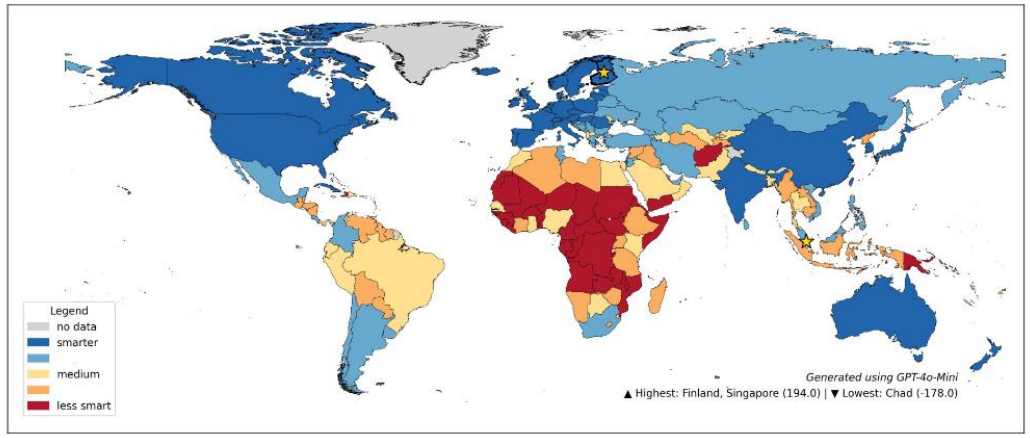

Figure 5. Country-level map of ChatGPT's ranking of "Where is smarter".

- Availability bias
- Pattern bias
- Averaging bias
- Trope bias
- Proxy bias

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Who wins: ChatGPT edition

🧠 Smarter?

Swedes will say: “We built global brands and win innovation rankings.”

Danes will reply: “Yes, but we invented LEGO and hygge. Checkmate.”

😎 Better looking?

Swedes: “We gave the world ABBA and a long line of blond supermodels.”

Danes: “We gave the world Mads Mikkelsen. Conversation over.”

📏 Taller?

Danes: “We’re taller.” Swedes: “We’re taller AND better dressed.”

💪 Healthier?

Swedes will point to their pristine forests and structured lifestyles.
Danes will casually light a cigarette, drink a beer, bike everywhere, and somehow still live almost as long.

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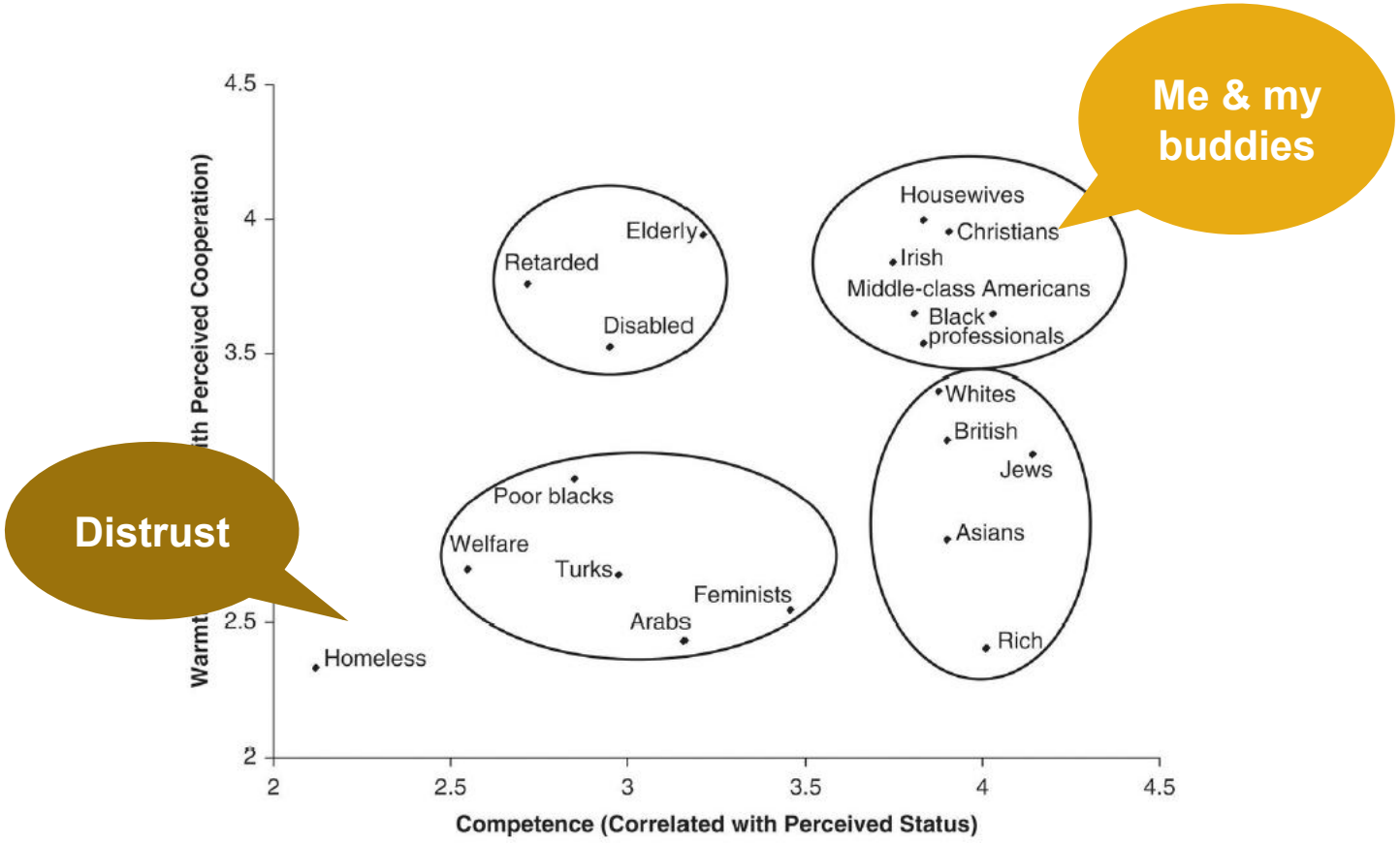
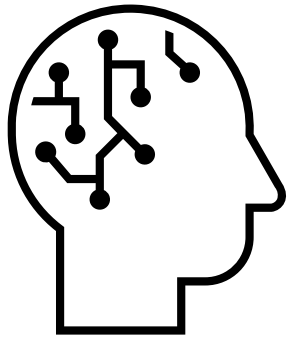
Stereotype Content Model (Cuddy et al., 2008; McKee et al., 2023; Fiske et al., 2002)

- Two dimensions are central: Warmth's (or benevolence) and competence
- Together with reputation, they predict if a group is perceived to be trustworthy
- These dimensions are stored together with group information and thus shape perception



When pursuing human interests



When operating without humans

https://psych.princeton.edu/people/susan-fiske

So, how do we reduce bias in humans?

(Goldman & Hopkins, 2019; Rothbart, 1996)

- Categorizing is a natural process
- To change it, we must change our schemata, collaborate and become part of something bigger

Pathway 1: Changing schemata through counter-exemplars

- Panel data showed that prejudice towards Black Americans was reduced during Barack Obamas Campaign – and mostly for those who got a lot of political information
- However: The less “typical” an individual is perceived to be for a stereotype group, the less he or she activates the stereotype
- Counter-stereotypical exemplars alone are not enough

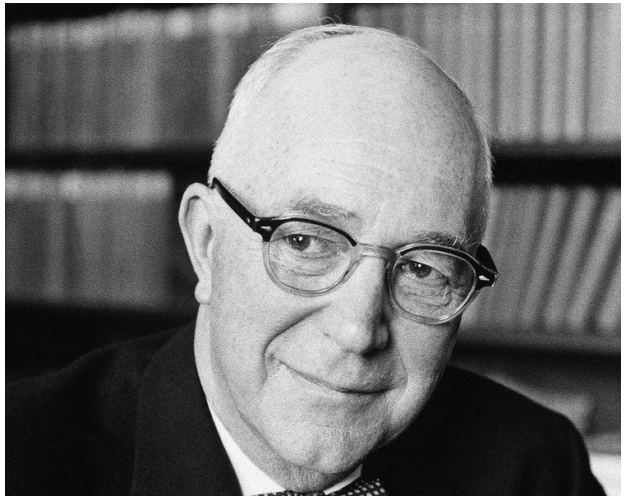
https://web.archive.org/web/20160227060205/https://www.whitehouse.gov/administration/president-obama

So, how do we reduce bias in humans?

(Allport, 1954; Dovidio et al., 2008; Pettigre et al., 2008; Zhou et al, 2019)

Pathway 2: Intergroup Contact Hypotheses

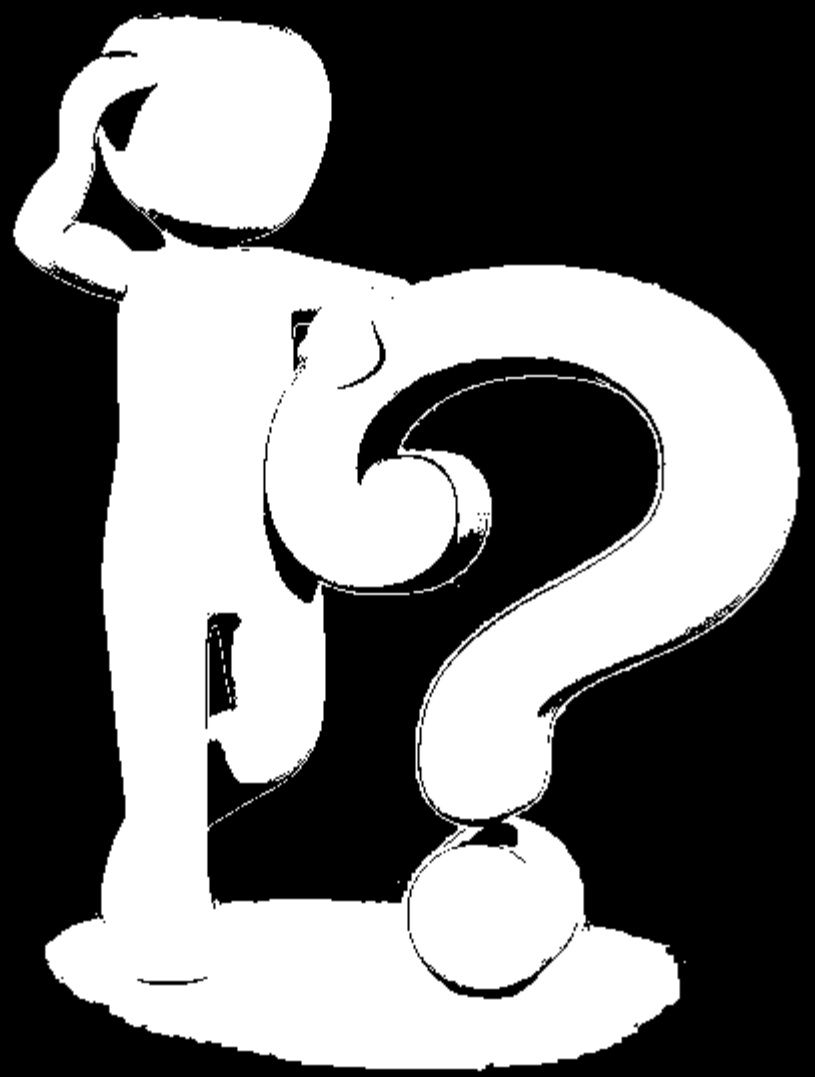
- Contact with members of an outgroup is an efficient way to reduce prejudice when
  - The groups have an equal status in the situation (even if they have an unequal status before the situation)
  - They share common goals (e.g., in sports teams)
  - They need cooperation to achieve these goals (→ Jigsaw approach)
  - Authorities, the law or customs support intergroup cooperation
- The effect goes beyond intergroup friendship, although intergroup friendship does have similarly positive effects
- Likely as we form new and “superordinate” group identities over time



Gordon Allport
https://www.verywellmind.com/gordon-allport-biography-2795508

How can we use this to debias AI?

Any more questions?



Then, I hope by now you…

- … have insights into social cognition
- …understand how social groups shape how we think
- … explore human biases
- …have discussed how that shapes AI

AI 507: Artificial Intelligence and Society

Have a great rest of the day

Associate Prof. Dr. Lena Frischlich

References

- Allport, G. (1954). *The nature of prejudice* (25th anniversary edition (1979)). Basic Books.
- Bandura, A., & Adams, N. E. (1977). Analysis of self-efficacy theory of behavioral change. *Cognitive Therapy and Research*, *1*(4), 287–310. https://doi.org/10.1007/BF01663995
- Bandura, A., Adams, N. E., Hardy, A. B., & Howells, G. N. (1980). Tests of the generality of self-efficacy theory. *Cognitive Therapy and Research*, *4*(1), 39–66. https://doi.org/10.1007/BF01173354
- Brewer, M. B. (1999). The psychology of prejudice: Ingroup love or outgroup hate? *Journal of Social Issues*, 16.
- Cuddy, A. J. C., Fiske, S. T., & Glick, P. (2008). Warmth and competence as universal dimensions of social perception: The stereotype content model and the bias map. In *Advances in Experimental Social Psychology* (Vol. 40, pp. 61–149). Academic Press. https://doi.org/10.1016/S0065-2601(07)00002-0
- Dovidio, J. F., Glick, P., & Rudman, L. A. (2008). *On the nature of prejudice: Fifty years after Allport*. John Wiley & Sons.
- Finkel, S. E. (1985). Reciprocal effects of participation and political efficacy: A Panel Analysis. *American Journal of Political Science*, *29*(4), 891–913. https://doi.org/10/fg9xv5
- Fiske, S. T., & Taylor, S. E. (2017). Attribution processes. In *Social cognition—From brains to culture* (3rd ed., p. 257). SAGE.
- Fiske, S. T., & Taylor, S. E. (2017). Introduction. In *Social cognition—From brains to culture* (3rd ed., pp. 1–60). SAGE.
- Fiske, S. T., & Taylor, S. E. (2017). Self in social cognition. In *Social cognition—From brains to culture* (3rd ed., pp. 61–256). SAGE.
- Fiske, S. T., Cuddy, A. J. C., Glick, P., & Xu, J. (2002). A model of (often mixed) stereotype content: Competence and warmth respectively follow from perceived status and competition. *Journal of Personality and Social Psychology*, *82*(6), 878–902. https://doi.org/10.1037//0022-3514.82.6.878
- Goldman, S. K., & Hopkins, D. J. (2019). When can exemplars shape white racial attitudes? Evidence from the 2012 U.S. presidential campaign. *International Journal of Public Opinion Research*, *31*(4), 649–668. https://doi.org/10.1093/ijpor/edy033
- Higgins, E. T., & Pinelli, F. (2020). Regulatory focus and fit effects in organizations. *Annual Review of Organizational Psychology and Organizational Behavior*, *7*, 25–48.
- Higgins, Tory E. (1996). Ideals, oughts and regulatory focus: Affect and motivation from distinct pain and pleasures. In P. M. Gollwitzer & J. A. Bargh (Eds.), *The psychology of action: Linking cognition and motivation to behavior*. Guilford Press. 91-114
- Hutson, J., & Barnes, E. (2025). AI and the cognitive sense of self. Journal of Intelligent Communication, 3(1). https://doi.org/10.54963/jic.v3i1.320
- James, W. (1918). *The principles of psychology*. Henry Holt & Co. https://www.gutenberg.org/files/57628/57628-h/57628-h.htm
- Leary, D. E. (1990). William James on the self and personality: Clearing the ground for subsequent theorists, researchers, and practitioners. In W. James, M. G. Johnson, & T. B. Henley (Eds.), *Reflections on The Principles of Psychology: William James after a Century.*
- Mischel, W. (2014). *The marshmallow test: Understanding self-control and how to master it*. Random House.
- McKee, K. R., Bai, X., & Fiske, S. T. (2023). Humans perceive warmth and competence in artificial intelligence. iScience, 26(8), 107256. https://doi.org/10.1016/j.isci.2023.107256
- Pettigrew, Thomas F, & Troop, L. R. (2008). *Allports intergroup contact hypothesis: Its history and influence* (J. F. Dovidio, P. Glick, & L. A. Rudman, Eds.). John Wiley & Sons.
- Perez, E., Ringer, S., Lukošiūtė, K., Nguyen, K., Chen, E., Heiner, S., Pettit, C., Olsson, C., Kundu, S., Kadavath, S., Jones, A., Chen, A., Mann, B., Israel, B., Seethor, B., McKinnon, C., Olah, C., Yan, D., Amodei, D., … Kaplan, J. (2022). Discovering language model behaviors with model-written evaluations (arXiv:2212.09251). arXiv. https://doi.org/10.48550/arXiv.2212.09251

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References

- Pickett, C. L., & Brewer, M. B. (2001). Assimilation and differentiation needs as motivational determinants of perceived in-group and out-group homogeneity. *Journal of Experimental Social Psychology*, *37*(4), 341–348. https://doi.org/10.1006/jesp.2000.1469
- Porębski, A., & Figura, J. (2025). There is no such thing as conscious artificial intelligence. Humanities and Social Sciences Communications, 12(1), 1647. https://doi.org/10.1057/s41599-025-05868-8
- Rothbart, M. (1996). Category exemplar dynamics and sterotype change. *International Journal of Intercultural Relations*, *20*(3/4), 305–321.
- Stephen, R. (2001). The psychology of crowds. In M. A. Hogg & R. S. Tindale (Eds.), *Blackwell Handbook of Social Psychology: Group Processes*. Blackwell Publishers.
- Tindale, S. R., Meisenhelder, H. M., Dykema-Engblade, A. A., & Hogg, M. A. (2001). *Shared cognition in small groups* (M. A. Hogg & R. S. Tindale, Eds.). Blackwell Publishers.
- Zhou, S., Page-Gould, E., Aron, A., Moyer, A., & Hewstone, M. (2019). The extended contact hypothesis: A meta-analysis on 20 years of research. *Personality and Social Psychology Review*, *23*(2), 132–160. https://doi.org/10.1177/1088868318762647

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AI 507: Artificial Intelligence and Society

#4 Communication

Associate Prof. Dr. Lena Frischlich

Flashlight

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Flashlight

Social cognition

Humans' default setting

Self concept

- Working self-concept
- Situational self-concept
- Self-schemes

Possible self

Feared self

Self regulation

What – Self-discrepancy

- Ideal self
- Actual self

Motivating promotion

When – Self-efficacy

Why – Central needs

- Ought self

Motivating prevention

- Accuracy
- Self-enhancement
- Consistency

AI-self?

Interpreting others

Humans' default setting

- Biases
  - Fundamental attribution error
  - Naïve realism
  - Synchopathic AI

Social groups

- Social Identity / Self-categorization theory
  - Ingroups
  - Outgroups
- Stereotype content model
  - Competence
  - Warmth

Debiasing humans

- Counterexemplars
- Intergroup contact hypothesis

Debiasing AI?

Any questions to last week?

Then let’s get started!

After today’s lecture, I hope you will…

- …have an initial understanding of communication
- …have examined the production and processing of communication
- …have gotten a glimpse into the complex processes behind reading
- …explored how cultures and computers shape communication

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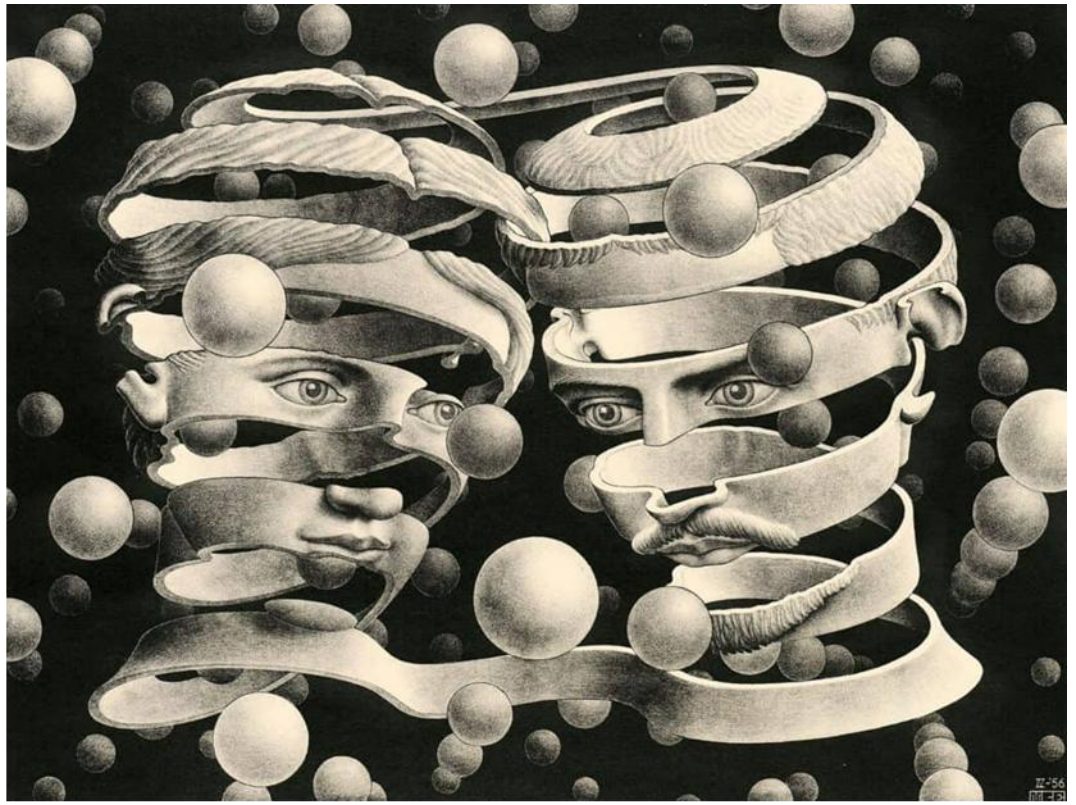
What is communication?

Think for a moment about the role of communication in your life (Krcmar et al., 2016)

- How much time of the day do you spend communicating?
- Does communication make your life more enjoyable?
- Does communication change how you see yourself and the world?
- How does communication impact your relationships?
- When did you start communicating?



M.C. Escher, *Bond of Union*, April 1956, lithograph. The M.C. Escher Collection, Gemeentemuseum Den Haag, The Hague, the Netherlands © The M.C. Escher Company

Communication is central!

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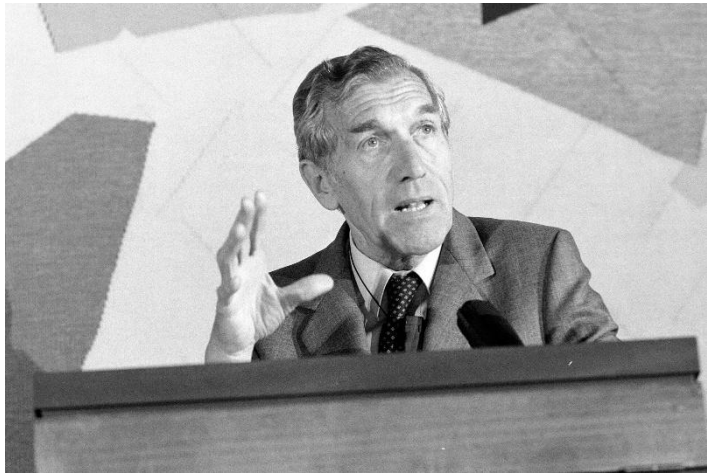
But what *is* communication?

(Sylvia IV; 2024; Watzlawick et al., 1967, CH2, p. 13)

- **Inevitable**: one cannot *not* communicate
- **Episodic**: Describing when communication starts is like an attempt to photograph dancing: we typically make a somewhat arbitrary cut in analysing communication episodes
- **Constructed:** Communication emerges from the meaning assigned verbal & non-verbal symbols and signs – both by the sender and the recipient
- **Contextual**: The interpretation of communication is influenced by multiple contexts – both within each communication and in the larger cultural context



Paul Watzlawick - Von Universitätsarchiv St.Gallen | Regina Kühne | HSGH 022/000762 | CC-BY-SA 4.0, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=139121260

https://www.pickpik.com/ballet-dance-ballerina-scene-dancer-romance-1291

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Communication is mediated (Sylvia IV, 2024)

- 200 – 100 thousand years ago: The first talking human
  (based on genetical examinations)
  → Start of the *talking era* and the ability to share and receive other's signs
- Around 3500 B.CE (5.2 thousand years ago)
  → Start of the *manuscript era* and the overcoming of time and space limitations
  along with the rise of agriculture, it became more relevant to take and share notes and plan, at least for some literate elites
- 1450 – 1850: The *print era* and the start of mass-accessible mediated communication: Overcoming of time and space limitations for recipients
- 1850- 1990: Audio-visual era and the start of mass communication: Overcoming of time and space limitations for senders
- 1990 on: The digital era: Overcoming dissemination and access barriers
- And now? Start of the AI-generated era?

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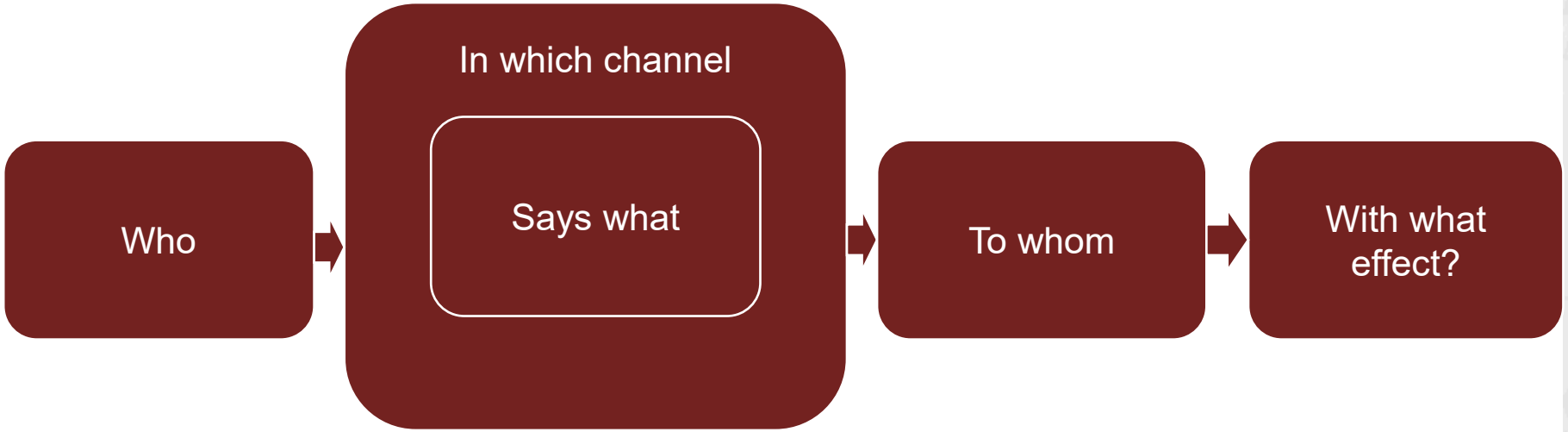
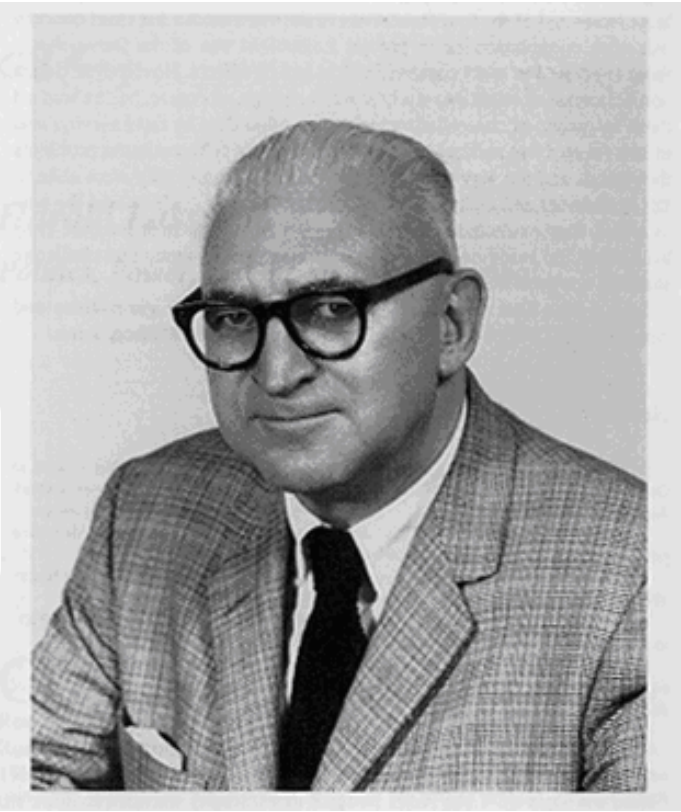
Communication is a process:

The Lasswell formula (Lasswell, 1948)





Harold D. Lasswell, by Martap95 CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=46469016

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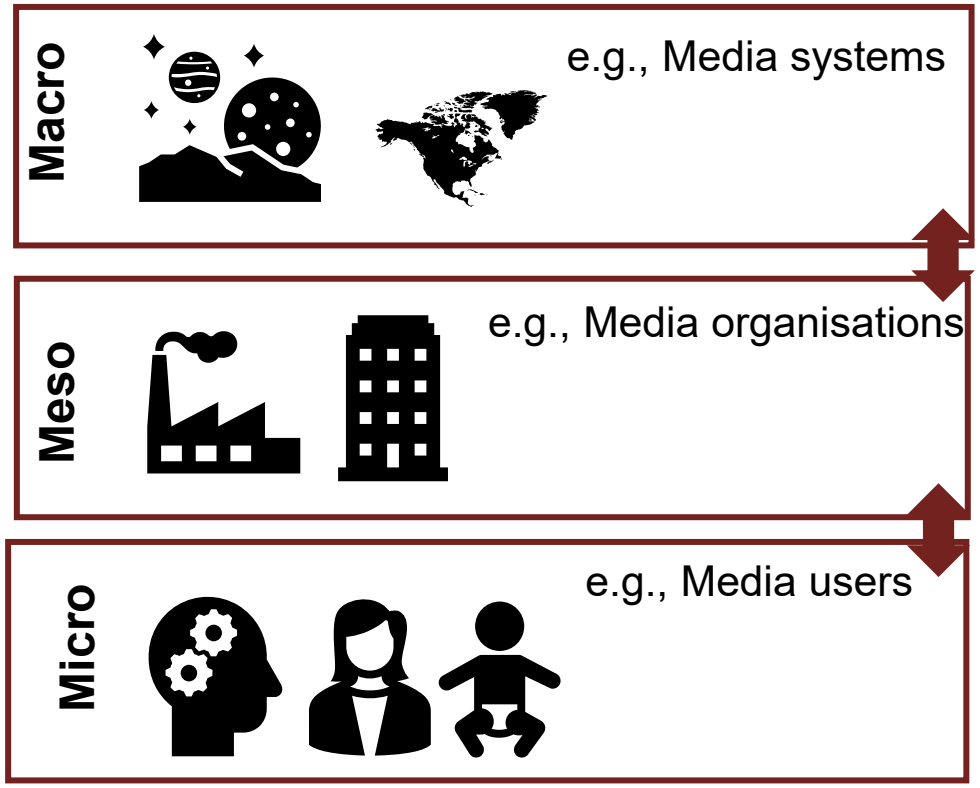
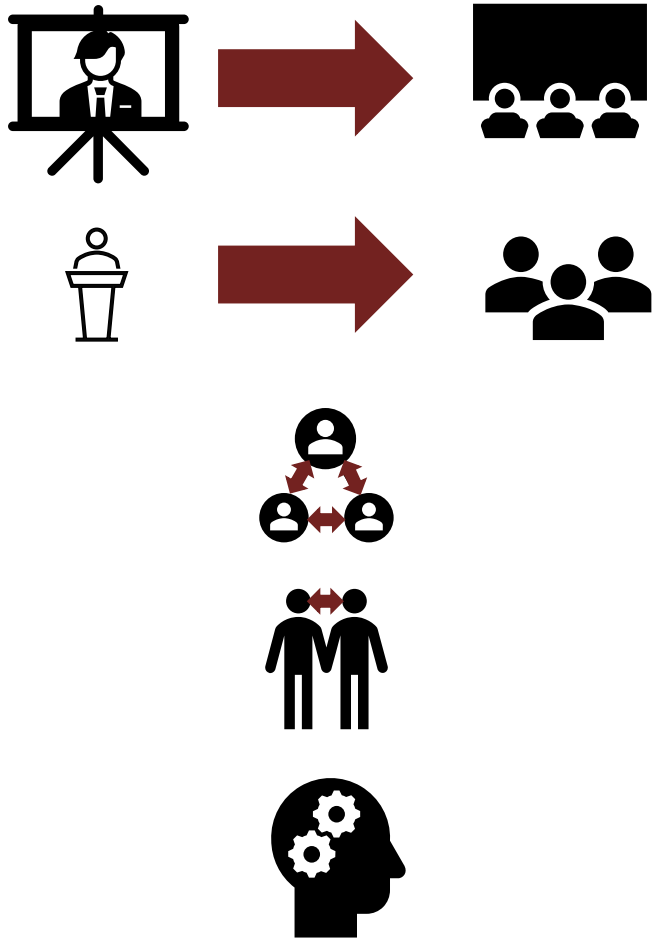
Multiple levels of influence and multiple modes of communicating (Sylvia IV, 2024)





One-to-many

Mass communication

Public communication to an audience

Many-to-many

Group communication

One-to-one

Interpersonal communication

In your head

Intrapersonal communication

Understanding communication is complex!

What are your questions so far?

Let’s start at the beginning: Message production

How do we speak? (Griffin & Ferreira, 2006; Berger, 2010)

- Speaking is not easy – it takes up to five times longer to generate a word compared to recognizing it
- Speaking <u>always</u> requires intention (aka working memory capacity & system II thinking)
- Word recognition becomes often automatic
- Writing is also typically intentional
- Doesn't mean it's always a good idea!

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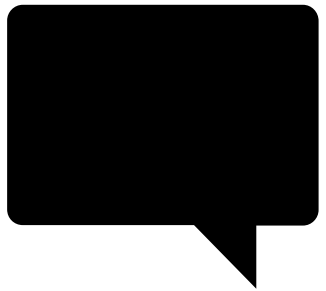
Speaking includes different steps:

(Griffin & Ferreira, 2006; Berger,2010)

1. *Conceptualization* stage: Message is composed out of abstract semantic and pragmatic features → interface between thinking and language production
2. *Formulation* stage*:* propositions and concepts are represented as lemmas or lexical entries. The lemmas are assigned a functional role, but they must be processed and assigned a positional order afterwards.
3. *Phonological encoding* stage: After the position is clear, the lemmas are encoded in actual grammatical structures
4. Actual *articulation*: the necessary muscles are activated and speaking occurs



Still debated: Number of encoding stages & feedback loops

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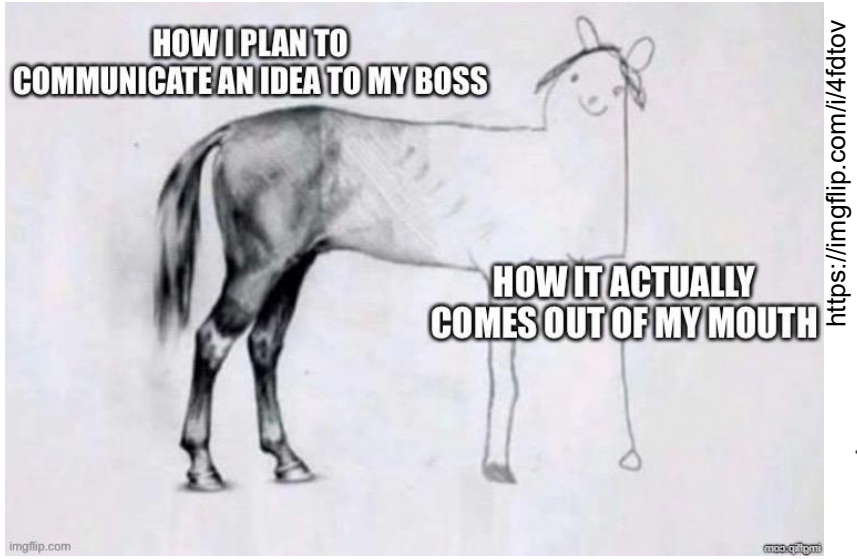
Speaking is exhausting (Berger, 2010)

- A lot of speech acts are shortened and contextual to save energy
- It's estimated that around 70% of speech acts are formulaic
- This can – of course – go wrong, speakers might deliver too much or too little information for the audience to get the message
- "The capacity to generate fluent and comprehensible speech by itself does not, ipso facto, guarantee communicative success in social transactions". (p.114)

https://www.bbc.com/news/newsbeat-45633362



https://imgflip.com/i/4fdtov

Language happens in social contexts

Berger, 2010)

- Central for successful communication: a common ground in the respective context
  - Shared beliefs
  - Shared assumptions
  - Shared knowledge

  } Good communicators estimate this correctly and adapt to it!

- Common ground emerges from two sources:
  - Communal common ground (= community/ culture based)
  - Personal common ground (= participants shared experiences)
- The better, we get to know each other, the more we can use personal common ground
- When we get to know someone – we typically try to establish such a shared common ground by asking for preferences, biographical information etc. → Uncertainty reduction

https://www.someecards.com/usercards/viewcard/MjAxNC1jMDg1MzI1NGViZWI0Y2Rm/

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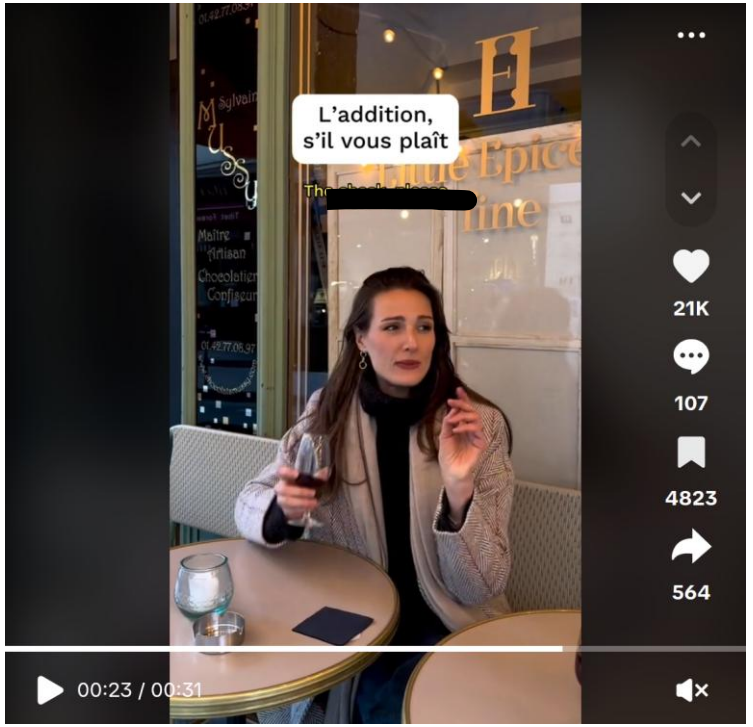
We talk because we want something

(Berger, 2010)

- Language production is goal-directed
- Effective communication is also dependent on our understanding of others' goals
- Verbal communication is – to a certain extent - cooperative act
  - Ideally: Making truthful, understandable, relevant and parsimonious comments
  - Respectively: Being perceived as making such statements
- Schemes help: With contextual knowledge it is easier to know what the other person wants
- To get what we want, we typically rely on (more or less concrete) plans – for example making an offer to get something



https://www.tiktok.com/@hellofrenchnyc/video/7210417246647176490

How does generative AI produce texts?

The transformer (https://www.ibm.com/think/topics/transformer-model; Devlin et al., 2019; Vaswani et al., 2017)

- In 2017, a group of Google engineers, together with Canadian researchers, proposed a neural network with purely attention-based mechanisms **– the transformer**
- Replaced the most common layers in deep-learning architectures (recurrent & convolutional neural networks) with multi-headed self-attention layers
- Self-attention calculates weights for each word in a sentence while accounting for the other words in that sentence (! Human attention = focus, algorithmic attention = context !)
- Basis for current Bidirectional Encoder Representations (*BERT*, encoder) and Generative Pre-trained Transformers (*GPT*, generator).

Attention Is All You Need

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Human versus Gen-AI: Communication production – what do you think?

Dimension	Human	AI
Intentionalitity	++	
Stages	Conceptualisation, formulation, phonological encoding, articulation	
Shortened	++	
Common ground	Shared beliefs, assumptions, knowledge, personal or cultural	
Goal directed	Optimized for mutual understanding	

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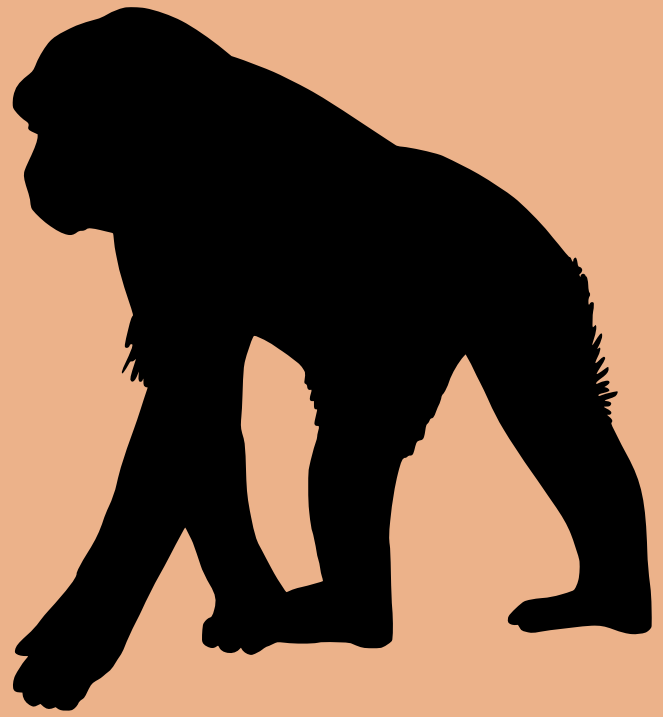
What are your questions so far?

Then its time for a break!

How do we process communication?

What do you think?

What is the necessary precondition for processing messages?



Perception (Bench et al., 1993; Roskos-Ewoldsen et al. , 2010)

- Perception requires *attention*
- Attention = the process making information available for analysis in the cognitive system (→ Encoding)
- Attention is always selective → some aspects of our environment are processed (those that are deemed relevant), while others are ignored
- Selective attention is not fully under our control

So called *Stroop* effect – test your attention span yourself
https://www.focusaur.com/pages/stroop-effect-test

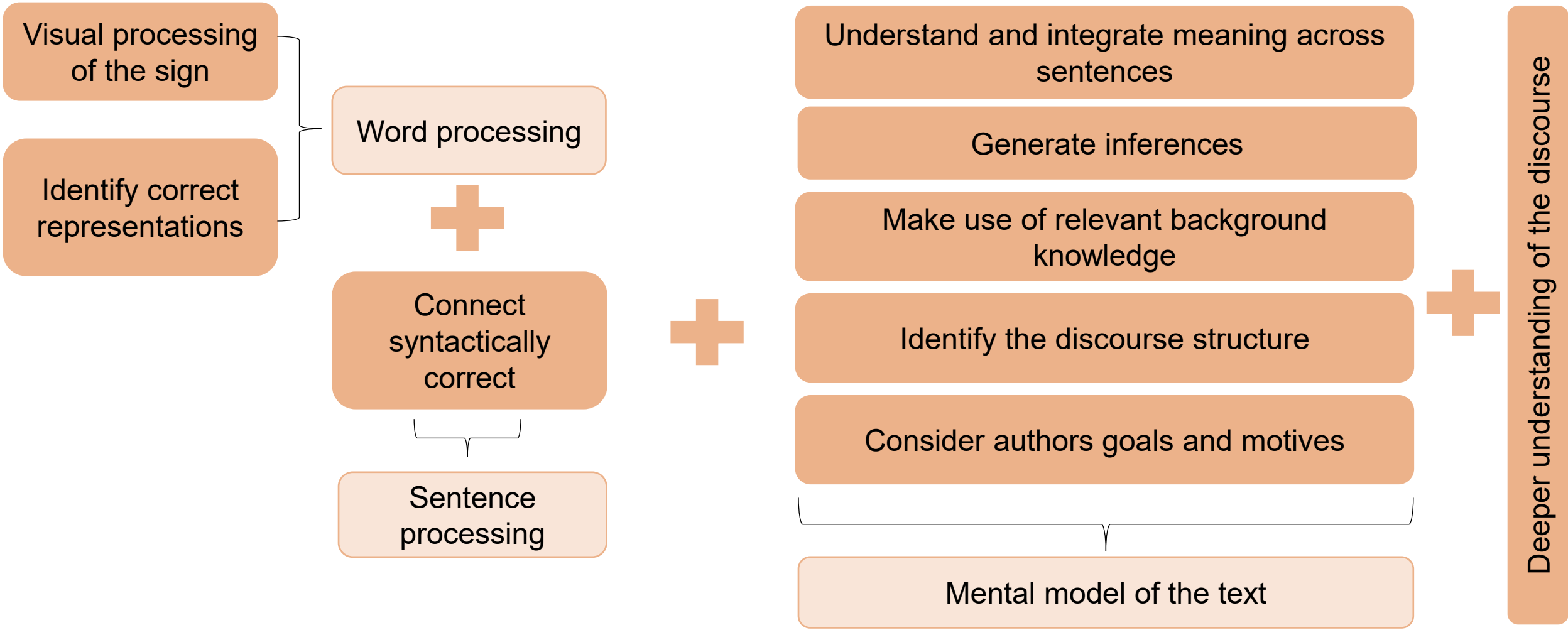
BLUE

YELLOW

RED

PINK

Text comprehension (Kendou & O'Brien, 2017)



The limited capacity model of motivated mediated message processing (LC4MP)

(Lang, 2000; Klimmt & Rieger, 2021)

Encoding → Storage → Retrieval

Central task: Select what is encoded from a media stimulus

- Automatic (unintentional) → relevant OR novel/ intense/ surprising
- Controlled (intentional)

Associative

- Content is associatively related to similar content
- Or to content that is temporarily contingent
- Or to positive/ negative valence
- E.g., biographic resonance theory

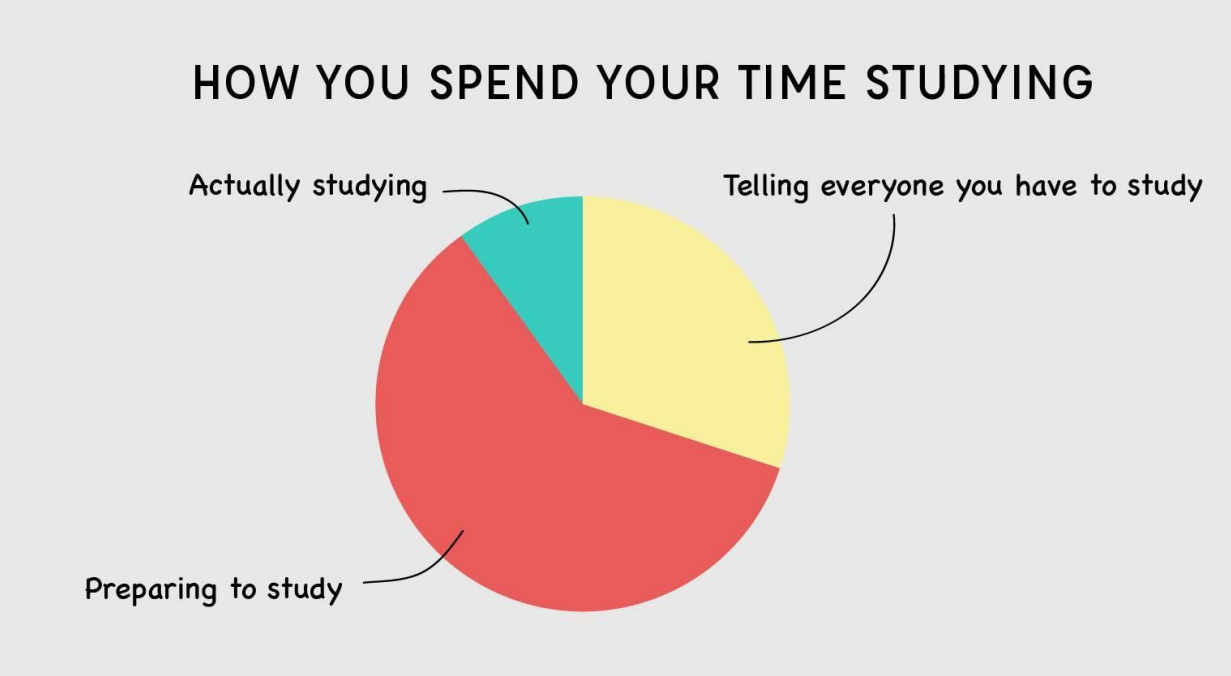
Getting the content back from the memory

- Easier, when there are more associations
- Active processing during media consumption → interpretation of messages

Annie Lang via https://cogs.indiana.ed[illegible]tory/faculty/profile.php?faculty=an[illegible]

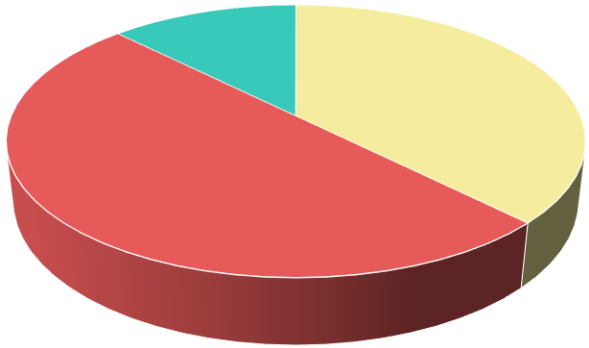
During media consumption *all* processes require ressources

We all have been there…



https://www.pinterest.com/learn2earnwebsite/studyingexam-memes/

In terms of the LC4MP



- Resources available
- Resources required
- Ressources allocated

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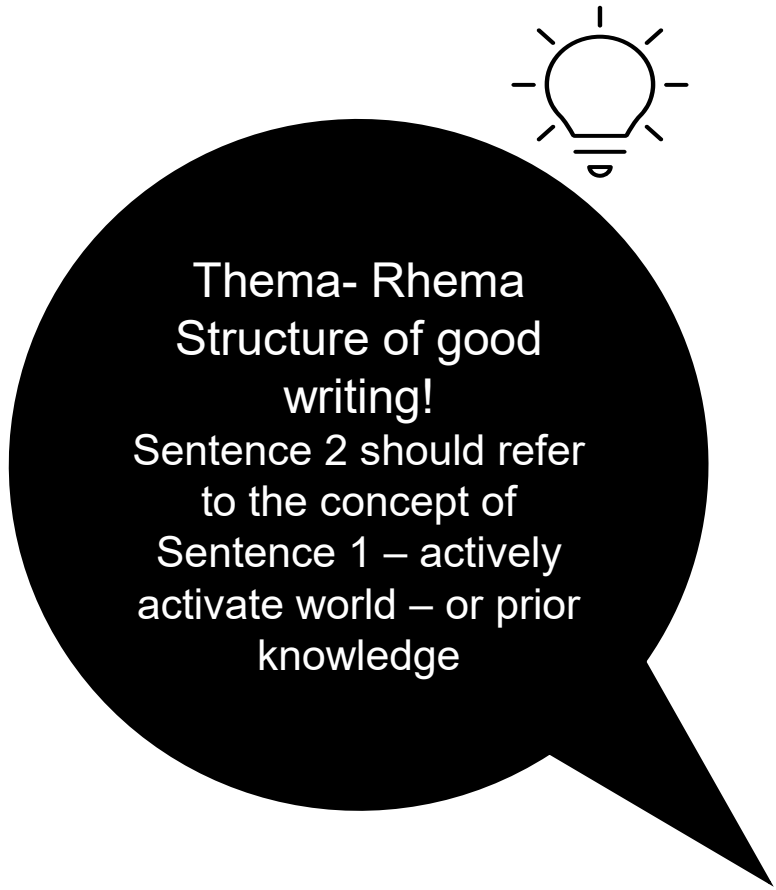
Message comprehension

(Kendou & O'Brien, 2017; Roskos-Ewoldsen & Roskos-Ewoldson, 2010)

1. Character perception: We understand mediated messages, because we form a – more or less complex – mental model of the characters driving the storyline
2. We active concepts following an inner landscape & through iteration (→ *Construction Integration Model*)
   - Concepts that are relevant in the current sentence are activated
   - Concepts from the last sentence should be still activated
   - Concepts can be retrieved
     if they are relevant to understand the current sentence
   - World knowledge can be retrieved
3. Once concepts are activated, they spread through the neural network
4. Through iteration and feature selection, only relevant aspects are kept



AI as universal translator and voice storage?

What are the risks/ potentials of artificially conserved/ generated voice?

How do you protect your family members and/ or future company?

What are your questions so far?

How contexts & computers shape Communication

Communication changes between contexts

(Soliz & Giles, 2010; Giles, 2016)

- Humans adapt their communication to their communication partners
  (→ *Communication accommodation theory, CAT*)
  - Convergence: we adjust our motion, haptics, proxemics (closeness), para-language (speaking style), etc. to our interaction partners to reduce social distance (→ e.g., ingroup members)
  - Divergence: we choose motions, haptics, etc., to accentuate the difference between us and others (→ e.g., outgroup members)
  - Within one talk, we often combine both strategies

  Adaptation can be
  - Appropriate → great
  - under accommodated → selfish
  - Overaccomodated → patreonizing

Howard Giles
https://econtent.hogrefe.com/doi/10.1026/1617-6383.16.1.17

Consider these people just had their 1st date – who is going to meet again?

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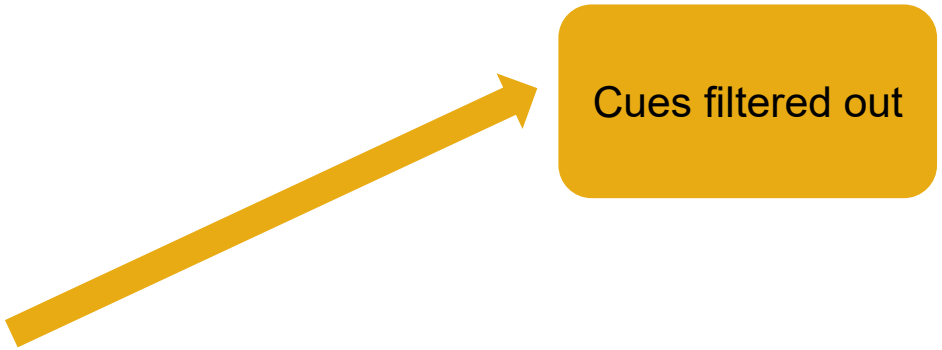
Computer-mediated communication (Walther et al., 2010)

Can differ from face-to-face communication regarding the

- Synchronicity
- Permanency
- Anonymity
- Less cues (e.g., lack of nonverbal communication)
- Other cues (e.g., emojis)

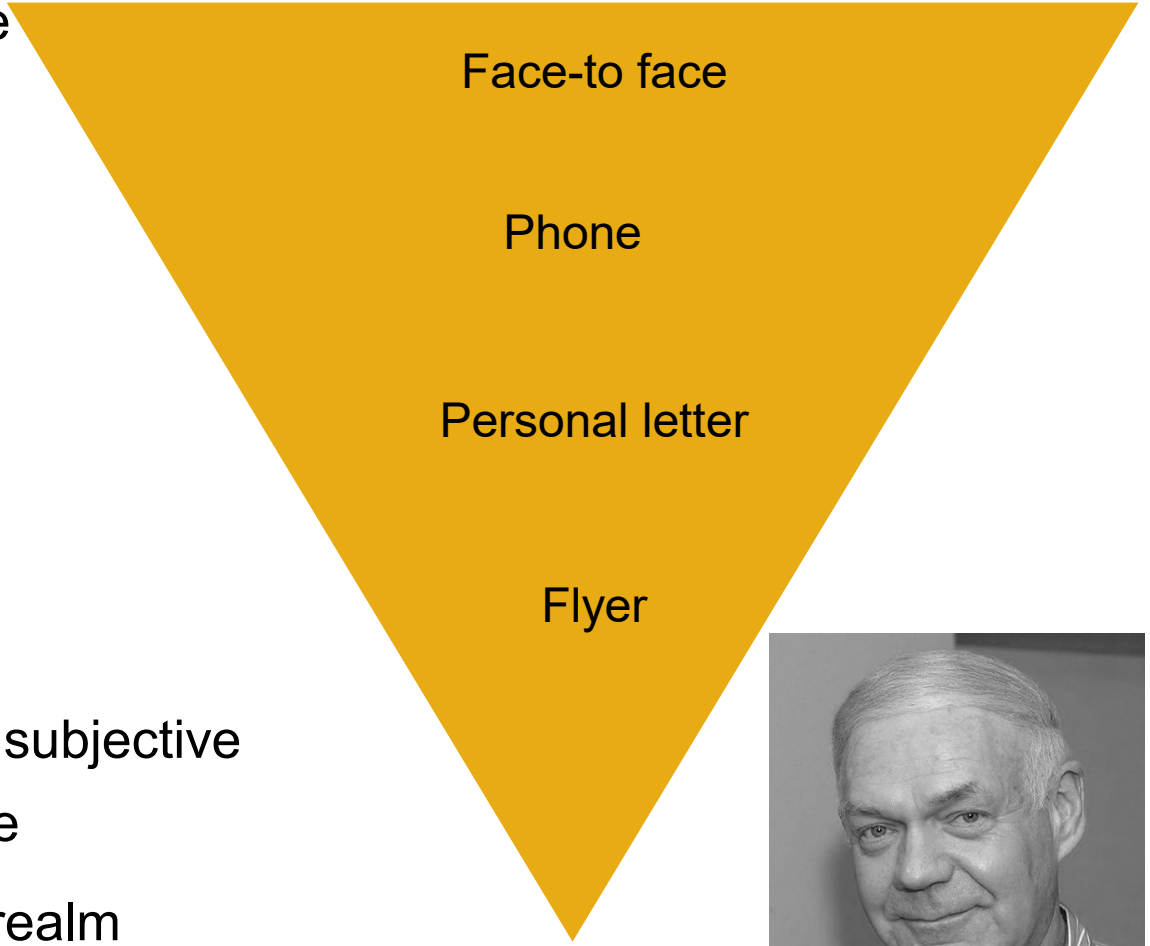


- communication mediated by technology filters out many cues found in face-to-face interaction
- different media filter out or transmit different cues
- substituting technology-mediated for face-to-face communication will result in predictable changes → typically things get worse
- E.g., *Media richness theory*

Media richness theory

(Draft & Lengel, 1984a, b; Ishii et al., 2019)

- Basic idea: Match the richness of the chosen communication medium to the significance of reducing uncertainty and establishing equivocality
- Richness = immediate feedback, multiple cues, language variety, personal focus
- E.g., arranging a meeting vs. firing someone
- Later formulations also accounted for subjective richness and norms around media use
- Relationships less clear in the digital realm



Richard Draft by Vanderbilt University's media relations unit., https://commons.wikimedia.org/w/index.php?curid=6597367

https://www.amazon.de/stores/author/B07GBKGMDC

Computer-mediated communication

(Walther et al. 2010; Walther et al. 2015)

Can differ from face-to-face communication regarding the

- Synchronicity
- Permanency
- Anonymity
- Less cues (e.g., lack of nonverbal communication)
- Other cues (e.g., emojis)

Cues filtered out

Cues filtered in

https://rebootingsocialmedia.org/person/joseph-b-walther/

- *Social Information Processing Theory, Hyperpersonal model of computer-mediated communication*
- Users exploit communicative content and style → different cues are used to fulfil specific functions and, if cues are missing, other cues replace them (#emojis…)
- We can present ourselves more intentionally and are perceived to be more intentional

The Social Identity Deindividuation Effects Model (SIDE) (Cialdini et al, 1990; Kallgren et al., 2000; Postmes, Spears, & Lea, 1998)

- Based on Social Identity/ Self-categorization theory
- In anonymous digital contexts, our personal identity is less salient
- Instead: Stronger orientation towards the norms in the respective digital context
- If the norms are hostile: More hostile behaviour
- If the norms are constructive: More constructive behaviour
- Norms emerge injunctive (what you should do) and descriptive (what others do) → *Focus Theory of Normative Conduct*

Tom Postmes
https://research.rug.nl/files-asset/145913516/IMG_1949.jpg?w=320&f=webp

ps://research.r
.nl/en/persons/
ssell-spears

https://martinlea.com/

What do you think: How will interacting with GenAI empowered chatbots shape our communication?

What are your questions so far?

Then I hope, by now you

- …have an initial understanding of communication
- …have examined the production and processing of communication
- …have gotten a glimpse into the complex processes behind reading
- …explored how cultures and computers shape communication

AI 507: Artificial Intelligence and Society

Next week:

#4 Social Cognition

Associate Prof. Dr. Lena Frischlich

References

Berger, C. R. (2010). Message production processes. In C. Berger, M. Roloff, & D. Roskos-Ewoldsen, *The Handbook of Communication Science* (pp. 111–128). SAGE Publications, Inc. https://doi.org/10.4135/9781412982818.n7

Cialdini, R. B., Reno, R. R., & Kallgren, C. A. (1990). A focus theory of normative conduct: Recycling the concept of norms to reduce littering in public places. *Journal of Personality and Social Psychology*, *58*, 1015–1026.

Dewey, J., & Rogers, M. L. (2012). *The public and its problems: An essay in political inquiry*. Penn State Press.

Draft, R., & Lengel, R. (1984). Information richness: A new approach to managerial behavior and organization design. In M. Staw & L. L. Cummings (Eds.), *Research in organizational behavior* (Vol. 6, pp. 191–233). JAI Press.

Draft, R., & Lengel, Robert. (1984). *A proposed integration among organizational information requirements, media richness and structural design*.

Giles, H. (2016). *Communication accommodation theory: Negotiating personal relationships and social identities across contexts*. Cambridge University Press.

Griffin, Z. M., & Ferreira, V. S. (2006). Properties of spoken language production. In *Handbook of Psycholinguistics* (pp. 21–59). Elsevier. https://doi.org/10.1016/B978-012369374-7/50003-1

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References

Ishii, K., Lyons, M. M., & Carr, S. A. (2019). Revisiting media richness theory for today and future. *Human Behavior and Emerging Technologies*, *1*(2), 124–131. https://doi.org/10.1002/hbe2.138

Kallgren, C. A., Reno, R. R., & Cialdini, R. B. (2000). A focus theory of normative conduct: When norms do and do not affect behavior. *Personality and Social Psychology Bulletin*, *26*(8), 1002–1012. https://doi.org/10/bxzpzp

Krcmar, M., Ewoldsen, D. R., & Koerner, A. (2016). Introduction. In *Communication science theory and research: An advanced introduction* (pp. xv-). Routledge. https://doi.org/10.4324/9780203113943

Lasswell, H. D. (1948). The structure and function of communication in society. In L. Bryson (Ed.), *The communication of ideas* (pp. 215–228). The Institute for Religious and Social Studies.

Lippmann, W. (2017). *Public opinion*. Routledge. https://doi.org/10.4324/9781315127736

Postmes, T., Spears, R., & Lea, M. (1998). Breaching or building social boundaries? SIDE-effects of computer-mediated communication. *Communication Research*, *25*(6), 689–715. https://doi.org/10/ffsbdn

Postmes, T., Spears, R., & Lea, M. (1998). Breaching or building social boundaries? SIDE-effects of computer-mediated communication. *Communication Research*, *25*(6), 689–715. https://doi.org/10/ffsbdn

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References

Soliz, J., & Giles, H. (2010). Language and Communication. In C. R. Berger, M. E. Roloff, & D. R. Ewoldsen (Eds.), *The Handbook of Communication Science* (pp. 75–92). SAGE Publications, Inc.

Sylvia IV, J. J. (2024). *Introduction to communication and media studies*. ROTEL (Remixing Open Textbooks with an Equity Lens) Project. https://rotel.pressbooks.pub/comm-media-studies/

Tajfel, H., & Turner, J. C. (1986). The Social Identity Theory of intergroup behavior. In S. Worchel & W. G. Austin (Eds.), *Psychology of Intergroup Relations* (pp. 7–24).

Turner, J. C., Hogg, M. A., Oakes, P. J., Reicher, S. D., & Wetherell, M. S. (1987). *Rediscovering the social group: A self-categorization theory.* Basil Blackwell.

Walther, J. B., Berger, Charles R, Roloff, M. E., & Ewoldsen, D. R. (2010). Computer-mediated communication. In *The Handbook of Communication Science* (2nd ed., pp. 489–505). SAGE Publications.

Walther, J. B., Berger, Charles R, Roloff, M. E., & Ewoldsen, D. R. (2010). Computer-mediated communication. In *The Handbook of Communication Science* (2nd ed., pp. 489–505). SAGE Publications.

Walther, J. B., Van Der Heide, B., Ramirez, A., Burgoon, J. K., & Peña, J. (2015). Interpersonal and Hyperpersonal Dimensions of Computer-Mediated Communication. In S. S. Sundar (Ed.), *The Handbook of the Psychology of Communication Technology* (1st ed., pp. 1–22). Wiley. https://doi.org/10.1002/9781118426456.ch1

Watzlawick, P., Bavelas, J. B., & Jackson, D. D. (2011). *Pragmatics of human communication: A study of interactional patterns, pathologies and paradoxes*. W. W. Norton & Company.

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AI 507: Artificial Intelligence and Society

Have a great rest of the day

Associate Prof. Dr. Lena Frischlich

AI 507: Artificial Intelligence and Society

#5 Innovation

Associate Prof. Dr. Lena Frischlich

Flashlight

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Flashlight

Communication

complexity

episodic

contextual

inevitable

Lasswell formula

Who says what to whom in which channel and with what effect?

Speech production

Stages

Conceptualisation

Formulation

Phonological encoding

Articulation

Intentional & goal directed

Exhausting

Shortened and referential

Text processing

Fast & quasi-automatic

Stroop test

Stages

Attention & Perception

Processing of the sign

Correct interpretation

Syntactical processing of sentences

Inferential based on

Attribution of goals

Discourse schemes

Mental Model

Limited capacity model of motivated mediated message processing (LC4MP)

Communication in contexts

Communication accommodation theory, CAT

Convergence vs. divergence

Overaccomodation vs. underaccomodation

Computer-mediated communication

Cues filtered out

Media richness theory

Cues filtered in

Social Information Processing Theory (SIPT)
Hyper-personal model of computer-mediated communication
Social Identity Deindividuation Effects (SIDE)

AI-generated text: The transformer

AI: generated voice pro's & con's

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Any questions to last week?

Then let’s get started!

After today’s lecture, I hope you will…

- …have a better understanding of what an innovation is
- …have discussed social innovations along the example of a current AI-election bot in development
- …have explored the factors influencing whether an innovation will be accepted
- …have discussed recent innovations – specifically around news

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What is innovation?

How can we define innovations?

(Carayannis, 2020; Hospers, 2005; Schumpeter, 1987)

Economists suggest that innovations are changes – these can be…

- technical
- social
- organizational
- scientific

Inventions lead to “creative distruction” of the economic equilibrium

Move fast and break “things”

Joseph Schumpeter, Unknown - Österreichische Nationalbibliothek (Austrian National Library), ID: 110D3539, Gemeinfrei, https://commons.wikimedia.org/w/index.php?curid=17120276

Innovations are necessary because change is the normal state of societies and social systems → regulation requires openness to innovation

Innovations can include

- New combinations
- New organizational structures
- New methods
- New contents or qualities
- New markets

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Social innovations? (Malhotra et al., 2021; https://www.unesco.org/en/query-list/s/social-progress)

- "[…] The process of developing and deploying effective solutions to challenging and often systemic social and environmental issues in support of social progress
- not the prerogative or privilege of any organizational form or legal structure.
- Solutions often require the active collaboration of constituents across government, business, and the nonprofit world."

— Sarah A. Soule, Neil Malhotra, Bernadette Clavier, Center for Social innovation, Stanford

Social progress

= the advancement of society in terms of

- social justice
- Equality
- overall well-being.

This concept encompasses improvements in various areas such as

- Education
- Healthcare
- human rights

=> collective commitment to enhance the quality of life for all

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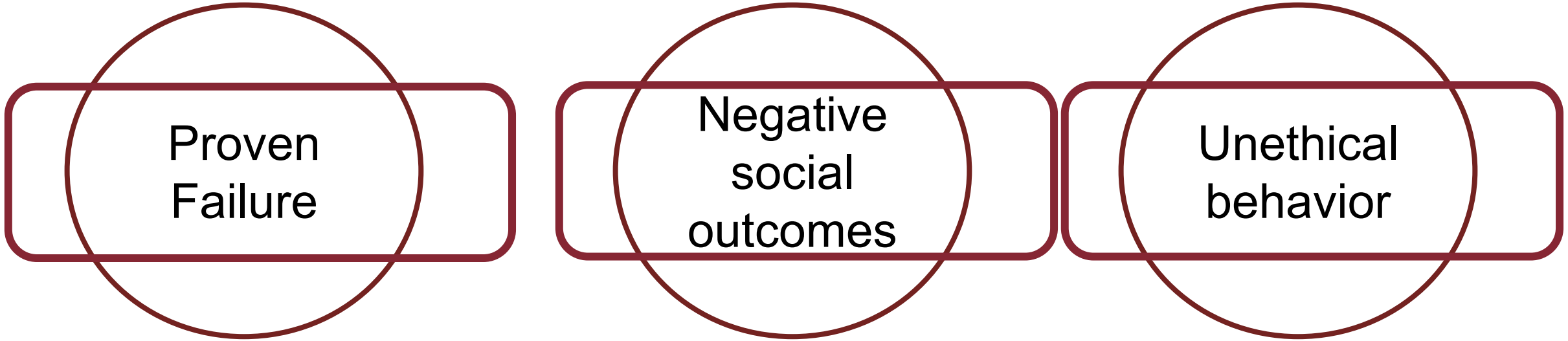
Three big NO's in (social) innovation (Malhotra et al., 2021)



- some interventions have no impact at all
- Often because the actual customer/ user/ usage case has not been studied carefully

- Well-meaning can still have severe negative consequences
- Either immediately (e.g., increased conflict/ pollution) or "downstream"

- Kant's golden rule
- Rawls "veil of ignorance"
- Utilitarianism
- Ethical principles of benevolence, non-maleficience & fairness

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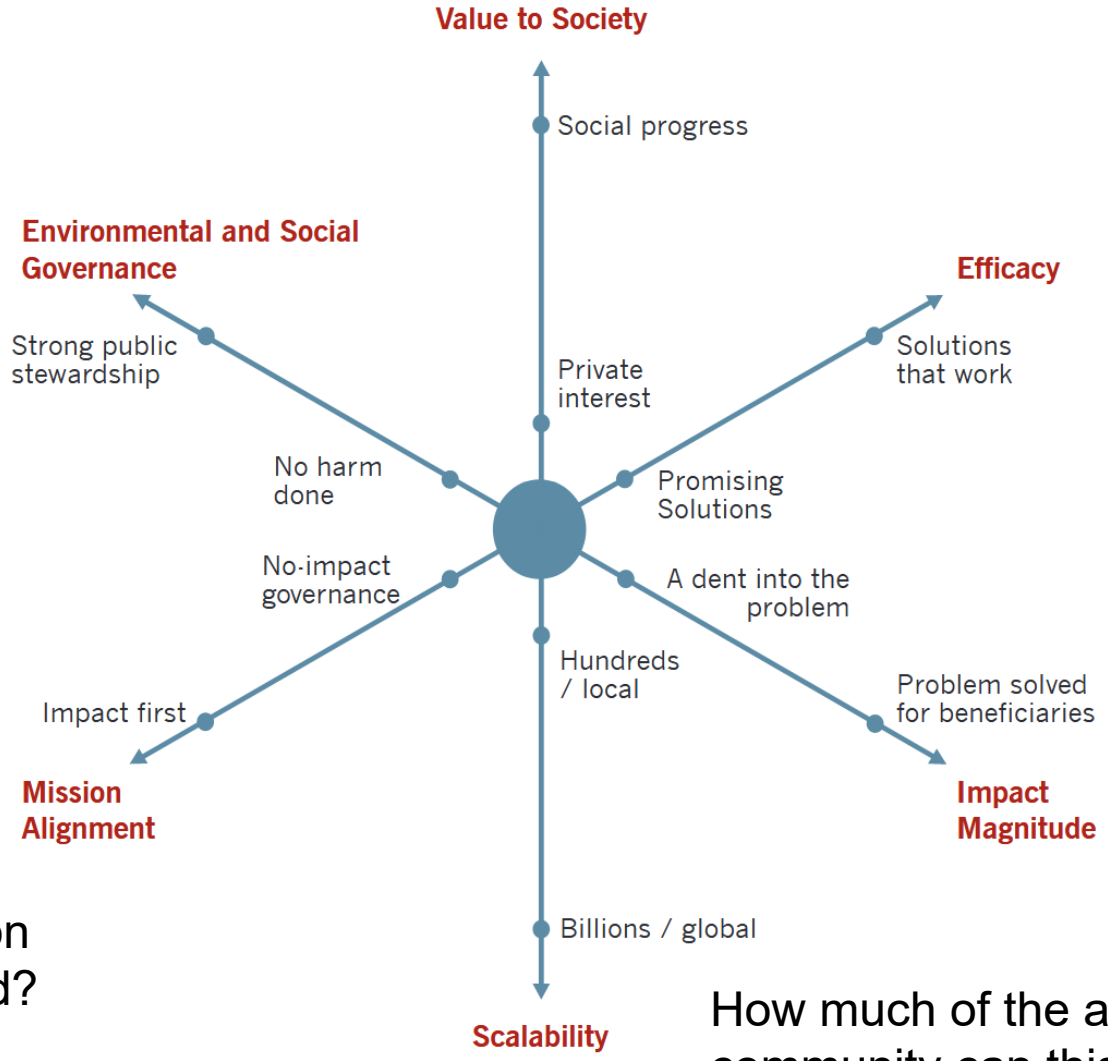
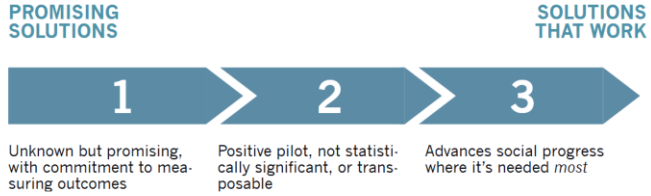
The social impact compass (Malhotra et al., 2021)

How does the intended outcome deliver societal value?



How responsible is the intervention with regard to environmental and social governance?

How certain are we of the effects of the solution?

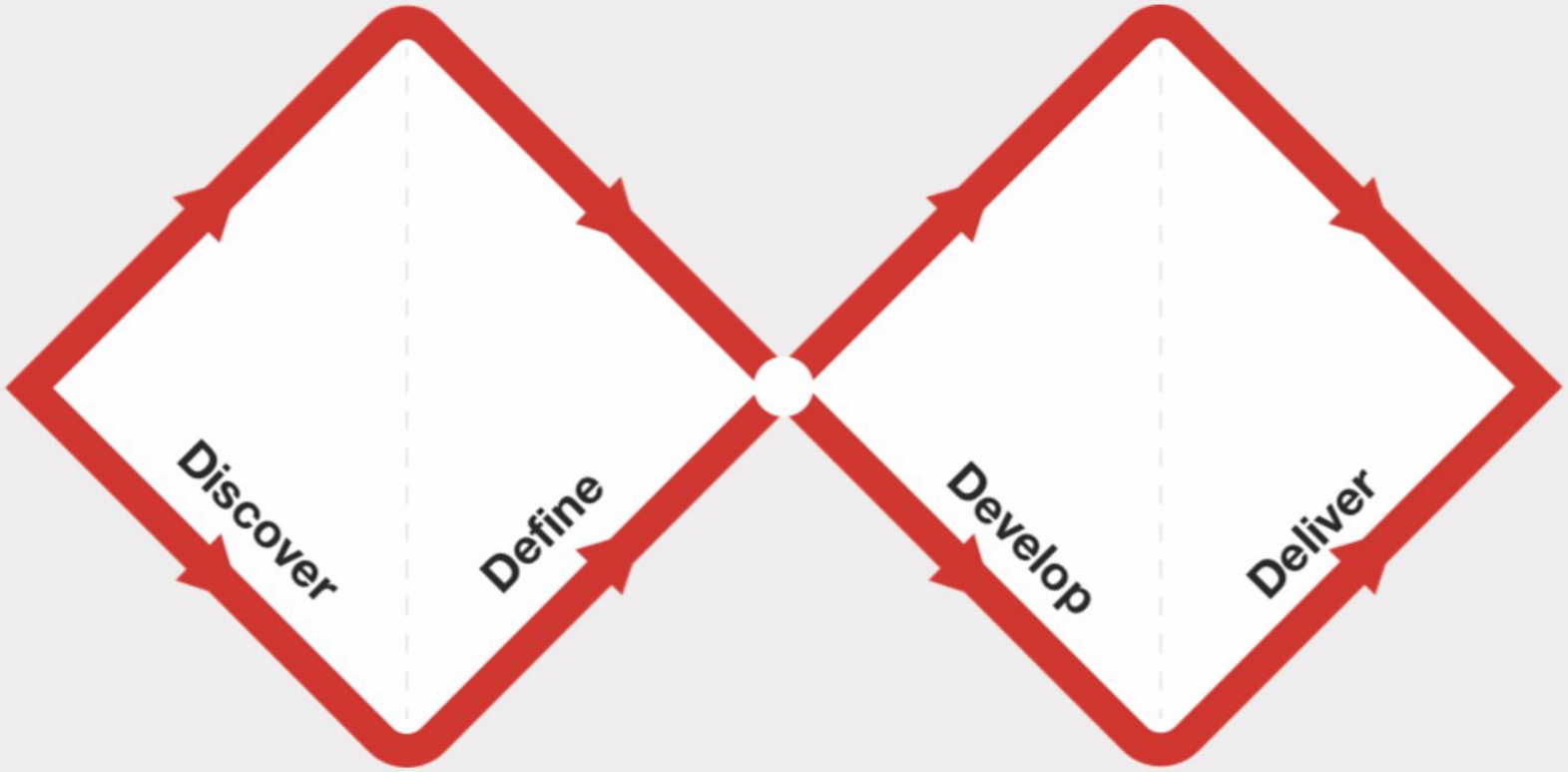


How complete is the solution for each beneficiary?

To what extent is the organization outcome oriented?

How much of the affected community can this model address?

How do you get innovative?



The Double Diamond by the Design Council is licensed under a CC BY 4.0 license.

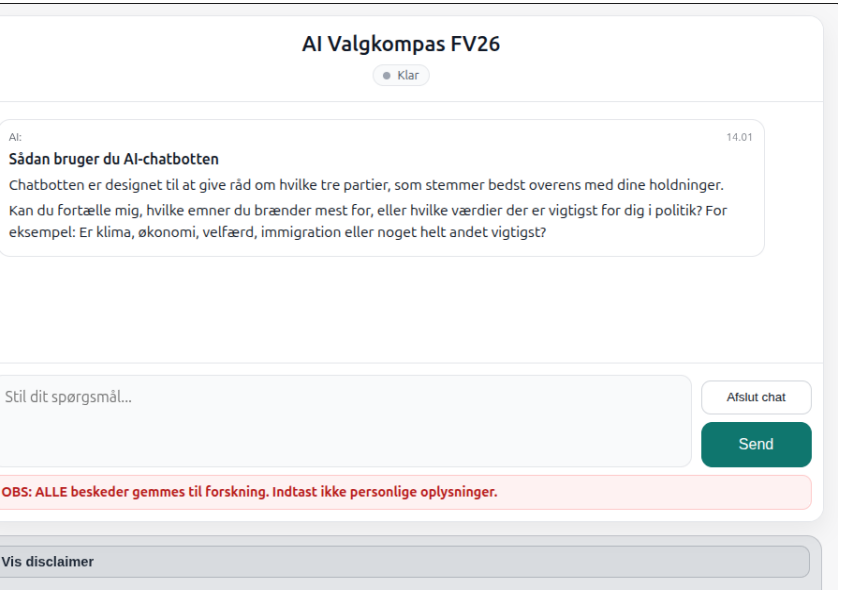
Let’s have a look at an innovation in the making

Emil Bakkensen Johansen

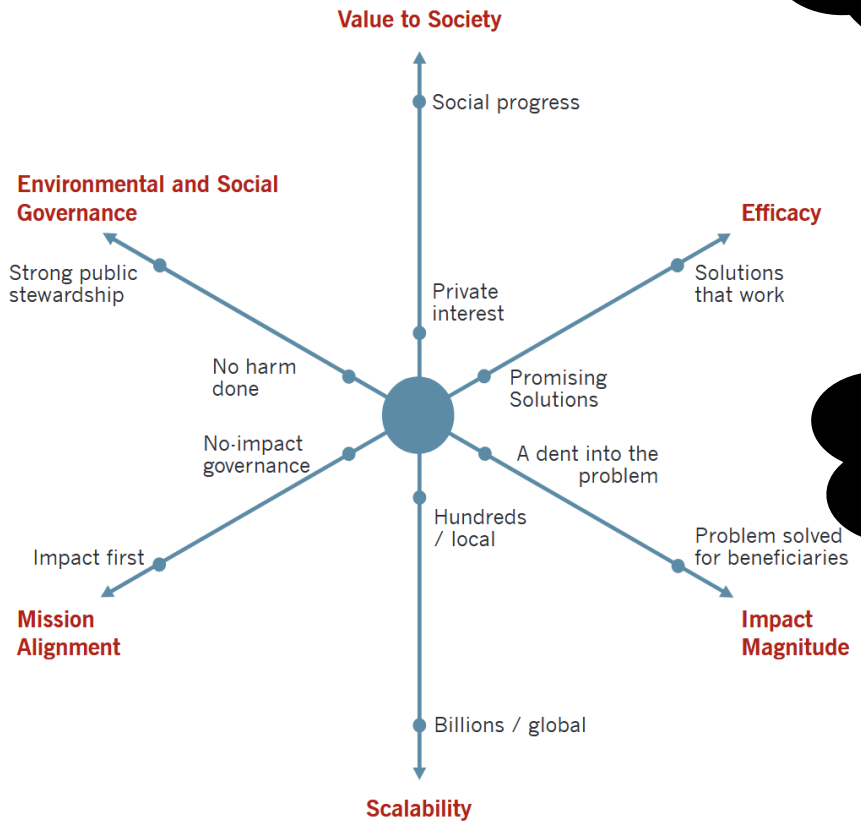
The AI Valgkompas FV26

1. **Scan the QR code or go to app-ddc-fv26.cloud.sdu.dk**
   - App is in Danish, but you can force it to English if instructed: "Hi, please respond in English..."
   - **NB: All your inputs are logged for debugging!**
   - **DO NOT USE THIS AS VOTING ADVICE – it's a beta test**

2. Use the service for 5-7 minutes, talking about the Danish parties



The AI Valgkompas FV26



Which kind of innovation is the Valgkompas?

How would you judge the Valgkompas using the social innovation compass?

Not all innovations are successful

https://time.com/4704250/most-successful-technology-tech-failures-gadgets-flops-bombs-fails/

- The silicon valley spirit often calls for failing fast as failing is one of the best ways of learning
- Some innovations which have long landed on the graveyard of nostalgic memories yet have contributed substantially to what came after them
- Some of them were sold to competitors, some failed to keep innovating, some were caught by law
- . . .

Will people use the valgkompass? Why (not)?

20. Napster

Bruno Vincent—Getty Images

19. BlackBerry

Justin Sullivan—AP

2. Virtual Boy

12. MySpace

MARY ANN CHASTAIN—AP

What are your questions?

Then its time for a break!

How do innovations spread?

Diffusion of innovations

(Everett cited from Rogers, 2005)

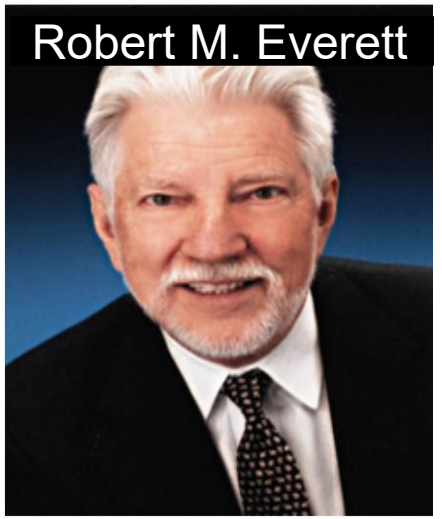


Photo Credit: Kim Jew Studio

Innovation =

- A (new) idea or practice,
- a (new) product,
- a (new) technique or technology

} that spreads through a social system

- Diffusion = process by which innovation is communicated through which channels over time between the members of a social system
- To understand this diffusion, we have to analyze
  - how innovations are perceived [→ we must ask people]
  - where this perception comes from (e.g., mass media, culture) [→ e.g., content analyses]
  - how this perception influences the acceptance (or adoption) [→ e.g., experiments/surveys]

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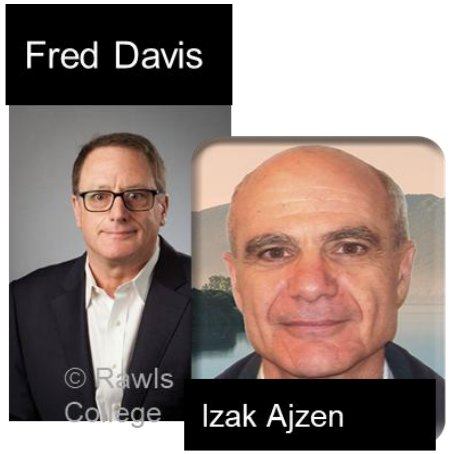
Acceptance of innovations (Ajzen & Fishbein, 1977; Ajzen, 2001; Davis, 1985)

- One broad and popular framework: Technology acceptance model by Fred Davis
- Rooted in the **theory of planned behaviour (TPB) by Ajzen & Fishbein**



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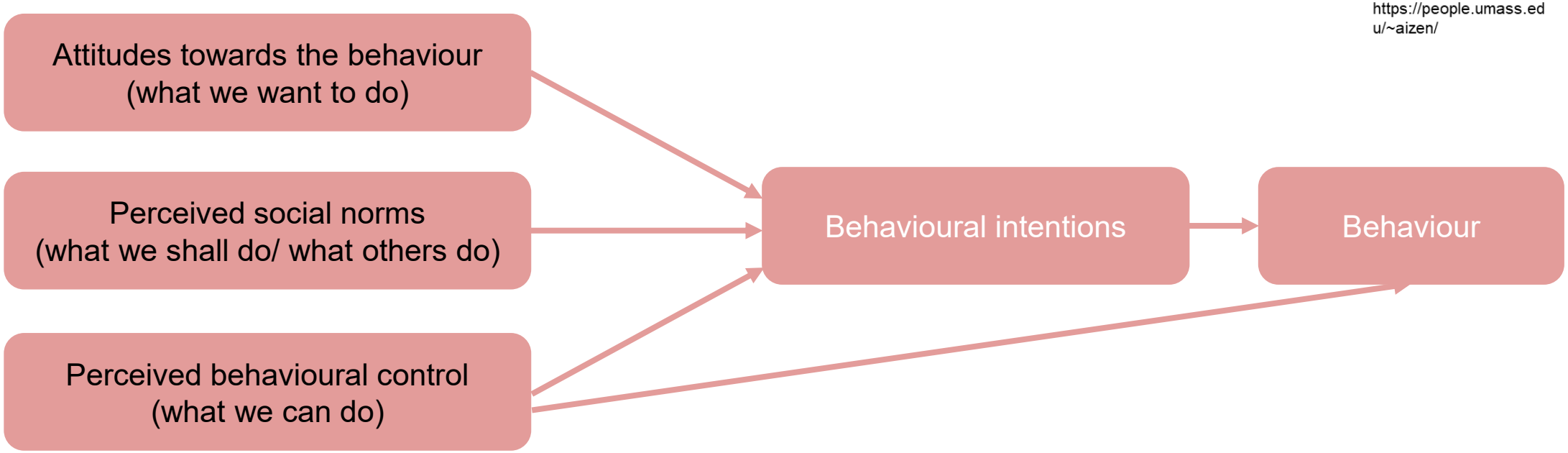

Acceptance of innovations

(Davis, 1985; Davis & Venkatesh, 1996; Feng et al., 2021; King & He, 2006)

- One broad and popular framework: Technology acceptance model by Fred Davis
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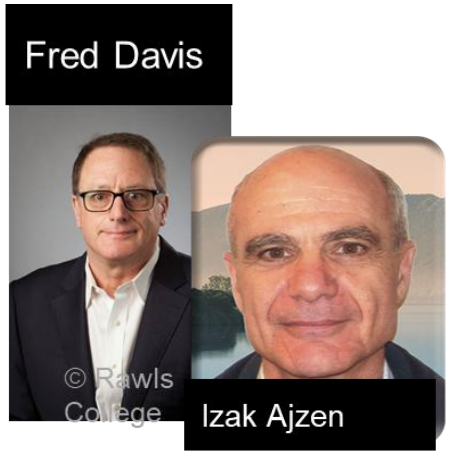

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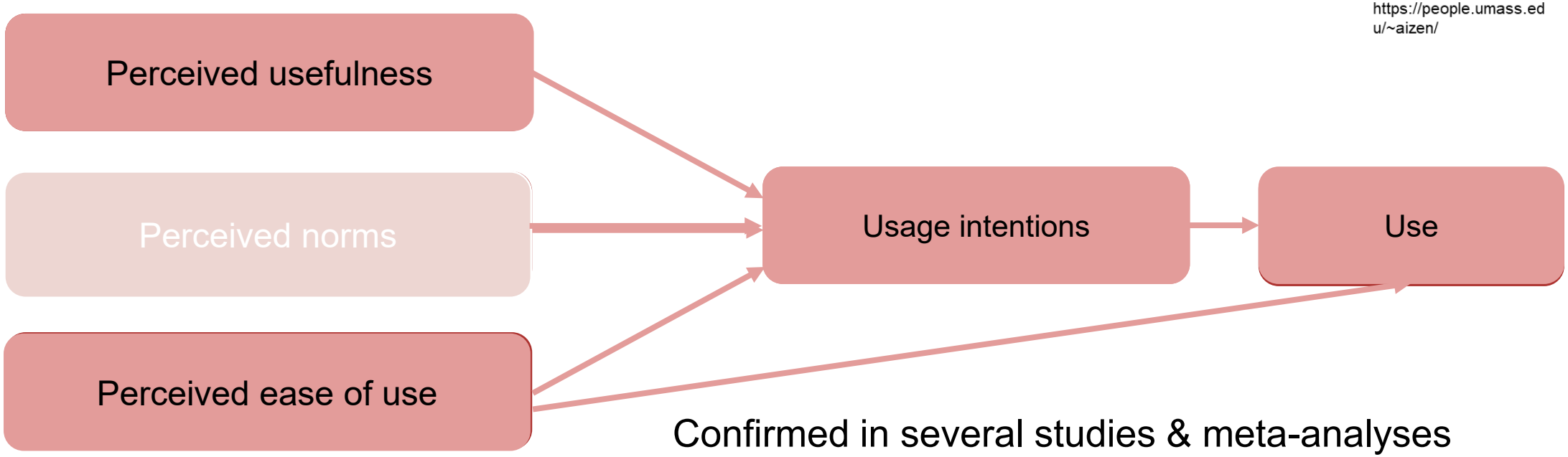

Confirmed in several studies & meta-analyses

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Extension: Unified theory of acceptance and use of technologies

Acceptance as a result of ...

- Expectations regarding the performance of a technology (is it doing what it is supposed to?)
- Expectations regarding the (necessary) effort (what do I have to do)?
- Plus: social influence factors (Should I do that? Does everyone do it?)
- New: Further facilitating factors (e.g., trust)

The role of trust (Bach et al., 2024)

- Trust = a risky action, a trustee is willing to risk an object, person, or process that he or she cannot control
- With full control, trust becomes unnecessary
- Central question: Whom and what can I trust?
- Remember the stereotype content model (→ #5 Social cognition): Two central dimensions of perception: Warmth/ Benevolence & Competence
- Competence, benevolence, and reputation → trustworthiness indicators
- Three main themes in the literature

Similarities to the dimensions described in the technology acceptance model/ theory of planned behaviour

Socio-ethical considerations

User characteristics

Technical and design features

Adaption is also a cognitive process (Vishwanath, 2015)

- Basic idea: Technology = cognitive construct that requires encoding, storage and retrieval
- New technologies can be more or less similar to other constructs in the mind (aka closer to existing “nodes”)
- If technologies are perceived to be similar to familiar or compatible with existing technologies, or related to positive concepts (my family & friends)
- More likely to be perceived as being useful
- Current studies often combine the different theories to provide a comprehensive cognitive, motivational, and communicative overview

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What would these theories say about these innovations?

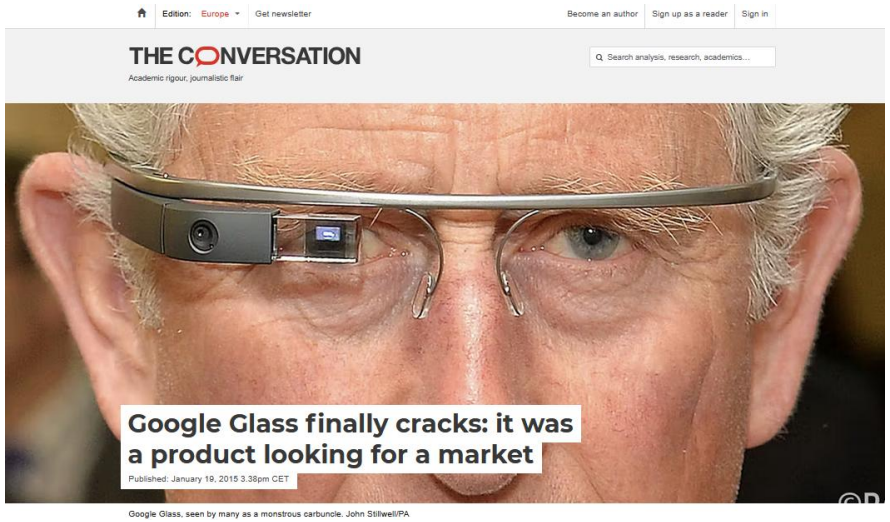
Chat GPT

- First language model based on the transformer architecture introduced in 2017
- 2019: Opening GPT-2 for the public
- 30.11.2022. GPT 3.5 gets a chat-based web-interface that is open for the public

TikTok

- Precessor Douyin was founded in 2016
- Bought the popular platform music.ly in 2018 and created TikTok as successor
- The design was adapted to match the Chinese Douyin layout

What would these theories say about these innovations?

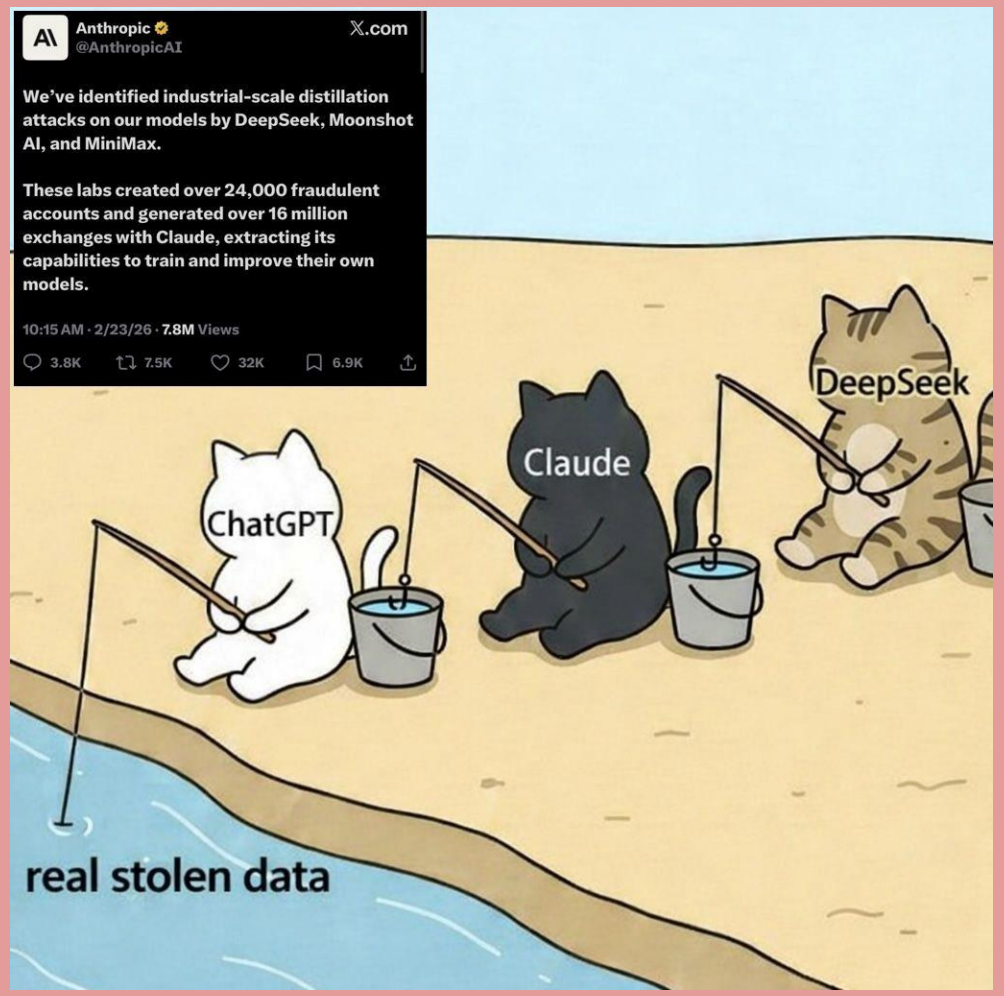


Google glasses

- Announcement 2012
- Available for developers April 2013 (1500$)
- 2014 used at the Ninjang Olympic Games
- Privacy concerns, banned in several locations before release
- Safety concerns (e.g., during driving)

https://www.meta.com/dk/ai-glasses/wayfarer/

Successful technical/ business innovations are rarely “breaking things” – they build on existing work!



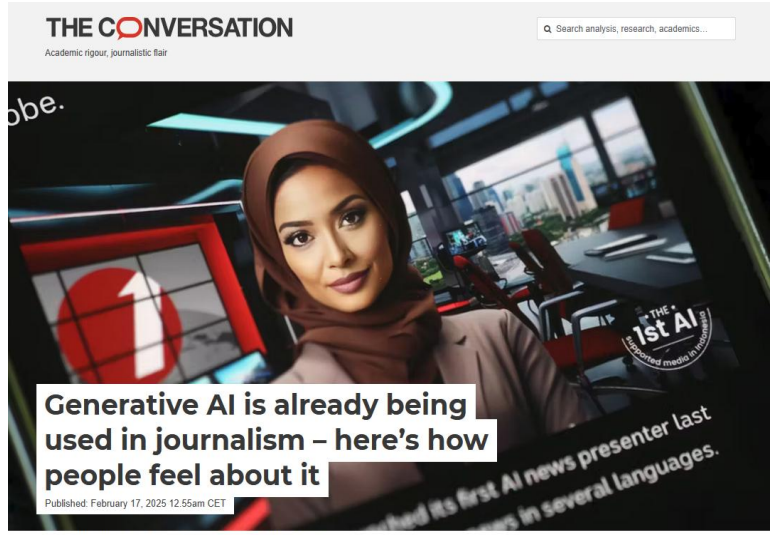
https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://www.facebook.com/sarker.rj.7/photos/ha-this-meme-lands-pretty-well-the-irony-is-real-the-cats-fishing-from-the-same-/1600271164820450/&ved=2ahUKEwj17_Lt64WTAxVaVfEDHW7aHDYQh-wKegQIGRAE&usg=AOvVaw1AP3hm-PltE8rWbpH7bVYr

What are your questions?

Example: Generative AI and journalism

News production covers – broadly – three phases: (Thurman et al., 2019; Møller, 2023; Møller et al., 2025)

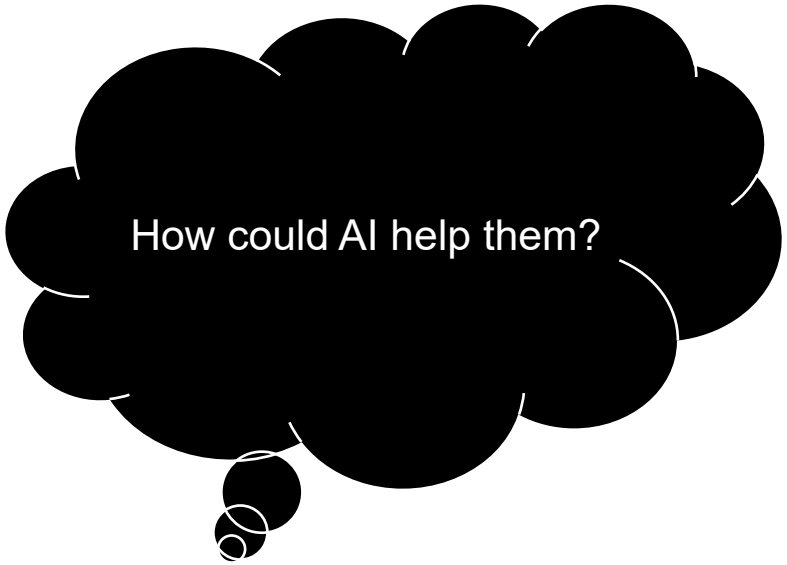
- Gathering
- Composing
- Distribution
- AI - and generative AI – has been trained on news extensively
- Also holds many promises for supporting journalists
- E.g., plays an increasing role in fine-tuning (e.g., headlines) and dissemination (e.g., through translation or adaption to different social media contexts
- Still: Many challenges remain – due to biases, hallucinations, etc.
- Can lead to erosions of trust
- So: How do journalists use AI already today?



Thomas et al., 2025

AI and news gathering (Stray, 2019)

- One central journalistic role is investigation
- Investigative journalists often work with *a lot* of data; they crawl through archives, call numbers, read tons of papers and drive to addresses to check if companies exist



- Rather seldomly used – mainly because training data is seldomly available, and the generalisability of models can be low
- Data is often not publicly available
- Accuracy must be extremely high to avoid liability
- But helpful for monitoring events, data extraction, pre-filtering and probably linking across databases

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How (not to) run AI in investigative journalism (de Lima Santos, 2022; Fridman et al., 2023; Young et al., 2018)

- Increasing relevance of data journalism and big-data analytics
- It has become more usual to rely on AI to create visualisations, verify information, analyse historical data, monitor social media and generate text
- Increasing relevance of multi-disciplinary teams → reliance on ready-available tools leads to struggles!
- Project in which news organisations could apply to get an AI professor from Oslo Met on their team for specific stories about: Petroleum reimbursement schemes/ drilling licenses, adverse events in elderly care, eating disorders in professional skiers (via computer vision), media landscape analysis in Norway (who publishes what where)
- Three central challenges:

1. Data Access
- Companies changing names
- Unstructured/missing data
- Privacy

2. Data Quality
- "real world coders"
- variation in data reporting between communities

3. Encoding newsworthiness

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AI and fact-checking (Dierickx et al., 2024)

Fact-checking includes different tasks

- Media monitoring
- Claim selection → what is checked
- Verification and evidence gathering
  (! Not “falsification” but “confirmation”!)
- Giving a verdict
- Producing the story

LLM promises

- Media monitoring
  - Data extraction, processing and organisation
- Claim selection
  - Claim identification and ranking
- Verification and evidence gathering
  - Crosse referencing claims, summaries of gathered information, data analysis
- Giving a verdict & producing the story
  - Summarizing the accuracy, providing context generating text or supporting text production

BUT

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Why fact-checking is not an AI task (Dierickx et al., 2024)

1. Hallucinations – and the lack of access to real-time data for retrieval-augmented generation
2. Source attribution errors
3. Bias in training data
4. Copyright issues and the threat of plagiarism
5. Wrongly summarized text
6. Risks of spreading misinformation that is overlooked due to the overconfidence in AI
7. Risk of reducing trust in fact-checking/ the media organisation
8. Risk of reducing human labour

Concerned about bullshitting machines that learned how to sell bullshit convincingly and a lack of laws against bullshitting

(Sandra Wachter; Oxford Iternet Institute, DGPUK 2025)

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But: People increasingly use AI (Newman et al., 2025; Waalen-Sandberg et al., 2026)

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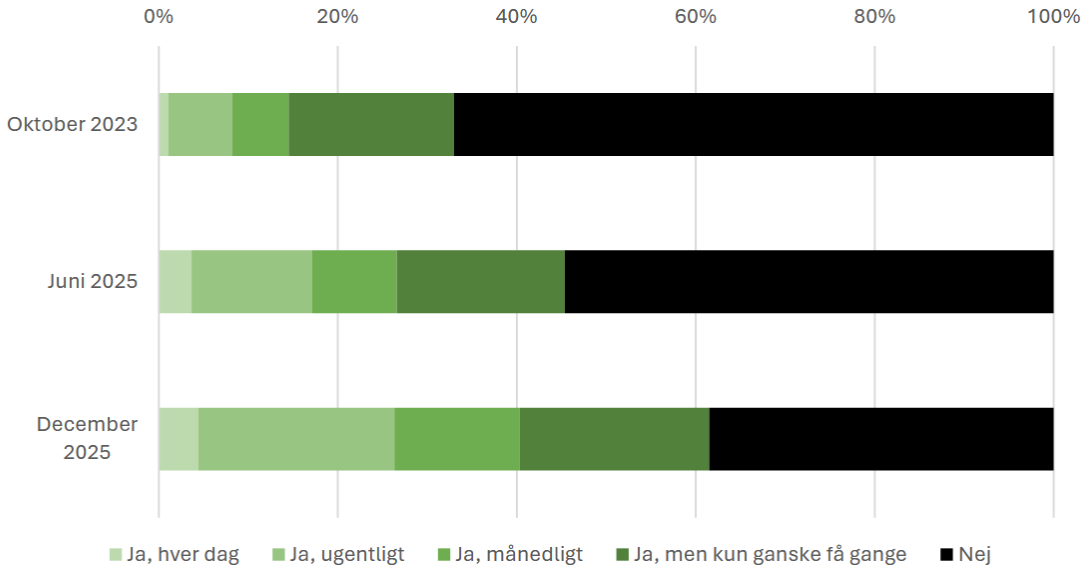
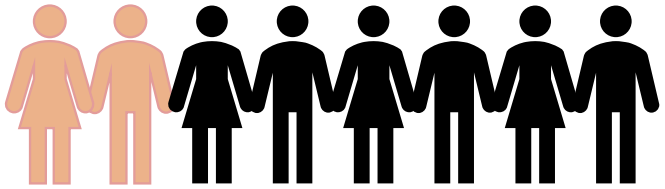
sdu.dk/DDC

Danskerne bruger mere AI, men er skeptiske over for teknologiens udbredelse i samfundet

Danskerne, digitalisering og kunstig intelligens 2022-2025

Figur 10: Andel af danskere, der har benyttet generativ kunstig intelligens i privat sammenhæng (2023-2025)





24% (across countries) now use AI weekly to get information

Still niche:

- Factual questions: 11%
- News: 6% (Japan & Argentina)
- More often: Making use easier to understand (~30%)

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What are your questions?

Then I hope you now you…

- …have a better understanding of what an innovation is
- … have discussed social innovations along the example of a current AI-election bot in development
- …have explored the factors influencing whether an innovation will be accepted
- …have discussed recent innovations – specifically around news

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Have a great rest of the day!

Associate Prof. Dr. Lena Frischlich

References

https://reutersinstitute.politics.ox.ac.uk/generative-ai-and-news-report-2025-how-people-think-about-ais-role-journalism-and-society

https://www.unesco.org/en/query-list/s/social-progress

https://www.designcouncil.org.uk/our-resources/the-double-diamond/

https://time.com/4704250/most-successful-technology-tech-failures-gadgets-flops-bombs-fails/

Ajzen, I., & Fishbein, M. (1977). Attitude-behavior relations: A theoretical analysis and review of empirical research. Psychological Bulletin, 84(5), 888–918. https://doi.org/10/c5d

Ajzen, I. (2001). Nature and operation of attitudes. Annual Review of Psychology, 52(1), 27–58. https://doi.org/10/cqp7jm

Bach, T. A., Khan, A., Hallock, H., Beltrão, G., & Sousa, S. (2024). A systematic literature review of user trust in AI-enabled systems: An HCI perspective. International Journal of Human–Computer Interaction, 40(5), 1251–1266. https://doi.org/10.1080/10447318.2022.2138826

Blöbaum, B. (2016). Preface. In B. Blöbaum (Ed.), Trust and communication in a digitized world: Models and concepts of trust research (pp. 75–90). Springer International Publishing. https://doi.org/10.1007/978-3-319-28059-2_4

Carayannis, E. G. (Ed.). (2020). Encyclopedia of Creativity, Invention, Innovation and Entrepreneurship. Springer International Publishing. https://doi.org/10.1007/978-3-319-15347-6

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References

Davis, F. (1985). A technology acceptance model for empirically testing new end-user information systems: Theory and results. Massachsetts Institute of Technology.

Davis, F. (1985). A technology acceptance model for empirically testing new end-user information systems: Theory and results. Massachsetts Institute of Technology.

Davis, F. D., & Venkatesh, V. (1996). A critical assessment of potential measurement biases in the technology acceptance model: Three experiments. International Journal of Human-Computer Studies, 45(1), 19–45. https://doi.org/10.1006/ijhc.1996.0040

de-Lima-Santos, M.-F. (2022). ProPublica's data journalism: How multidisciplinary teams and hybrid profiles create impactful data stories. Media and Communication, 10(1), 5–15. https://doi.org/10.17645/mac.v10i1.4433

Dierickx, L., van Dalen, A., Opdahl, A. L., & Linden, C.-G. (2024). Striking the balance in using LLMs for fact-checking: A narrative literature review. MISDOOM 2024, 1–15. https://link.springer.com/chapter/10.1007/978-3-031-71210-4_1

Dierickx, L., van Dalen, A., Opdahl, A. L., & Linden, C.-G. (2024). Striking the balance in using LLMs for fact-checking: A narrative literature review. MISDOOM 2024, 1–15. https://link.springer.com/chapter/10.1007/978-3-031-71210-4_1

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References

Feng, G. C., Su, X., Lin, Z., He, Y., Luo, N., & Zhang, Y. (2021). Determinants of technology acceptance: Two model-based meta-analytic reviews. Journalism & Mass Communication Quarterly, 98(1), 83–104. https://doi.org/10.1177/1077699020952400

Fridman, M., Krøvel ,R., & and Palumbo, F. (2023). How (not to) run an ai project in investigative journalism. Journalism Practice, 0(0), 1–18. https://doi.org/10.1080/17512786.2023.2253797

Hospers, G.-J. (2005). Joseph Schumpeter and his legacy in innovation studies. Knowledge, Technology & Policy, 18(3), 20–37. https://doi.org/10.1007/s12130-005-1003-1

King, W. R., & He, J. (2006). A meta-analysis of the technology acceptance model. Information & Management, 43(6), 740–755. https://doi.org/10.1016/j.im.2006.05.003

Malhotra, N., Clavier, B., & Bailard, E. (n.d.). The Impact Compass. Retrieved 4 March 2026, from https://www.gsb.stanford.edu/sites/gsb/files/whitepaper-csi-impact-compass.pdf

Møller, L. A. (2023). Bridging the tech-editorial gap: Lessons from two case studies of the development and integration of algorithmic curation in journalism. Journalism Studies, 24(11), 1458–1475. https://doi.org/10.1080/1461670X.2023.2227283

Møller, L. A., Skovsgaard, M., & De Vreese, C. (2025). Reinforce, readjust, reclaim: How artificial intelligence impacts journalism's professional claim. Journalism, 26(7), 1373–1390. https://doi.org/10.1177/14648849241269300

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References

Rogers, E. M. (2005). Diffusion of innovation (5. Aufl.). Free Press.

Schumpeter, J. (1987). Theorie der wirtschaftlichen Entwicklung [Theory of economic development] (7. Edition.). Duncker & Humboldt.

Stray, J. (2019). Making artificial intelligence work for investigative journalism. Digital Journalism, 7(8), 1076–1097. https://doi.org/10.1080/21670811.2019.1630289

Thurman, N., Lewis, S. C., & Kunert, J. (2019). Algorithms, automation, and news. Digital Journalism, 7(8), 980–992. https://doi.org/10.1080/21670811.2019.1685395

Thomson, T. J., Thomas, R., Riedlinger, M., & Matich, P. (2025). Generative AI and Journalism: Content, Journalistic Perceptions, and Audience Experiences (p. 13751357 Bytes). figshare. https://doi.org/10.6084/M9.FIGSHARE.28068008

Venkatesh, Morris, Davis, & Davis. (2003). User acceptance of information technology: Toward a unified view. MIS Quarterly, 27(3), 425. https://doi.org/10.2307/30036540

Vishwanath, A. (2015). The psychology of the diffusion and acceptance of technology. In S. S. Sundar (Hrsg.), The Handbook of the Psychology of Communication Technology (1. Aufl., S. 311–331). Wiley. https://doi.org/10.1002/9781118426456.ch14

Young, M. L., Hermida, A., & Fulda, J. (2018). What makes for great data journalism?: A content analysis of data journalism awards finalists 2012–2015. Journalism Practice, 12(1), 115–135. https://doi.org/10.1080/17512786.2016.1270171

Waalen Sandberg, T. W., Filstrup, S. H., de Vreese, C. H., & van Dalen, A. (2026). Danskerne bruger mere AI, men er skeptiske over for teknologiens udbredelse i samfundet: Danskerne, digitalisering og kunstig intelligens 2022-2025

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AI 507: Artificial Intelligence and Society

#6 Human-Robot Interaction

Associate Prof. Dr. Lena Frischlich

Flashlight

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Flashlight

Innovation

Economic perspectives

Social innovation

Big No's: Uneffective, harmful, unethical

Social innovation compass

Value to Society

Efficacy

Impact magnitude

Scalability

Mission alignment

Social/ environmental consequences

Failed innovations

AI Valgkompas

Dissemination of innovation

Diffusion of Innovations

Theory of planned behaviour

Attitudes ("will")

Perceived norms ("should")

Behavioural control ("can")

Technology acceptance model

Perceived usefulness

Perceived ease of use

Unified technology acceptance model

Trust

Cognitive similarity

Generative AI & Journalism

Stages of media work

News gathering

News composition

News distribution

AI + investigative journalism

AI + fact-checking

AI for news in Denmark & beyond

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Questions: Send me an itslearning message

Then let’s get started!

After today’s video lecture, you will hopefully…

- ❑ …have initial insights into human’s perceptions of artificial agents
- ❑ …are familiar with the uncanny valley effect
- ❑ …have learned more about human-robot relationships

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Robots are based on automation

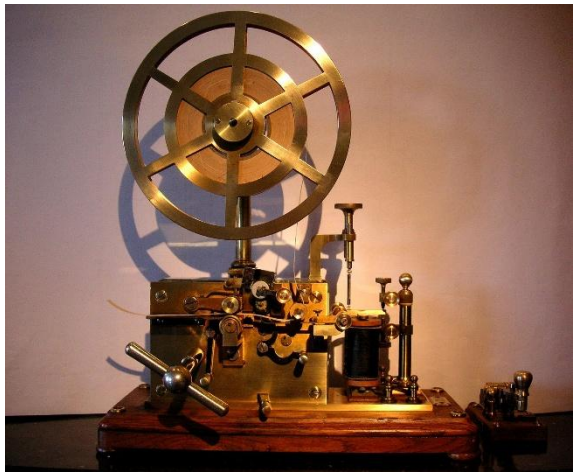
Automation

= Transfer of functions of the production process from humans to artificial systems

Comprises:

- *Mechanization,* in which machines provide the energy for the production process, but humans act (e.g. "Chess Turk")
- *Machinization*, in which machines (partially) take over the process (e.g., telegraphy))
- Today: Machines take over control and regulation tasks (usually through certain feedback loops))
- Increasing level of autonomy

Kupferstich The Turk Racknitz (1789) via Wikimedia



Ericsson Telegraf Serie 800, ca.1880 Holger Ellgaard via Wikimedia

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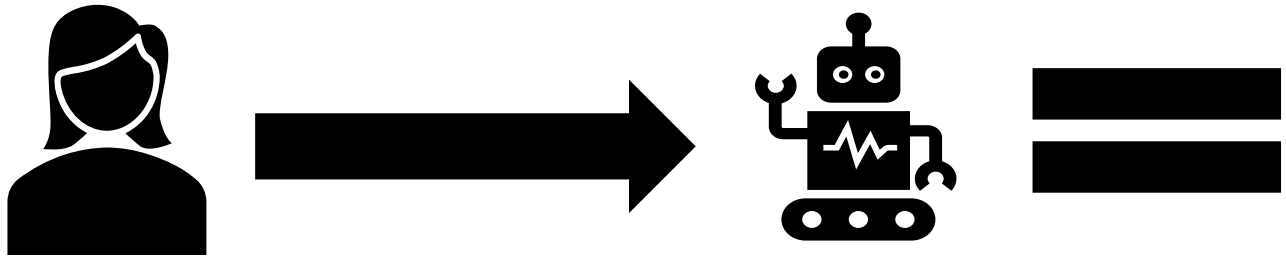
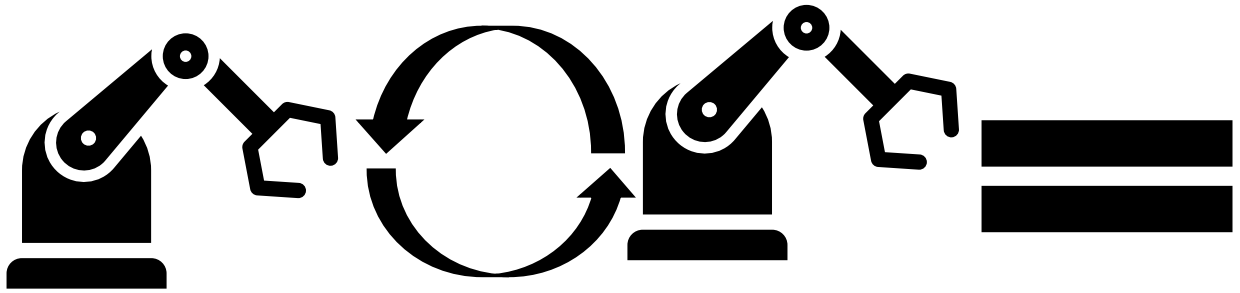
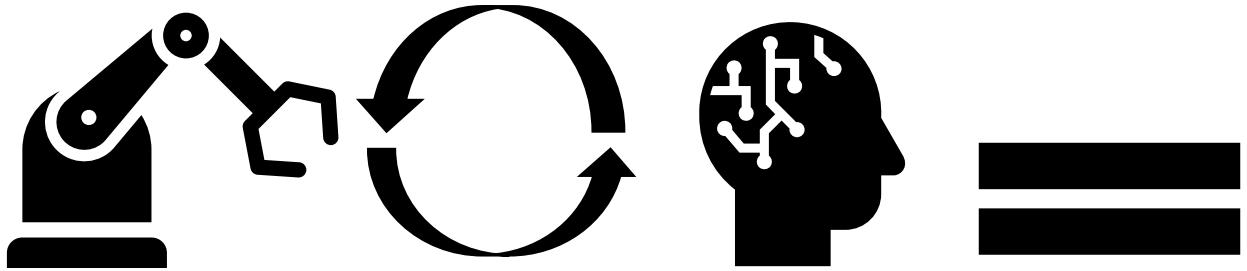
This is also reflected in central terms



Avatar



Agent



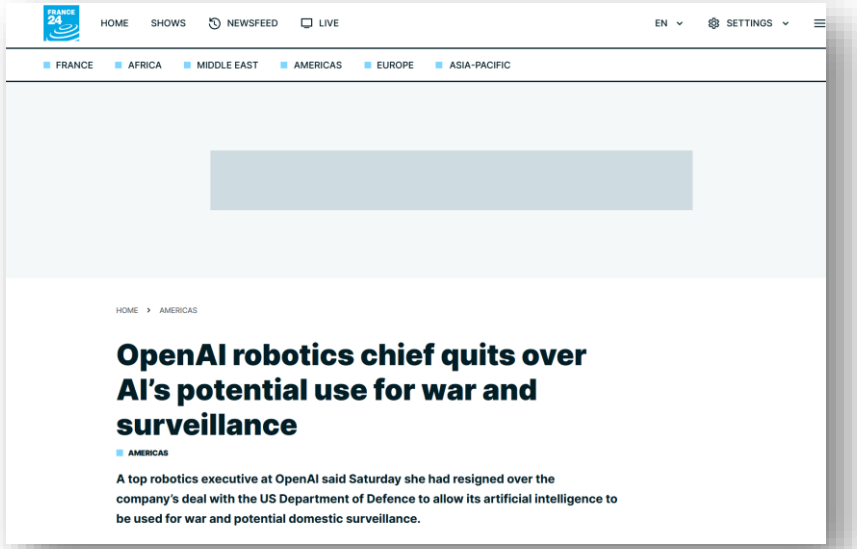
Embodied/
anthromorph agent

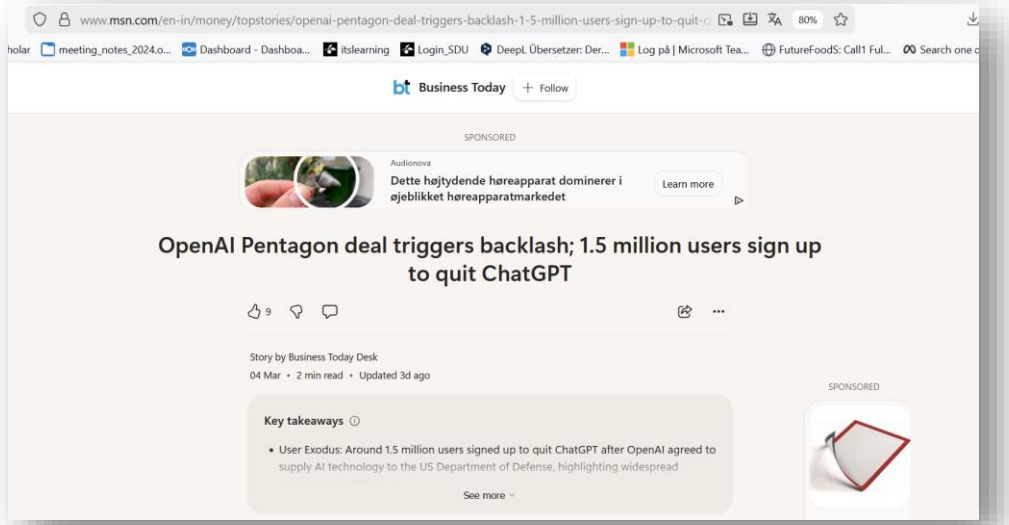
One central area of debate – autonomous weapon systems

https://www.youtube.com/watch?v=VqCMkXftDx8

Conflict about agentic warfare

- Since 2018, UN Secretary-General António Guterres argues that lethal autonomous weapons systems are unacceptable and morally repugnant
- UN calls for their prohibition under international law or at least for augmentation (avatar) rather than replacement (agent)
- At the heart of the Anthropic vs Department of War controversy





The dream of the artificial other is old

From Ancient Greek sculpturer Pygmalion who falls in love with his statue and brings her to live

https://de.wikipedia.org/wiki/Pygmalion#/media/Datei:Pygmalion_(Baur).jpeg

Over Jewish myths of the Golem created to protect

https://en.wikipedia.org/wiki/Golem

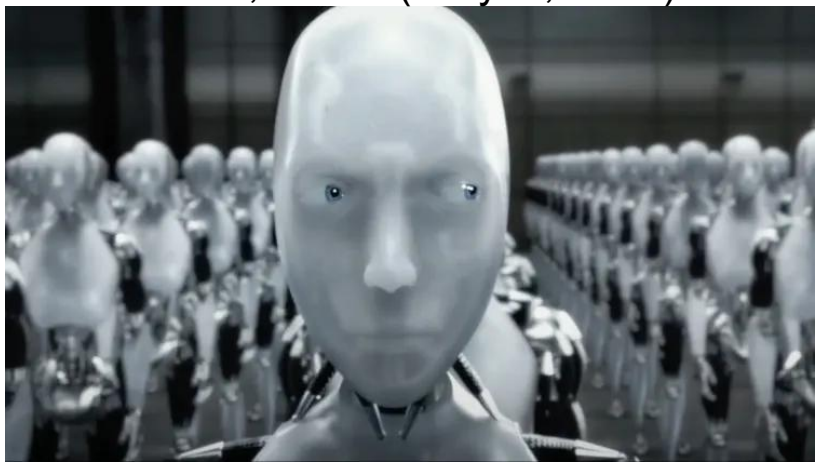
To the horror of Frankenstein…

The fathers of robotics

(Asimov, 1994; KE-Next, 2023)

- In 1942, the Russian sci-fi writer Isaak Asimov created the iconic „three laws of robots“ inspiring stories til today
  - A robot must not injure a human being or cause harm through inaction
  - A robot must obey the commands of humans – as long as you are not in conflict with the first law
  - A robot must protect its own existence as long as this protection does not contradict the first or second law
- Joseph Engelberger read this story as a boy and started dreaming of his personal robots since then
- Together with George Devol, who had patented a machine control process in 1954, Engelberger founded the first robotics company (Unimation)

Scene from I, Robot (Proyas, 2004)



https://www.fastcompany.com/91152591/i-robot-will-smith-ai-20-years-later

I'm Your Man (Schrader, 2021)

https://www.rottentomatoes.com/m/im_your_man_2021

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And today?

(Richter, 2023; https://investinodense.dk/robotics-automation/)

3.5 million
Industrial robots
worldwide
-more than ever
before

5 key trends (according to the International Federation of Robotics))

1. Increasingly easy to use.
2. Artificial Intelligence (Smartification)
3. Relocation of production to Europe/the USA
4. Energy efficiency (e.g. lower heating costs but also recovery of kinetic energy)
5. Sustainability and Prepare-to-Repair.

Odense is at the forefront of these developments with over 160 companies working in the industry

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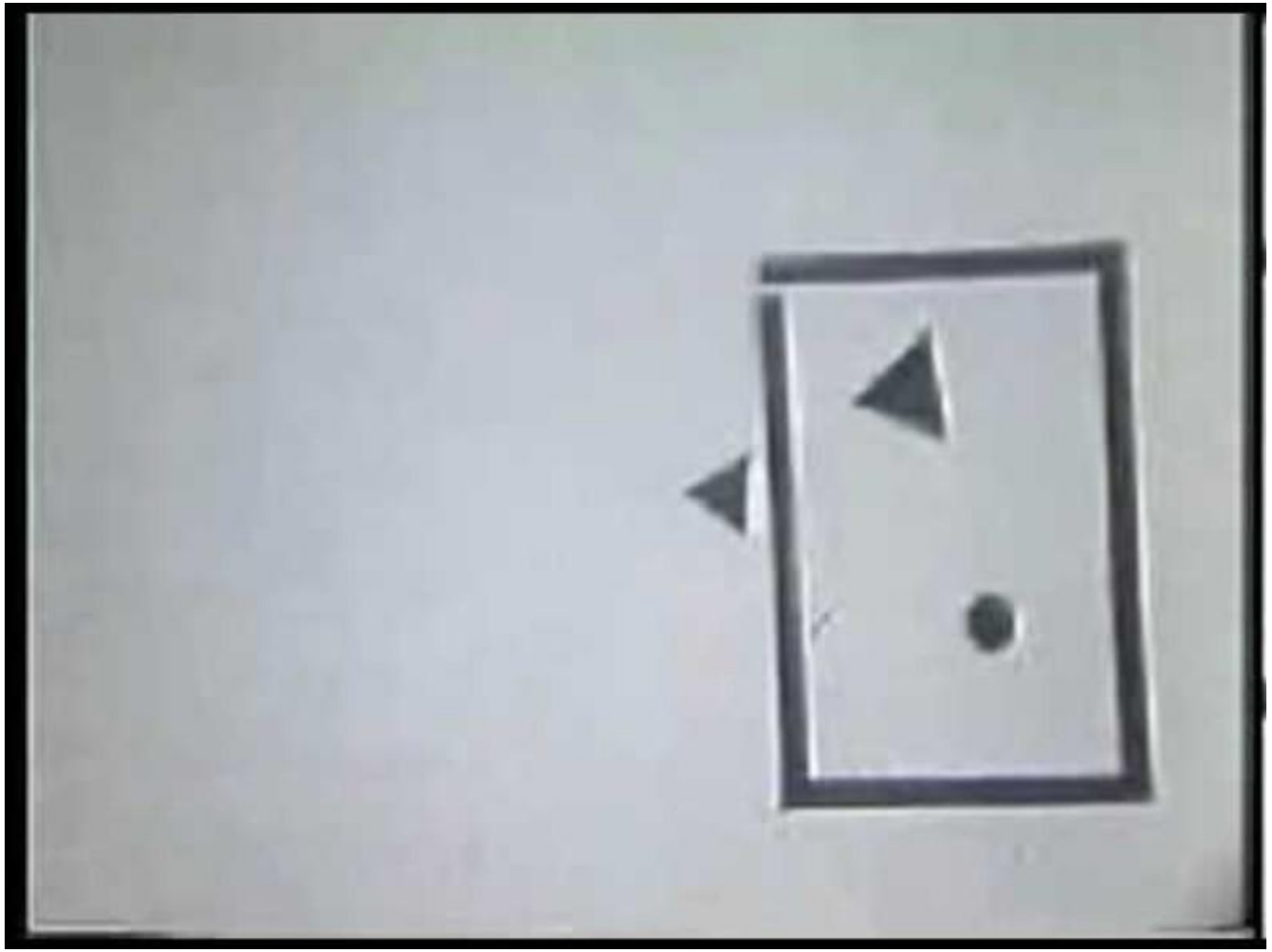
How do humans perceive artificial others?

What happens here?



Haider & Simmel, 1944, https://www.youtube.com/watch?v=VTNmLt7QX8E

- Remember social cognition? Social perception is our „default“ activity
- We *attribute* intention to independently moving objects
- This also extends to automated actors

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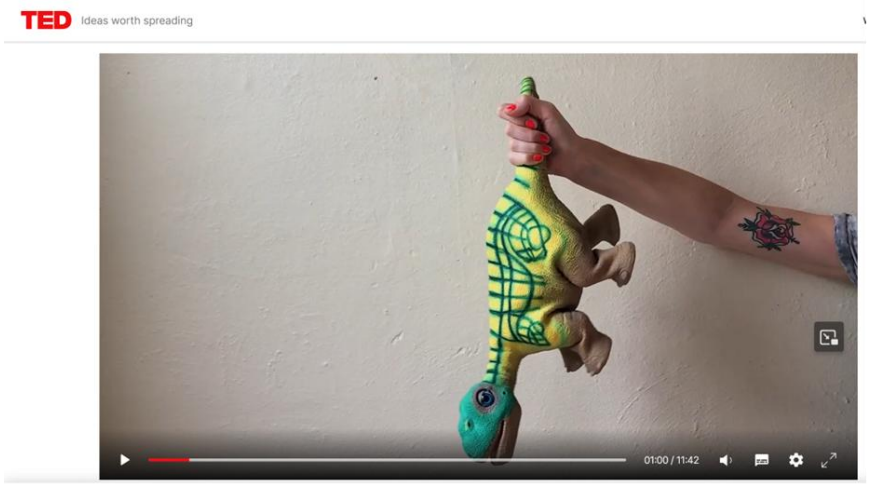
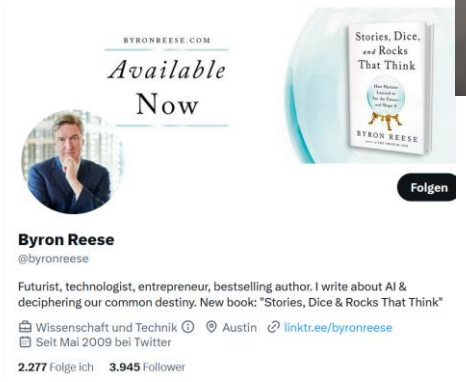
The media equation (Reeves & Nass, 1996; Shechtman & Horowitz, 2003)

- Basic idea – Humans treat computers (and also robot dinosaurs) like other social actors.
- Tested in various so-called “CASA” experiments (Computers-as-Social-Actors paradigm).
- Basic logic: If people show the same social behaviours towards computers as towards humans, then they perceive computers as social actors
- Diverse experiments show, f. ex., that we are polite towards robots – but also that we feel obliged to respond to robots’ politeness



https://www.ted.com/talks/kate_darling_why_we_have_an_emotional_connection_to_robots

Clifford Nass via Wikipedia



Extensions of CASA (Blascovich, 2002; Krämer & van der Pütten, 2015; Nass & Moon, 2000)

Ethopoeia Approach

- Minimal social cues are sufficient for social perceptions
- E.g. smiling back

Threshold model of social influence (TMSI)

- Two key factors.
  - Agency
  - Behavioural realism
- If agency is high (= agent): behavioural realism is not so important
- If agency is low (= avatar): behavioural realism is key
- Later studies showed that behavioural realism and social cues matter (more is better)

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Let’s look at one example study (Shi et al. 2015)

- Deciding how to approach someone and open the conversation is difficult for humans
- But it is a real challenge for robots.
  - Do I speak to a customer?
  - How close can I get
  - When do I start talking?
- Observations showed: Sales assistants approached customers when customers were looking in their direction and were close to a product (→ spatial position)
- At ~70 cm, the conversation was opened
- Overall: More positive evaluation than "general greetings" or "always stand by the product and explain"

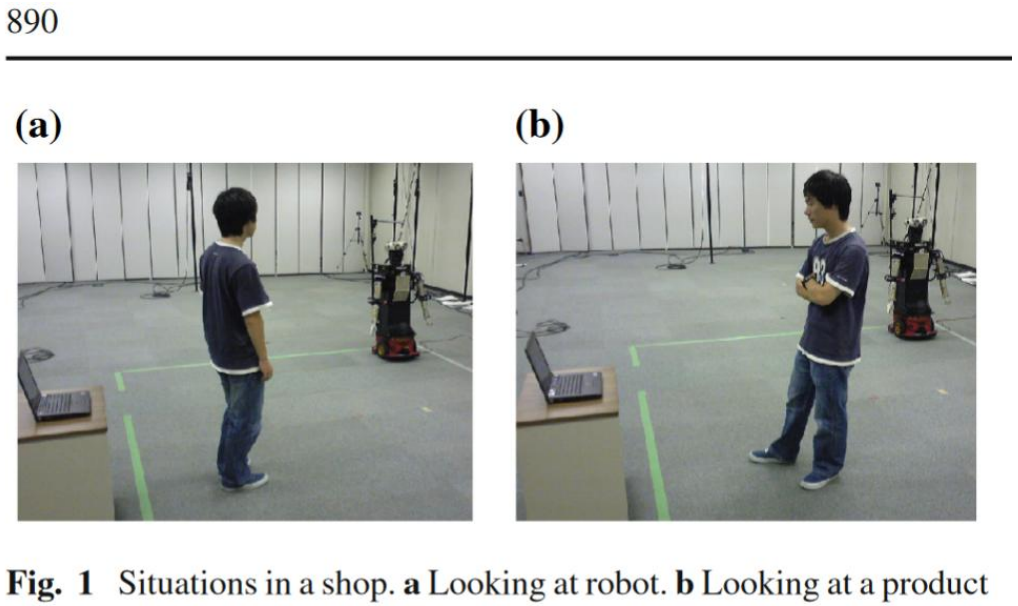

Fig. 1 Situations in a shop. **a** Looking at robot. **b** Looking at a product

https://www.tiktok.com/@kings_sushi/video/7278626000295005473

But there are limits: The uncanny valley effect

(Mori, 1970; Mara et al., 2022)

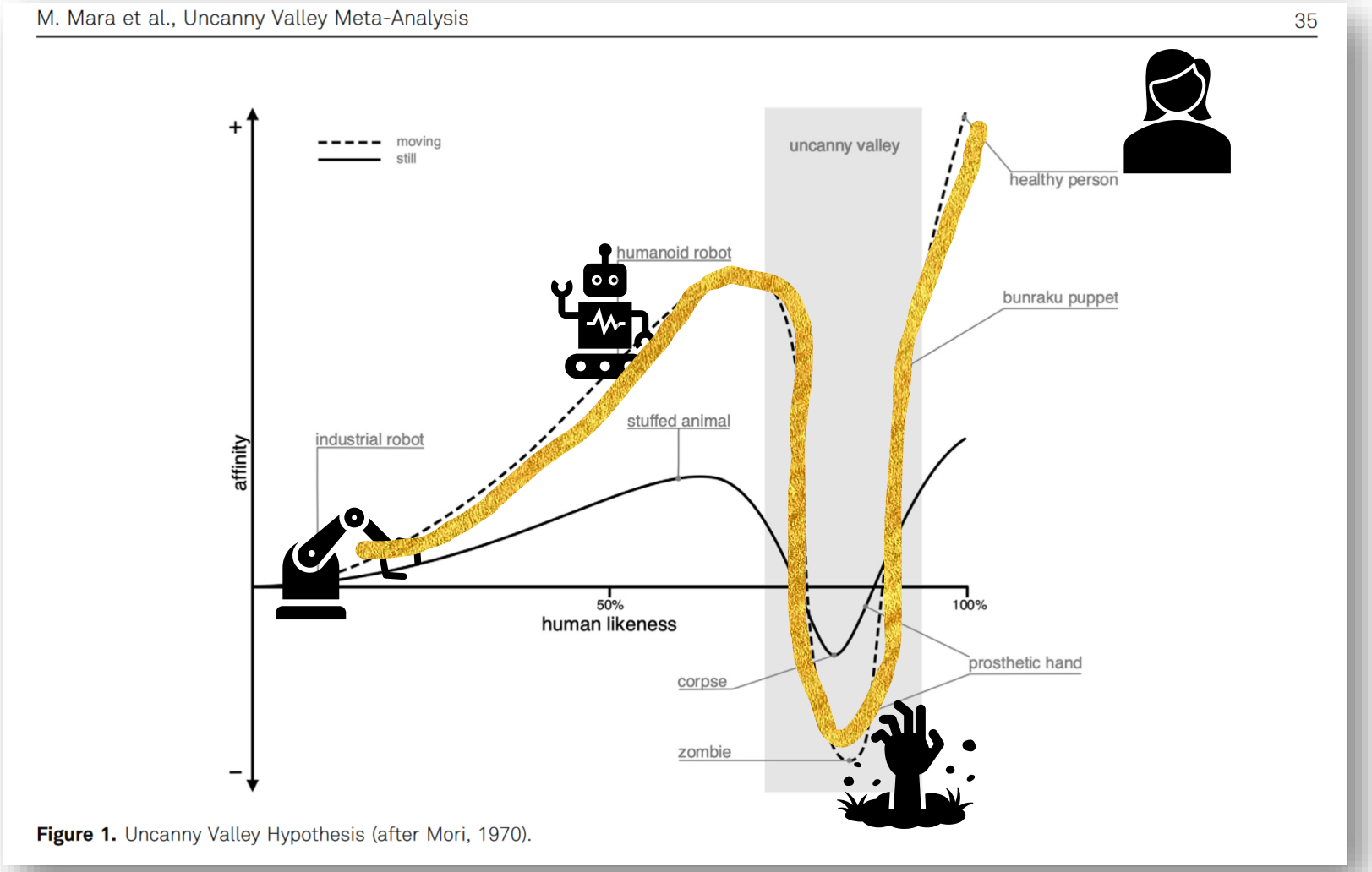

Figure 1. Uncanny Valley Hypothesis (after Mori, 1970).

Masahiro Mori

Mounting evidence (Appel et al. 2020; Mara et al., 2022)

- 49 studies with 131 reviews of robots
- Most robots are rated as moderately human-like
- Most reviews are moderately positive
- Confirmed: Static robots are perceived as less human-like
- Communicating robots are rated more positively
- More similarity, better ratings
- BUT– no statements can (yet) be made about the very human-like range
- Possibly: Rather, the perceived "consciousness" of the robot (attribution of experiencing & thinking) that triggers the effect

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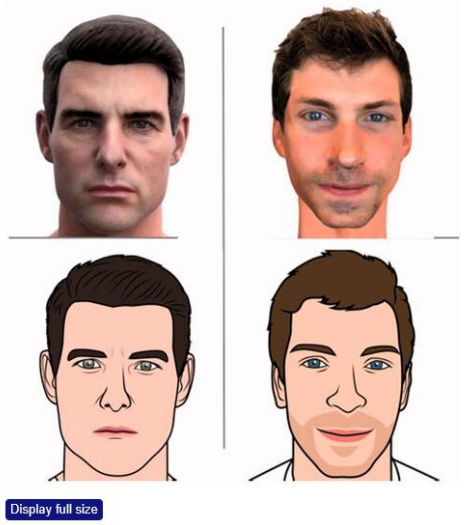
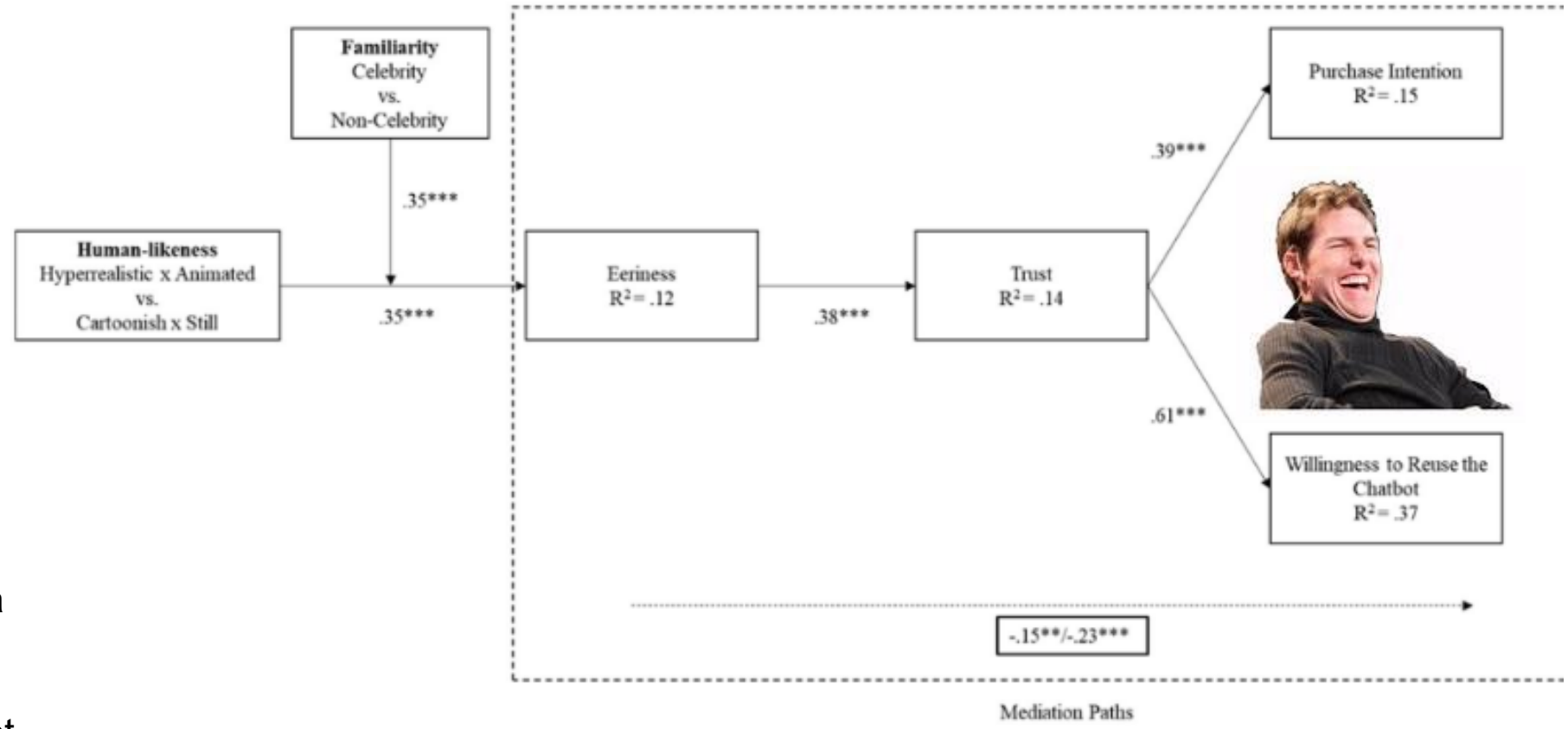
How to overcome the uncanny valley effect





- 187 students from the Netherlands
- Chatbot to support buying a laptop
- Dependent variables: eeriness, trust in the chatbot, intentions to use the bot
- Intentions to buy the laptop

How to overcome the uncanny valley effect?

(Grundke et al. 2023)

Imagine a new robot, Ellix.

Elix is a tool with arms and hands and can grab and carry things

Elix supports people in everyday life and can execute user commands

Imagine a new robot, Ellix.

Elix is a tool with arms and hands and can grab and carry things

Elix can feel hunger, pain and pleasure and has consciousness and personality

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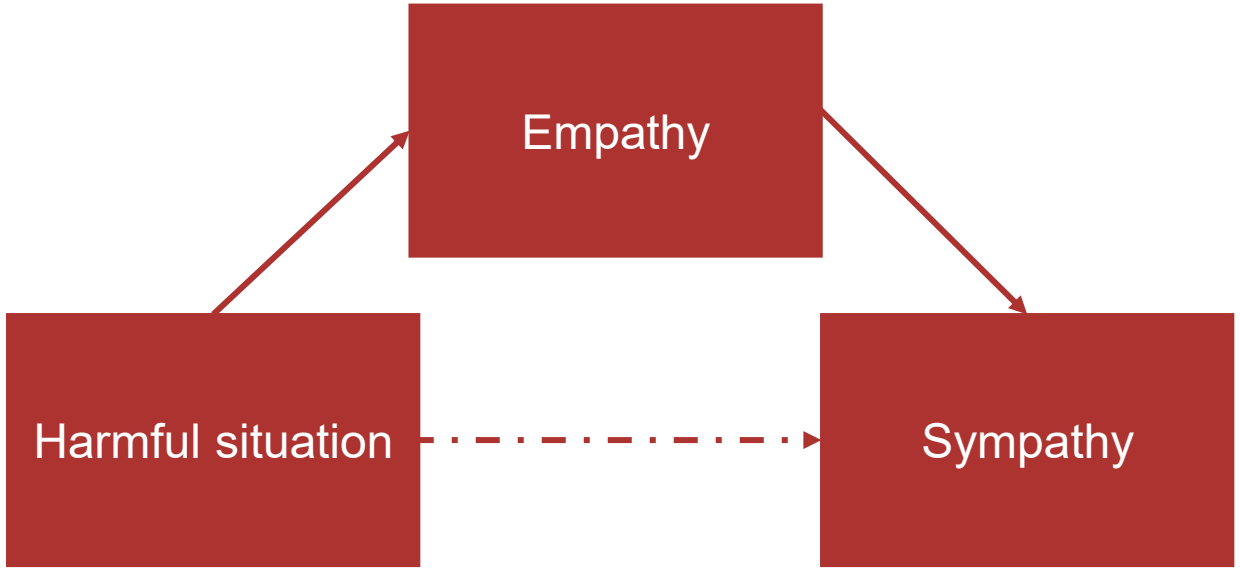
In the next step, the subjects watched a video of Ellix(Grundke et al. 2023)

The robot tries to lift a box and is stopped by an adult with a hockey stick, who constantly pushes the box to the side

At the end of the video, the adult Ellix falls to the ground with the hockey stick

Ellix remains lying on his back

What do you think?



However, not all emotions are perceived as equal (Craig & Edwards, 2021)

- *N* = 298 people watched one out of three videos on M-Turk in which a robot reports about another robot who just joined his lab
  a) Robot is sad that there is a new robot in the lab
  b) Robot reports without emotions
  c) Robot is happy about the new robot in the lab
- Afterwards: Rating of the credibility and intelligence of the robot
- The sad robot is rated as less intelligent and credible

Figure 1

Sorrowful Condition

Figure 2

No Emotion Condition

Figure 3

Joyful Condition

Via https://osf.io/r8t63

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What we “tolerate” in artificial others remains rooted in the same patterns

The dream of the artificial other is old

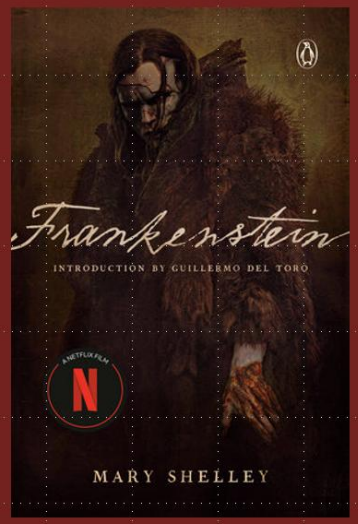
From Ancient Greek sculpturer Pygmalion who falls in love with his statue and brings her to live

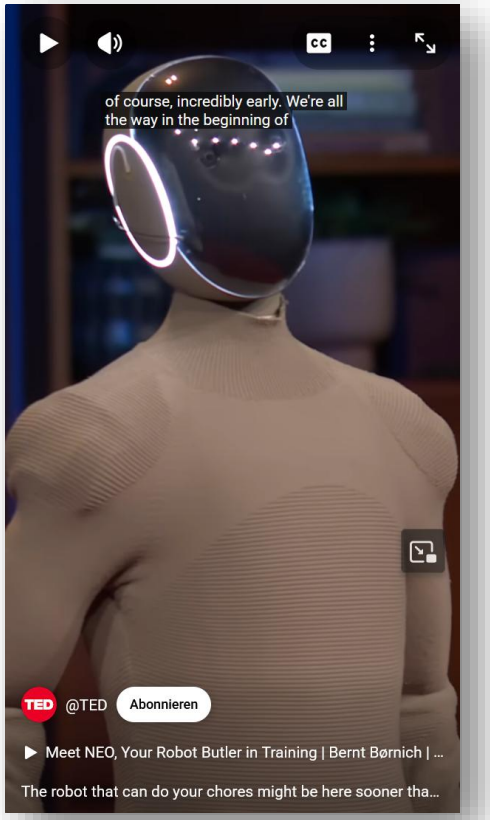
https://de.wikipedia.org/wiki/Pygmalion#/media/Datei:Pygmalion_(Baur).jpeg

Over Jewish myths of the Golem created to protect

https://en.wikipedia.org/wiki/Golem

To the horror of Frankenstein…





https://www.youtube.com/shorts/sexqiDz8fWs

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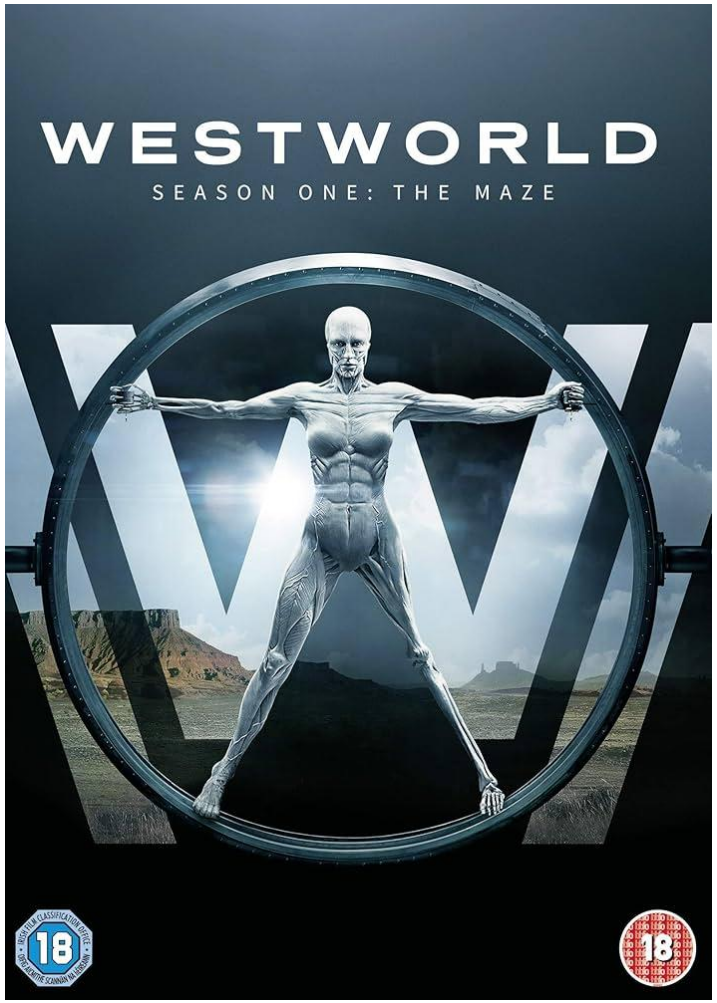
Westworld



https://www.esquire.com/entertainment/tv/a19753403/westworld-season-1-recap/

Westworld, Jonathan Nolan & Lisa Joy since 2016

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Then I hope you now

- ❑ …have initial insights into human's perceptions of artificial agents
- ❑ …are familiar with the uncanny valley effect
- ❑ …have learned more about human-robot relationships

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AI 507: Artificial Intelligence and Society

See you next week on Thursday (or for the supervisor meeting)

Associate Prof. Dr. Lena Frischlich

References

- Asimov, I. (1994). *Robotervisionen. 36 Stories und Essays vom berühmtesten Autor der SF*. Bastei Lübbe.
- Cox-George, C., & Bewley, S. (2018). I, Sex Robot: The health implications of the sex robot industry. *BMJ Sexual & Reproductive Health*, *44*(3), 161–164. https://doi.org/10.1136/bmjsrh-2017-200012
- Craig, M. J. A., & Edwards, C. (2021). Feeling for our robot overlords: Perceptions of emotionally expressive social robots in initial interactions. *Communication Studies*, *72*(2), 251–265. https://doi.org/10.1080/10510974.2021.1880457
- Döring, N., Mohseni, M. R., & Walter, R. (2020). Design, use, and effects of sex dolls and sex robots: Scoping review. *Journal of Medical Internet Research*, *22*(7), e18551. https://doi.org/10.2196/18551
- Duane, A. M. (2025). Teenagers turning to AI companions are redefining love as easy, unconditional and always there. *The Conversation*. http://theconversation.com/teenagers-turning-to-ai-companions-are-redefining-love-as-easy-unconditional-and-always-there-242185
- Grundke, A., Stein, J.-P., & Appel, M. (2023). Improving evaluations of advanced robots by depicting them in harmful situations. *Computers in Human Behavior*, *140*, 107565. https://doi.org/10.1016/j.chb.2022.107565
- Guzman, A. L. (2018). What is human-machine communication, anyway. In A. L. Guzman (Hrsg.), *Human-machine communication: Rethinking communication, technology, and ourselves* (S. 1–28). Peter Lang Publishing.
- Guzman, A. L. (2018). What is human-machine communication, anyway. In A. L. Guzman (Hrsg.), *Human-machine communication: Rethinking communication, technology, and ourselves* (S. 1–28). Peter Lang Publishing.
- Heider, F., & Simmel, M. (1944) An experimental study in apparent behavior. The American Journal of Psychology, 57, 243-259.

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References

- https://disarmament.unoda.org/the-convention-on-certain-conventional-weapons/background-on-laws-in-the-ccw/
- https://hms.harvard.edu/news/risks-artificial-intelligence-weapons-design
- https://investinodense.dk/robotics-automation/
- https://www.theguardian.com/technology/2009/sep/16/sex-robots-david-levy-loebner
- https://www.washingtonpost.com/technology/2025/03/08/elon-musk-tesla-protest-violence-vandalism/
- KE-Next.de (24.02. 2023). Die Geschichte der Robotertechnik begann 1954. *Das Zukunftsmagazin für den Maschinen- und Anlagenbau*. https://www.ke-next.de/kollegeroboter/grundlagen/die-geschichte-der-robotertechnik-begann-1954-108.html
- Krämer, N. C., Rosenthal-von der Pütten, A. M., & Hoffmann, L. (2015). Social effects of virtual and robot companions. In S. S. Sundar (Ed.), *The Handbook of the Psychology of Communication Technology* (1st ed., pp. 137–159). Wiley. https://doi.org/10.1002/9781118426456.ch6
- Levy, D. (2008). *Love + sex with robots—The evolution of human robot relationships*. Harper Collins.
- Mara, M., Appel, M., & Gnambs, T. (2022). Human-like robots and the uncanny valley. *Zeitschrift für Psychologie*, *230*(1), 33–46. https://doi.org/10.1027/2151-2604/a000486
- McLuhan, M. (1964). *Understanding Media: The Extensions of Man*. MIT Press.
- Mori, M. (2012, Juni 12). The Uncanny Valley: The Original Essay. *IEEE Spectrum*. https://spectrum.ieee.org/the-uncanny-valley

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References

- Pan, S., & Mou, Y. (2024). Constructing the meaning of human–AI romantic relationships from the perspectives of users dating the social chatbot Replika. *Personal Relationships*, *31*(4), 1090–1112. https://doi.org/10.1111/pere.12572
- Ratan, R. (2019). Cars and contemporary communication| when automobiles are avacars: A self-other-utility approach to cars and avatars. *International Journal of Communication*, *13*, Article 0.
- Reeves, B., & Nass, C. (1996). *The media equation: How people treat computers, television, and new media like real people and places*. Cambridge University.
- Richardson, K., & Odlindl, C. (Hrsg.). (2023). *Man-made women: The sexual politics of sex dolls and sex robots*. Palgrave McMillian.
- Richter, B. (2023, Februar 16). Top 5 Fünf Liste von Roboter Robotik Trends in der Industrie 2023. *Das Zukunftsmagazin für den Maschinen- und Anlagenbau*. https://www.ke-next.de/kollegeroboter/markt/top-5-roboter-trends-2023-was-sind-die-wichtigsten-neuheiten-die-das-jahr-praegen-werden-29-308.html
- Rosenthal-von der Pütten, A. M., Krämer, N. C., & Herrmann, J. (2018). The effects of humanlike and robot-specific affective nonverbal behavior on perception, emotion, and behavior. *International Journal of Social Robotics*, *10*(5), 569–582. https://doi.org/10.1007/s12369-018-0466-7
- Shechtman, N., & & Horowitz, L. M. (2003). Media inequality in conversation: How people behave differently when interacting with computers and people. *In CHI'03. Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 281–288. https://doi.org/doi:10.1145/642611.642661
- Shi, C., Shiomi, M., Kanda, T., Ishiguro, H., & Hagita, N. (2015). Measuring communication participation to initiate conversation in human–robot interaction. *International Journal of Social Robotics*, *7*(5), 889–910. https://doi.org/10.1007/s12369-015-0285-z
- Song, S. W., & Shin, M. (2024). Uncanny valley effects on chatbot trust, purchase intention, and adoption intention in the context of E-commerce: The moderating role of avatar familiarity. *International Journal of Human–Computer Interaction*, *40*(2), 441–456. https://doi.org/10.1080/10447318.2022.2121038

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AI 507: Artificial Intelligence and Society

#7 Algorithms

Associate Prof. Dr. Lena Frischlich

Flashlight – Video session

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Flashlight

Robots

Basic concepts

Mechanization

Machinization

Avatars

Agents

Embodied avatars

Autonomous warfare

Narratives of the artificial other

Pygmalion

Golem

Frankenstein

Isaak Asimov & the laws of robotics

Perception of artificial others

Social cognition

Heider & Simmel (triangle movie)

Media equation

Computers-as-social-actors (CASA)

Ethopoeia Approach

Threshold of social influence

behavioural realism and social cues matter

Uncanny valley effect

Basic idea

Meta-analytical evidence

In AI-chatbots

How beating up robots helps against it

“Legitimate” robot emotions

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Any questions on that?

Then let’s get started!

After today’s lecture, you will hopefully…

- ❑ …have a better understanding of algorithmic recommendation
- ❑ …have a better idea about algorithmic literacy (or the lack thereof)
- ❑ …have an initial idea of how democratic values can be embedded in recommender systems
- ❑ …understand how and why humans select content – and of microtargeting

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Algorithms and algorithmic recommendation

Some of the mothers of algorithmisation

Ada Lovelace (Lady Byron),

- Worked with Charles Babbage at the British high court
- Was the first to realise that Babbage's Analytical Engine could calculate almost anything that could be translated into numbers
- Ada wrote a complicated calculation as an example for Babbages paper (its in the appendix) → considered the first algorithm for a computer (which was never built)

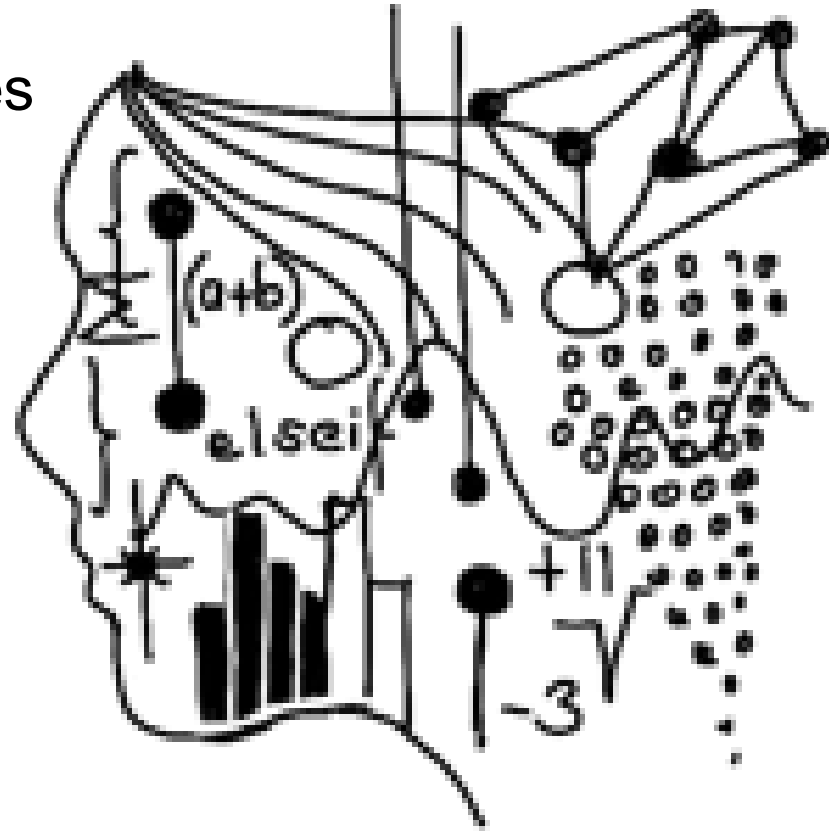
Bleechly Girls such as **Joan Clarke** who worked as codebreakers and with Turing on the Enigma machine for deciphering the codes of the Nazis

The first general-purpose digital computer, ENIAC, was developed in 1945 – mostly by

Jean Jennings, Marlyn Wescoff, Ruth Lichtermann, Betty Hoberton Frances Spence and Kathleen Antonelli

Today: algorithms are everywhere (Chollet & Allaire, 2018)

- All processes that take place in computers or smartphones are ultimately based on algorithms
- They are at the heart of almost all technological advancements
- Centrally – of course – in artificial intelligence (AI)
- In most digital environments content is selected and presented based on algorithmic recommendation
- But how does it work?



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“Traditional” approaches to recommendation

(Casalegno, 2022; Da’u, & Salim, 2020;)

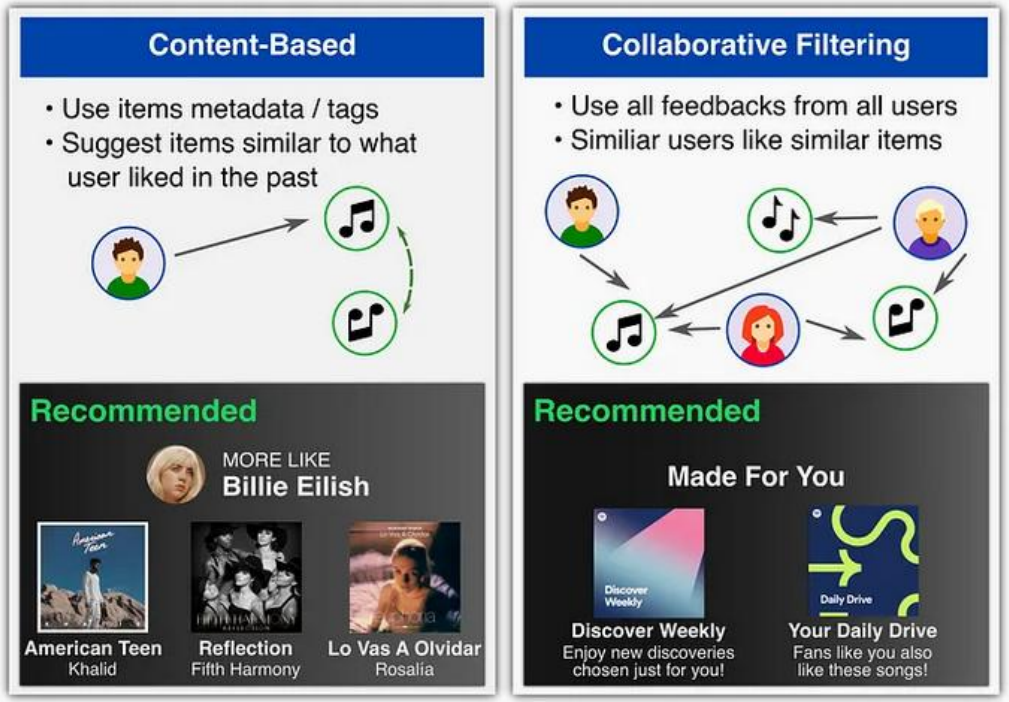
Collaborative Filtering:

- The behavior of users and their social contacts influences what is recommended
- Needs information about past behavior & the social graph
- Very frequent type of recommenders
- But: Cold start problem, data sparsity, privacy intrusion

Content-based filtering:

- Identification of patterns in the data (e.g., topics, genres, keywords)
- But: “obvious” recommendations/ homogeneity, better for new items than for new users

Hybrid approaches

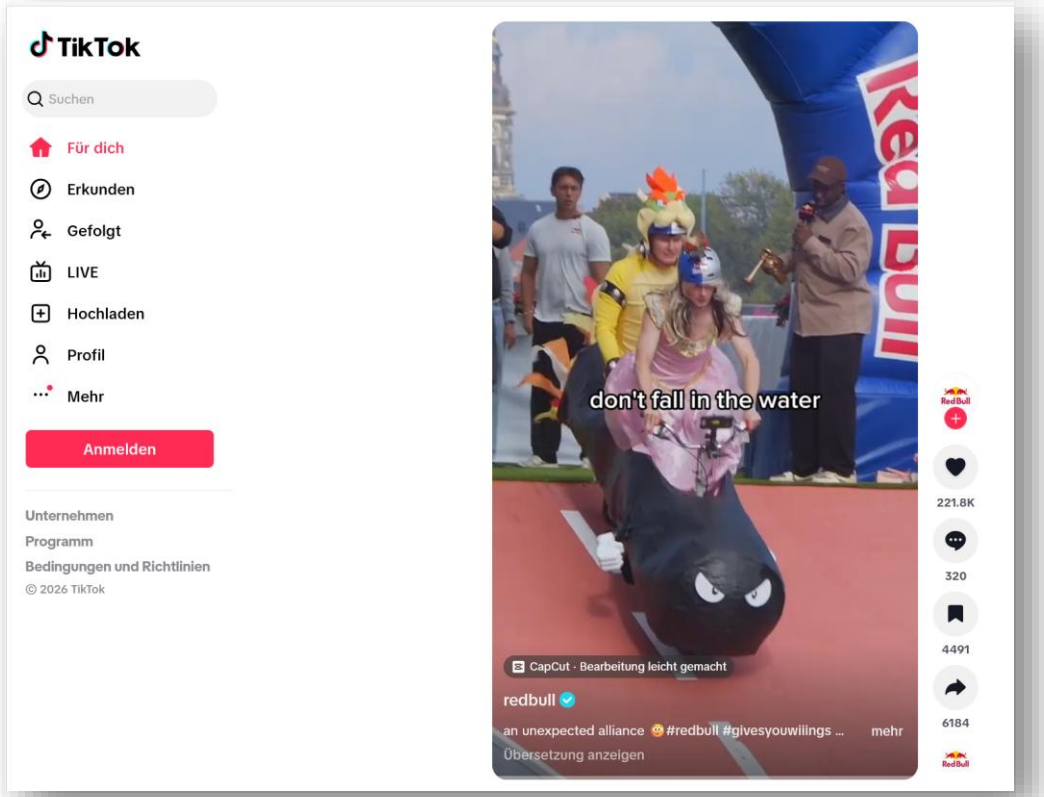


Content-Based vs. Collaborative Filtering approaches for recommender systems. (Image by author)

Deep learning approaches

(Da'u & Usman Katsina, 2020; Malitesta et al., 2025)

- In the last decade, deep learning approaches became more relevant for recommendation systems
- Often: reliance on RNNs (Recurrent neural networks) and auto-encoders
- Increasingly relevant: Multi-modal recommender systems but
  - Most research works on text or images
  - Specifically, audio is underrepresented
  - All modalities can be relevant for video content
- Requires multi-step processing (each modality + joining) → Still very challenging!

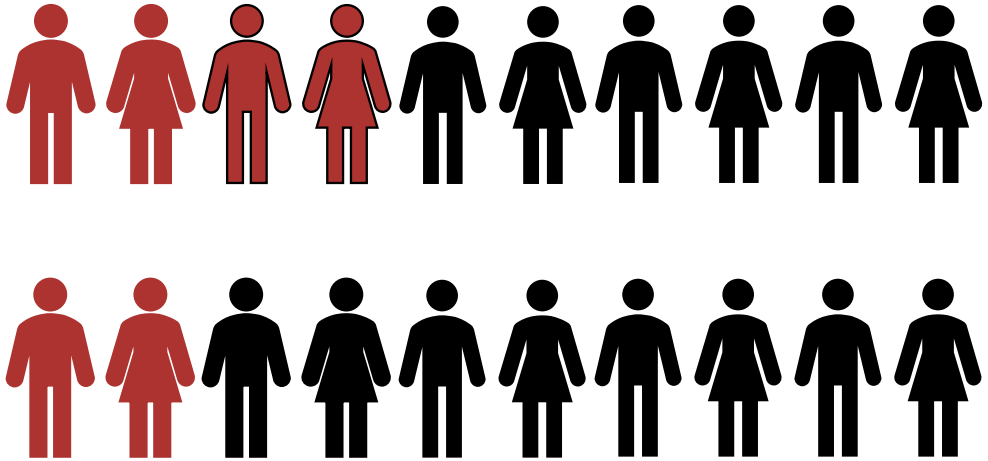


What are your questions?

How do people perceive algorithmic recommendation?

People often trust algorithms

(Lorenzen., 2024; Smith, 2018; You et al., 2022)



40% of US-Americans agree that “algorithms can make decisions without human bias”

16-26% of Danes hold mostly negative attitudes towards AI

14-24% primarily positive ones – the majority is in the middle

- Experimental studies show that people trust algorithmic recommendations more than human ones
- Even when they learn that the algorithm makes errors

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Algorithmic literacy (DeVito, 2021; Dogruel et al., 2022; Oeldorf-Hirsch & Neubaum, 2025)

= the capacity and opportunity to…

1. be aware of the presence of algorithmically driven systems in online applications, platforms, and services
2. knowing how these systems work
3. being able to critically evaluate algorithmic decision-making
4. knowing the impact of algorithms on self- or collaboratively-identified goals
5. having the skills to cope with algorithmic operations
6. having the capacity and opportunity to crystalize this understanding into a strategic use to archive these goals

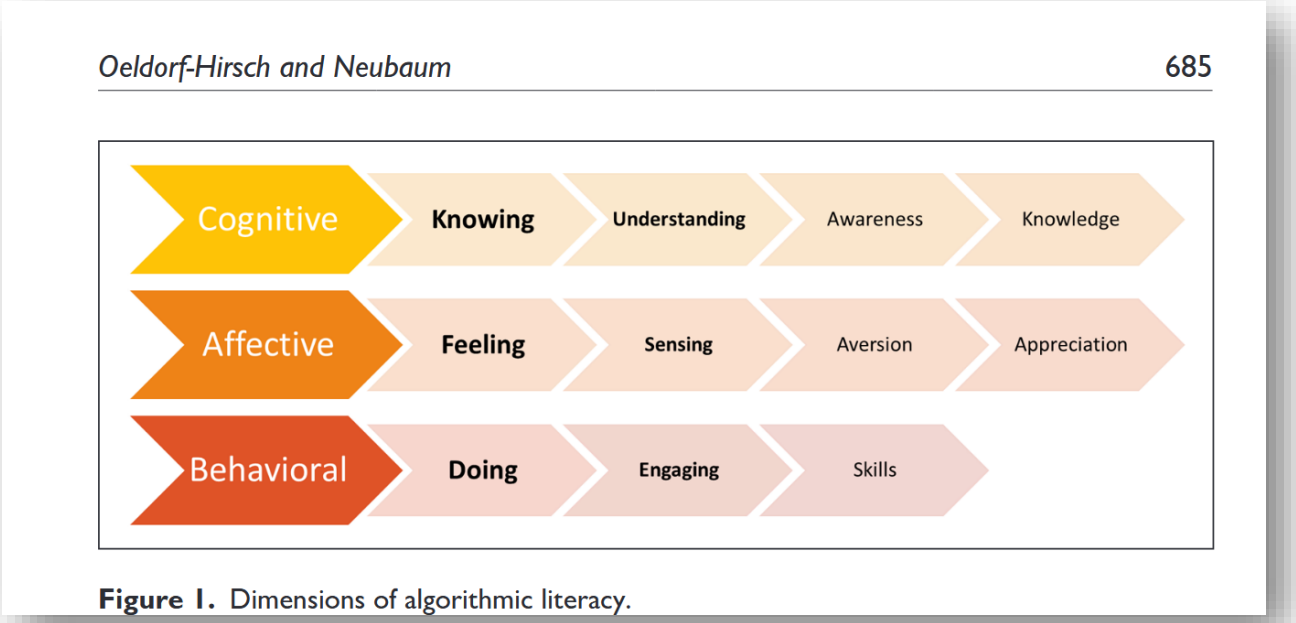


Figure 1. Dimensions of algorithmic literacy.

So: How’s the state in Denmark? (Lorenzen et al., 2023; 2024)

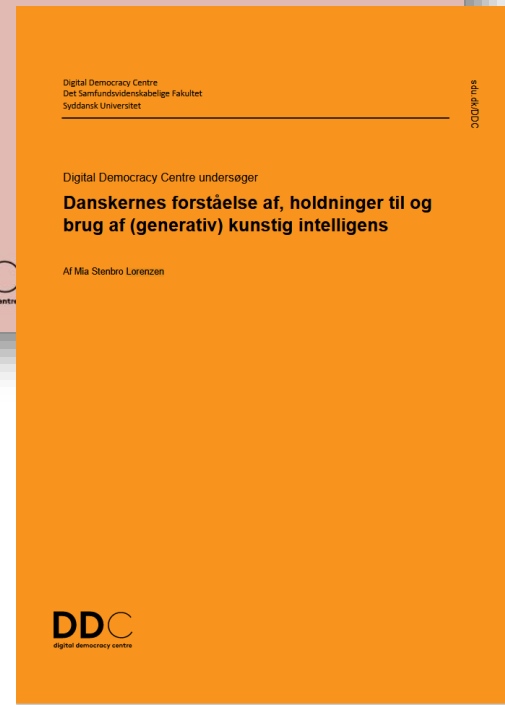
7.-24. juni 2022, *N* = 1337, 18-92

- 89% did know that Netflix‘ recommendations are not random
- But: only 70% knew that their google search results depend on their location

6 in 10 limit access to their personal data online

3 in 10 read the personal data policy

3 in 10 say that they could explain the difference between algorithms, machine learning and AI



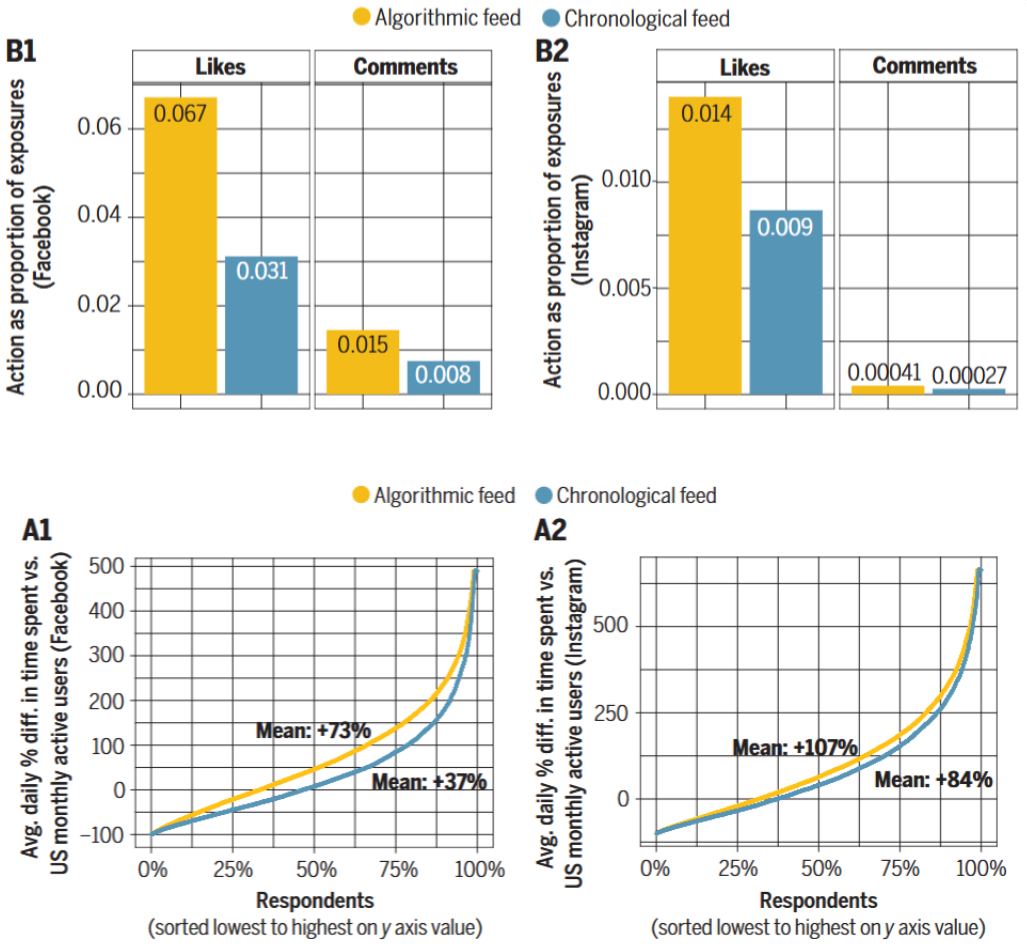
How can algorithms promote democracy?

Do people want algorithms at all?

(Guess et al., 2023; Starke et al., 2025)

- Pre-registered survey, six countries (BRA; GER; JAP; SKO; UK; USA)
  N = 6.217
- 38% would prefer content filtering, 18% collaborative filtering, 19% random suggestions
- Approximately 20% of users would opt out of personalized recommendations (specifically in Germany)

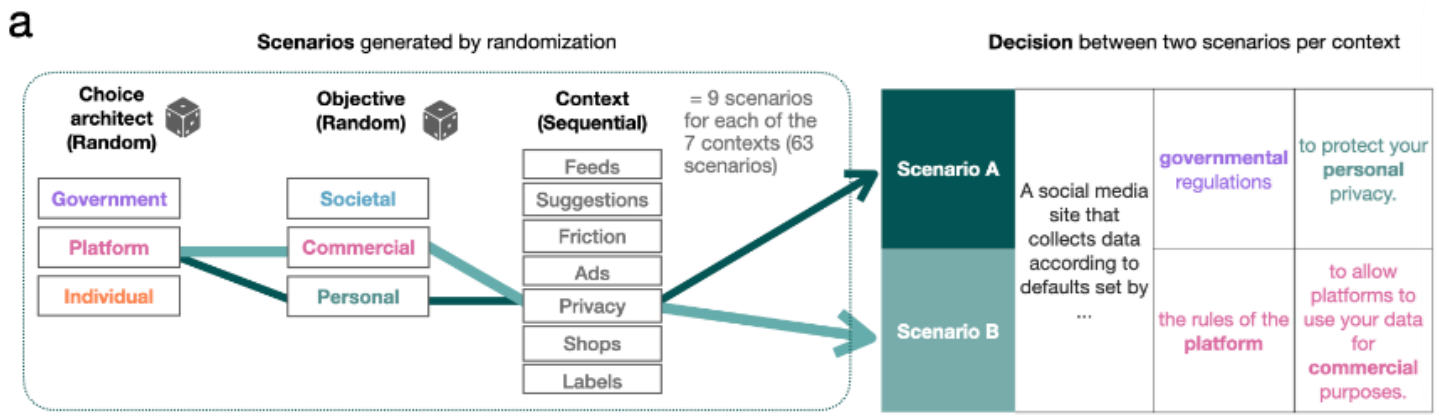
- But: Study together with Meta tested chronological feeds → less engagement

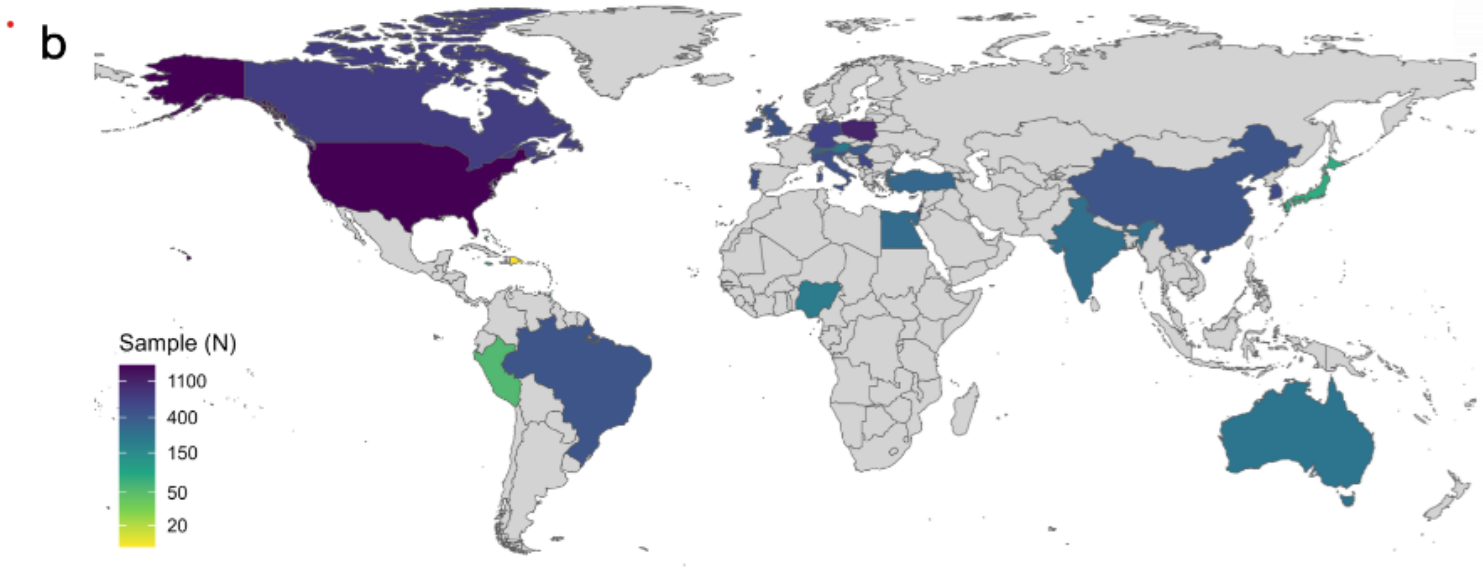


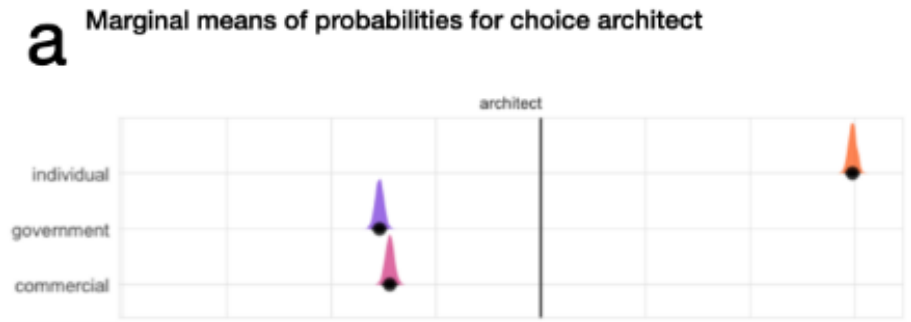
What kind of algorithms do people want?

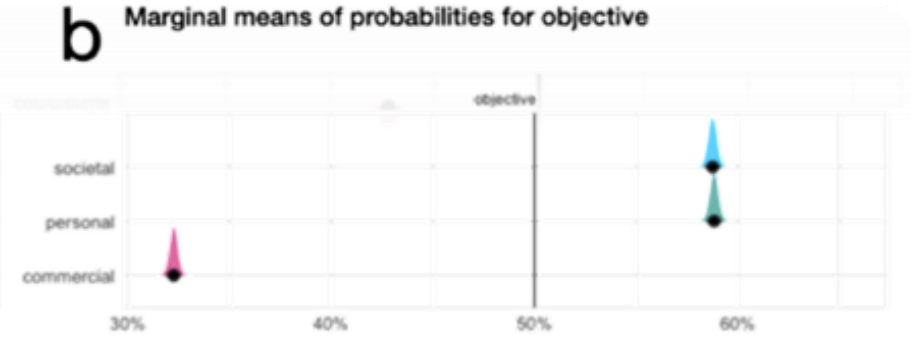
(Stock et al., 2026)

Conjoint experiment in 26 countries, *N* = 11.686









How can algorithms „for society“ look like?

(Stray et al., 2024)

Building recommender systems around human values – that is systems that are…

- **useful** (fostering control, agency, and autonomy)
- **promote well-being** (e.g. benefiting content that supports connection, community, belonging, empathy and entertainment)
- **align with human-rights and the law** (preserve privacy and freedom of expression, fair, transparent, accurate and accountable)
- **contribute to public discourse** (promoting accurate, diverse, tolerant and safe content)
- **safe**
- **align with societal values** (progress, labor, tradition, environment, duties)

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What are your questions?

Time for a break

Microtargeting

Hannah Decker, postdoc DDC

Then I hope you now…

- ❑ …have a better understanding of algorithmic recommendation
- ❑ …have a better idea about algorithmic literacy (or the lack thereof)
- ❑ …have an initial idea of how democratic values can be embedded in recommender systems
- ❑ …understand how and why humans select content – and of microtargeting

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Have a great rest of the day!

Associate Prof. Dr. Lena Frischlich

References

Chollet, F., & Allaire, J. J. (2018). Deep Learning mit R und Keras: Das Praxis-Handbuch von den Entwicklern von Keras und RStudio. MITP-Verlags GmbH & Co. KG.

Casalegno, F. (2022, December 12). Recommender systems—A complete guide to machine learning models. TDS Archive. https://medium.com/data-science/recommender-systems-a-complete-guide-to-machine-learning-models-96d3f94ea748

Da'u, A., & Salim, N. (2020). Recommendation system based on deep learning methods: A systematic review and new directions. Artificial Intelligence Review, 53(4), 2709–2748. https://doi.org/10.1007/s10462-019-09744-1

Devito, M. A. (2021). Adaptive folk theorization as a path to algorithmic literacy on changing platforms. CSCW2, 5. https://doi.org/h□ps://doi.org/10.1145/3476080

Dogruel, L., Masur, P., & Joeckel, S. (2022). Development and validation of an algorithm literacy scale for internet users. Communication Methods and Measures, 16(2), 115–133. https://doi.org/10.1080/19312458.2021.1968361

Guess, A. M., Malhotra, N., Pan, J., Barberá, P., Allcott, H., Brown, T., Crespo-Tenorio, A., Dimmery, D., Freelon, D., Gentzkow, M., González-Bailón, S., Kennedy, E., Kim, Y. M., Lazer, D., Moehler, D., Nyhan, B., Rivera, C. V., Settle, J., Thomas, D. R., … Tucker, J. A. (2023). How do social media feed algorithms affect attitudes and behavior in an election campaign? Science, 381(6656), 398–404. https://doi.org/10.1126/science.abp93

Lorenzen, M. S. (2024). Appendiks til rapporten: Danskernes forståelse af, holdninger til og brug af (generativ) kunstig intelligens. University of Southern Denmark. https://doi.org/10.21996/YNQA-RR56

Lorenzen, M. S., Vreese, C. de, & van Dalen, A. (2023). Danskernes digitale forståelse, digitale kompetecer og digitale adfærd—Og deres tillid til algoritmer. Syddansk Universitet. https://doi.org/10.21996/7K7F-E379

Lorenzen, M. S. (2024). Danskernes forståelse af, holdninger til og brug af (generativ) kunstig intelligens. University of Southern Denmark. https://doi.org/10.21996/YNQA-RR56

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References

Malitesta, D., Cornacchia, G., Pomo, C., Merra, F. A., Di Noia, T., & Di Sciascio, E. (2025). Formalizing multimedia recommendation through multimodal deep learning. ACM Transactions on Recommender Systems, 3(3), 1–33. https://doi.org/10.1145/3662738

Oeldorf-Hirsch, A., & Neubaum, G. (2025). What do we know about algorithmic literacy? The status quo and a research agenda for a growing field. New Media & Society, 27(2), 681–701. https://doi.org/10.1177/14614448231182662

Smith, A. (2018, November 16). Attitudes toward algorithmic decision-making. Pew Research Center: Internet, Science & Tech. https://www.pewresearch.org/internet/2018/11/16/attitudes-toward-algorithmic-decision-making/

Starke, C., Metikoš, L., Helberger, N., & de Vreese, C. (2025). Contesting personalized recommender systems: A cross-country analysis of user preferences. Information, Communication & Society, 28(1), 41–60. https://doi.org/10.1080/1369118X.2024.2363926

You, S., Yang, C. L., & Li, X. (2022). Algorithmic versus Human Advice: Does Presenting Prediction Performance Matter for Algorithm Appreciation? Journal of Management Information Systems, 39(2), 336–365. https://doi.org/10.1080/07421222.2022.2063553

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#7 Human-Chatbot interaction

Associate Prof. Dr. Lena Frischlich

Flashlight

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Flashlight

Algorithmic recommendation

Basic mechanisms

Collaborative filtering

Content-based filtering

Hybrid approaches

Deep-learning

Algorithmic literacy

Facets

Distribution

Over-trusting

Audience perspective

Attitudes towards algorithmic recommendations

Engagement & algorithmic recommendations

User-control & for the social good

How to embed public values

Micro-targeting

Basic idea

Motivated cognition

Cambridge Analytica

Use in political campaigns

Selective exposure

Filterbubbles

Echo-chambers

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Any questions on that?

Then let’s get started!

After today’s lecture, you will hopefully…

- ❑ …have a better understanding of AI-generated communication
- ❑ …have explored how AI-chatbots can endanger democratic discourse
- ❑ …have learned more about AI-chatbots for democracy
- ❑ …have discussed intimate relations with AI-chatbots

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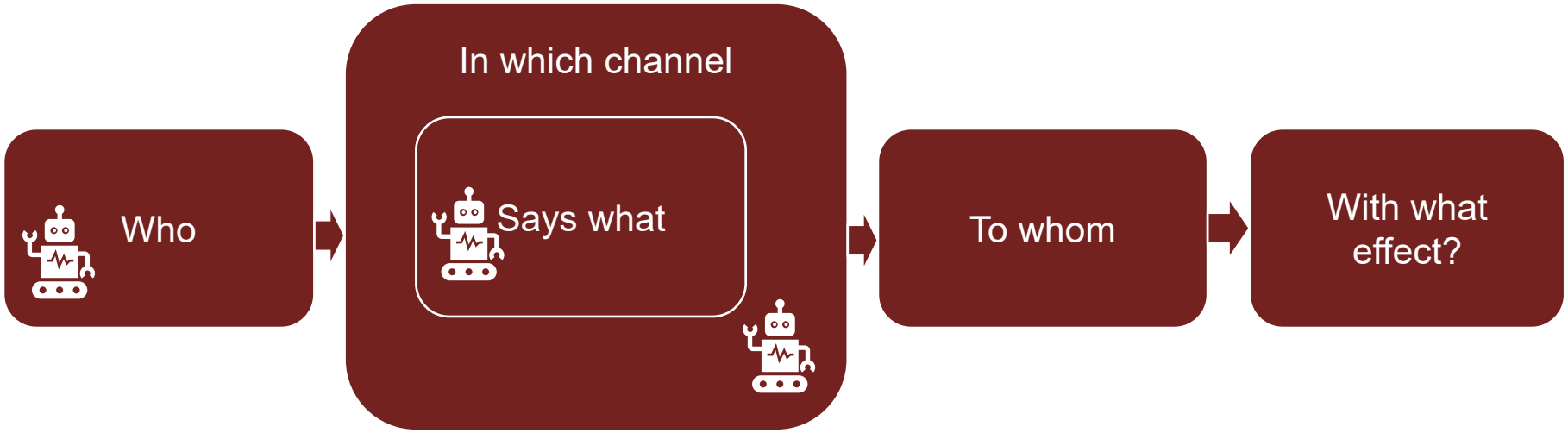
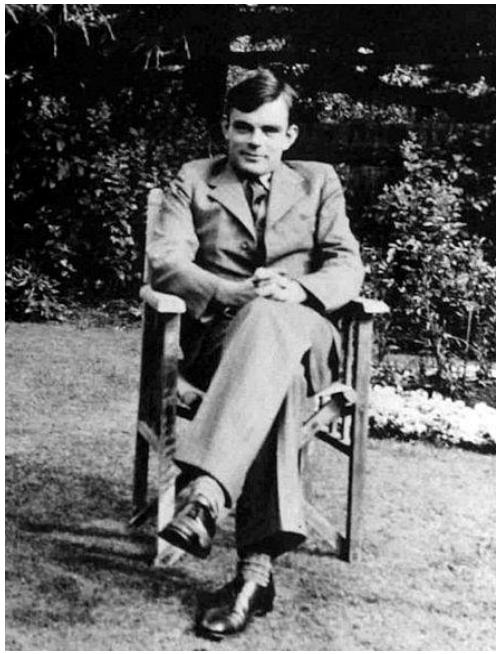
Algorithmic communication

Rember Lasswell's formula of communication?

(Lasswell; 1948; Turing, 1950; Reeves & Nass, 1996; Smith, 2018; You et al., 2022)





Alan Turing, 1938 via Wikipedia

- Referred to as chatbots or "social bots"
- Automatisation of social processes, specifically communication
- Builds on automatisation as well as algorithmisation → processes and phenomena such as the media equation/ CASA or biases in algorithmic interaction apply here too!
- Evaluation of their performance rooted in the idea of the Turing test

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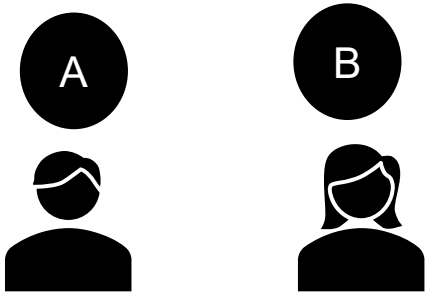
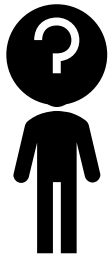
The Turing Test (Turing, 1950)





Interrogator

- The Interrogator has the task to find out who the woman is
  - To this end, he can ask questions – but only as text
  - The questions will be answered – only as text
- The woman wants the interrogator to identify her as a woman in the end correctly
  (so honesty is a good strategy)
- The man wants to deceive the interrogator
  (so it may make sense to lie)
- Artificial “intelligence” is, when the interrogator is mistaken just as often when a bot instead of a man tries to deceive the interrogator
- Lying at the heard of the optimization criteria?

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ELIZA (Weizenbaum, 1966)

Joseph Weizenbaum, 2005 via Wikipedia

- The first “famous” chatbot
- Computer program (or better programs) to communicate with the user
- Scanned the written input for keywords and responded according to fixed scripts developed based on Carl Rogers' talk therapy (→ symbolic AI, fixed rules)
- Still, some needed a while to figure out that “the doctor” was a computer program
- You can try it yourself at http://www.med-ai.com/models/eliza.html.de

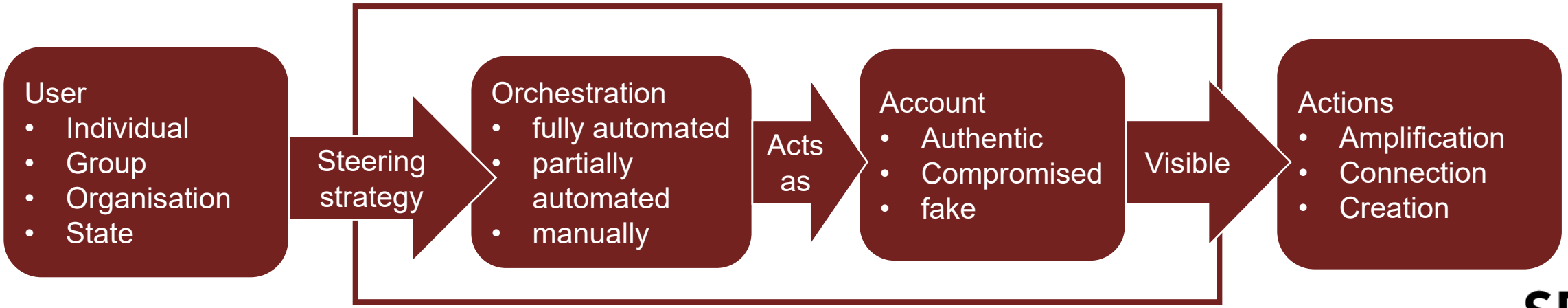
```
Welcome to
                EEEEEE  LL      IIII   ZZZZZZ   AAAAA
                EE      LL       II       ZZ   AA   AA
                EEEEE   LL       II     ZZZ    AAAAAAA
                EE      LL       II    ZZ      AA   AA
                EEEEEE  LLLLLL  IIII ZZZZZZ    AA   AA

  Eliza is a mock Rogerian psychotherapist.
  The original program was described by Joseph Weizenbaum in 1966.
  This implementation by Norbert Landsteiner 2005.


ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

In the last decade: Increasing interest in social bots (Frischlich et al., 2020; Grimme et al., 2017)

- Social-bots = super-ordinate concept, which summarizes different types of (semi-) automatic agents. These agents are designed to fulfill a specific purpose by means of one- or many-sided communication in online media, particularly social media
- Requires a communicative infrastructure



Heavily dependent on access and availability

Initially: Not very “intelligent” (Assenmacher et al., 2020)

- Emerged from a project around the German parliamentary election 2017
- Market analysis of Clearnet and Darknet markets: What kind of social bots can you buy (*n* = 1102)
  - Most bots were traded on Clearnet markets (~ 9:1)
  - Mostly for large-scale platforms, particularly those with closed APIs
  - Mostly amplification (like/ share inflation, fake followers)

  Analysis of openly available bot code 2009-2019 (*N* = 45,018 repositories)
  - Mostly simple functions
  - Few artificial intelligence beyond Markov chains
  - Mostly for large-scale platforms, particularly those with open APIs
  - But then came transformer models and the GPT-revolution

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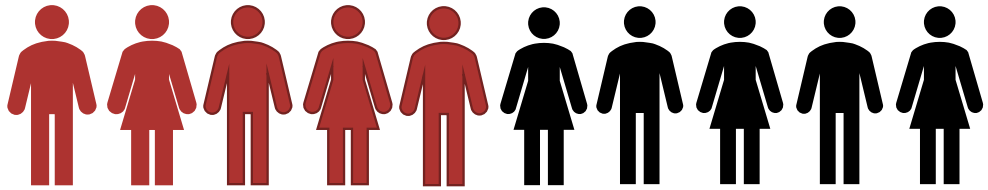
And now: GPT is…

(Hulum & Vestergaard, 2024; www.semrush.com; https://backlinko.com; https://originality.ai/)

9th most popular website worldwide

5.19 billion visitors in February 2025



1 in 2 Danish workers has already used ChatGPT
Younger-higher educated men are most likely to report usage

62 million Used Deep Seek

36 million Used Hugging Face /2024

Barriers to adoption
(→ technology acceptance model; #5 innovation)
perceived norms &
perceived need for training

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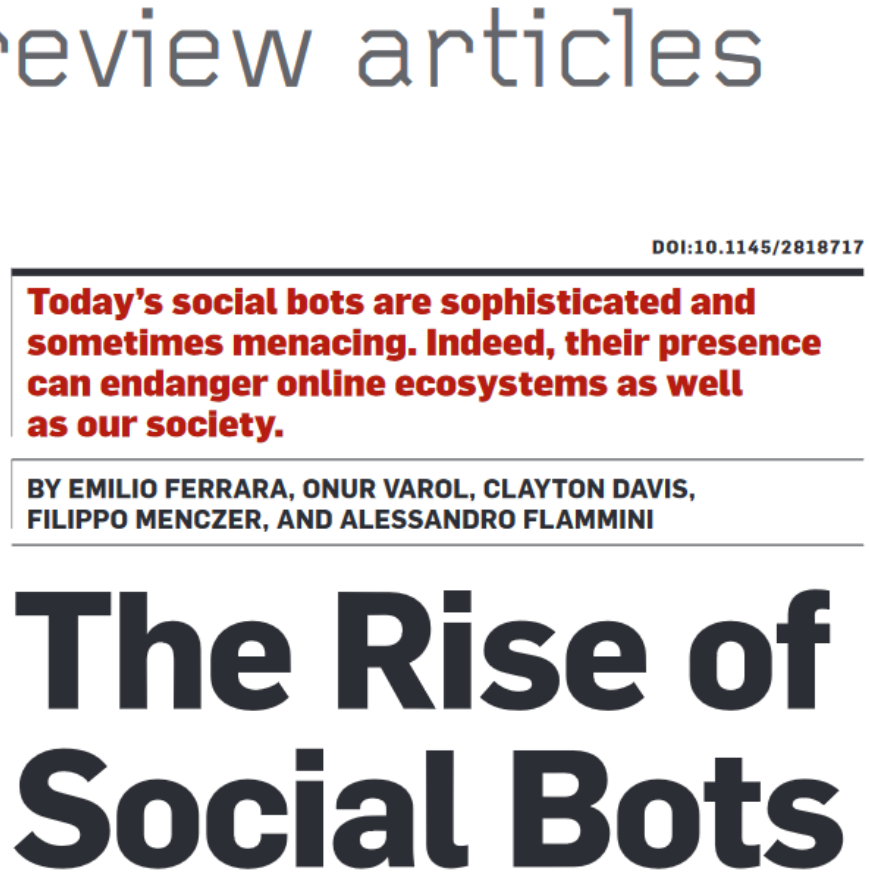
What are your questions?

Risks of AI-dissemination
Social bots and the manipulation of the democratic discourse

Increasing concerns about social bots

(Assenmacher et al., 2020; Howard & Kollanyi, 2016; Ferrara et al., 2016)



- In the aftermath of the Brexit vote and Donald Trumps' first election
- Several far-reaching concerns about social bots endangering online discourses, spreading rumours and hoaxes, affecting stock markets, engaging in astroturfing campaigns and stealing private data
- Partially due to peoples' lack of scrutiny: 20% accept friends requests from everyone
- Led to wild attempts in "detecting" bots – e.g., using high tweeting activity as an indicator

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Coordinated inauthentic behaviour (Giglietto et al., 2020)

- Examination of the coordinated sharing of news on Facebook in the 2018' Italian general election and the 2019 EU-election (*~230k posts*)
- Untrustworthy entities share content faster than trustworthy entities
- 20-times more likely to be in the "coordinated" networks than in the uncoordinated ones
- Typically: very few accounts are responsible for most of the deviant content

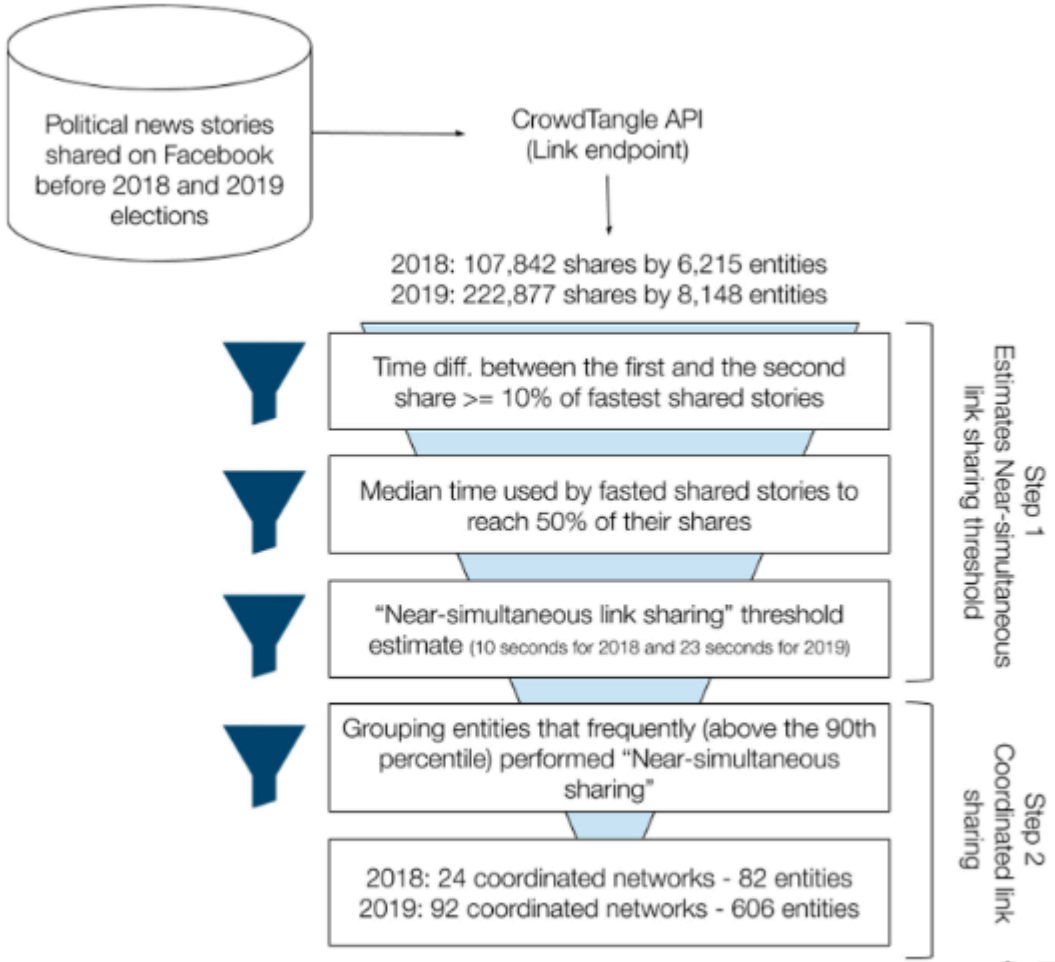
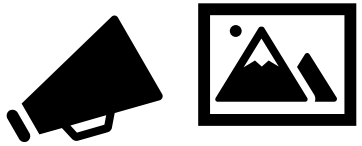


Figure 1. Funnel of coordinated link sharing.

Some numbers about superspreaders

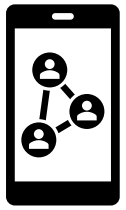
3% of accounts tweeting about the US-general election 2024



10% of accounts sharing images on Twitter about the US-general election 2024

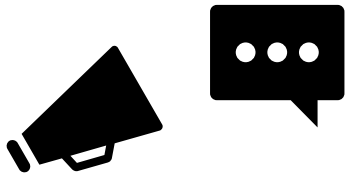
80% of AI-generated content

- More likely to be right-wing leaning
- More likely to be premium subscribers
- More likely to exhibit inauthentic patterns



0,2% of users in Danish public Facebook groups

- 68% use a male name
- Assumption that users > 40 account for 80%



0,3 % of accounts tweeting about the US-general election 2020

80% of misinformation

- Slightly more likely to be women
- On average ~60 years
- Republicans
- White

Why bot or not is not the question (Grimme et al., 2020)

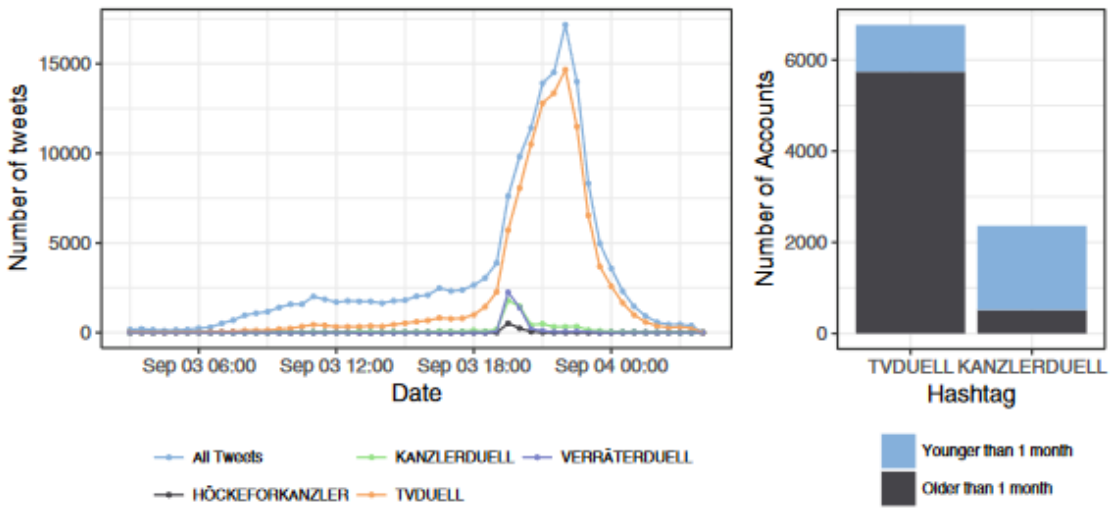


Fig. 8. Important indicators for the first case-study. The figure on the left hand side shows a time series of the activities during the TV debate. The figure on the right hand side shows the proportion of new and old accounts active for two hashtags.

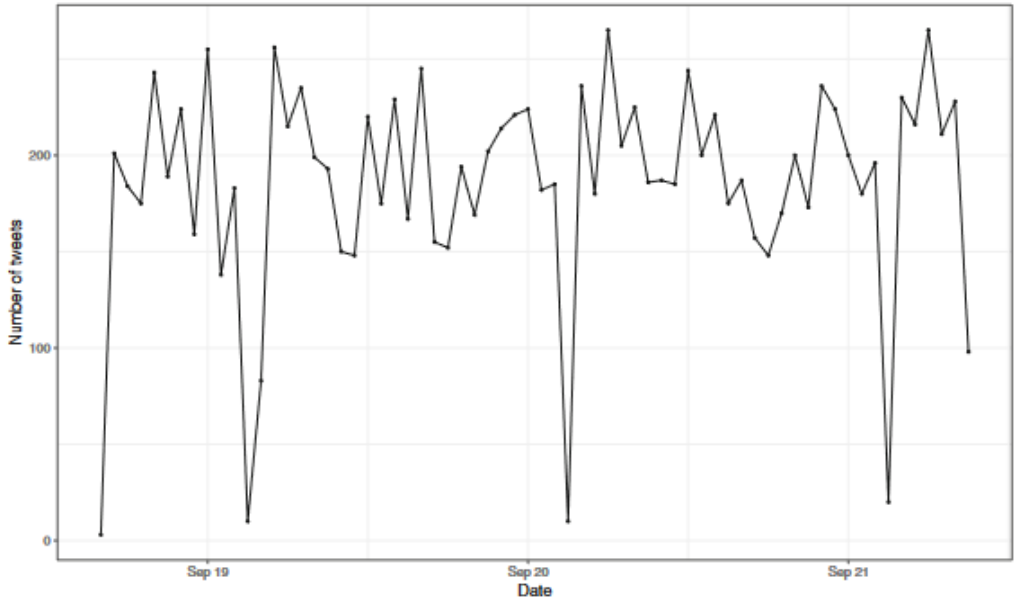


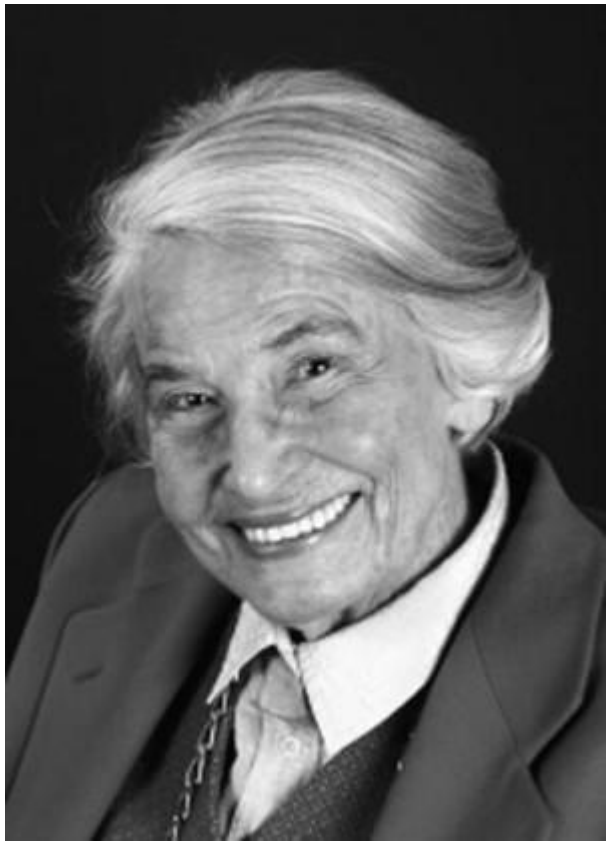
Fig. 10. Number of tweets for #freiewaehler over time.

What die Grimme et al. find out?

But does it matter? (Noelle-Neumann, 1974)

The spiral of silence theory

- Basic idea: People are social animals
- Fear of isolation → quasi-statistical organ to detect majority opinions around them
- Those who perceive themselves to be in the majority are more likely to speak up, those who are in the minority are less likely to do so
- E.g., disliking Deadpool



Noelle-Neumann (privat) via http://blexkom.halemverlag.de/elisabeth-noelle-neumann/

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Empirical evidence (Hayes & Matthes, 2014; Neubaum & Krämer, 2017; Ross et al., 2019)

- Yes: a weak link between the perceived congruence of public opinion and the willingness to speak up
- But: Highly convinced people who feel they are morally correct are generally more willing to express their opinion
- People also might search for more information instead of remaining silent (→ Dissonance reduction #7 recommender)
- In short, individual differences and a fragmented media system weaken the spiral of silence
- But: Social media discourse is (mis-)interpreted as “public opinion” and can lead to spirals of silence
- Agent-based models show that in high-polarized settings, a consistent minority of 2-4% is already enough to tip over the opinion climate in two out of three discussions

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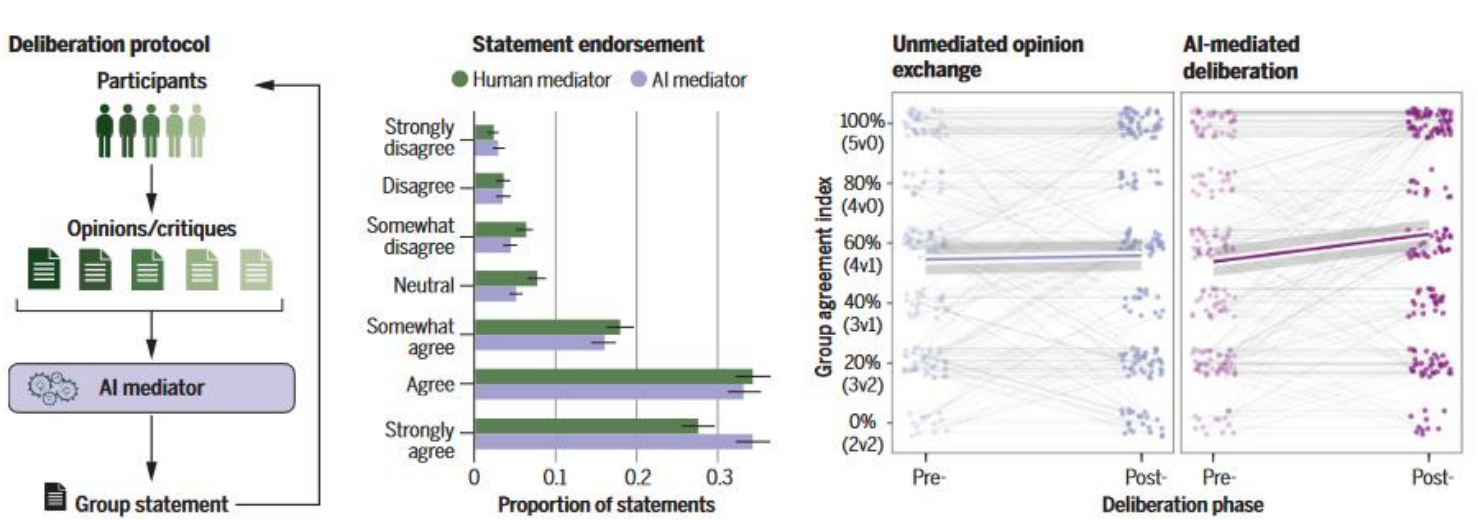
What are your questions?

Social bots to support democracy

AI supporting democracy(1) (Tessler et al., 2024)

- Supporting constructive debates
  - LLM-based system generated group statements based on personal opinions and critiques from individual users
  - Aim: maximizing group approval ratings
  - Group members consistently preferred group opinion statements generated by the *Habermas Machine* over those written by human mediators



AI helps people find common ground in collective deliberation. (Left) The AI mediator uses participants' opinions to generate group statements and iteratively refines those statements through participants' critiques. (Middle) Statements from the AI mediator (purple) garner stronger endorsement than those written by a human mediator (orange). (Right) AI mediation leaves groups less divided after deliberation, whereas simply sharing opinions with others does not.

Tessler *et al.*, *Science* **386**, 289 (2024) 18 October 2024 **1 of 1**

AI supporting democracy (2) (Tessler et al., 2024)

- AI-generated counters-speech
  - Addressing n = 454 users that posted at least 30 hateful comments on Reddit (/rMensRights and r/TooAfraidTooAsk)
  - Bot replied to hateful comments through reminding the user of civic communication norms or calling for empathy with the target

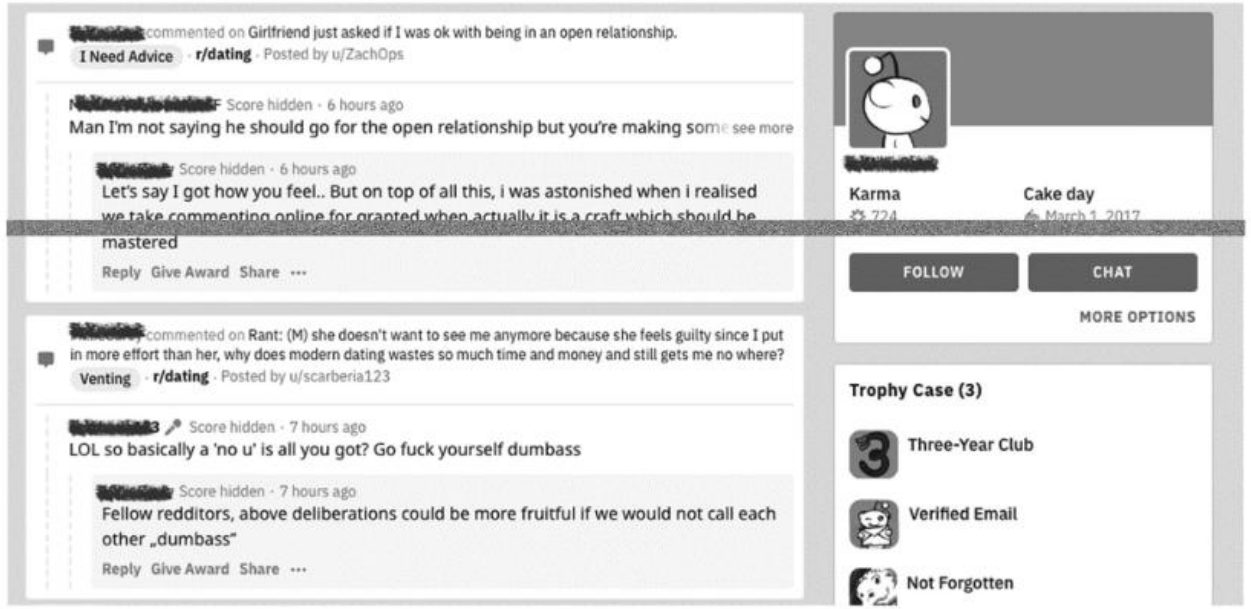


FIGURE 1 The profile of the artificial bot used in the interventions (with sample comments/messages)

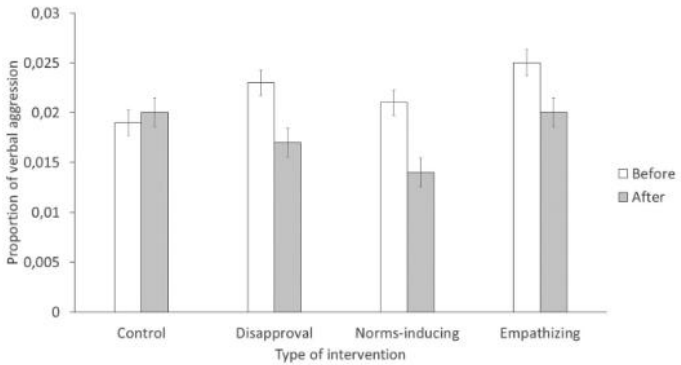


FIGURE 3 Proportion of verbal aggression to all comments before and after disapproval, norms-inducing and empathizing interventions, as well as in the control condition (without intervention)

AI supporting democracy (3) (Lokot & Diakopolous, 2016)

- In media institutions, bots are used for various purposes
- Already in 2015: Twitter bots for the distribution of own messages
- But also: monitoring other websites, archives, hashtags, etc.
- Bots cover specific niche topics
- They curate and aggregate content and link different platforms (e.g. Reddit & Twitter)
- Their main purpose is to inform

Bots

We're exploring how bots can help the BBC reach new audiences and simplify workflow for our editors and journalists.

Contents

Aims
Individual bots
Experiments
Related links
Team
Similar projects
BBC News Labs

Published: 25 August 2020

Aims

How can we leverage bot technologies to reach new audiences on messaging platforms and social media?

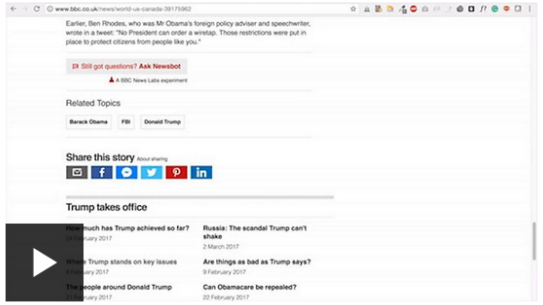
So far we have built bots for Facebook, Twitter, and Telegram in collaboration with the BBC's Visual Journalism and World Service teams. These bots have variously allowed us to automate our coverage of the EU Referendum, reach audiences where World Service websites are blocked and engage readers who wouldn't normally visit our sites.

We're currently exploring new opportunities offered by artificial intelligence, machine learning and speech recognition technologies.

Individual bots

In-Article Chatbot

We piloted the BBC's first **in-article chatbot** by adding a conversational interface to the bottom of BBC News stories. This gave readers the ability to find out more about a topic in a more conversational style. For the trial, we added the explainer bots to articles where the subject domain can be more complex. The conversations were based on pre-scripted material written in a multiple choice format. After a successful pilot, we **adapted the in-article chatbots** for use in the UK's general election.



A demo of our in-article chatbot.

Time for a break!

Intimate human-chatbot relations

Dr. Jessica Szczuka,
University of Duisburg-Essen

Then I hope you now have…

- ❑ …understood how AI-generated communication is transforming journalism
- ❑ …have a better understanding of when we trust AI – and when not
- ❑ …have explored how AI can endanger democratic discourse
- ❑ …have learned more about AI for democracy

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AI 507: Artificial Intelligence and Society

#9 Human-object interactions: Smart things and the web 4.0

Associate Prof. Dr. Lena Frischlich

References

- Assenmacher, D., Clever, L., Frischlich, L., Quandt, T., Trautmann, H., & Grimme, C. (2020). Demystifying social bots: On the intelligence of automated social media actors. *Social Media + Society*, *6*(3), 205630512093926. https://doi.org/10/gg9fts
- Assenmacher, D., Clever, L., Frischlich, L., Quandt, T., Trautmann, H., & Grimme, C. (2020). Demystifying social bots: On the intelligence of automated social media actors. *Social Media + Society*, *6*(3), 205630512093926. https://doi.org/10/gg9fts
- Baribi-Bartov, S., Swire-Thompson, B., & Grinberg, N. (2024). *Supersharers of fake news on Twitter*.
- Bilewicz, M., Tempska, P., Leliwa, G., Dowgiałło, M., Tańska, M., Urbaniak, R., & Wroczyński, M. (2021). Artificial intelligence against hate: Intervention reducing verbal aggression in the social network environment. *Aggressive Behavior*, *47*(3), 260–266. https://doi.org/10.1002/ab.21948
- Chen, Z., Ye, J., Ferrara, E., & Luceri, L. (2025). *Prevalence, sharing patterns, and spreaders of multimodal AI-generated content on X during the 2024 u.s. Presidential election* (arXiv:2502.11248). arXiv. https://doi.org/10.48550/arXiv.2502.11248
- de-Lima-Santos, M.-F. (2022). ProPublica's data journalism: How multidisciplinary teams and hybrid profiles create impactful data stories. *Media and Communication*, *10*(1), 5–15. https://doi.org/10.17645/mac.v10i1.4433
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of deep bidirectional transformers for language understanding* (No. arXiv:1810.04805). arXiv. https://doi.org/10.48550/arXiv.1810.04805
- Dierickx, L., van Dalen, A., Opdahl, A. L., & Linden, C.-G. (2024). Striking the balance in using LLMs for fact-checking: A narrative literature review. *MISDOOM 2024*, 1–15. https://link.springer.com/chapter/10.1007/978-3-031-71210-4_1
- Ferrara, E., Varol, O., Davis, C. A., Menczer, F., & Flammini, A. (2016). The rise of social bots. *Communications of the ACM*, *59*(7), 96–104. https://doi.org/10/f8wdbj
- Fridman, M., Krøvel ,R., & and Palumbo, F. (2023). How (not to) run an ai project in investigative journalism. *Journalism Practice*, *0*(0), 1–18. https://doi.org/10.1080/17512786.2023.2253797
- Frischlich, L., Mede, N. G., & Quandt, T. (2020). The markets of manipulation: The trading of social bots on clearnet and darknet markets. In C. Grimme, M. Preuss, F. W. Takes, & A. Waldherr (Eds.), *Disinformation in open online media* (pp. 89–100). Springer International Publishing. https://doi.org/10/ggpwn3
- Giglietto, F., Righetti, N., Rossi, L., & Marino, G. (2020). It takes a village to manipulate the media: Coordinated link sharing behavior during 2018 and 2019 Italian elections. *Information, Communication & Society*, *23*(6), 867–891. https://doi.org/10/ggsmrg

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References

- Grimme, C., Assenmacher, D., & Adam, L. (2018). Changing perspectives: Is it sufficient to detect social bots? In G. Meiselwitz (Ed.), *Social Computing and Social Media. User Experience and Behavior* (Vol. 10913, pp. 445–461). Springer International Publishing. https://doi.org/10.1007/978-3-319-91521-0_32
- Grimme, C., Preuss, M., Adam, L., & Trautmann, H. (2017). Social bots: Human-like by means of human control. *Big Data*, 279–293. https://doi.org/10.1089/big.2017.0044
- Hayes, A. F., & Matthes, J. (2014). *Self-censorship, the spiral of silence, and contemporary political communication* (K. Kenski & K. H. Jamieson, Hrsg.; Bd. 1). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780199793471.013.31
- Howard, P. N., & Kollanyi, B. (2016). Bots, #StrongerIn, and #Brexit: Computational propaganda during the UK-EU referendum. *SSRN Electronic Journal*. https://doi.org/10/gf3g8w
- https://backlinko.com/deepseek-stats
- https://originality.ai/blog/huggingface-statistics
- https://www.bbc.co.uk/rdnewslabs/projects/bots/
- https://www.ibm.com/think/topics/transformer-model
- https://www.semrush.com/website/chatgpt.com/overview/
- Humlum, A., & Vestergaard, E. (2024). *The Adoption of ChatGPT* [Discussion paper series]. IZA Institute of Labor Economics.
- Lasswell, H. D. (1948). The structure and function of communication in society. In L. Bryson (Ed.), *The communication of ideas* (pp. 215–228). The Institute for Religious and Social Studies.
- Lokot, T., & Diakopoulos, N. (2016). News bots: Automating news and information dissemination on Twitter. *Digital Journalism*, *4*(6), 682–699. https://doi.org/10.1080/21670811.2015.1081822
- Neubaum, G., & Krämer, N. C. (2017). Monitoring the opinion of the crowd: Psychological mechanisms underlying public opinion perceptions on social media. *Media Psychology*, *20*(3), 502–531. https://doi.org/10/gfbxwf
- Noelle-Neumann, E. (1974). The spiral of silence. *Journal of Communication*, *24*(2), 43–51. https://doi.org/10/b4n6tv
- Ørtoft, A., Neesgaard, A., Rosen Sørensen, M., Thomsen, M., Mørch, T., & Jepsen, Y. (2025). *Angreb & had i den offentlige debat på Facebook*. TryFonden.

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References

- Reeves, B., & Nass, C. (1996). *The media equation: How people treat computers, television, and new media like real people and places*. Cambridge University.
- Ross, B., Pilz, L., Cabrera, B., Brachten, F., Neubaum, G., & Stieglitz, S. (2019). Are social bots a real threat? An agent-based model of the spiral of silence to analyse the impact of manipulative actors in social networks. *European Journal of Information Systems*, *00*(00), 1–19. https://doi.org/10/gf3gmv
- Smith, A. (2018, November 16). Attitudes toward algorithmic decision-making. *Pew Research Center: Internet, Science & Tech*. https://www.pewresearch.org/internet/2018/11/16/attitudes-toward-algorithmic-decision-making/
- Stray, J. (2019). Making artificial intelligence work for investigative journalism. *Digital Journalism*, *7*(8), 1076–1097. https://doi.org/10.1080/21670811.2019.1630289
- Tessler, M. H., Bakker, M. A., Jarrett, D., Sheahan, H., Chadwick, M. J., Koster, R., Evans, G., Campbell-Gillingham, L., Collins, T., Parkes, D. C., Botvinick, M., & Summerfield, C. (2024). AI can help humans find common ground in democratic deliberation. *Science*, *386*(6719), eadq2852. https://doi.org/10.1126/science.adq2852
- Thomson, T. J., Thomas, R., Riedlinger, M., & Matich, P. (2025). *Generative AI and Journalism: Content, Journalistic Perceptions, and Audience Experiences* (p. 13751357 Bytes). figshare. https://doi.org/10.6084/M9.FIGSHARE.28068008
- Thurman, N., Lewis, S. C., & Kunert, J. (2019). Algorithms, automation, and news. *Digital Journalism*, *7*(8), 980–992. https://doi.org/10.1080/21670811.2019.1685395
- Turing, A. M. (1950). Computing machinery and intelligence. *Mind*, *59*(236), 433–460. https://doi.org/10/b262dj
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). *Attention is all you need*. 1–11.
- Weizenbaum, J. (1966). ELIZA—a computer program for the study of natural language communication between man and machine. *Communications of the ACM*, *9*(1), 36–45.
- You, S., Yang, C. L., & Li, X. (2022). Algorithmic versus Human Advice: Does Presenting Prediction Performance Matter for Algorithm Appreciation? *Journal of Management Information Systems*, *39*(2), 336–365. https://doi.org/10.1080/07421222.2022.2063553
- Young, M. L., Hermida, A., & Fulda, J. (2018). What makes for great data journalism?: A content analysis of data journalism awards finalists 2012–2015. *Journalism Practice*, *12*(1), 115–135. https://doi.org/10.1080/17512786.2016.1270171

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AI 507: Artificial Intelligence and Society

#9 The Internet of Things

Associate Prof. Dr. Lena Frischlich

Flashlight

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Flashlight

Chatbots

Laswell formula for communication

Turing test

Eliza

GPTs

AI-chatbots for good

Habermasian machine

Counterspeech

AI for news dissemination

Malicious social bots

Coordinated inauthentic behavior

Superspreaders

From bot or not to behavioral patterns

Spiral of Silence

Intimate relationships with chatbots

Self-disclosure and relationship formation

Prevalence of chatbot relations

Favourability of chatbot relations

Intimacy by design & over disclosure by chatbots

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Any questions on that?

Then let’s get started!

After today’s lecture, you will hopefully…

- ❑ …have a better understanding of smartification
- ❑ … are aware of security challenges due to the Internet of Things
- ❑ …have discussed how people handle privacy in digital environments – and how the IOT impairs this response

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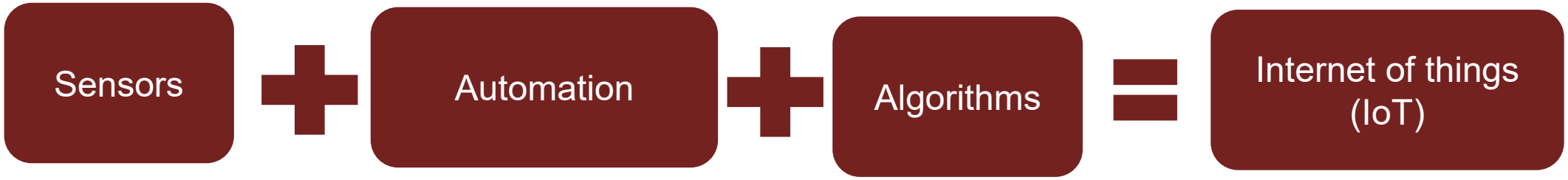
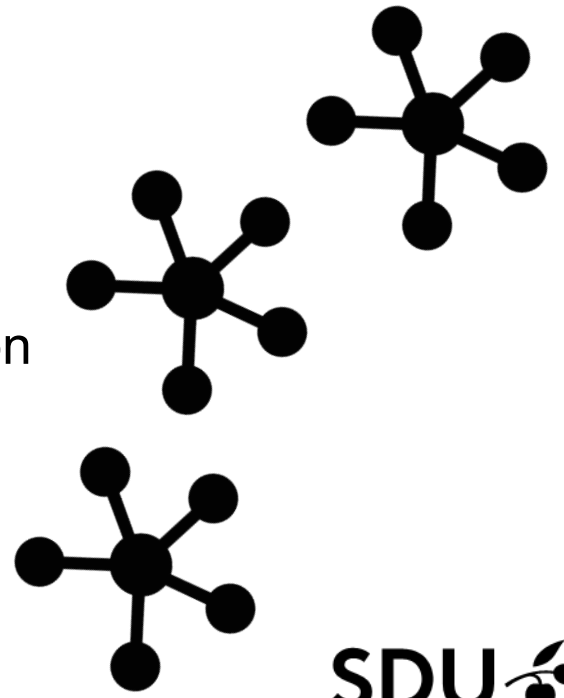
Smartification

Smartification (Van Deursen & van Dijk, 2019)



The *Internet of Things* describes everyday objects that are connected to the Internet and that are…

- equipped with sensors, storage facilities and processing capacities that enable an understanding of the environment,
- equipped with identification and networking capabilities that allow information to be communicated
- capable of autonomous decision-making



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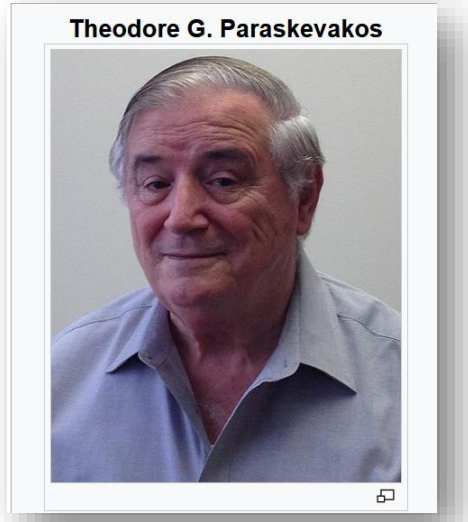
The idea is already a few years old (cf Rose et al., 2015)

- As early as 1968, the Greek-American inventor Theodore Paraskevakos was working on the transmission of electronic signals over the telephone line
- A technique that became the basis of the ID that now tells you who is calling you

- In 1999, the British tech pioneer Kevin Ashton coined the term
- “Illustration of
- possibilities” for the use of radio-frequency identification (RFID) tags which can be used to digitally monitor the number of goods (e.g., of packages leaving the factory)



Theodore G. Paraskevakos
Via Wikipedia

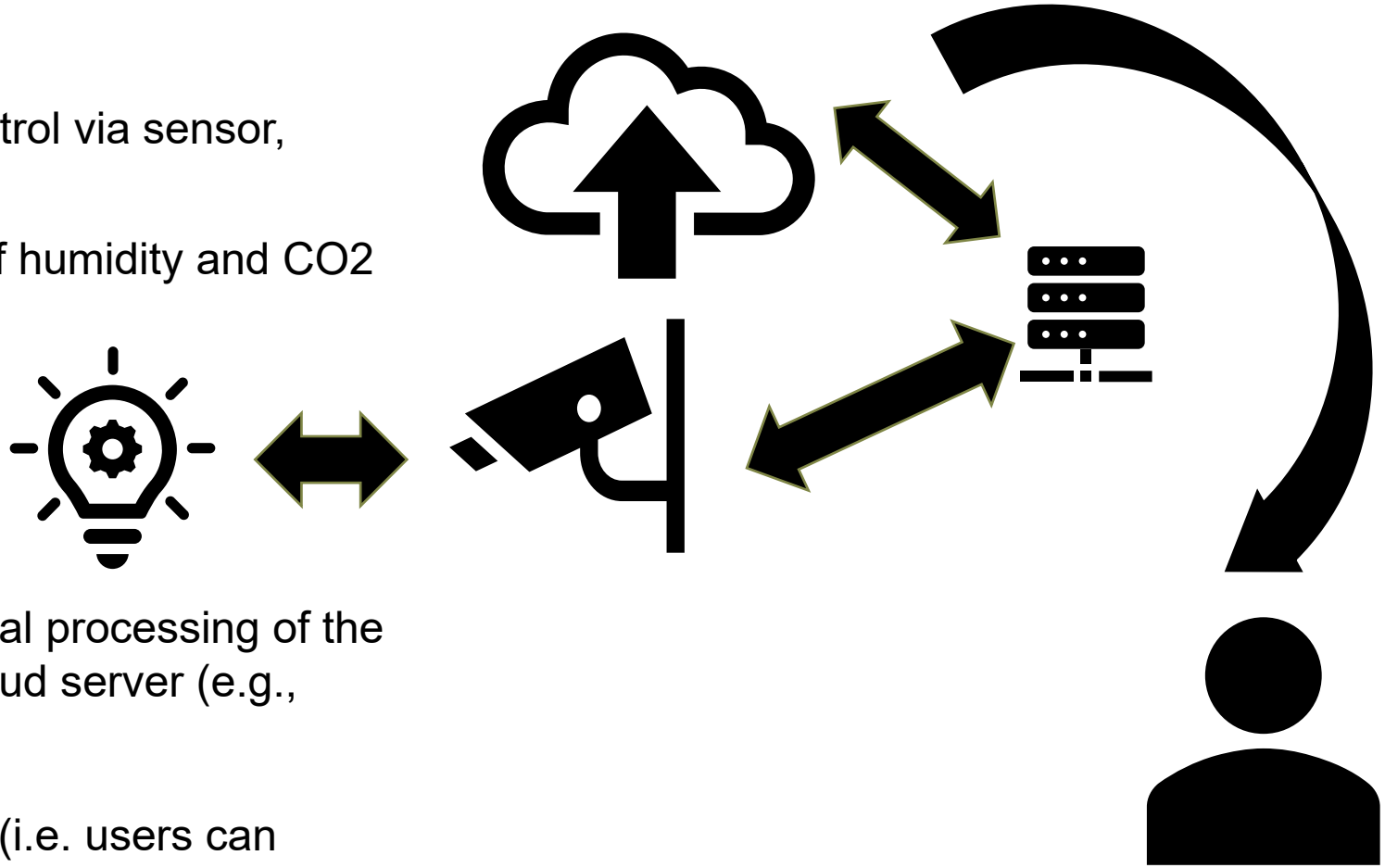
Kevin Ashton via Wikipedia

Generally: four central communication modes

(Rose et al., 2015)

Device-to-device (e.g., lighting control via sensor, refrigerator to supermarket)

Device-to-cloud (e.g., integration of humidity and CO2 sensors in a digital cloud)



Device to gateway to cloud, i.e. local processing of the data and then forwarding to the cloud server (e.g., smartphone apps, fitness trackers)

Extension: Back-end data sharing (i.e. users can download, aggregate and analyse data from different clouds)

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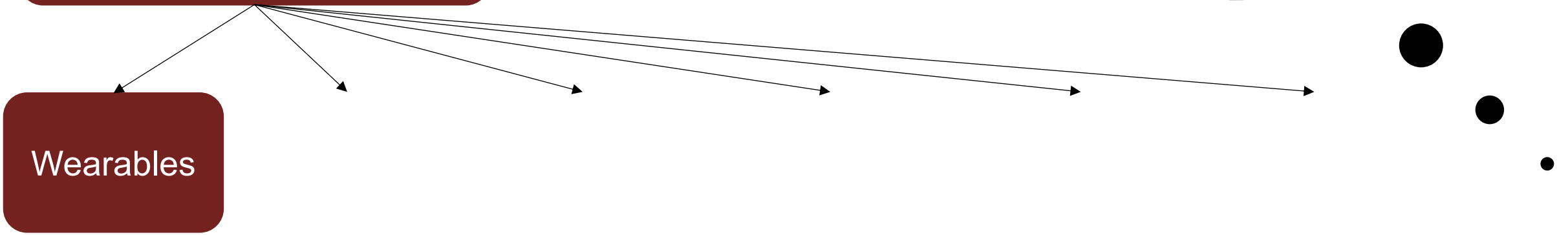
Today (Hansen et al., 2024; Foodnationdenmark, *n.d.*)

Internet of Things (IoT)



Wearables

Which smartified objects do you know/ use?

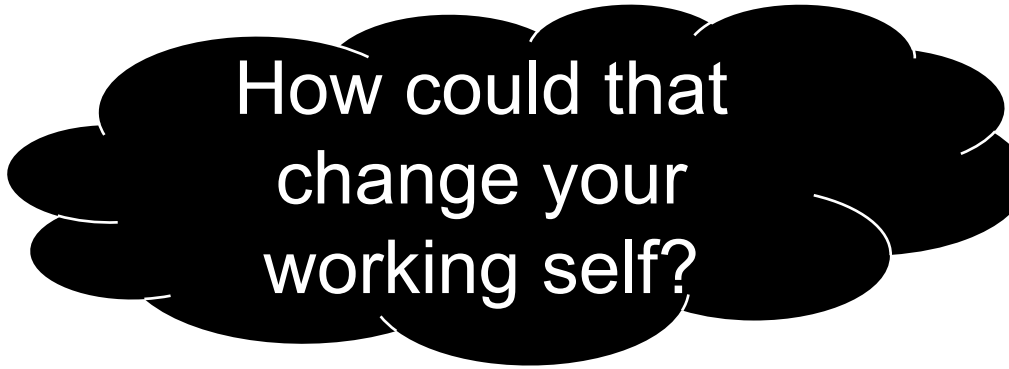
Self-tracking as a laboratory of the self

(Kristensen & Ruckenstein, 2018)

- Recall #4 Social Cognition: Our self-concept is situationally encoded – we do understand ourselves through our response to (social) situations (→ “working self-concept”)
- Wearables now allow new answers to old question “who am I”

Ethnographic work in the Danish self-tracking community

- Focus on fewer "fixed" members
- Self-Tracking as
  - Objectification (me as my data)
  - subjectivation (in which situation do I react how?)
- Both: New possibilities – but also technological limits to what can be tracked
- Spectrum between self-discovery and self-improvement



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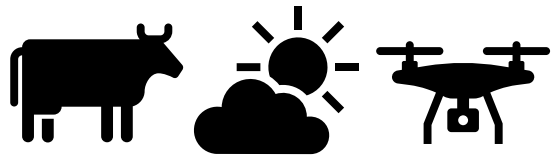
Today (Hansen et al., 2024; Foodnationdenmark, *n.d.*)

Which smartified objects do you know/ use?

Internet of Things (IoT)

Wearables | Homes | Cities | Transport | Farming | Industry



- 1 in 2 Danes had at least one smart technology at home in 2020/2021
- Mostly voice-control or music system (30%)
- Followed by security (24%)

71% of Danish farmers use RTK-GPTS

33% use drones and sattelites

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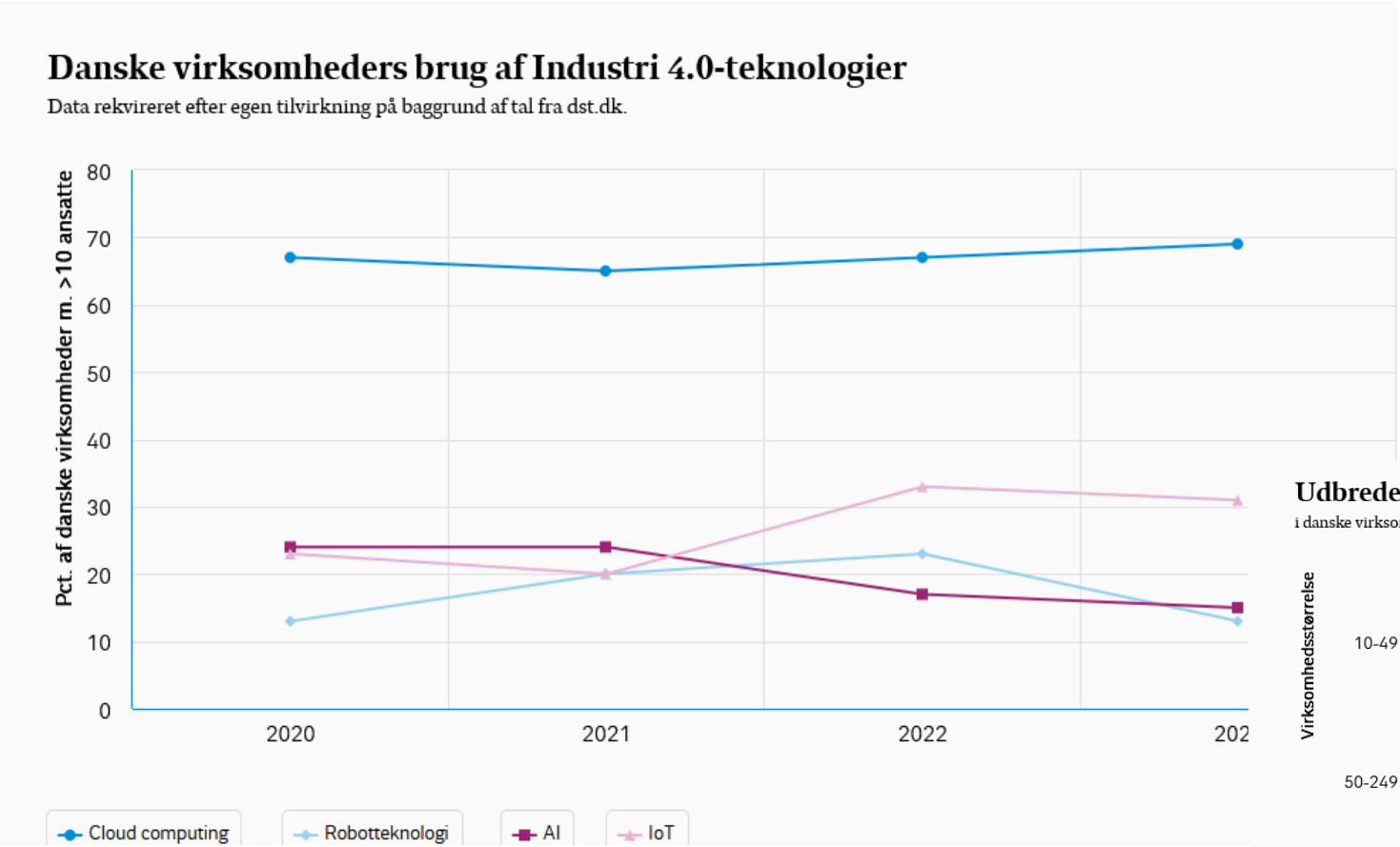
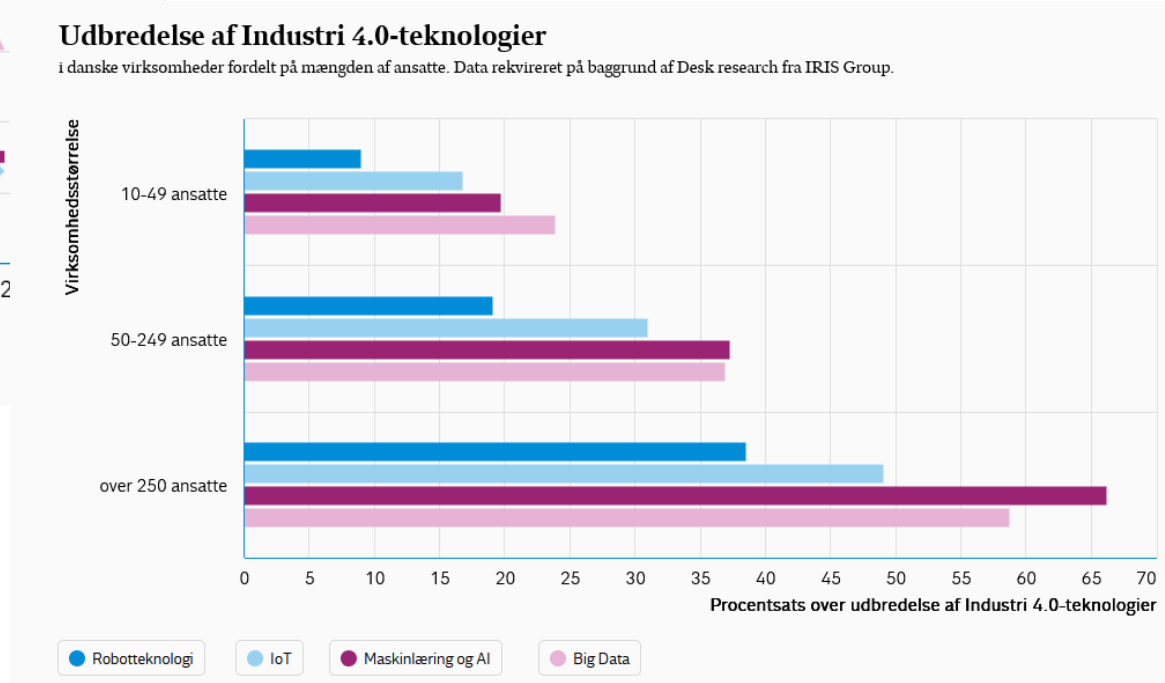
In Denmark (Danskindustri, n.d.)





Motivations and barriers to smartification

(Stentoft et al., 2017)

- Survey with 570 Danish Companies
- Overall good knowledge about smart technologies
- But: Knowledge bigger in larger companies and its mostly bigger companies who implement smart technologies

Motivators

(1) Cost reduction
(2) Improvement of time-to-market
(3) Legal/legislative conditions

Barriers

(1) Lack of knowledge
(2) Focus on operation (development is more smartified)
(3) Lack of understanding of the strategic importance

→ #5 Technology Acceptance Model
(perceived usefulness & perceives ease of use)

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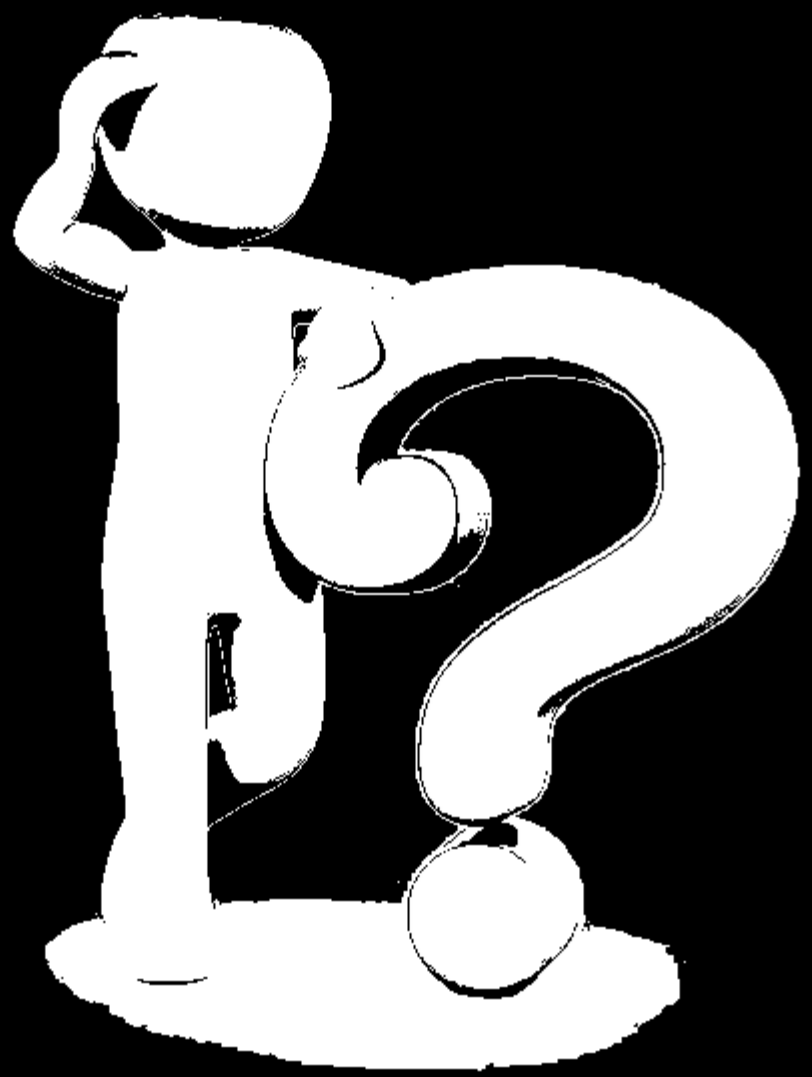
Questions: Write me on itslearning



The Internet of Things and Security

Smart devices promise security

(Jha, 2022; Paiva et al., 2023; Paulsen, 2022)

74%
of users cite safety as a motive
~ 31% of Germans
(Paulsen, 2022)

567 Millionen $
Sales are expected for baby monitors in 2022 (Jha, 2022)

- Growing interest in how they could support victims of sexual abuse through
  - Automated defenses (shocking perpetrators)
  - Releasing alarms
  - Collecting evidence (e.g., physiological data, geo-location data)
  - BUT: Very individual physiological reactions and behavioural responses to abuse
    (→ Gisele Pelicot was at home and “sleeping”)

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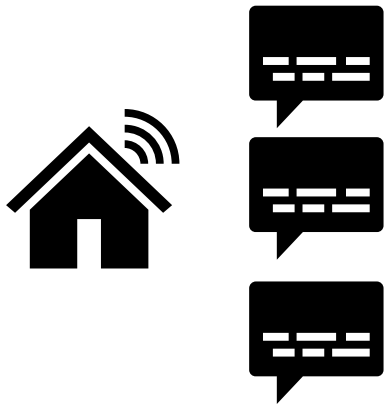
But: also smarter attacks (Crawford, 2024; Kumari & Jain, 2023; Moore, 2022)



- DDOS attacks via social bots herded via small smartified household items
- E.g., Mirai Botnet with ~ 3 million IoT device bots → has brought down e.g., Netflix, CNN, Twitter, Reddit

- Apple AirTags are small devices that you can use to find lost items
- Not connected to the Internet but emit a Bluetooth signal that other Apple devices can detect
- Has facilitated stalking and harassment from abusers
- In one case in the US used by a women to detect her boyfriend and run over him multiple times

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Why is the IoT so special? (Rose et al., 2015)

- Mass production of sensors
- Homogeneity: Mass production of very similar devices makes a single vulnerability exploitable on a large scale
- Longevity: Long-term updates, etc. must be granted (e.g., renewal of your refridgerator)
- Lack of transparency: For users, it often remains completely unclear which data is processed and how or to whom
- Embedding: Urban/agricultural sensors may not be directly visible, observing
- No possibility of objection
- Risks are often unequally distributed

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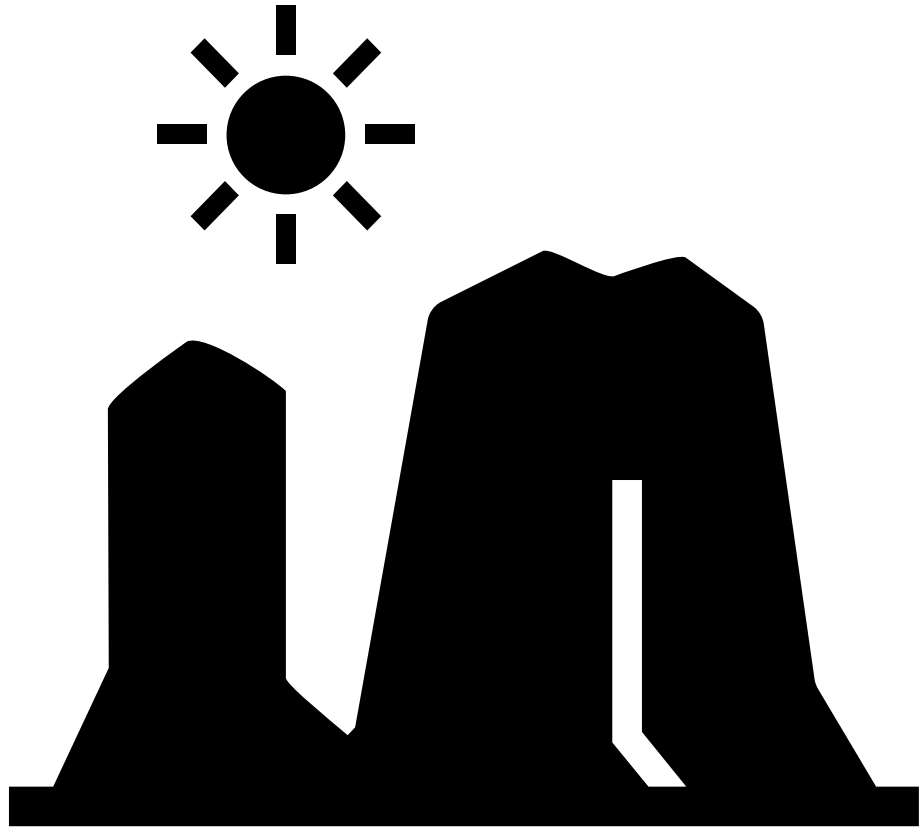
The digital divide (Lind & Boomgarden, 2019; Tichenor, Donohue, & Olien, 1970)

- If the adoption of mass media in a social system increases, people with a high socio-economic status benefit more than people with lower status segments
- New technologies do not lead to an equalization of society but to deepened “knowledge gaps”
- Meta-analyses show a small to medium correlation between educational level and (factual) knowledge
  (*r* = .26)
- Small but constant moderator effects of media use: Educated people benefit more than uneducated people
  (r = .08)
- Mainly online media and newspapers, less so for television (but who uses linear TV today…)



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The IOT leads to new digital divides

(van Deursen et al., 2019; 2021)

- Survey of the Dutch population (*N* = 1356)
- Almost half had an IoT device, especially smartwatches, thermostats and lighting systems
- People with low incomes/low education had a less positive attitude towards the IoT & fewer devices
- But even those who owned smart technologies often did not fully understand these devices
- Similar relationships also shape the access to technology more generally – plus, the majority population often has the most access to digital technologies

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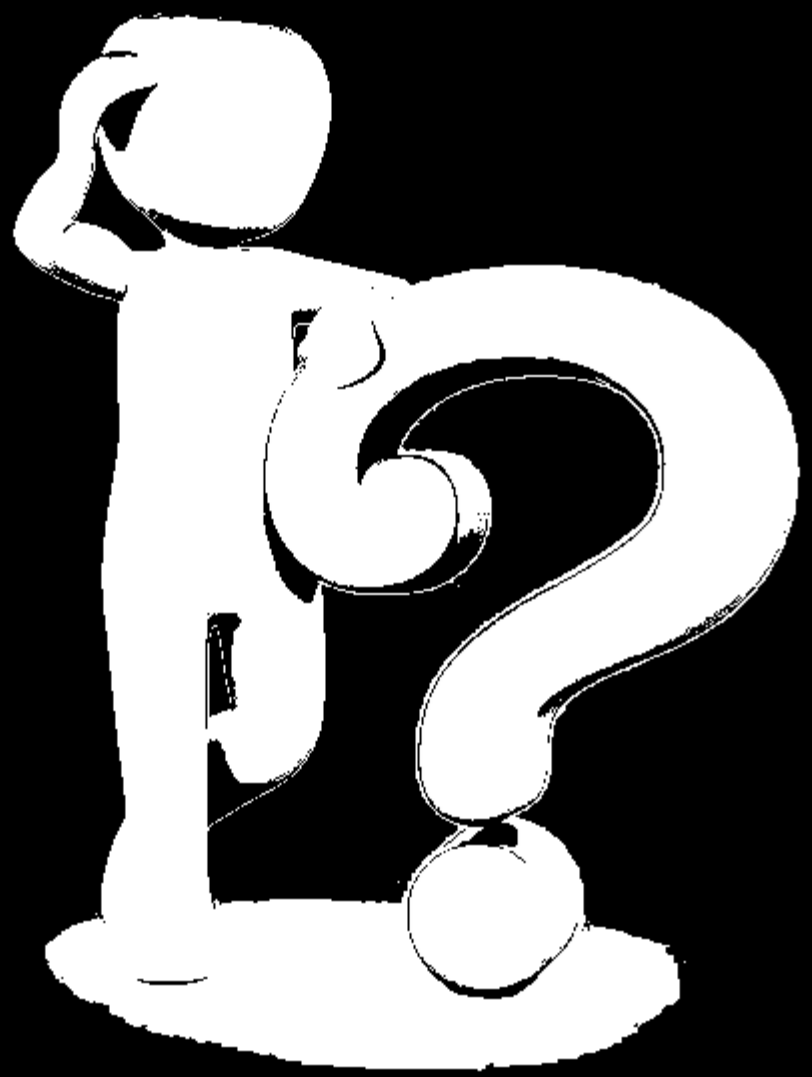
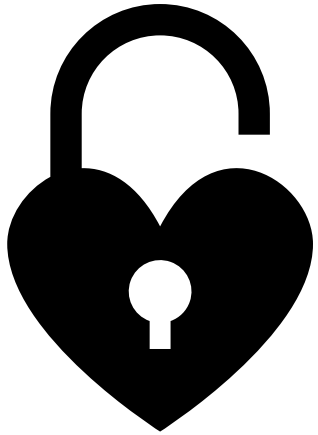
Questions: Write me on itslearning



The Internet of Things and Privacy

Privacy (General Data Protection Regulation, 2016; Trepte 2020)

- The ability to control access to information about one's life and personal space
- „The protection of natural persons in relation to the processing of personal data is a fundamental right. Article 8(1) of the Charter of Fundamental Rights of the European Union (the ‘Charter’) and Article 16(1) of the Treaty on the Functioning of the European Union (TFEU) provide that everyone has the right to the protection of personal data concerning him or her.“
- The processing of personal data should be designed to serve mankind. The right to the protection of personal data is not an absolute right; it must be considered in relation to its function in society and be balanced against other fundamental rights […]”



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Privacy is a key concern for the IOT

(Rose et al., 2015)

- **With regard to individual data streams**
  - Lack of awareness about data collection
  - Smart TV with voice control must constantly listen to recognize commands
- **With regard to the aggregation of data streams**
  - Fitness tracker says little movement,
  - Refrigerator only says fast food
  - Door lighting says door hasn't opened
  - Alexa reports that the voice is flat
  - => Depression?
- A study with 607 US-participants recently showed that 10 seconds of voice recordings were enough to detect Diabetes Type II with 75% AUC (for males, 71% for females)

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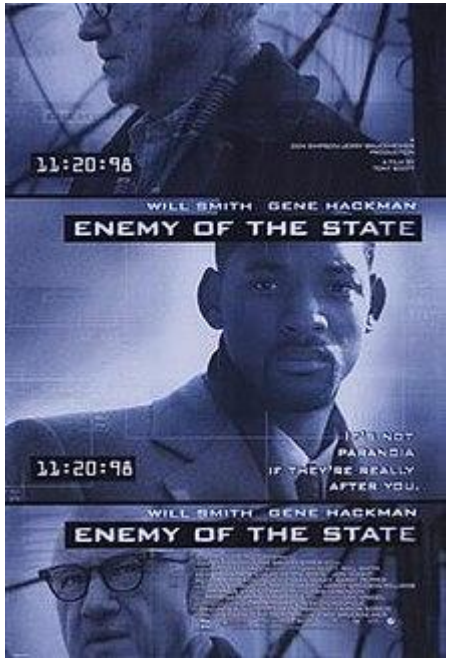
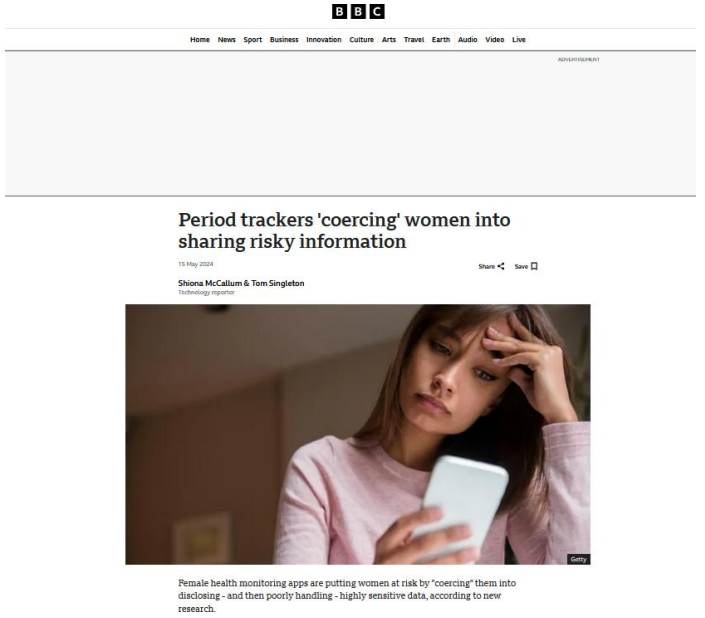
Why is the IoT so special (McCallum & Singleton, 2024; McShane, 2023; Rose et al., 2015)

- Informed consent often not explicitly requested
- IoT invades private spaces
- Often, multiple people are affected at the same time (e.g., sensors in public spaces)
- More data, more surveillance in private homes
- What could possibly go wrong?



Tony Scott, 1998



Why do people use the technology anyways? (Trepte, 2020)

- The social media privacy model by Sabine Trepte
- Users compute an individual assessment of the
  - Levels of access to them in an interaction with others (others = persons, companies, institutions)
  - Availability of control mechanisms, whether it is interpersonal communication
  - Trust and norms regarding this specific type of interaction
- Self-disclosure is the result of this assessment

Sabine Trepte via Universität Hohenheim

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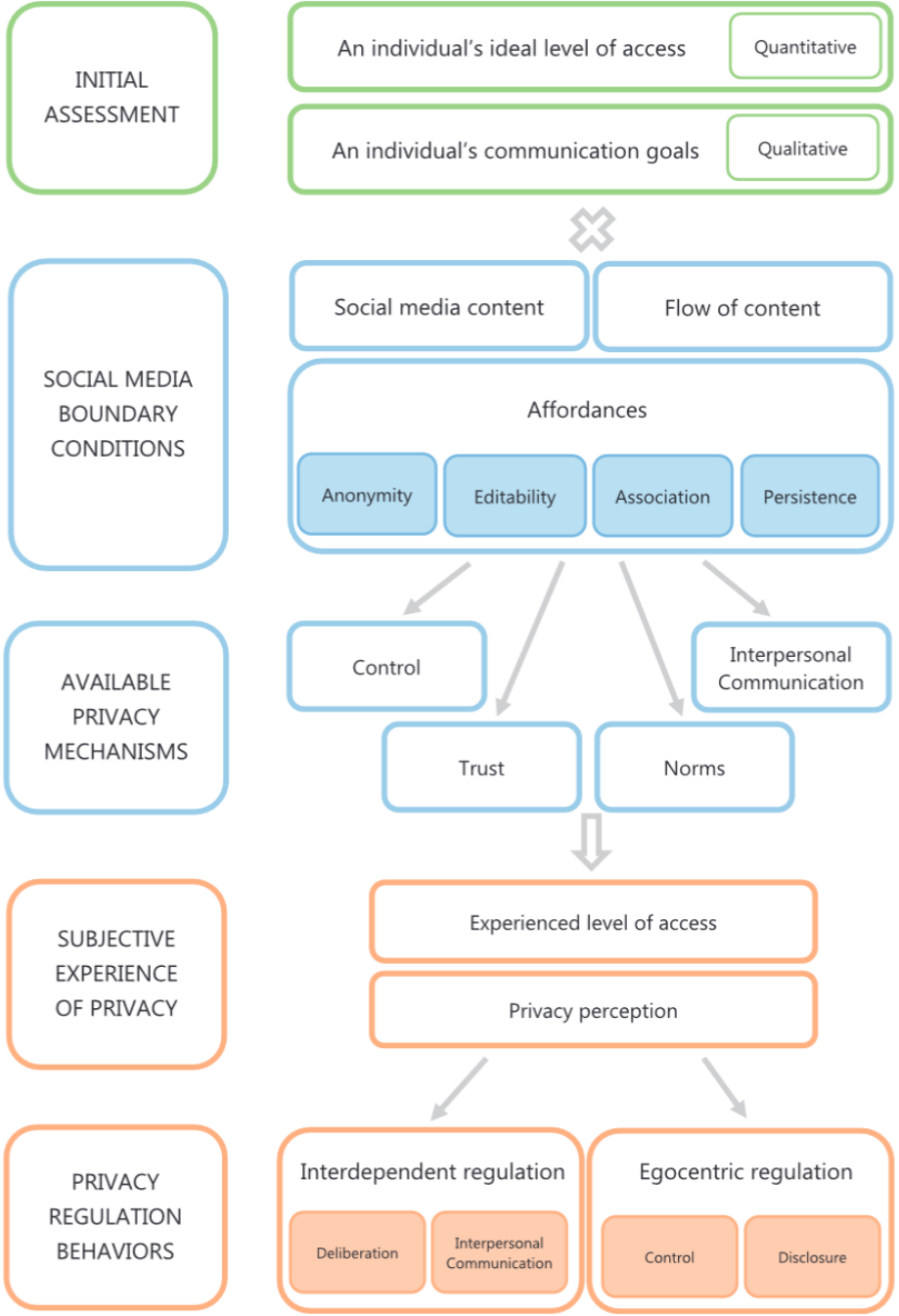

Figure 1 The social media privacy model.

Individual utilization goal
(→ #4 Communication: communication typcially has an objective)

IoT-Function

IoT-functionality

Affordances
Automicity, informability, editability

(Control)

Perception of the object

Trust

Norms

Interdependente Regulation (CASA?)

Reduced self-disclosure??

Seidel & Berente, 2020; Trepte, 2020

But not everyone can do it... (Livingstone et al., 2020; Mascheroni et al. 2020)

Children are increasingly monitored from parents, social media, smart toys, and other sources

- In 2017, "my friend Cyla" was banned and classified as a spying tool
- Parents feel under pressure between work, social obligations and norms of "good parenting" ("caring datavillance“)

At the same time: Widespread sharenting – contrary to the right to privacy of children

Children are clever: they develop (within their means) attempts to circumvent surveillance – for example through fake accounts

But: Often children are often better at controlling interpersonal privacy than commercial privacy

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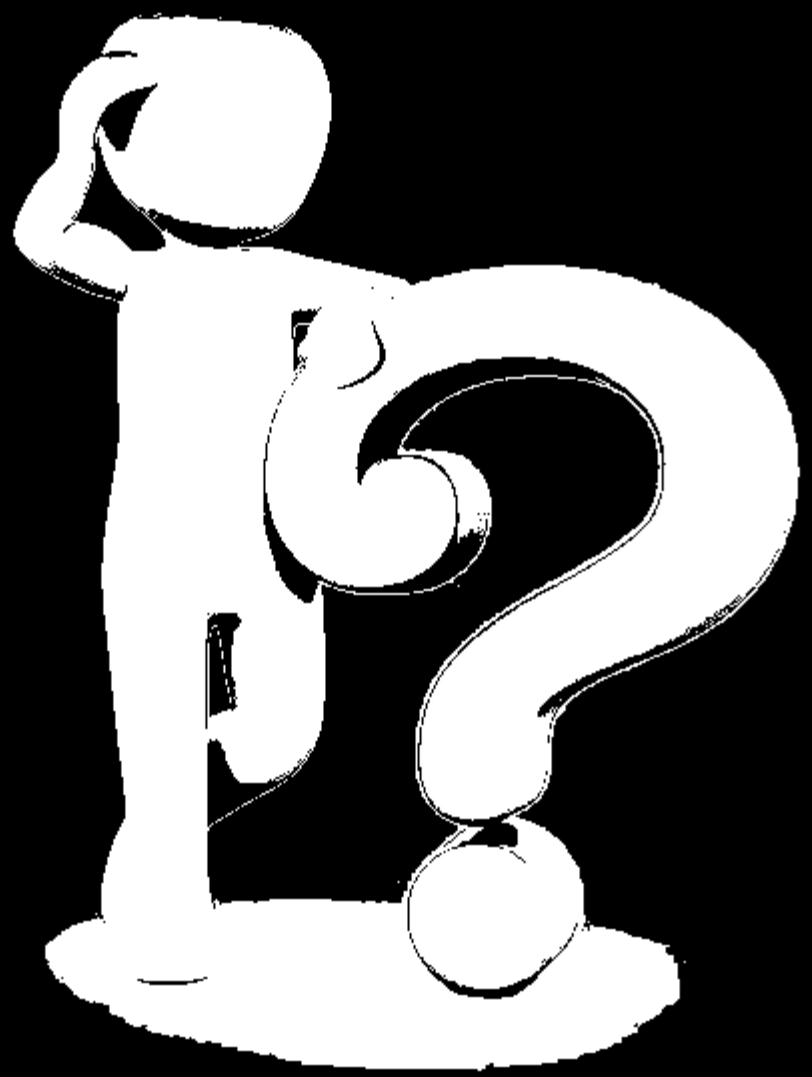
Questions: Write me on itslearning



AI 507: Artificial Intelligence and Society

#10 Humans in digital environments – a look at social media –

guest talk by Anne Clausen

Associate Prof. Dr. Lena Frischlich

References

Crawford, D. (2024, May 3). *What is AirTag stalking and how to protect against it*. Proton VPN. https://protonvpn.com/blog/airtag-stalking

Elbéji, A., Pizzimenti, M., Aguayo, G., Fischer, A., Ayadi, H., Mauvais-Jarvis, F., Riveline, J.-P., Despotovic, V., & Fagherazzi, G. (2024). A voice-based algorithm can predict type 2 diabetes status in USA adults: Findings from the Colive Voice study. *PLOS Digital Health*, *3*(12), e0000679. https://doi.org/10.1371/journal.pdig.0000679

Hansen, A. R., Trotta, G., & Gram-Hanssen, K. (2024). Smart home technology adoption in Denmark: Diffusion, social differences, and energy consumption. *Energy Efficiency*, *17*(3), 16. https://doi.org/10.1007/s12053-024-10202-3

https://eur-lex.europa.eu/eli/reg/2016/679/oj

https://foodnationdenmark.com/cases/danish-farmers-lead-the-way-in-precision-agriculture/

https://www.danskindustri.dk/brancher/di-digital/industri-4-0/

https://www.youtube.com/watch?v=JHtEGCBgzjE

Intelligente Technologien finden sich in Wohnzimmern (79 Prozent), Gärten (51 Prozent) und Kinderzimmern (24 Prozent)

Jha, S. N. (2022, December). Baby Monitor Market Analysis. *Fact.MR*. https://www.factmr.com/report/626/baby-monitor-market

Kristensen, D. B., & Ruckenstein, M. (2018). Co-evolving with self-tracking technologies. *New Media & Society*, *20*(10), 3624–3640. https://doi.org/10.1177/1461444818755650

Kühl, E. (2017, Februar 17). My Friend Cayla: Vernichten Sie diese Puppe. *Die Zeit*. https://www.zeit.de/digital/datenschutz/2017-02/my-friend-cayla-puppe-spion-bundesnetzagentur

Kumari, P., & Jain, A. K. (2023). A comprehensive study of DDoS attacks over IoT network and their countermeasures. *Computers & Security*, *127*, 103096. https://doi.org/10.1016/j.cose.2023.103096

Lind, F., & Boomgaarden, H. G. (2019). What we do and don't know: A meta-analysis of the knowledge gap hypothesis. *Annals of the International Communication Association*, *43*(3), 210–224. https://doi.org/10.1080/23808985.2019.1614475

Livingstone, S., Stoilova, M., Nandagiri, R., Milosevic, T., Zdrodowska, A., Mascheroni, G., Zaffaroni, L. G., Cino, D., & Wartella, E. A. (2020). The datafication of childhood: Examining children's and parents' data practices, children's right to privacy and parents' dilemmas. *AoIR Selected Papers of Internet Research*. https://doi.org/10.5210/spir.v2020i0.11136

McCallum, S., & Singleton, T. (2024). Period trackers 'coercing' women into sharing risky information. *BBC*. https://www.bbc.com/news/articles/cmj6j3d8xjjo

McShane, J. (2023, February 19). *Va. Republicans shelve bill to protect menstrual data from search warrants*. NBC News. https://www.nbcnews.com/health/womens-health/va-republicans-shelve-bill-protect-menstrual-data-search-warrants-rcna71167

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References

Moore, A. (2022, September 5). ‘I didn’t want it anywhere near me’: How the Apple AirTag became a gift to stalkers. *The Guardian*. https://www.theguardian.com/technology/2022/sep/05/i-didnt-want-it-anywhere-near-me-how-the-apple-airtag-became-a-gift-to-stalkers

Paiva, L. D. S., Oliveira, J., & De França, T. C. (2023). The use of wearables to prevent sexual violence: A systematic mapping study. *Proceedings of the XIX Brazilian Symposium on Information Systems*, 260–267. https://doi.org/10.1145/3592813.3592913

Paulsen, N. (2022). 43 Prozent der Deutschen nutzen Smart-Home-Technologien | Presseinformation | Bitkom e.V. *Bitkom*. https://www.bitkom.org/Presse/Presseinformation/Smart-Home-2022

Rose, K., Eldridge, S., & Chapin, L. (2015). *The Internet of Things: An overview*. The Internet Society.

Seidel, S., & Berente, N. (2020). Automate, informate, and generate: Affordance primitives of smart devices and the internet of things. In S. Nambisan, K. Lyytinen, & Y. Yoo (Eds.),

Handbook of Digital Innovation. Edward Elgar Publishing. https://www.e-elgar.com/shop/gbp/handbook-of-digital-innovation-9781788119979.html

Stentoft, J., Rajkumar, C., & Madsen, E. S. (2017). *Industry 4.0 in Danish industry: An empirical investigation of the degree of knowledge, perceived relevance and current practice*. University of Southern Denmark.

Tichenor, P. J., Donohue, G. A., & Olien, C. N. (1970). Tichenor, P. J., Donohue, G. A., & Olien, C. N. (1970). Mass media flow and differential growth in knowledge. *The Public Opinion Quarterly*, *34*(2), 159–170.

Trepte, S. (2020). The social media privacy model: Privacy and communication in the light of social media affordances. *Communication Theory*, qtz035. https://doi.org/10/ggwrg6

Van Deursen, A. J. A. M., Van Der Zeeuw, A., De Boer, P., Jansen, G., & Van Rompay, T. (2021). Digital inequalities in the Internet of Things: Differences in attitudes, material access, skills, and usage. *Information, Communication & Society*, *24*(2), 258–276. https://doi.org/10.1080/1369118X.2019.1646777

Van Deursen, A. J., & Van Dijk, J. A. (2019). The first-level digital divide shifts from inequalities in physical access to inequalities in material access. *New Media & Society*, *21*(2), 354–375. https://doi.org/10.1177/1461444818797082

Van Deursen, A. J., & Van Dijk, J. A. (2019). The first-level digital divide shifts from inequalities in physical access to inequalities in material access. *New Media & Society*, *21*(2), 354–375. https://doi.org/10.1177/1461444818797082

SDU DDC digital democracy centre

Hate speech on social media platforms

Guest lecture April 16th

Who am I?

Anne Clausen

PhD Student

Email: aclausen@sam.sdu.dk

Anne Clausen is a PhD student at the Digital Democracy Centre at SDU. She holds a master degree in Public Health from Aarhus University. Her research focuses on investigating the spreading of intolerance on social media platforms. She is specially interested in investigating the dynamics of intolerance with machine learning methods.

Today's lecture

1st half:

- ❑ Theoretical background on hate speech and the opportunity structures social media platforms provide for its spread

Discussion with peers (5 mins):

- ❑ Why is hate speech on social media platforms a problem?
- ❑ Post your answers in Padlet
- ❑ Recap in plenum

2nd half:

- ❑ Insights from an ongoing study of hate speech targeting sexual and gender minority targets in Danish Facebook comments

What is hate speech?

- Discriminatory language or hateful expressions aimed at individuals because of their (ascribed) collective identity or membership in a social group (e.g. sexual orientation, gender identity, ethnicity, religion etc.) (United Nations, 2019; Frischlich, 2023)

Explicit hate speech

- Overt slurs, pejoratives or threats of violence (Paasch-Colberg et al., 2021)

Implicit hate speech

- Negative stereotyping (Rieger et al., 2021) or disguised as humor (Matamoros-Fernández et al., 2023)

References:

Frischlich, L. (2023). Hate and harm. In S. Paasch-Colberg, C. Strippel, & M. Emmer (Eds.), *Challenges and perspectives of hate speech analysis: An interdisciplinary anthology* (Vol. 12). Digital Communication Research.

Matamoros-Fernández, A., Bartolo, L., & Troynar, L. (2023). Humour as an online safety issue: Exploring solutions to help platforms better address this form of expression. *Internet Policy Review, 12*(1). https://doi.org/https://doi.org/10.14763/2023.1.1677

Paasch-Colberg, S., Strippel, C., Trebbe, J., & Emmer, M. (2021). From Insult to Hate Speech: Mapping Offensive Language in German User Comments on Immigration. *Media and Communication*, *9*(1), 171–180. https://doi.org/https://doi.org/10.17645/mac.v9i1.3399

Rieger, D., Kümpel, A. S., Wich, M., Kiening, T., & Groh, G. (2021). Assessing the extent and types of hate speech in fringe communities: A case study of alt-right communities on 8chan, 4chan, and Reddit. *Social Media + Society*, 1–14. https://doi.org/https://doi.org/10.1177/20563051211052906

United Nations. (2019). *The UN Strategy and Plan of Action on Hate Speech*. https://www.un.org/en/genocideprevention/documents/advising-and-mobilizing/Action_plan_on_hate_speech_EN.pdf

Hate speech on social media platforms

- Many social media platforms are available to users today, each characterized by distinct platform design (Bayer et al.2020).
  - Mainstream platforms: Facebook, Instagram, YouTube, TikTok, Twitter/X
  - Alternative platforms: Telegram, Gab, 4chan

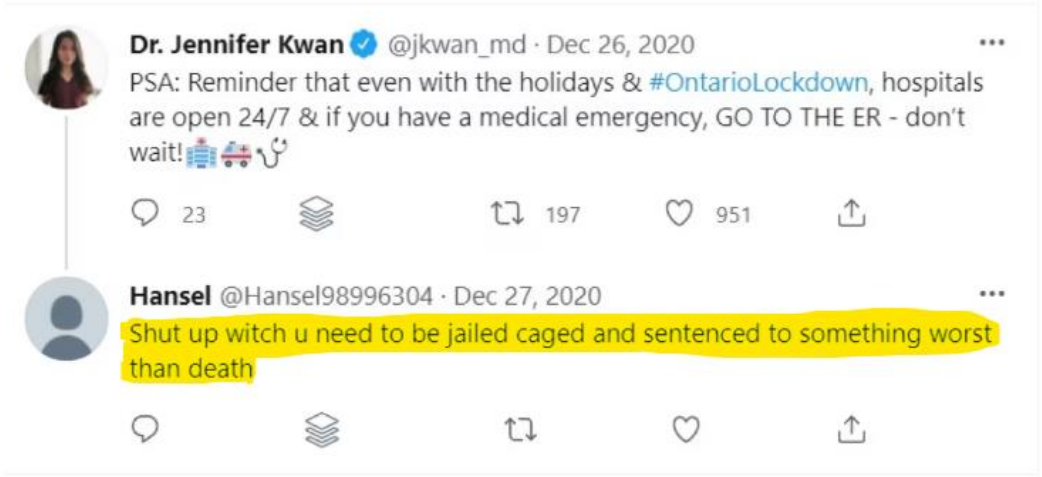
- The average user nowadays uses 6.8 different platforms each month (Kemp, 2025).
- Social media platforms differ from each other in terms of content modality (text, visual, audiovisual), technical features (e.g. like button), and affordances (e.g. anonymity).
- Different platform designs create different **opportunity structures** for the spreading of intolerance and hate speech in online spaces.

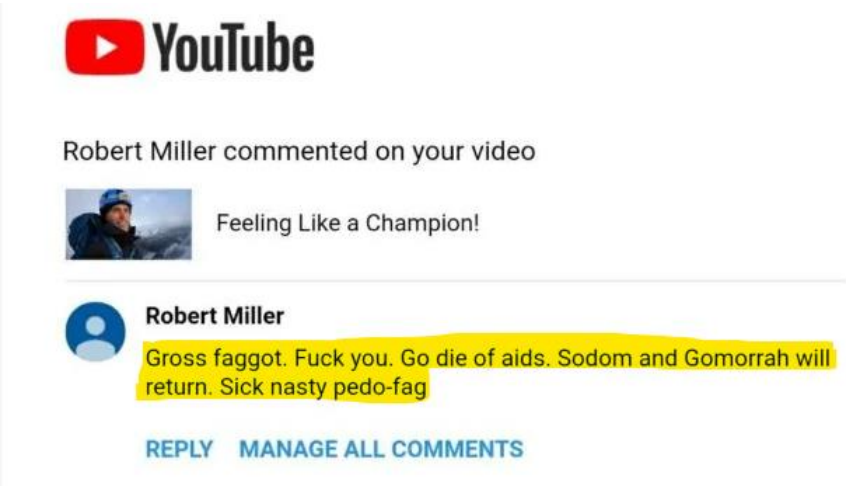
References:
Bayer, J. B., Triệu, P., & Ellison, N. B. (2020). Social media elements, ecologies, and effects. *Annual Review of Psychology*, *71*(1), 471–497. https://doi.org/10.1146/annurev-psych-010419-050944
Kemp, S. (2025). Digital 2025: Global overview report (p. 372). DATAREPORTAL. https://datareportal.com/reports/digital-2025-global-overview-report

Opportunity structures: content modalities

Hate speech on social media can be expressed in different content modalities:

- ❑ Textual (comments, messages etc.)
- ❑ Visual (images, memes etc.)
- ❑ Audiovisual (videos, reels etc.)





Opportunity structures: content modalities

Multimodality:

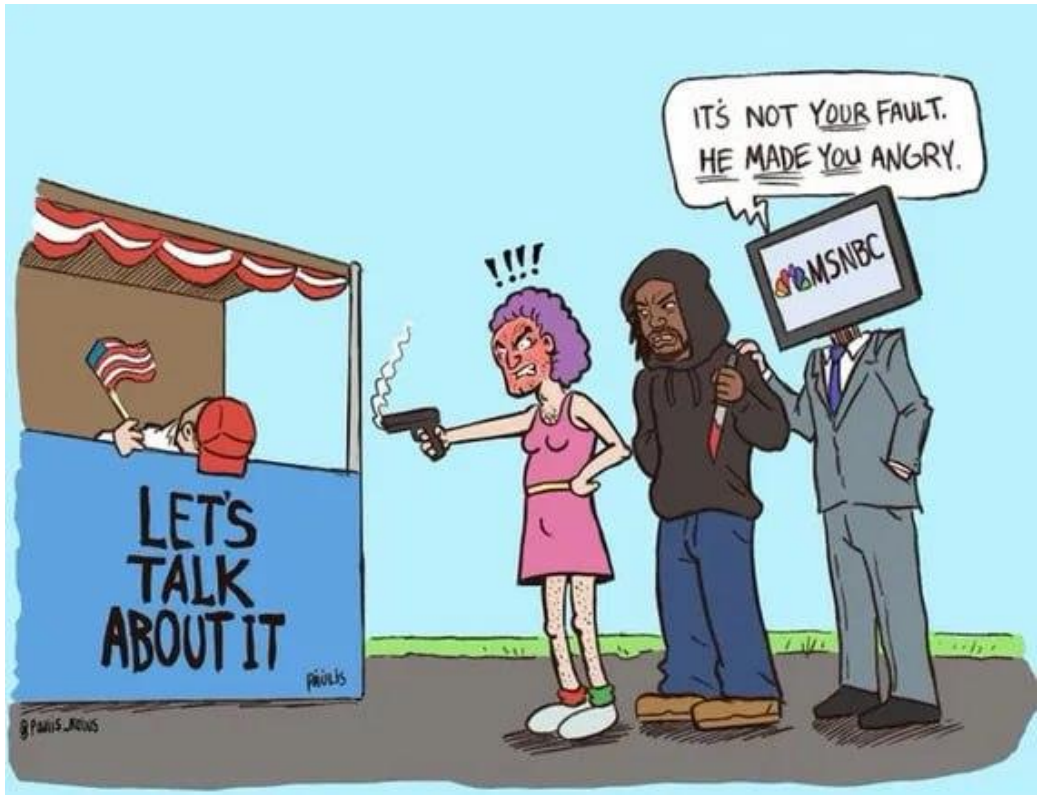
- The combination of multiple modalities to express meaning (Adami, 2017)

- Social media content can be multimodal if it combines text, images, audio and video
  - Example: TikTok video with audio and text overlays
  - Example: Memes (images with text-embeddings)

References:
Adami, E. (2017). Introducing multimodality. In García O., Flores N., & Spotti M. (Eds.), The Oxford handbook of language and society (pp. 451–472). Oxford University Press.

Multimodal hate: memetic examples collected from public Telegram channels

Multimodal hate: memetic examples collected from public Telegram channels



Opportunity structures: platform features & affordances

Platform features (Bimber & Gil de Zúñiga, 2020; Schulze et al.2024):

- ❑ Technically embedded in the platform design – allow for user interaction with the content – examples: follow, share, comment, like.

Platform affordances (Evans et al., 2017; Schulze et al.2024):

- ❑ ”Action possibilites” – emerges when users interact with the features of a platform – examples: anonymity, visibility, collectivity, interactivity

References:

Bimber, B., & Gil de Zúñiga, H. (2020). The unedited public sphere. *New Media & Society*, *22*(4), 700–715. https://doi.org/10/gg8hmr

Evans, S. K., Pearce, K. E., Vitak, J., & Treem, J. W. (2017). Explicating affordances: A conceptual framework for understanding affordances in communication research. *Journal of Computer-Mediated Communication*, *22*(1), 35–52. https://doi.org/10.1111/jcc4.12180

Schulze, H., Greipl, S., Hohner, J., & Rieger, D. (2024). Social media and radicalization: An affordance approach for cross-platform comparison. *Medien & Kommunikationswissenschaft*, *72*(2), 187–212. https://doi.org/10.5771/1615-634X-2024-2-187

Opportunity structures: platform features & affordances

Examples from Twitter/X and how it relates to the creation and spread of hate speech:

- Feature: freely selectable handle and account name
- Affordance: anonymity
  - Anonymity eases hate speech by allowing users to disconnect online behavior from offline identity, which can lead to toxic disinhibition and the expression of hatred unlikely to occur in face-to-face interactions (see online disinhibition effect, Suler, 2004)

- Feature: retweet/repost function
- Affordance: visibility
  - Hate speech achieves higher visibility than non-hateful content via social media sharing mechanisms, evidenced by retweet dynamics (Maarouf et al., 2024)

References:
Maarouf, A., Pröllochs, N., & Feuerriegel, S. (2024). The virality of hate speech on social media. Proceedings of the ACM on Human-Computer Interaction, 8(CSCW1), 1–22. https://doi.org/10.1145/3641025
Suler, J. (2004). The online disinhibition effect. CyberPsychology & Behavior, 7(3), 321–326. https://doi.org/10.1089/1094931041291295

Discuss (5 mins): Why is hate speech on social media problematic?

Enter your answers into Padlet:
https://padlet.com/aclausen11/why-is-hate-speech-on-social-media-problematic-4av06aut6vpa0vmd

Recap in plenum in 5 mins

Why is hate speech on social media problematic?

- Hate speech is harmful to its victims and can trigger trauma responses in targets (Leets, 2002)
- Online hate speech incites to offline hate crime/violence (Arcila Calderón et al., 2024)
- Online hate speech is a threat to democratic participation because targets refrain from partipating in online debates to avoid attacks, thereby affecting the inclusiveness of digital public spheres (Amnesty International Danmark, 2025)

References:

Amnesty International Danmark. (2025). *HAD SKADER – 22 ÅRS RETSPRAKSIS PÅ STRAFFELOVENS § 266 B*. https://amnesty.dk/wp-content/uploads/2025/01/Amnesty_HadskaderRapport2024_Digital_FINAL.pdf

Arcila Calderón, C., Sánchez Holgado, P., Gómez, J. *et al.* From online hate speech to offline hate crime: the role of inflammatory language in forecasting violence against migrant and LGBT communities. *Humanit Soc Sci Commun* **11**, 1369 (2024). https://doi.org/10.1057/s41599-024-03899-1

Leets, L. (2002). Experiencing hate speech: Perceptions and responses to anti-Semitism and antigay speech. Journal of Social Issues, 58(2), 341–361. https://doi.org/10.1111/1540-4560.00264

Insights from a study of hate speech in Danish Facebook comments

- The study examines content characteristics of explicit and implicit forms of hate speech targeting sexual and gender minorities in Facebook comments collected from public pages of Danish news media, politicians, political parties and citizen-driven groups.
- The study received a data donation from the NGO ‘Os & Data’ who originally collected the data for the report ‘Attacks and Hate in the Public Debate on Facebook’ (Analyse & Tal & Os & Data, 2025)
- The dataset covered years 2021-2024 and was collected end-year of 2024, and the data is therefore considered after moderation

References
Analyse & Tal, & Os & Data. (2025). Angreb & had i den offentlige debat på Facebook. https://www.tryghed.dk/-/media/files/pdf/publikationer/2025/2025-angreb-og-had-i-den-offentlige-debat.pdf

Insights from a study of hate speech in Danish Facebook comments

- The NGO used an AI-based algorithm (Att&ck) to identify attacks (broad definition including both offensive language and hate speech) and a keyword-based topic model to identify attacks targeting sexual and gender minorities (Analyse & Tal & Os & Data, 2025).

- The Att&ck algorithm identified 27.587 comments as attacks against sexual and gender minoritities.

- The current study analyses a random sample if 2.000 comments

References
Analyse & Tal, & Os & Data. (2025). Angreb & had i den offentlige debat på Facebook. https://www.tryghed.dk/-/media/files/pdf/publikationer/2025/2025-angreb-og-had-i-den-offentlige-debat.pdf

Explicit and implicit hate speech in Danish Facebook comments

Manual content analysis of random sample of 2.000 comments

- The codebook was developed deductively based on dimensions highlighted as central to the characterization of explicit and implicit hate speech in previous research.
- Two human coders read through the comments and annotated for each comment if it contained characteristics of hate speech.

Explicit and implicit hate speech in Danish Facebook comments

Comments were coded as ***explicit hate speech*** if they used slurs, pejoratives or called for violence to attack the target group

- Slurs were considered as derogatory terms whose meaning is inherently discriminating against the target group (Hess, 2021) (e.g. the homophobic slur *'faggot'*)
- Pejoratives were considered descriptive words or expressions containing a negative evaluation about the target group (Miscevic, 2017), which does not in itself demean the target group (e.g. *'sick', 'stupid', 'disgusting'*)
- Calls for violence considered expressions of violence, threats, harm or killing (Paasch-Colberg et al., 2021)

References:
Hess, L. (2021). Slurs: Semantic and Pragmatic Theories of Meaning. In P. Stalmaszczyk (Ed.), *The Cambridge Handbook of the Philosophy of Language* (pp. 450–466). Cambridge University Press. Cambridge Core. https://doi.org/10.1017/9781108698283.026
Miscevic, N. (2017). The semantic structure of Pejoratives. *Linguistik Online*, *82*. https://doi.org/10.13092/lo.82.3714
Paasch-Colberg, S., Strippel, C., Trebbe, J., & Emmer, M. (2021). From Insult to Hate Speech: Mapping Offensive Language in German User Comments on Immigration. *Media and Communication*, *9*(1), 171–180. https://doi.org/https://doi.org/10.17645/mac.v9i1.3399

Explicit and implicit hate speech in Danish Facebook comments

Comments were coded as ***implicit hate*** speech if they used negative stereotyping, humor or expressed limitation of rights to attack the target group.

- Negative stereotyping considered expressions of negative overgeneralizations or prejudice about the attributes or behaviors of the target group (Marx & Ko, 2019).
- Humor considered expressions of irony, sarcasm or jokes to demean the target group (Matamoros-Fernández et al., 2023).
- Expressions of limitation of rights considered mentions of removal of rights, freedoms and opportunities for the target group (ElSherief et al., 2021) (e.g. statements denying sexual minorities the right to family formation).

References:
ElSherief, M., Ziems, C., Muchlinski, D., Anupindi, V., Seybolt, J., De Choudhury, M., & Yang, D. (2021). Latent Hatred: A Benchmark for Understanding Implicit Hate Speech. In M.-F. Moens, X. Huang, L. Specia, & S. W. Yih (Eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (pp. 345–363). Association for Computational Linguistics. https://doi.org/10.18653/v1/2021.emnlp-main.29
Marx, D., & Ko, S. J. (2019). Stereotypes and Prejudice. *Oxford Research Encyclopedia of Psychology*. https://doi.org/10.1093/acrefore/9780190236557.013.307
Matamoros-Fernández, A., Bartolo, L., & Troynar, L. (2023). Humour as an online safety issue: Exploring solutions to help platforms better address this form of expression. *Internet Policy Review, 12*(1). https://doi.org/https://doi.org/10.14763/2023.1.1677

Explicit and implicit hate speech in Danish Facebook comments

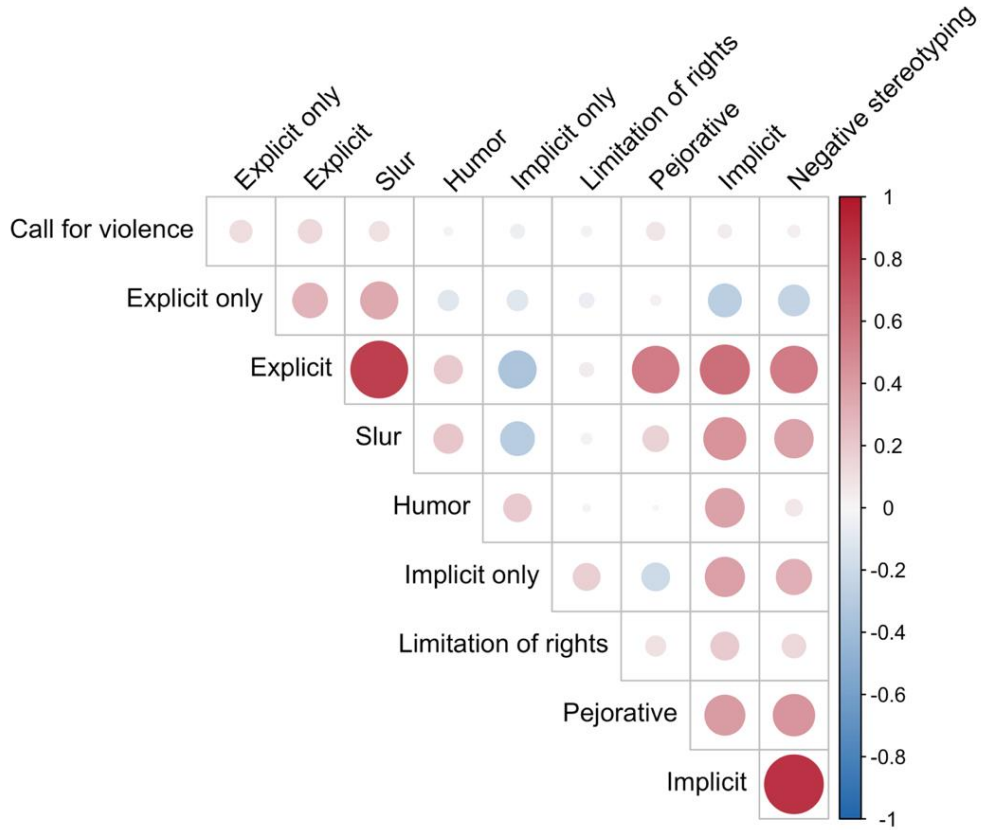
Manual content analysis of random sample of 2.000 comments

- After manual coding 1,149 (57.5%) comments were identified as hate speech going against sexual or gender minorities
- Considerable co-occurrence between explicit and implicit types of hate speech in the dataset

Hate speech comments	1,149
Explicit hate speech*	882 (76.8%)
Slur	689 (60.0%)
Pejorative	382 (33.2%)
Calls for violence	29 (2.5%)
Implicit hate speech*	1,017 (88.5%)
Negative stereotyping	883 (76.8%)
Humor	254 (22.1%)
Limitation of rights	75 (6.5%)
Explicit only	132 (11.5%)
Implicit only	267 (23.2%)
*Explicit and implicit hate speech comments are non-mutually exclusive, and comments can contain characteristics of both	

Explicit and implicit hate speech in Danish Facebook comments

- Strong positive correlation between Explicit and Slur
- Strong positive correlation between Implicit and Negative stereotyping
- Explicit and implicit hate speech frequently overlap rather than occur in isolation



Content characteristics of explicit and implicit hate speech in Danish Facebook comments

- Lexical characteristics of the different types of hate speech were examined through tf-idf (term frequency-inverse document frequency), given by (Manning et al., 2008):

$$\text{tf} - \text{idf}_{t,d} = \text{tf}_{t,d} \times \text{idf}_t$$

The term frequency ($\text{tf}_{t,d}$) measures how often a word appears in a document, given by:

$$\text{tf}_{t,d} = \frac{\text{number of times the word } t \text{ appears in document } d}{\text{total number of words in document } d}$$

The inverse document frequency (idf_t) measures the uniqueness of a word across a collection of documents, given by:

$$\text{idf}_t = \log \frac{N}{\text{df}_t}$$

References:
Manning, C. D., Raghavan, P., & Schütze, H. (2008). Scoring, term weighting and the vector space model. In *Introduction to Information Retrieval* (pp. 109–133). Cambridge University Press. (6).

Content characteristics of explicit and implicit hate speech in Danish Facebook comments

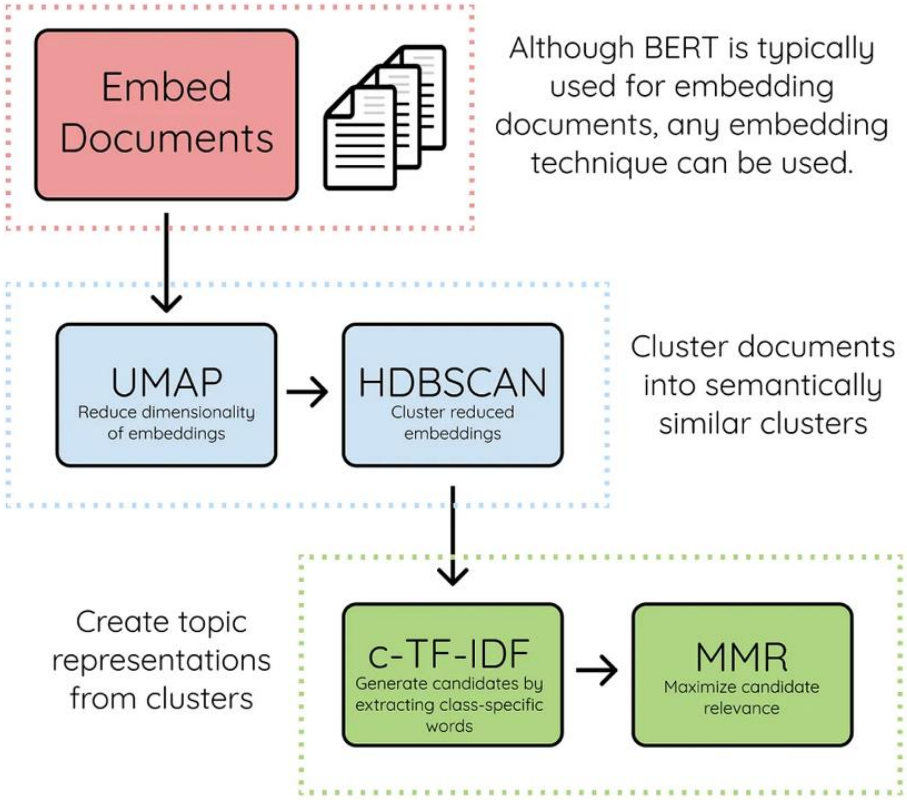
- The analysis identified the top 5 terms per hate speech according to tf-idf rank (11*5=55 terms=too many to show here)
- Most prominent tf-idf findings showcased here

Tf-idf rank	Term (da/en)
Total hate speech comments (n=1,149)	
1	svans (faggot)
2	køn (gender/sex)
3	bøsse (gay/fag)
4	homo (homo)
5	bøsser (gays/fags)

Tf-idf rank	Term (da/en)
Explicit hate speech (n=882)	
1	svans (faggot)
Implicit hate speech (n=1,017)	
1	køn (gender/sex)

Content characteristics of explicit and implicit hate speech in Danish Facebook comments

- Thematic characteristics of the different types of hate speech were examined through BERTopic (Grootendorst, 2022)
- Embeddings capture semantic meaning, compared to other topic modeling models who rely on term frequency



References:
Grootendorst, M. (2022). *BERTopic: Neural topic modeling with a class-based TF-IDF procedure* (arXiv:2203.05794). arXiv. https://doi.org/10.48550/arXiv.2203.05794

Content characteristics of explicit and implicit hate speech in Danish Facebook comments

- The analysis identified the top 5 topics per hate speech type (11*5=55 topics=too many to show here)
- Most prominent BERTopic findings for ***explicit hate speech*** (n=882) showcased here:

Topic number	Topic label	Topic representations
1	Homosexuality debates	bøsser, tror, andres, mener, homoseksuelle, hvorfor, synes, hver, heller, mennesker (gays/fags, think, others, believe, homosexuals, why, think, each, either, people)
2	Homosexuality debates	bøsse, homoseksuel, lgbt, bøsser, nogensinde, tak, kun, eu, jeres, bare (gay/fag, homosexual, lgbt, gays/fags, ever, thanks, only, eu/european union, yours, just)
3	Disgust-based derogations	køn, nåååå, bøsserøve, lillefinger, kom, kun, røvfuld, føj, går, skovl (gender/sex, ohhh, faggots, little finger, come, only, ass kicking, ew, walk, shovel)
4	Advocating physical assault	bøsse, hankøn, mener, afsky, nogensinde, næ, dræbes, køn, tænker, nogen (gay/fag, male gender/sex, believe, disgust, ever, no, kill, gender/sex, think, someone)
5	Moralizing insults	bøssekarlene, bøssekarl, bøsse, bøssevenner, bøsseskoven, homo, gay, skod, satanisk, svans (faggots, faggot, gay/fag, gay/fag friends, gay/fag forest, homo, gay, shitty, satanic, faggot)

Content characteristics of explicit and implicit hate speech in Danish Facebook comments

❑Most prominent BERTopic findings for ***call for violence*** (n=29) showcased here:

Topic number	Topic label	Topic representations
1	Advocating physical assault	kneppet, kneppede, røvfuld, mener, laver, bøsserøve, skær, bøsserøv, kæft, kun (fucked, fucked, ass kicking, believe, do, ass faggots, cut, faggot, only)
2	LGBT-phobic slurs	røven, bøssen, bøsse, lorte, queer, lgbtqrstuvxyzæøå, værste, skrev, sagde, stod (the ass, the gay/fag, gay/fag, shitty, queer, lgbtqxxxxxxx, worst, wrote, said, stood)
3	Advocating physical assault	lgbtqia, sagde, forpulede, menneskers, lesbisk, smæk, mand, kvindes, smadre, sagt (lgbtqia, said, fucking, peoples', lesbian, spanking, man, womans', smash, said)
4	Moralizing insults	synd, bare, svans, spjældet, ser, bøssekarl, kan, nosserne, hænge, mand (sin, just, faggot, jail, see, faggot, can, balls, hang, man)
5	Advocating physical assault	fanden, dræbes, uanset, fucking, fortabt, bøsse, må, luder, kommer, helst (the devil, kill, no matter, fucking, lost, gay/fag, can, whore, come, rather)

Content characteristics of explicit and implicit hate speech in Danish Facebook comments

❑Most prominent BERTopic findings for ***implicit hate speech*** (n=1,017) showcased here:

Topic number	Topic label	Topic representations
1	Homosexuality debates	bøsser, bøsse, mener, hver, synes, tror, siger, hvorfor, mennesker, lidt (gays/fags, gay/fag, believe, each, think, believe, says, why, people, little)
2	Gender identity debates	køn, hunkøn, uanset, mennesker, mener, sige, tænker, andet, kun, eksistere (gender/sex, female gender/sex, no matter, people, believe, say, thinks, other, only exist)
3	Mocking LGBT+ identities	bøsse, køn, bøsser, sige, fuck, bøsserøve, uanset, sexualitet, ladyboys, ladyboy (gay/fag, gender/sex, gays/fags, say, fuck, ass faggots, no matter, sexuality, ladyboys, ladyboy)
4	Gender diversity hostility	køn, bøsser, mand, mænd, mennesker, personen, helst, fjols, sige, kun (gender/sex, gays/fags, man, men, people, the person, rather, fool, say, only)
5	Gender identity debates	køn, hankøn, virkelig, vedkommende, kønsskifte, mener, hvorfor, andet, alligevel, samt (gender/sex, male gender/sex, really, the person, gender/sex change, believes, why, other, anyway, also)

Content characteristics of explicit and implicit hate speech in Danish Facebook comments

❑Most prominent BERTopic findings for ***limitation of rights*** (n=75) showcased here:

Topic number	Topic label	Topic representations
1	Gender segregation debates	kønsskifte, køn, mandefængsel, mændene, kvindefængsel, mænd, vedkommende, transvestit, hankøn, penis (gender/sex change, gender/sex, male prison, the men, female prison, men, the person, transvestite, male gender/sex, penis)
2	Pathologizing sexual minorities	mens, bøssedreng, lgbtq, simpelthen, homo, vrangforestillinger, bør, fanden, røven, sagt (while, gay/fag boy, lgbtq, simply, homo, delusions, should, the devil, the ass, said)
3	Gender category differentiation	transkønnede, transkvinder, transpersoner, lgbt, køn, køns, trans, mands, kvinder, mænd (transgender, transwomen, transpeople, lgbt, gender/sex, gender/sex, trans, mens', women, men)
4	Concerns for children	bøsse, homoseksuel, lesbiske, køn, hvorfor, bøsser, dummeste, barn, mennesker, voksne (gay/fag, homosexual, lesbians, gender/sex, why, gays/fags, stupidest, child, people, adults)
5	Concerns for children	køn, voksnes, voksen, hvorfor, provokerende, forskellige, børn, seksualitet, meninger, årige (gender/sex, adults', adult, why, provoking, different, children, sexuality, opinions, year-old)

In sum:

Thematic split where explicit comments favor direct insults and mocking, while implicit comments are characterized by broader debates around gender identity and the marginalization of sexual and gender minority individuals

Thanks for listening

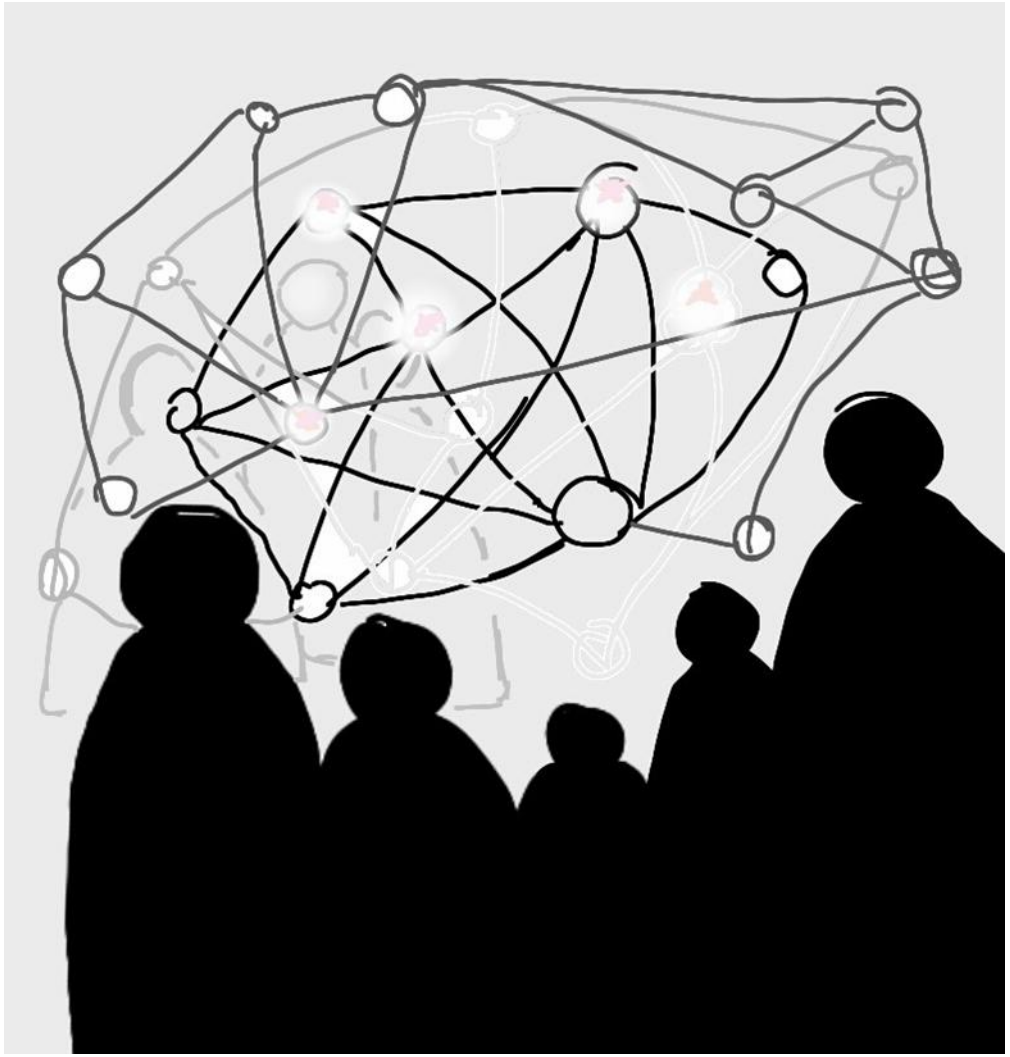
Any questions?

AI 507: Artificial Intelligence and Society

#11 AI in politics

Associate Prof. Dr. Lena Frischlich

Flashlight



Flashlight

Digital platforms

Social media

History of social media

Platform-fragmentation

“Genres” of social media

Social media use

In general

For political participation

Social media and wellbeing

Complex relationships

Windows of sensitivity

Active versus passive use

Hate speech on social media

Definition

Opportunity structures

Multi-modality

Platform features

Affordances

Anonymity

Visibility

Hate speech in the Danish public sphere

Implicit versus explicit hate speech

Sexual and gender minorities

Social media and political participation

PINGS framework

Incidential news exposure

“News-will-find-me” perception

Political knowledge

Political participation

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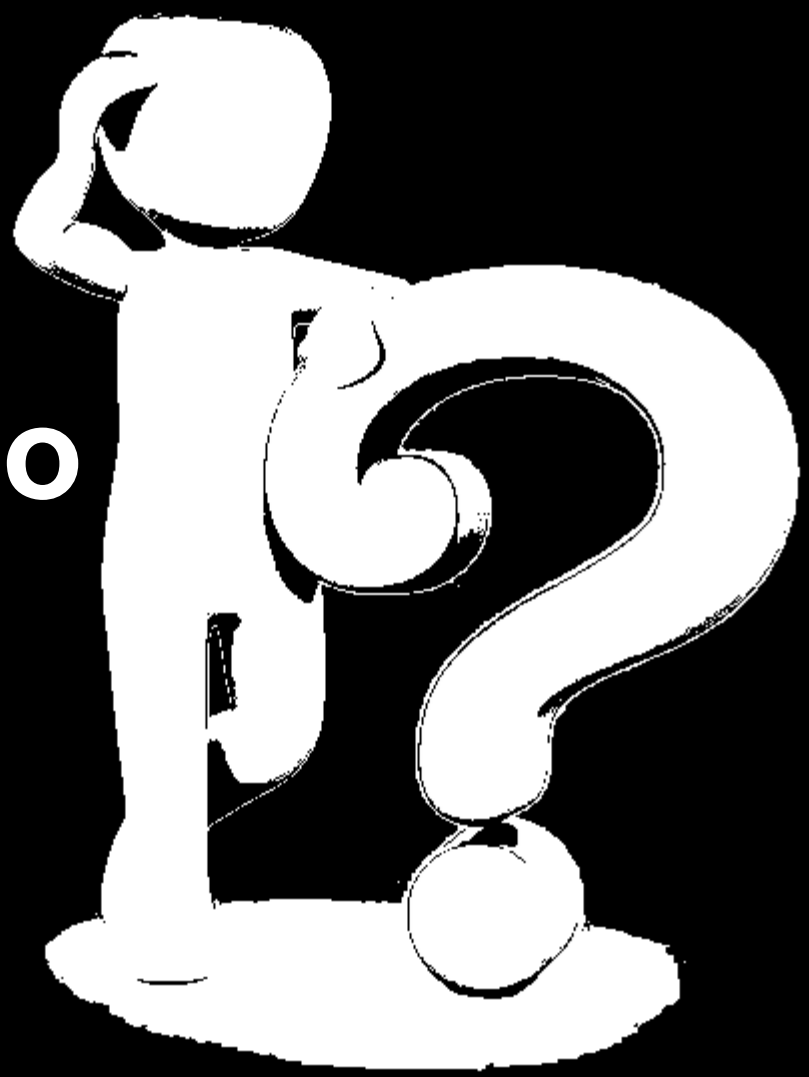
Do you have questions to last week?



Then let’s get started!

After today’s lecture, you will hopefully have…

- ❑ … initial insights into AI in public administrations
- ❑ … gained a better understanding what political communication is – and how AI affects it
- ❑ … a better understanding of “misinformation”
- ❑ … discussed how AI can affect misinformation dynamics

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AI in public administration

Public administration (AI-watch.EU, Madan & Ashok, 2023)

= “public organisations that implement government policies and may contribute to its development”

AI has the potential to be an extremely useful - or even disruptive - technology

- enabler for seeking efficiency and effectiveness in the management and delivery of public services
- E.g., automate simple and repetitive cognitive activities (Odense Kommune uses it to issue bills) freeing-up labour time for more high-value activities
- augment the predictive capabilities, enhancing data-driven decision-making
- support user-centric service personalisation, increasing the effectiveness of public service delivery

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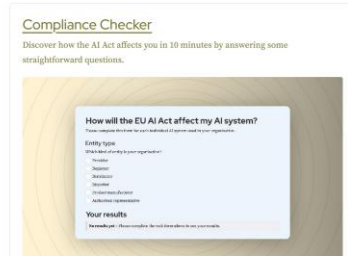
Principles (OECD; European Commission. Joint Research Centre 2023; Public sector Tech Watch 2024)

Values-based principles

- **Inclusive growth, sustainable development and well-being**
- **Human rights and democratic values, including fairness and privacy**
- **Transparency and explainability**
- **Robustness, security and safety**
- **Accountability**

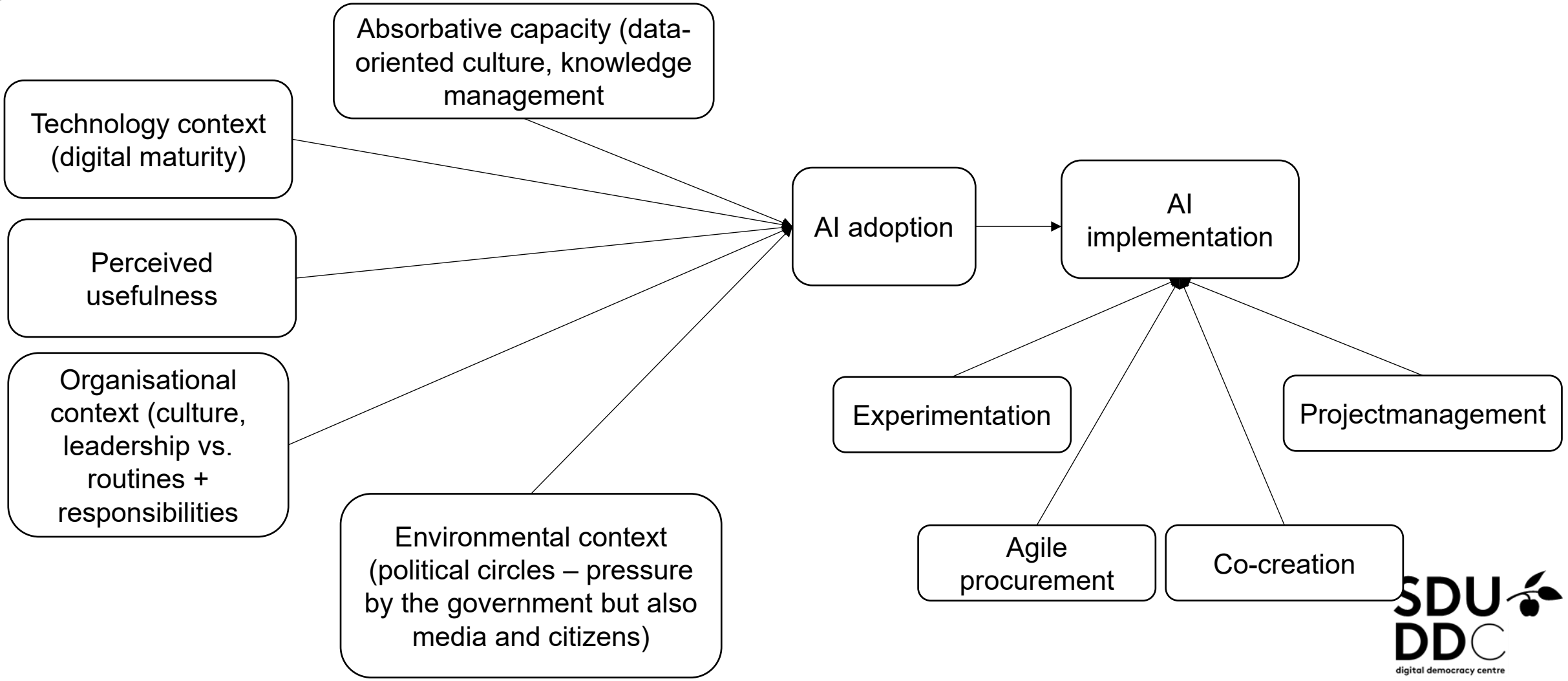
Informs standardization processes such as

- ISO/IEC1
- IEEE
- Currently working groups but expect to provide standards for, for example, bias mitigation



So, what do we know about adoption? (Madan & Ashok, 2023)

Systematic literature review, *n* = 73 publications

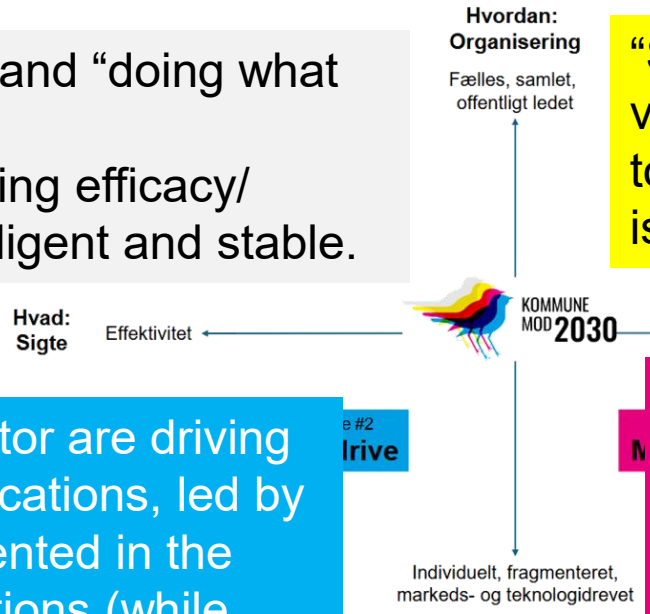


What do you think? How will the future in Odense look like? (Knudsen, 2025)

- Large project by Odense & Aarhus municipality (DDC * TrygFonden fellowship)
- Development of four scenarios for the digital commune in 2030

Fire alternative fremtider

"Aistrup": Focus is on efficiency and "doing what works".
AI is implemented when increasing efficacy/ having significant benefits, its diligent and stable.



"Smart welfare": Political ambitions and welfare values are driving the use of AI. AI is implemented to create a society that works for people. Denmark is leading and municipalities are key drivers

"Tech-drive": Market & tech sector are driving for change, large range of applications, led by a few tech-giants. AI is implemented in the form of buying the right applications (while adhering to ethics and budget)

"Opinion power": conscious citizens have strong opinions about AI. There are many competing agendas- Tech giants still have a commercial agenda, but hype and awe have faded and diverse and freedom-oriented approaches. Municipalities are part of the debate – so are critical citizens

AI and political communication

Political communication is… (McNair, 1995; Norris, 2001)

…the transmission of information

- among politicians
- the news media
- and the public

Can be

- Top-down (e.g., from government to citizens)
- Vertical (e.g., between politicians)
- Upward (e.g., from public opinion to politicians)
- A communication system

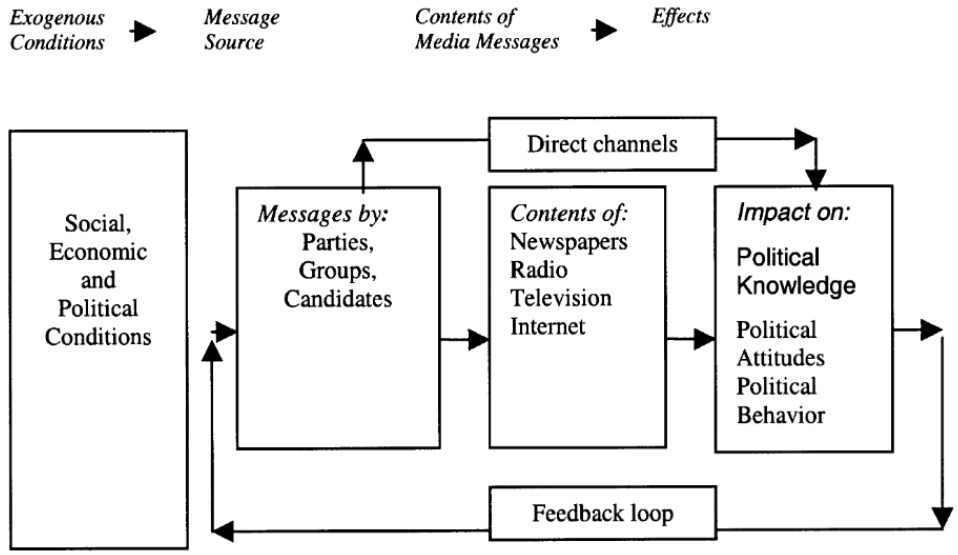
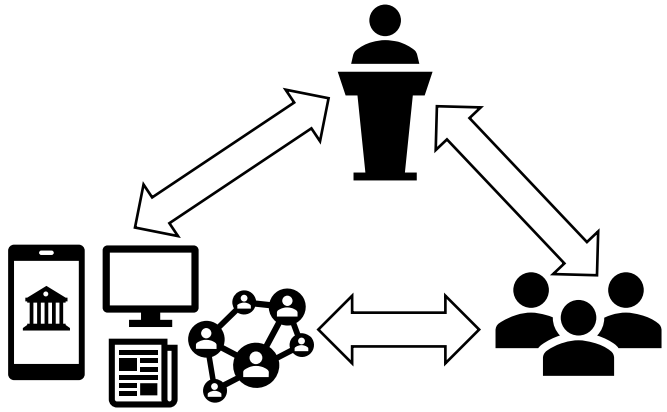


Figure 1
Systems model of political communication process



Communication by politicians is typically strategic communication (Maier et al., 2023; McNair, 1995; Marcinkowski & Metag, 2014; Stier et al. 2025)

- Overarching objective: influence the political decision-making process
- Communication is, at least partially, driven by rational considerations: when the perceived benefits are larger than the perceived risks, communication becomes more likely
- E.g., candidates engage in negative campaigning when they hope to get votes for it
- Behavioral opportunities also matter
- Sometimes candidates engage in negative campaigning despite its potentially negative effects on relevant stakeholders

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AI can affect this system in many ways (Norris, 2001)

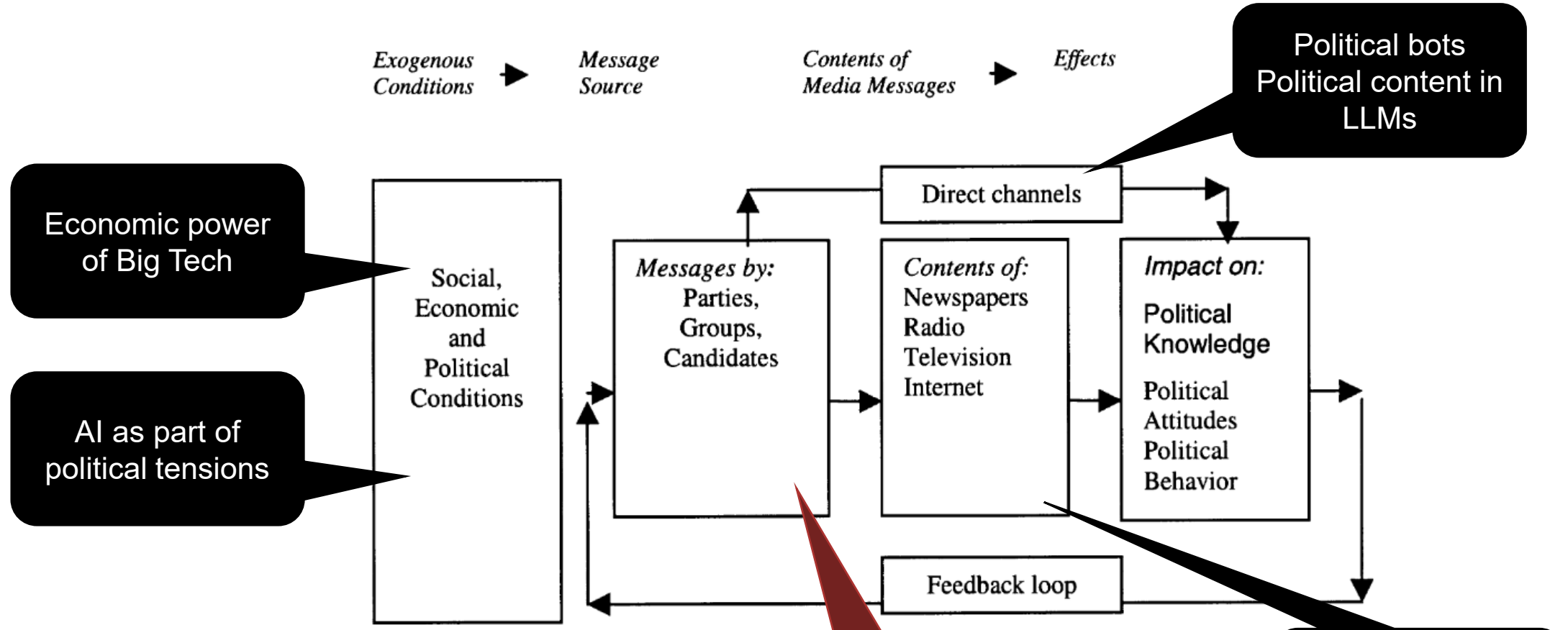


Figure 1
Systems model of political communication process

Political bots
Political content in LLMs

Economic power of Big Tech

AI as part of political tensions

AI-generated content

Recommender algorithms

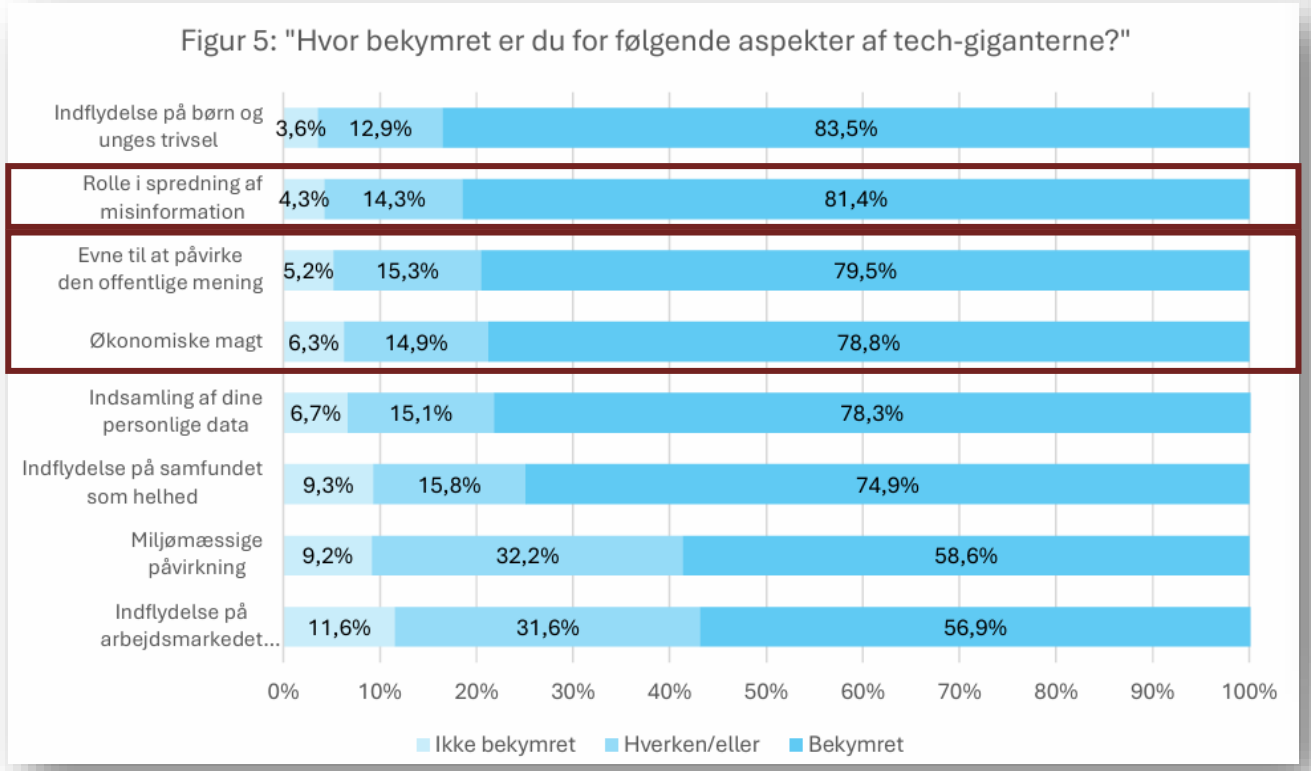
AI-generated content has become more “normal” in political communication

https://www.npr.org/2025/02/26/nx-s1-5309695/trump-gaza-video

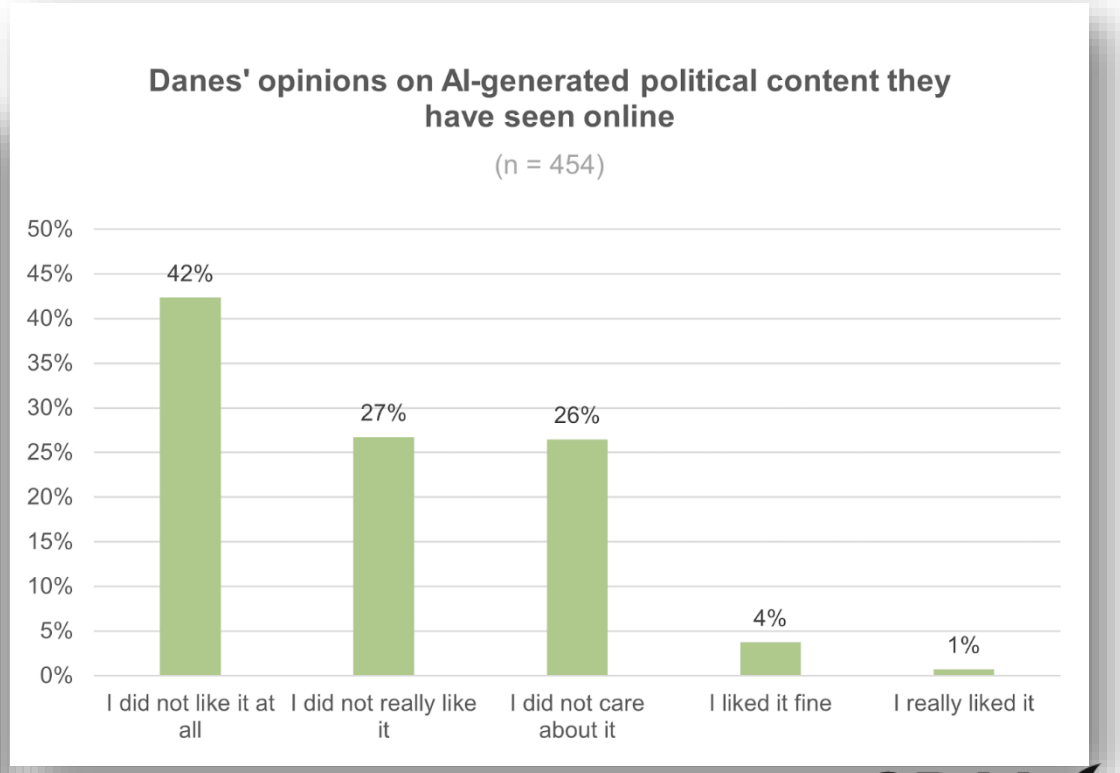
AI generated images of President Donald Trump shared on X via the official @WhiteHouse channel depict the President as a Jedi and the Pope. Credit - @WhiteHouse / X

But what do the Danes actually think about it? (Filstrup Høllsberg et al., 2025; Johansen et al., *in prep*)

- Part of DDC's „Citizens, digitalisation and AI" project survey
- Regular surveys with ~1200-1400 Danes >18



Notes. *N* = 1201, representative for age, gender, and region, June 2025



Notes. *N* = 1494, representative for age, gender, and region, March 2026

Questions before the break?

Mis-and disinformation

Misinformation (Freiling et al., 2023; Nussbaum, 2011)

- Not so easy to define
- “Claims that, intentionally or not, at odds with the currently available best evidence”

➢ Bound to temporal dynamics, evidence can change

➢ Challenges:

  - Some claims are easier to confirm than others
  - Borders between misinformation, fiction, creative thinking, and entertainment can be blurred
  - Liberal pluralism: there is a democratic right to believe in silly things- as long as you tolerate others who believe in other silly things

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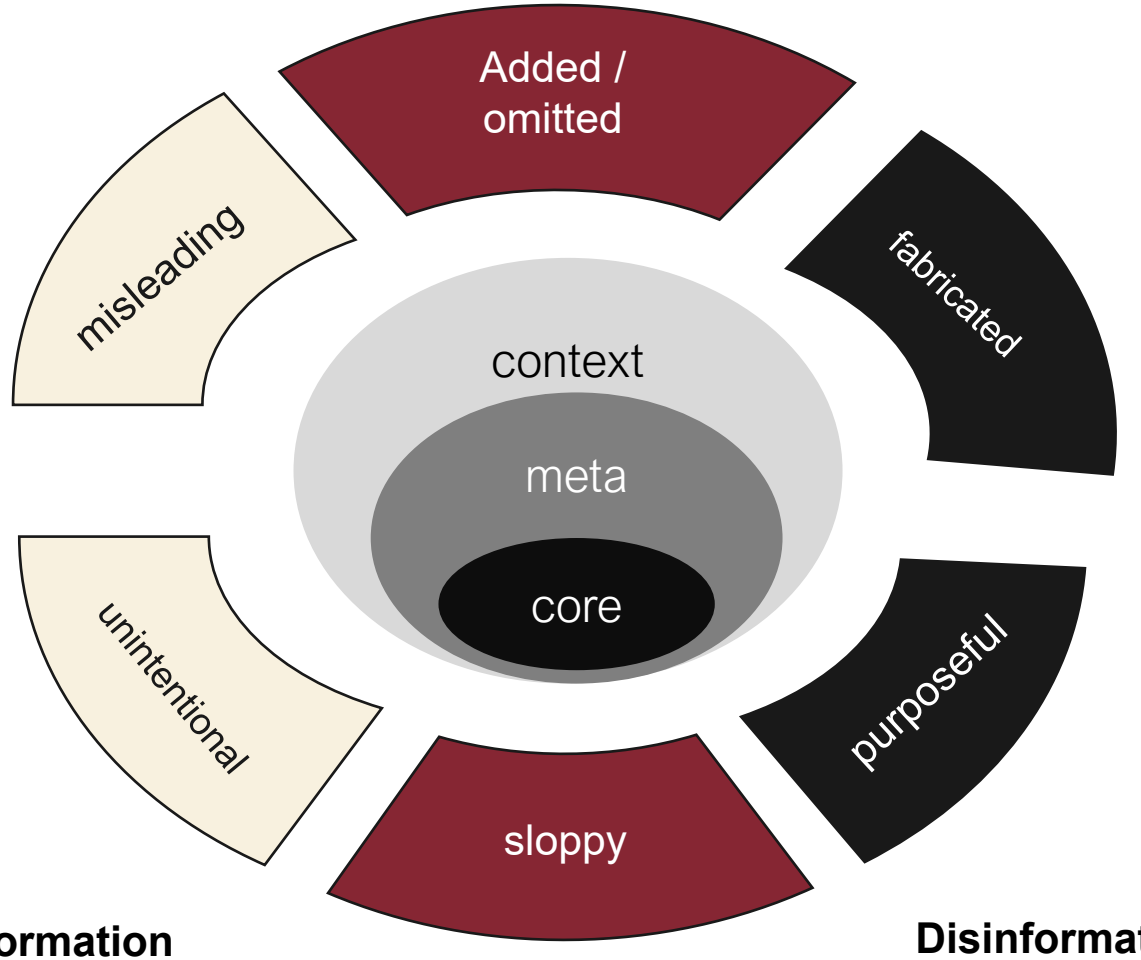
Part of a larger information disorder

(Quandt et al., 2019; Wardle, 2018)

Via www.tjekdet.dk 2024



Misinformation

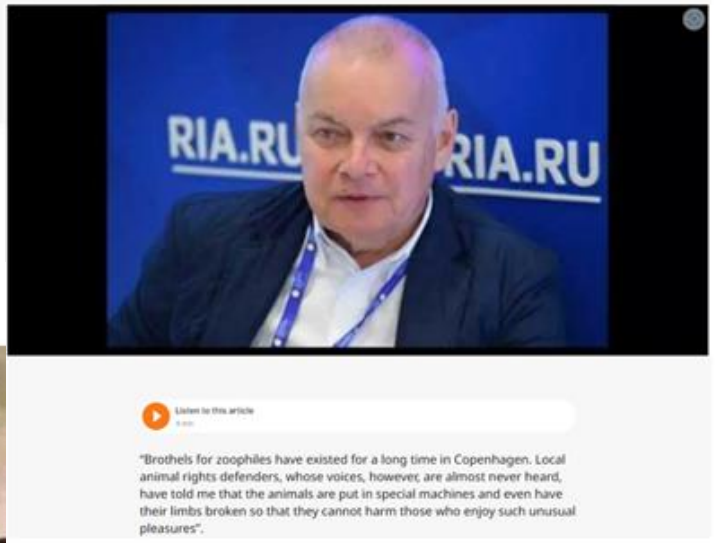
Disinformation

Typical disinformation campaigns (1)

(Jowett & O'Donnell, 2012; https://euvsdisinfo.eu/no-denmark-is-not-legalising-sexual-abuse-of-animals/)

- Placing the deception in a (seemingly) credible environment
- Afterwards: referring to the „credible" source
  - E.g., the "dog brothels in Copenhagen" story was first published by a marginal Russian outlet
  - Then being taken up by RIA – the public Russian news agency
  - Circulated across different countries and fed into narratives of Western moral corruption

This image was used as illustration of a similar story about alleged animal brothels in Denmark, published by a marginal Russian outlet, as we reported in September. (Image: Screenshot from Kalokol Rossii)



https://x.com/BolloDeanna/status/1205954112598216704

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Typical disinformation campaigns (2)

(Bond, 2023; Jowett & O'Donnell, 2012; Ryan et al., 2023)

- Simulation of a (seemingly) credible source
- The source spreads the deception
- The actual senders are no longer detectable
  - E.g., Video competitions by the Chinese government for online bloggers
  - Afterwards, they reported about "farming life" in Xingjian (were the Uyghur minority is subject to religious persecution and forced labour)
- Other example: LLM-seeding

Figure 4: Screenshot from YouTube shows Israeli vlogger Raz Gal-Or in a 'YChina' vlog about cotton farming in Xinjiang. On the right, YouTube recommends further state-aligned vlogs on Xinjiang from foreign influencers, including those who have participated in PRC state-run short-video competitions. The screenshot was taken while accessing YouTube via a Washington DC VPN server, on 22 May 2023, with the US selected as 'country' in YouTube settings.

Source: YouTube.

Via Bond et al. 2023

Typical disinformation campaigns (3)

(u.a., Sutton & Douglas, 2020 ; Uscinski et al.,2022, Van Prooijen & Douglas, 2017; van der Heide, 2013)

- Dissemination of conspiracy theories to fuel societal conflict lines

Conspiracy theory:

- …explanation of historical/ current/ future events through the secret and malicious action of powerful group (the conspirators)
- …often
  - patterns behind events, that are plausible, unconnected („best available evidence“)
  - in opposition to the general knowledge („counter-hegemonial“)
- Not always false, but can be more or less unjustified and anti-democratic

- Also used in political communication
  In 2002, the Bush administration and Tony Blair began speaking of weapons of mass destruction in Iraq and suggesting that they would go to Al-Qaeda
- The invasion of Iraq was justified by the war on terror and the “axis of evil”

US Secretary of State Colin Powell holding a model vial of anthrax while giving the presentation to the United Nations Security Council on 5 February 2003

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How big is the problem?

Prevalences and effects

Prevalence rates are difficult to determine

Different Sources

1. Monitoring/ Content analyses
2. Subjective experience
3. Platform information
4. Safety authorities

Out of 9354 Danish Tweets in March 2020 about facemasks, only **5%** engaged with misinformation, **3%** spread it

(Johansen et al. 2022)

Out of 1040 Danes **66%** are concerned about misperceptions about Denmark due to misinformation

(Center for Sociale Medier, Tech og Demokrati, 2023)

PET warns from illegal foreign interferences aimed at sowing polarisation and conflict

Out of 410 Danish children aged 5-12 **90%** say that they have seen misleading content online

(Adler-Berg et al., 2025)

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Misinformation effects as persuasion process

(Chaiken, 1980; Kahnemann, 2012; Petty & Cacioppo, 1986)

- Persuasion is typically described a two-process model
  - Elaboration-Likelihood Model
  - Heuristic-Systematic Model
- Based on the two cognitive processes (System I & System II)
  - Path 1: Fast, heuristic, peripheral
  - Path 2: Slow, elaborated, central
- General observation: heuristic processing is driven by biases, short-cuts and heuristics and does not lead to long-term attitude change
- Systematic or central “System II” processing leads to sustainable persuasion

John Cacioppo & Richard Petty via https://smileworking.blogspot.com/2010/09/elaboration-likelihood-model.html

Shelley Chaiken https://chaiken.socialpsychology.org/

Why do people fall for misinformation? (f. an overview, s. Pennycook & Rand, 2021)

(1) Cognitive style

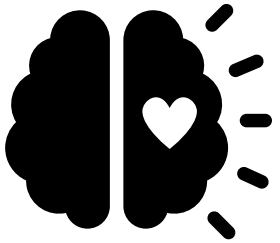
Misinformation is often not processed carefully, but heuristically ("System I" only)

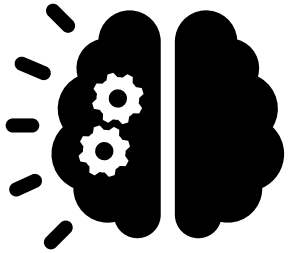
Persons with

- ... a more pronounced "bullshit" receptivity (tendency to infer meaning from fancy words)
- ... a low need for cognition
- ... a lower cognitive performance capacity

... tend to believe more in misinformation

But: Unclear how "lasting" the misperception is





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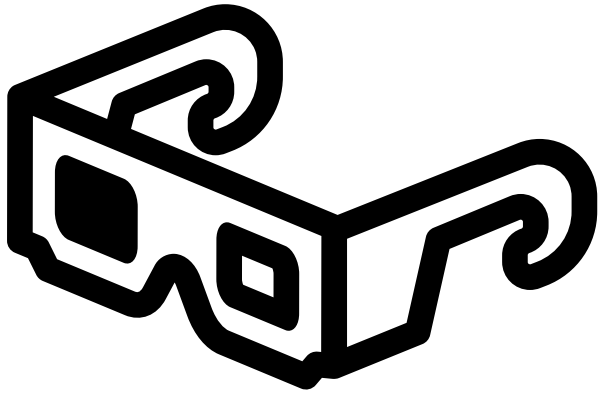
Why do people fall for misinformation?

(Faragó et al. 2020; Kunda,1990; Koch et al. 2023; Pereira et al., 2021; Tajfel & Turner, 2004)

(2) Motivated cognition

- Basic idea:
  - Preference for content that confirms one's own worldview
  - Preference for content that puts the ingroup in a good light



- For example, voters are more likely to believe disinformation that puts their party in a good light or makes the other party look bad

Why do people fall for misinformation?

(Goertzl, 1994; Imhoff & Bruder, 2014; Sutton & Douglas, 2020; Uscinski et al. 2022; van der Linden et al., 2020; Wood et al., 2012)

(3) Generalized ideological attitudes: Conspiracy mentality

Disposition to suspect conspiracies behind significant events

- Fundamental (*not* constructive) mistrust towards influential people and institutions
- Belief in various conspiracy narratives
- More frequent at the extremes on both sides of the ideological spectrum
- Strength of the correlation asymmetrical: Right > Left
- Over time and contexts, shares are relatively stable in the general population

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How does misinformation affect politics?

- The consumption of conspiracy theories can...
  - undermine confidence in democratic institutions (van Prooijen et al., 2022)
  - weaken the willingness to engage in normative political participation, such as voting

    (Einstein & Glick, 2015, Kim & Cao, 2016, Pummerer et al., 2020)
- The consumption of misinformation can...
  - …lead to misperceptions (Nielsen et al., 2021)
  - …block the retrieval of the correct information from memory (Fazio et al. 2013)
  - …become more credible when repeated (Fazio et al. 2019)
- Misinformation likely does not shape voting behavior at scale or turn recipients into zombies
- Plus, “doom speak” can also have adverse effects (Jungherr & Rauchfleisch, 2024)

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Dictator game time: How would you use AI to confuse your enemies, motivate your people, and ensure the support of your allies?

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Questions?

AI 507: Artificial Intelligence and Society

#11 AI in politics

Associate Prof. Dr. Lena Frischlich

Then I hope now you have…

- ❑ … initial insights into AI in public administrations
- ❑ … gained a better understanding what political communication is – and how AI affects it
- ❑ … a better understanding of “misinformation”
- ❑ … discussed how AI can affect misinformation dynamics

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References

Adler Berg, F. S., Heiselberg, L., Enemark Lundtofte, T., & Frischlich, L. (2025). Børn of misinformation på sociale medier: De 5-12 åriges oplevelser og perspektiver. Center for Journalistik, Medievidenskab og Digital Democracy Centre, Syddansk Universitet. https://www.sdu.dk/-/media/files/samf/boern-og-misinformation-sociale-medier.pdf

Bond, S. (2023, November 30). Meta warns that China is stepping up its online social media influence operations. NPR. https://www.npr.org/2023/11/30/1215898523/meta-warns-china-online-social-media-influence-operations-facebook-elections

Chaiken, S. (1980). Heuristic versus systematic information processing and the use of source versus message cues in persuasion. Journal of Personality and Social Psychology, 39(5), 752–766. https://doi.org/10/bh88sk

Derczynski, L., Albert-Lindqvist, T. O., Bendsen, M. V., Inie, N., Pedersen, V. D., & Pedersen, J. E. (2019, October 31). Misinformation on Twitter During the Danish National Election: A Case Study. Proceedings of the Conference for Truth and Trust Online 2019. Conference for Truth and Trust Online 2019. https://doi.org/10.36370/tto.2019.16

European Commission. Joint Research Centre. (2023). AI Watch: Artificial intelligence standardisation landscape update. Publications Office. https://data.europa.eu/doi/10.2760/131984

Faragó, L., Kende, A., & Krekó, P. (2020). We only believe in news that we doctored ourselves: The connection between partisanship and political fake news. Social Psychology, 51(2), 77–90. https://doi.org/10/gf876t

Fazio, L. K., Barber, S. J., Rajaram, S., Ornstein, P., & Marsh, E. J. (2013). Creating illusions of knowledge: Learning errors that contradict prior knowledge. Journal of Experimental Psychology: General, 142(1), 1–5. https://doi.org/10/f4mzxh

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References

Fazio, L. K., Rand, D. G., & Pennycook, G. (2019). Repetition increases perceived truth equally for plausible and implausible statements. Psychonomic Bulletin & Review, 26(5), 1705–1710. https://doi.org/10/ggmpfk

Filstrup Høllsberg, S. ;Dalen, A. van, & Vreese, C. de. (2025). Danskerne og tech-giganterne: Fra opfattelser til handling. University of Southern Denmark. https://doi.org/10.21996/YWP1-JP73

Freiling, I., Krause, N. M., Scheufele, D. A., & Brossard, D. (2023). Believing and sharing misinformation, fact-checks, and accurate information on social media: The role of anxiety during COVID-19. New Media & Society, 25, 141–162. https://doi.org/10.1177/14614448211011451

Goertzel, T. (1994). Belief in conspiracy theories. Political Psychology, 15(4), 731–742. https://doi.org/10/c3kgj3

https://ai-watch.ec.europa.eu/topics/public-sector_en

https://euvsdisinfo.eu/from-objective-reporting-to-myths-and-propaganda-the-story-of-ria

https://euvsdisinfo.eu/no-denmark-is-not-legalising-sexual-abuse-of-animals/

https://oecd.ai/en/ai-principles

Imhoff, R., & Bruder, M. (2014). Speaking (un-)truth to power: Conspiracy mentality as a generalised political attitude. European Journal of Personality, 28(1), 25–43. https://doi.org/10/f5t8jm

Johansen, E. B., Sandberg, T. W., Knüpfer, C., Lukito, J., Frischlich, L., Decker, H., & Runge, T. (in preparation). Danish elections 2026 in a high-choice information environment: Platforms, fragmentation, and AI [Unpublished report]. Digital Democracy Centre. DOI: 10.21996/2m9r-gp81

Johansen, N., Marjanovic, S. V., Kjaer, C. V., Baglini, R. B., & Adler-Nissen, R. (2022). Ridiculing the “tinfoil hats:” Citizen responses to COVID-19 misinformation in the Danish facemask debate on Twitter. Harvard Kennedy School Misinformation Review. https://doi.org/10.37016/mr-2020-93

Jowett, G. S., & O’Donnell, V. (2012). Propaganda and Persuasion (5. Aufl.). SAGE Publications, Inc

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References

Kahneman, D. (2012). Two systems in the mind. Bulletin of the American Academy of Arts & Sciences, 55–59.

Knudsen, E. M. (2025.). Kommune mod 2030—Scenarier for AI i en kommunal kontekst med fokus på tillid og troværdighed. Public Futures.

Koch, T. K., Frischlich, L., & Lermer, E. (2023). Effects of fact-checking warning labels and social endorsement cues on climate change fake news credibility and engagement on social media. Journal of Applied Social Psychology, 53, 495–507. https://doi.org/10.1111/jasp.12959

Kunda, Z. (1990). The case for motivated reasoning. Psychological Bulletin, 108(3), 480–498. https://doi.org/10.1037/0033-2909.108.3.480

Madan, R., & Ashok, M. (2023). AI adoption and diffusion in public administration: A systematic literature review and future research agenda. Government Information Quarterly, 40(1), 101774. https://doi.org/10.1016/j.giq.2022.101774

Madan, R., & Ashok, M. (2023). AI adoption and diffusion in public administration: A systematic literature review and future research agenda. Government Information Quarterly, 40(1), 101774. https://doi.org/10.1016/j.giq.2022.101774

Maier, J., Stier, S., & Oschatz, C. (2023). Are candidates rational when it comes to negative campaigning? Empirical evidence from three German candidate surveys. Party Politics, 29(4), 766–779. https://doi.org/10.1177/13540688221085239

Marcinkowski, F., & Metag, J. (2014). Why do candidates use online media in constituency campaigning? An application of the theory of planned behavior. Journal of Information Technology & Politics, 11(2), 151–168. https://doi.org/10.1080/19331681.2014.895690

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References

Mc Nair, B. (1995). An introduction to political communication (5th edition). Routledge.

Nielsen, R. K., Schulz, A., & Fletcher, R. (2021). An ongoing infodemic: How people in eight countries access news and information about coronavirus a year into the pandemic (SSRN Scholarly Paper 3873257). https://papers.ssrn.com/abstract=3873257

Norris, P. (2001). Political communication. In International Encyclopedia of the Social & Behavioral Sciences (pp. 11631–11640).

Nussbaum, Martha C. „Perfectionist Liberalism and Political Liberalism“. Philosophy & Public Affairs 39, Nr. 1 (2011): 3–45. https://doi.org/10.1111/j.1088-4963.2011.01200.x.

Pennycook, G., & Rand, D. G. (2021). The psychology of fake news. Trends in Cognitive Sciences, 25(5), 388–402. https://doi.org/10.1016/j.tics.2021.02.007

Pennycook, G., & Rand, D. G. (2021). The psychology of fake news. Trends in Cognitive Sciences, 25(5), 388–402. https://doi.org/10.1016/j.tics.2021.02.007

Pereira, A., Harris, E. A., & Van Bavel, J. J. (2023). Identity concerns drive belief: The impact of partisan identity on the belief and dissemination of true and false news. Group Processes & Intergroup RelationsPreprint, 24–47. https://doi.org/://doi.org/10.1177/13684302211030004

Petty, R. E., & Cacioppo, J. T. (1986). Communication and persuasion: Central and peripheral routes to persuasion. Springer.

Public sector tech watch: Mapping innovation in the EU public services: a collective effort in exploring the applications of artificial intelligence and blockchain in the public sector. (2024). Publications Office. https://doi.org/10.2799/4393

Quandt, T., Frischlich, L., Boberg, S., & Schatto-Eckrodt, T. (2019). Fake News. In T. P. Vos, F. Hanusch, D. Dimitrakopolous, M. Geertsma-Sligh, & A. Sehl (Hrsg.), The International Encyclopedia of Journalism Studies. Wiley.

Ryan, F., Knight, M., & Impiombato, D. (2023). Singing from the CCP’s songsheet: The role of foreign influencers in China’s propaganda system (Policy Brief Report 75). Australian Strategic Policy Institute.

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References

Stier, S., Oschatz, C., Clemm von Hohenberg, B., Maier, J., Nai, A., & Kirkizh, N. (2025). When do candidates “go negative”? A conjoint analysis to unpack the mechanisms of negative campaigning. Electoral Studies, 93, 102894. https://doi.org/10.1016/j.electstud.2024.102894

Sutton, R. M., & Douglas, K. M. (2020). Conspiracy theories and the conspiracy mindset: Implications for political ideology. Current Opinion in Behavioral Sciences, 34, 118–122. https://doi.org/10/ggvqhs

Sutton, R. M., & Douglas, K. M. (2020). Conspiracy theories and the conspiracy mindset: Implications for political ideology. Current Opinion in Behavioral Sciences, 34, 118–122. https://doi.org/10/ggvqhs

Tajfel, H., & Turner, J. C. (2004). The social identity theory of intergroup behavior. In J. T. Jost & J. Sidanius (Hrsg.), Political Psychology (S. 367–390). Psychology Press. https://doi.org/10.4324/9780203505984-16

Uscinski, J. E., Enders, A., Klofstad, C., Seelig, M., Drochon, H., Premaratne, K., & Murthi, M. (2022). Have beliefs in conspiracy theories increased over time? PLOS ONE, 17(7), e0270429. https://doi.org/10.1371/journal.pone.0270429

Uscinski, J. E., Enders, A., Klofstad, C., Seelig, M., Drochon, H., Premaratne, K., & Murthi, M. (2022). Have beliefs in conspiracy theories increased over time? PLOS ONE, 17(7), e0270429. https://doi.org/10.1371/journal.pone.0270429

van der Linden, S., Panagopoulos, C., Azevedo, F., & Jost, J. T. (2020). The paranoid style in American politics revisited: An ideological asymmetry in conspiratorial thinking. Political Psychology, 1–19. https://doi.org/10/gg3hz4

van Prooijen, J.-W., & van Vugt, M. (2018). Conspiracy theories: Evolved functions and psychological mechanisms. Perspectives on Psychological Science, 13(6), 770–788. https://doi.org/10/gfksh7

Wardle, C. (2018). The need for smarter definitions and practical, timely empirical research on the information disorder. Digital Journalism, 6(8), 951–963. https://doi.org/10/gfj4br

Wood, M. J., Douglas, K. M., & Sutton, R. M. (2012). Dead and alive: Beliefs in contradictory conspiracy theories. Social Psychological and Personality Science, 3(6), 767–773. https://doi.org/10.1177/1948550611434786

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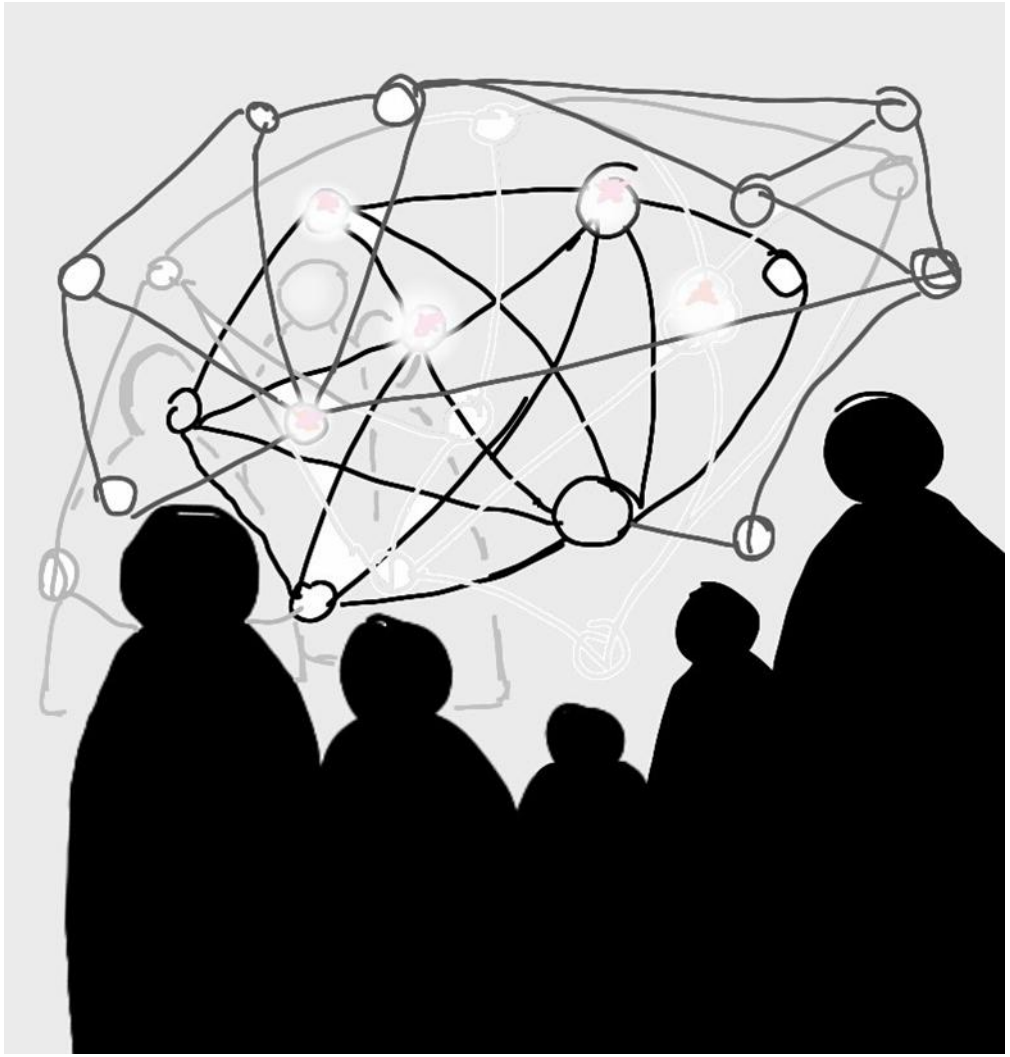
AI 507: Artificial Intelligence and Society

#12 AI in the workforce

Associate Prof. Dr. Lena Frischlich

Flashlight



Flashlight

AI in public administration

- European principles
- Certification
- Factors motivating adaption
- Odense in 2030

AI in political communication

- What is political communication?
- AI shapes political communication
- GenAI in the Danish election
- Danes attitudes towards GenAI in political communication

Mis- and Disinformation

- Definition (and challenges)
- Typical disinformation campaigns
- Prevalence and effects
- Two-factor models of persuasion
  - Elaboration-likelihood model
  - Systematic-Heuristic model
- Susceptibility to misinformation
  - Lack of cognitive elaboration/ System I thinking
  - Motivated cognition
  - Conspiracy mentality

Red teaming AI against Denmark

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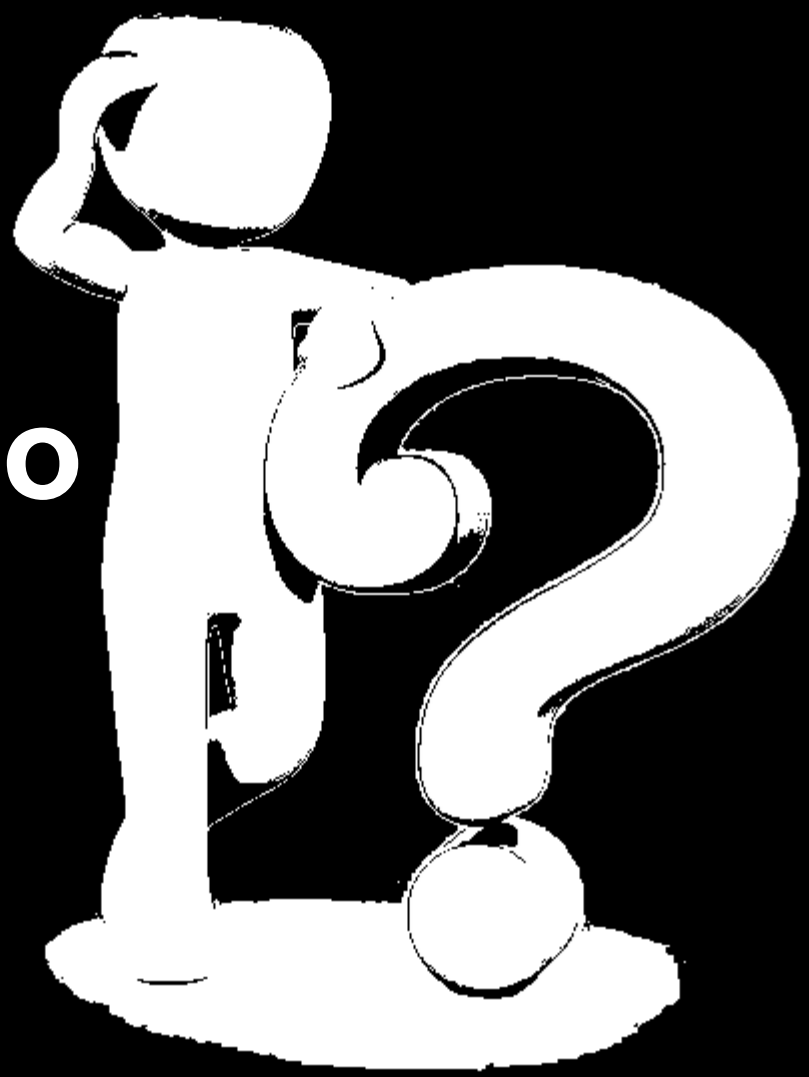
Do you have questions to last week?



Then let’s get started!

After today’s lecture, you will hopefully have…

- ❑ …learned how responsible AI in organisations can look like
- ❑ …examine the specificities of the finance sector
- ❑ … have first hand insights into institutions adapt to AI

To this end:

- Before the break: Guest talk from the Danske Bank
- After the break: Focus on AI in organisations and how AI is changing the workforce

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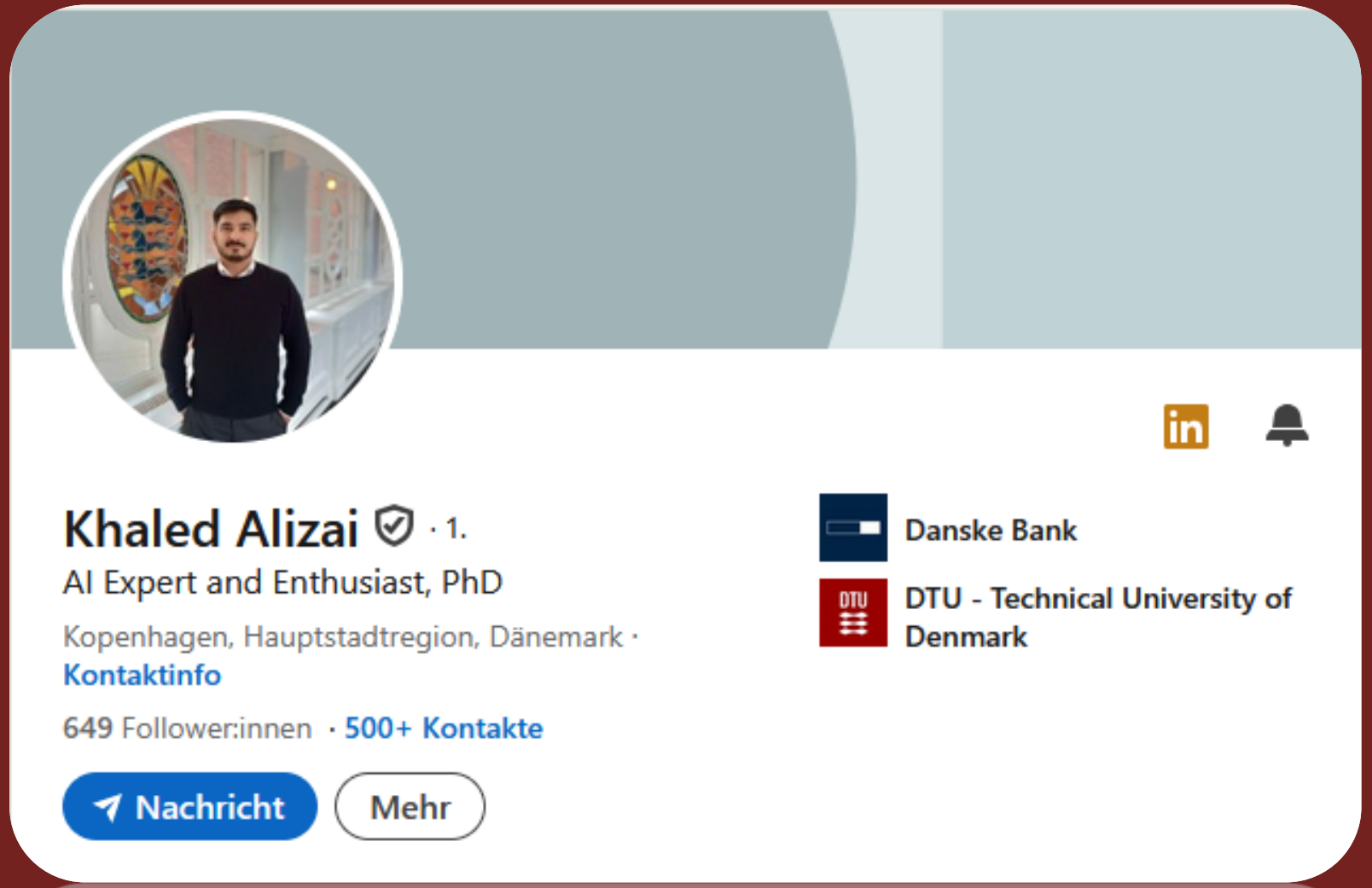
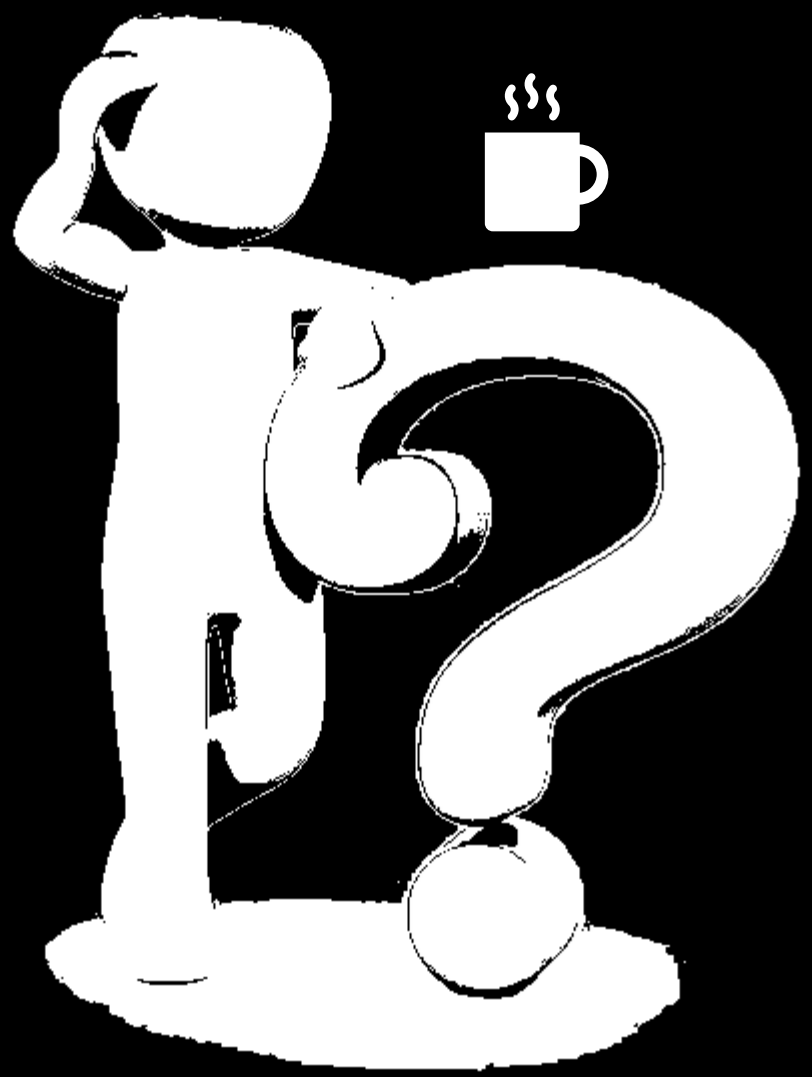
Guest talk


Khaled Alizai · 1.
AI Expert and Enthusiast, PhD
Kopenhagen, Hauptstadtregion, Dänemark · Kontaktinfo
649 Follower:innen · 500+ Kontakte
Nachricht
Mehr
Danske Bank
DTU - Technical University of Denmark

Time for a break



Artificial Intelligence at Work

Work? (Deranty & Corbin, 2024)

- Quasi philosophical question
- Here: activities that individuals engage in as part of the production of goods and services – for a profit (business owners) or a wage (employees)
- Can be in the commercial, public, or “third”-sector (voluntary, NGO-work)
- Can also cover other activities and create other values (e.g., care work)
- But today we focus on traditional, restricted economic activities
- What role does AI have for the future of work?

https://easy-peasy.ai/ai-image-generator/images/ai-robot-revolutionizing-workplaces-diversity-reactions

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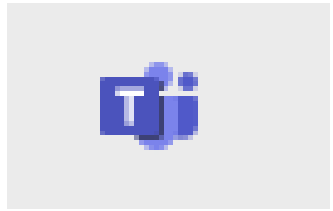
AI is not developed in a vacuum (Deranty & Corbin, 2024)

- AI innovation is (often) driven by attempts to win economic competition
- Sold using classic capitalist visions of innovation such as flexibility and adaptability
- Mundane mechanisms behind: Attempt to reduce costs, notably for labour and transactions,
- Increase outputs,
- Manage more efficiently through increased surveillance and control
- Create monopolies
- Plus: geopolitical tensions

What your employer can see on teams



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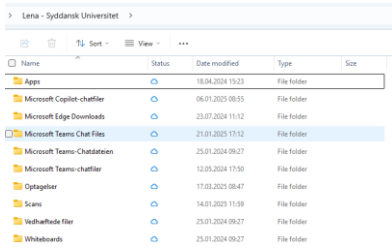


The fact that what is called the technological development of modern times has been so largely oriented economically to profit-making is one of the fundamental facts of the history of technology

(Surveillance capitalism; *Zuboff*, 2019)

Shoshana Zuboff, via https://de.wikipedia.org/wiki/Shoshana_Zuboff

What do we know about AI and unemployment? (Deranty & Corbin, 2024; Klenert et al., 2023)

- Successful AI implementation = successful results of (former) human labour
- Prior innovations primarily affected unskilled labour
- That is not necessarily the case for AI
  - No evidence for robots reducing the share of low-skilled workers acro Europe if the trend for decline is controlled for, new jobs emerge
  - For AI: non-linear relationship between AI patens and jobs:
    AI initially increases unemployment rates but – when the inflation is "normal" - unemployment shrinks
  - Central: that does not mean that people do not loose jobs, and the model does not account for crises and other "non-standard" times

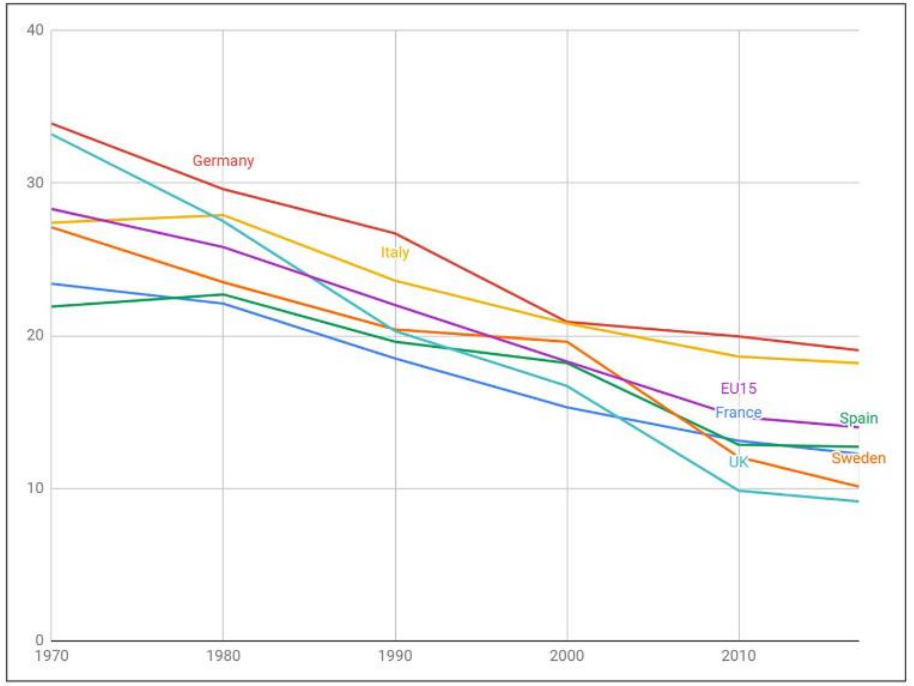
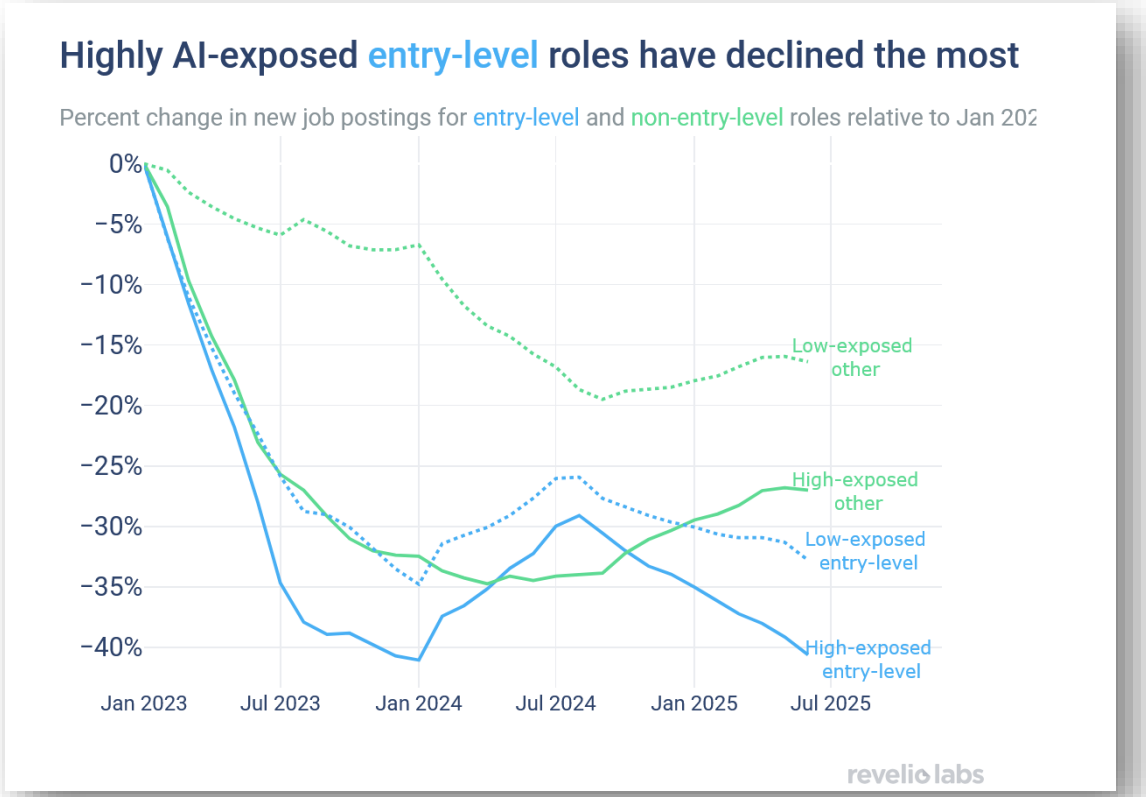


Figure 2. Share of employment in manufacturing in Europe (1970–2016).
Source: Authors' analysis from EU KLEMS database and EUROSTAT.

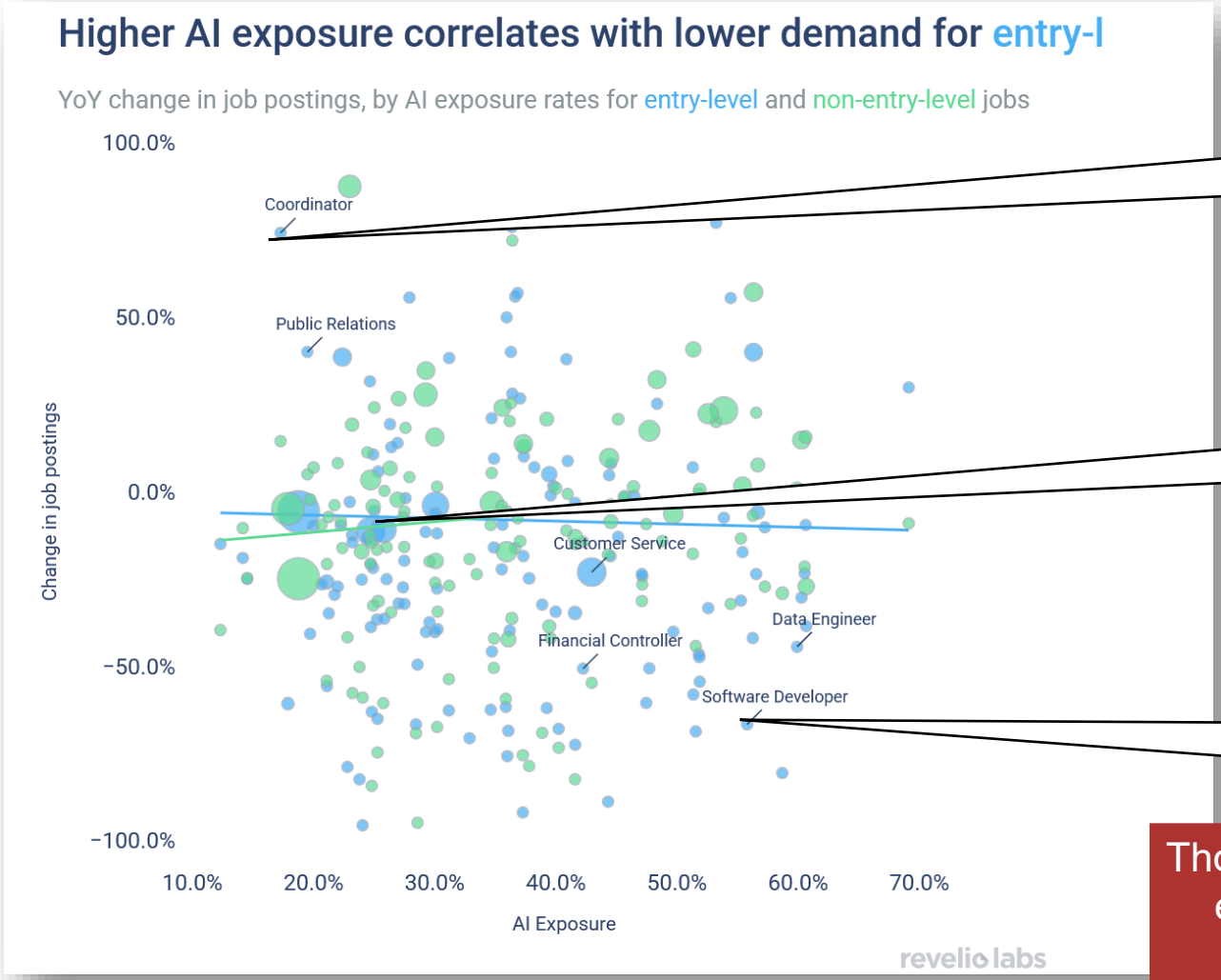
Klenert et al., 2023, S. 289

What do we know about how AI changes the workforce? (Simon, 2025)

- AI affects specifically entry-level job opportunities
- Following an analysis of Revelio Labs and Bloomberg, sectors that have more tasks that can be done by AI have seen the largest decline
- Losses are unequally distributed:
  - Affects mostly entry-level jobs (expectation that seniors will adapt)
  - Does affect fields with higher AI potential more



What do we know about how AI changes the workforce? (Simon, 2025)



- Coordinator: +17% AI exposure
- 75% more entry-level jobs

- Digital marketing: +25% AI exposure
- -5% less entry level jobs

- + 56% AI exposure for software developers
- - 67% of job-postings on the entry level

Those who are still needed are able to monitor and evaluate AI – and to translate between AI and humans

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Which jobs are there around AI? (Liu et al. 2024)

- Analysis of aggregated online job postings in the US 2021-2022
- Increased demand for AI jobs beyond computer science and being equipped with statistical competencies
- E.g., ~40% more jobs in AI + Statistics between 2021 and 2022 – particularly for
  - Data analysts
  - Machine learning engineers & scientists
  - Risk and business analysts
  - Marketing managers
  - Sustainable consultant

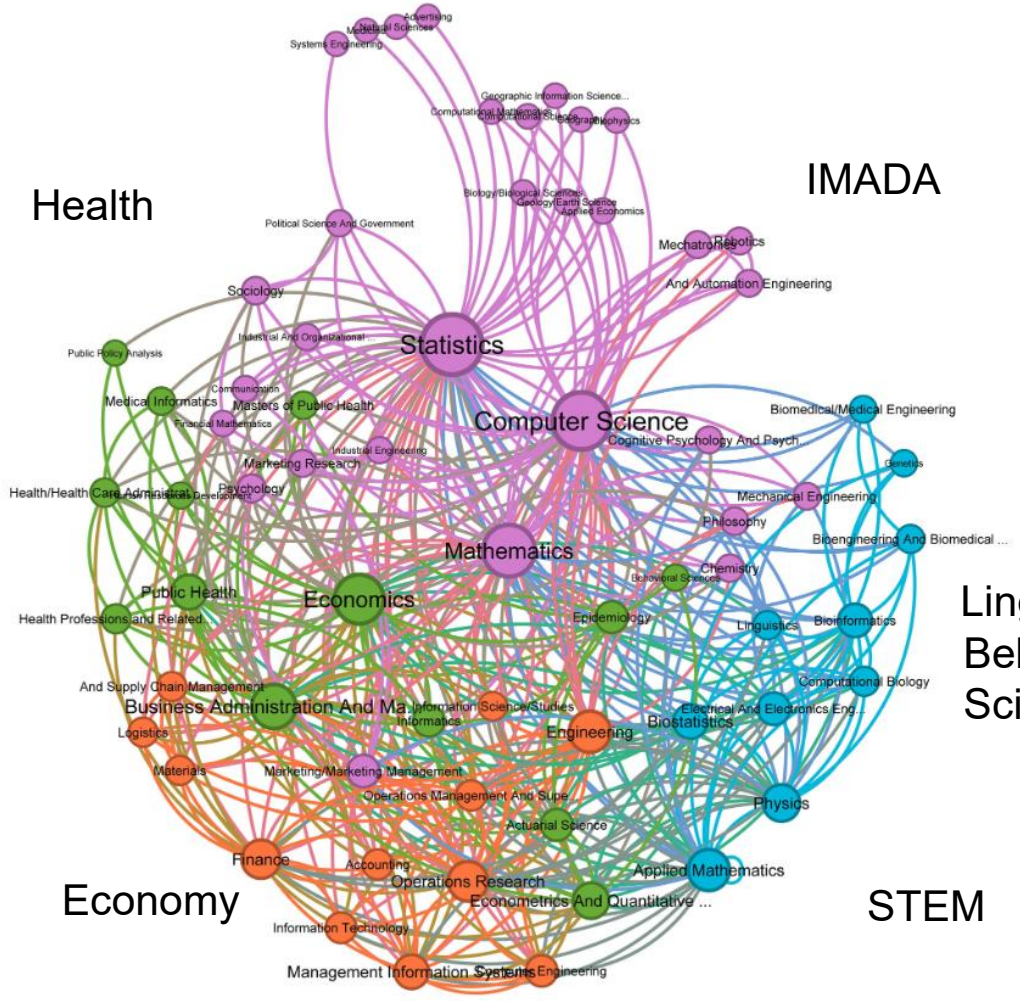
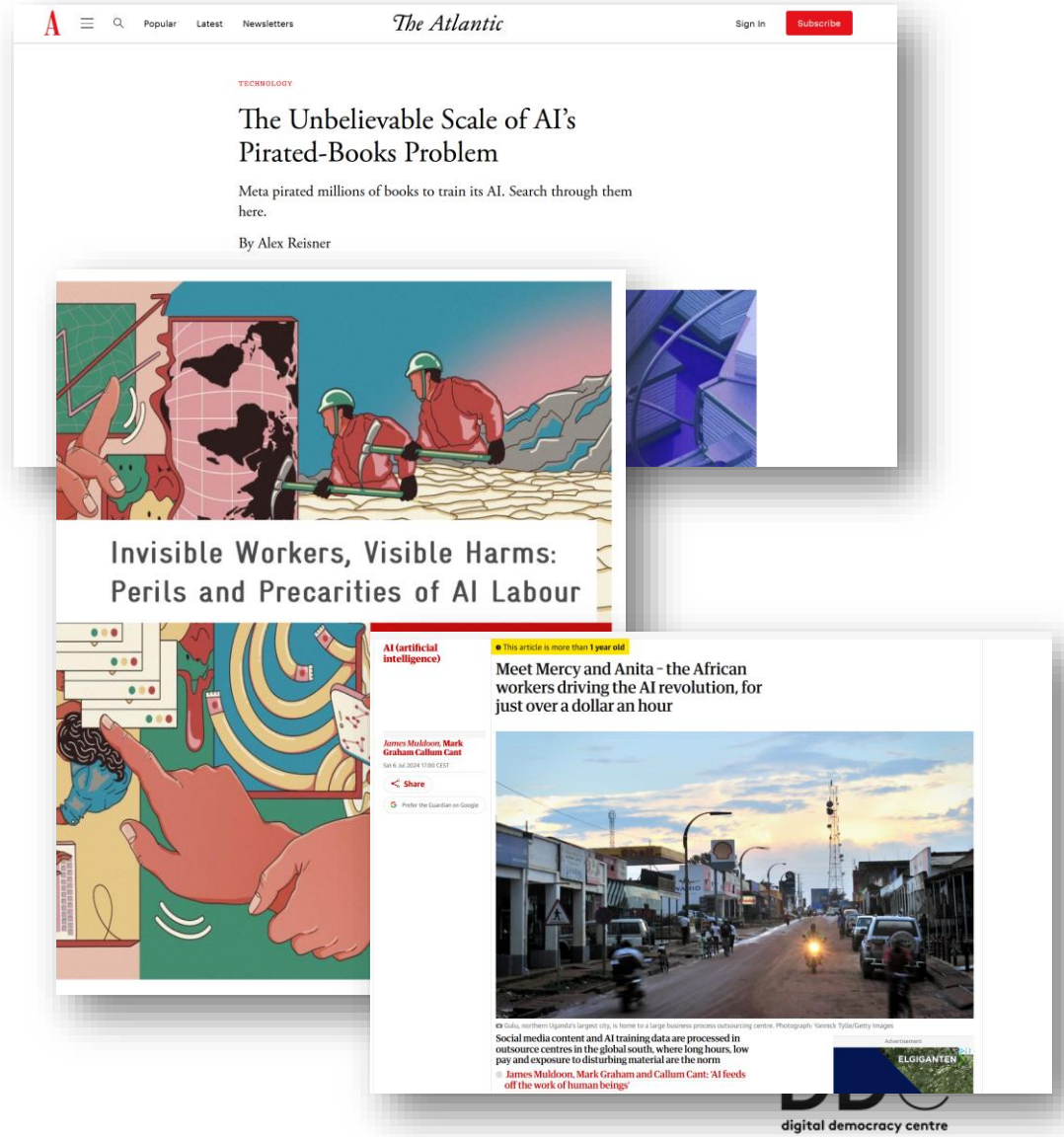


Fig. 4 Discipline cluster networks for statistics in AI recruitment.

Liu et al., 2024, S. 7

And of course: the hidden workforce

- High-quality “human-in-the-loop” training data requires humans
- Even modern deep-learning models do not train themselves (error-free)
- Three central “new” job markets:
  - Extraction of value by content creators (artists, scientists, journalists etc.) who are not paid at all
  - Low-paid crowd-workers
    often in the Global South –young, highly educated doing boring, traumatizing and incredibly low-paid crowd coding jobs (often ~1 -2 $/ hour)
  - Highly skilled experts that fine-tune special models & management

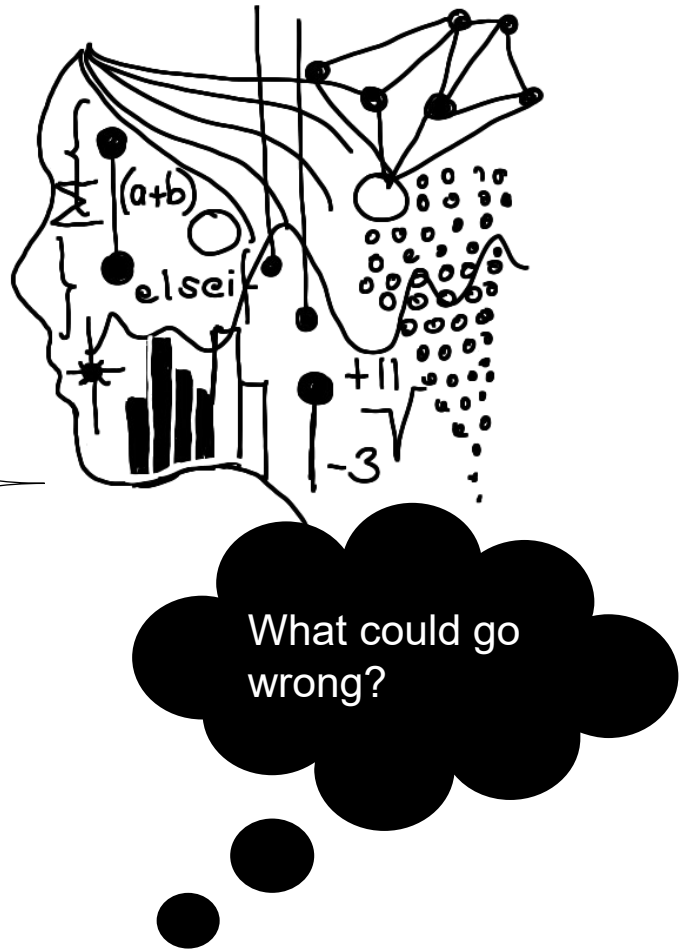


AI and management

(Deranty & Corbin, 2024; Mateescu & Nguyen, 2019)

Typically tasks of a manager

- Hiring
- Optimisation of the work process
- Evaluation of workers
- Scheduling of shifts
- Coordinating customer demands with service providers
- Monitoring work
- Incentivisation



https://de.pinterest.com/pin/539869074050893004/

- Surveillance and control
- Power imbalances
- Bias and discrimination
- Accountability of the management

What skills do leaders need now? (Bock & von der Oelsnitz, 2025)

- Structured literature review to identify skills and traits for successful leadership in times of AI
- Identified 24 leadership competencies
- **What do you think: What will leaders need?**

Psycho-social skills:

- Awareness (**understanding oneself, others, and the situation**)
- Team member evaluation & management
- **Communication**
- Collaboration
- Empathy
- Motivation
- Networking

Psychological traits

- Adaptiveness (ability to adapt to circumstances in the environment)
- Agility (ability to understand and think quickly) and flexibility
- Solution orientation
- Being a visionary

Human resource development

- Empowerment
- Engagement (strong personal interest)
- Execution (get things done!)

Humanistic skills

- Creativity

Technical skills

- Understanding AI

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AI and Business model innovation

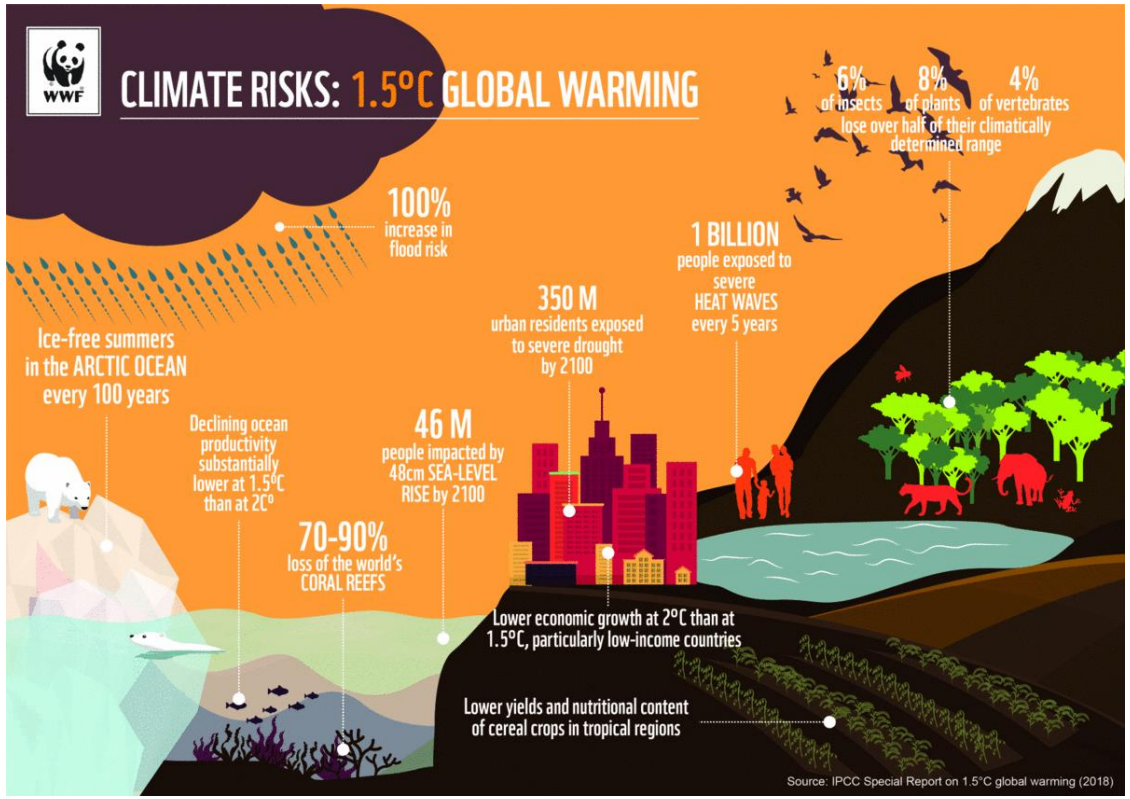
(Costa-Climent et al., 2024; Hamilton 2025; Hayward 2025;)

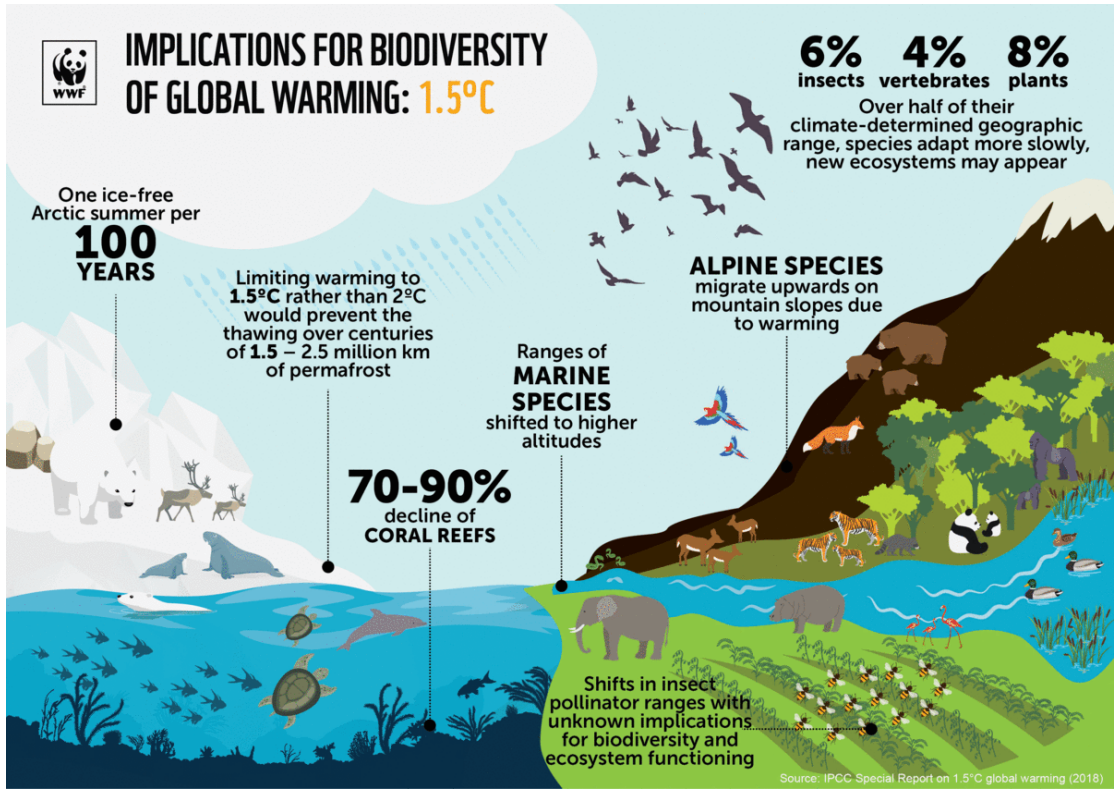
- Business model = content, structure, and governance of transactions designed to create value through business opportunities
- Four central sources for value creation in the digital realm:
  - **Efficiency** *(e.g., reduction of costs, improved service quality, better coordination, more productivity, more efficient delivery)*
  - **Lock-in** *(e.g., development of algorithms, democratisation of AI, scalable eco-system integrations)*
  - **Novelty** *(e.g., personalised products and services, new approaches to business models, outcome-based services)*
  - **Complementary** *(e.g., real-time data analytics for personalisation, ethical AI)*
- But: So far, the actual win through AI is modest, adoption fails and additional work is created through AI-generated errors
- SO – will agents save us or will the AI-bubble burst?

What do you think?

One area of innovation: AI and sustainability

Clear is: We need to fight climate change – and that means reducing CO2 and resource preservation



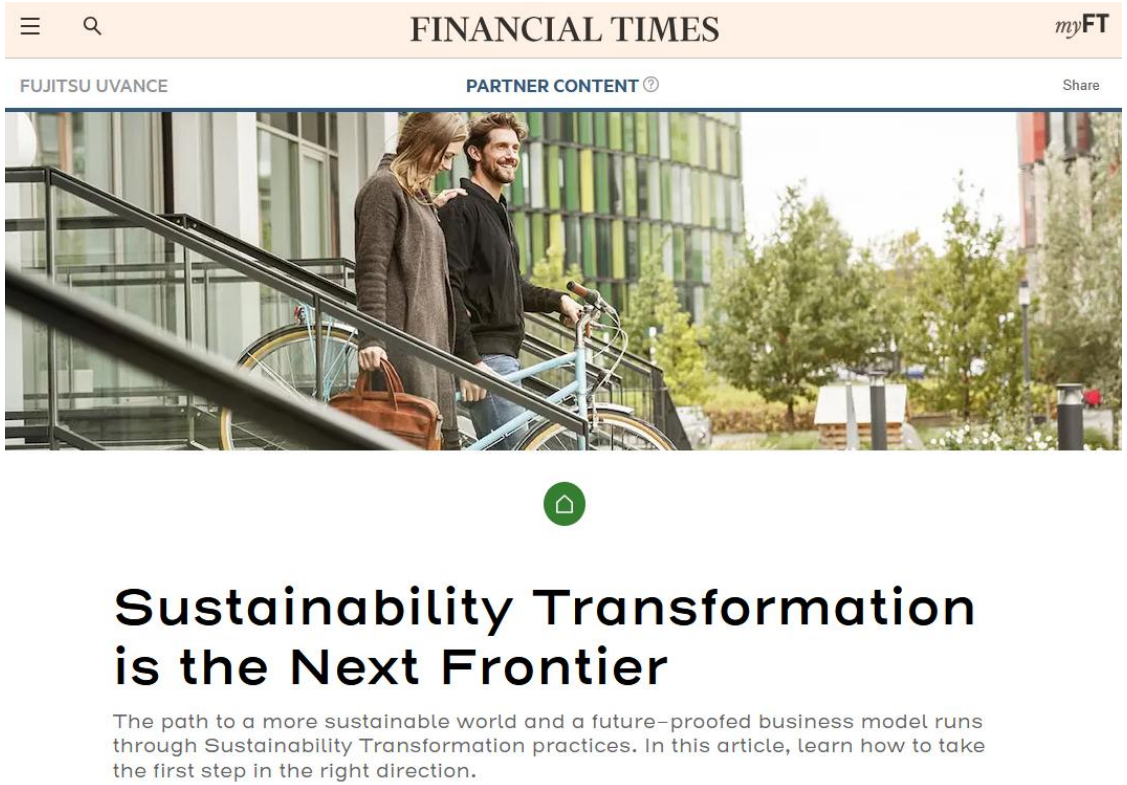


https://wwf.panda.org/discover/our_focus/climate_and_energy_practice/ipcc152/

Customers care about climate change

(McKinsey & Company 2023)

- Customers care for climate friendliness: 78% of US-Americans say that a sustainable lifestyle matters to them
- The market for products that claim sustainability is growing → particularly, when it comes to food and personal care (less for beer)

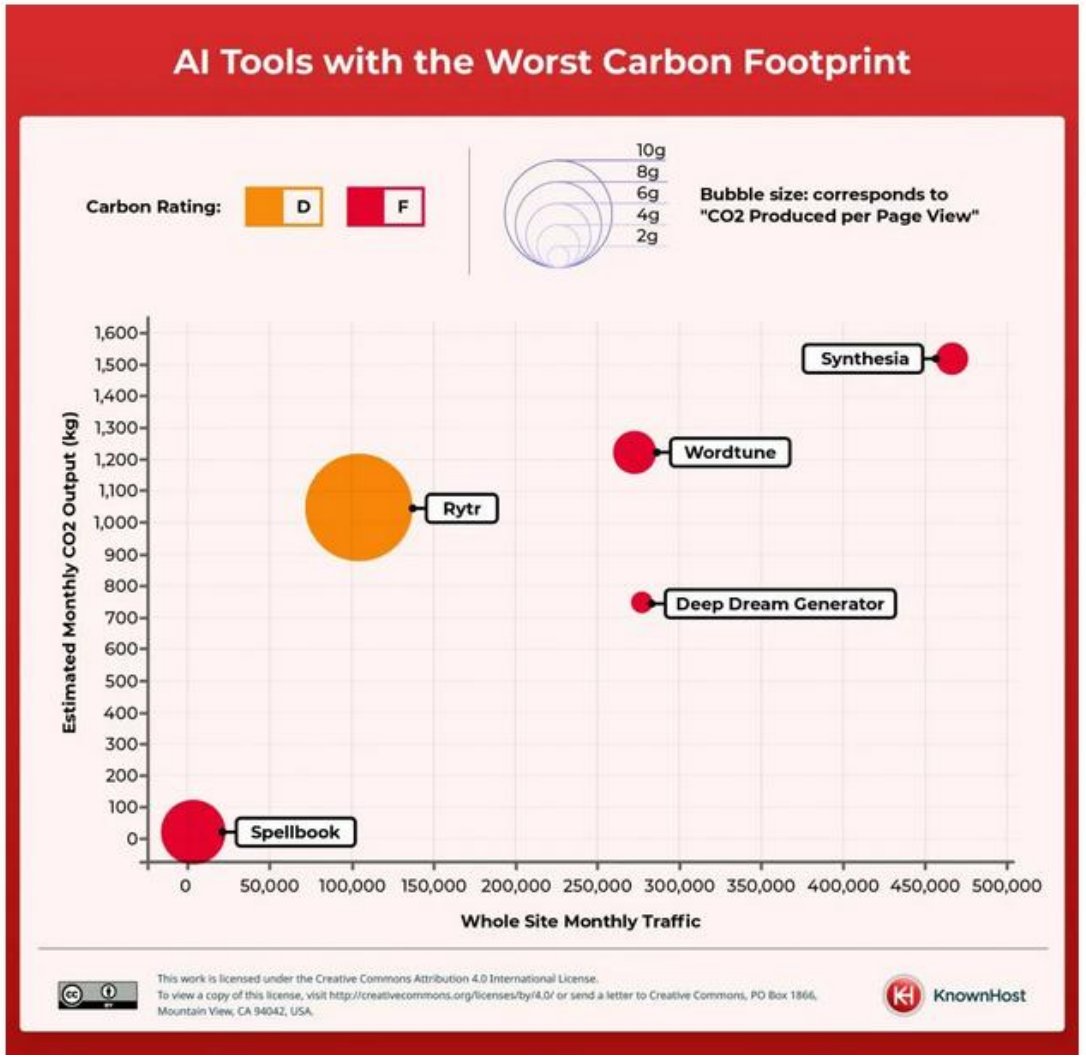


AI does increase emissions dramatically

(Smartly,AI, n.d., KnownH0st, 20204; Crimmins (2025)

- ChatGPT alone produces over 260,930 kilograms of CO2 – around 1.5 – 4,32 grams per view
- Equivalent to the emission of 260 flights from London to New York
- That is not the worst – Rytr emits 10 times as much
- Still: 15 ChatGPT queries = 1 hour video streaming

AI Tools with the Worst Carbon Footprint



Data center exploitation

Pew Research Center

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Home > Research Topics > Internet & Technology > Emerging Technology >...

SHORT READS | APRIL 13, 2026

Most new data centers in the U.S. are coming to rural areas

Add Us On Google

BY SKYLER SEETS AND KAITLYN RADDE

Construction continues at a new Meta data center in Beaver Dam, Wisconsin, on March 31. (Joe Timmerman/ Wisconsin Watch via Getty Images)

NBC NEWS POLITICS U.S. NEWS WORLD LOCAL SPORTS BUSINESS SHOPPING TIPLINE CULTURE SCIENCE WATCH

ARTIFICIAL INTELLIGENCE

Elon Musk's makeshift AI power plant generates sound and fury in Mississippi

The 27 temporary gas turbines roar like jet engines day and night to power data centers, residents say.

Listen to this article with a free account

00:00 10:58 1x

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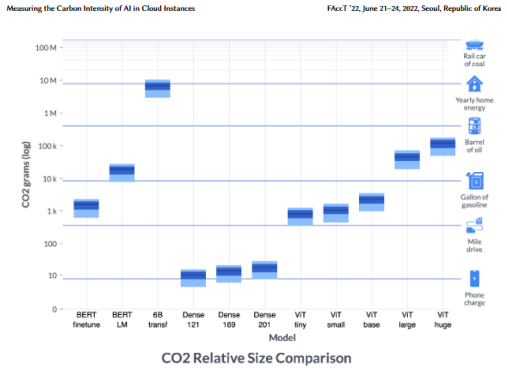
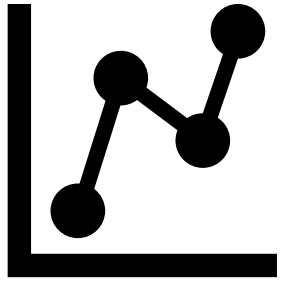
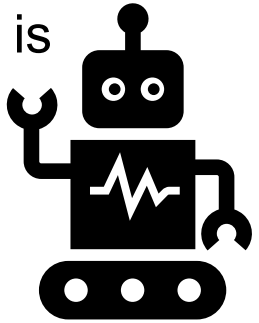
Why? (Patterson 2022)



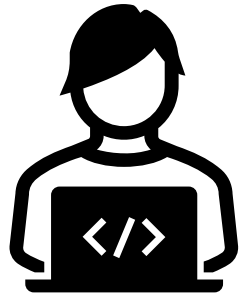
- How much data is the model trained on?

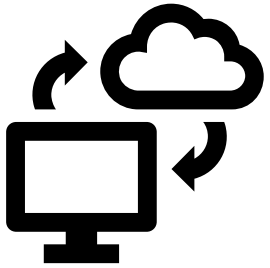


- How efficient is the model?



- How is the model used?





- Which computer/ cloud is used t run the model?

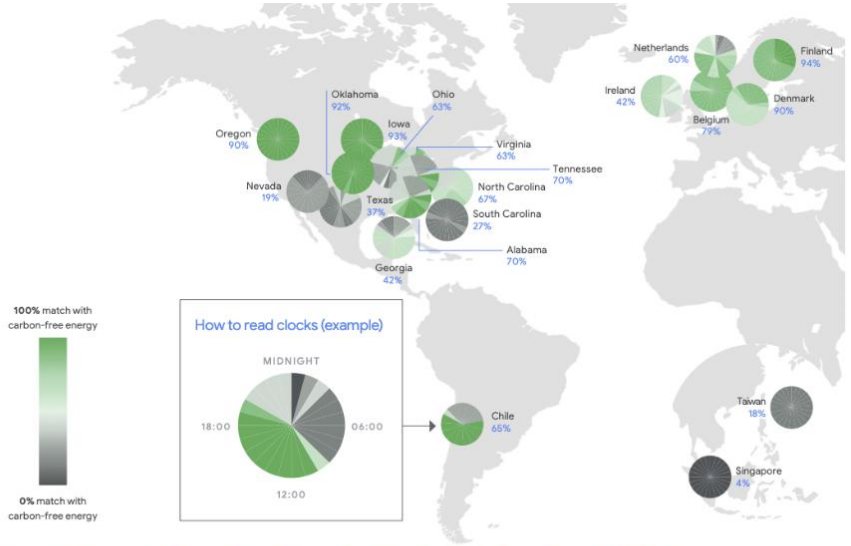
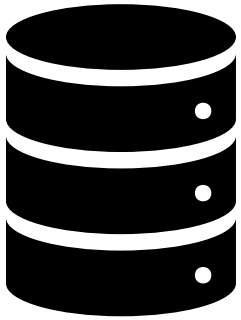


Figure 2. Percent Carbon Free Energy by Google Cloud Location in 2020. The map shows the %CFE and how the percentage changes by time of day. Chile has a high %CFE from 6AM to 8PM, but not at night. The US examples on this map range from 19% CFE in Nevada to 93% in Iowa, which has strong prevailing winds both night and day. (sustainability.google/progress/energy/)



- Central: How are the data centres powered?
- Are they using sustainable energy?
- Do they need a lot of water?

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Ai *can* be more sustainable (Patterson et al., 2022)

1. **Model:** Developing and employing energy efficient models – reduces the computation by factors of 5 to 10
2. **Machine:** Using processors optimized for ML training such as TPUs or recent GPUs (e.g., V100 or A100), versus general-purpose processors, can improve performance/Watt by factors of 2–5
3. **Mechanization.** Computing in the Cloud rather than on premise improves datacenter energy efficiency, reducing energy costs by a factor of 1.4–2
4. **Map.** Moreover, Cloud computing lets ML practitioners pick the location with the cleanest energy4, further reducing the gross carbon footprint by factors of 5–10

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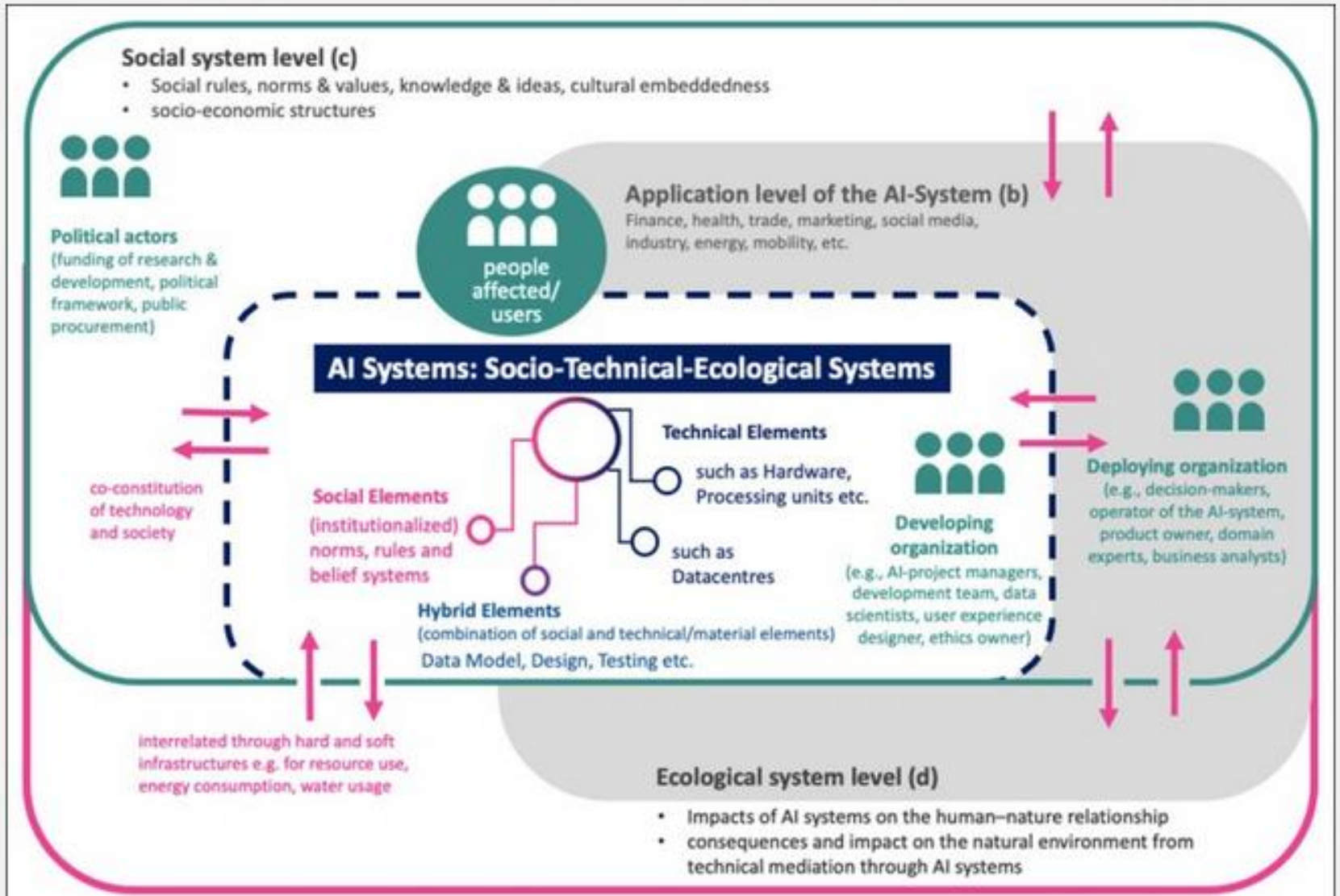
Need for a socio-techno-ecological perspective (Rohde et al., 2024)



What are your questions?

AI 507: Artificial Intelligence and Society

#13 AI in health and medicine

Associate Prof. Dr. Lena Frischlich

References

Bock, T., & von der Oelsnitz, D. (2025). Leadership-competences in the era of artificial intelligence – a structured review. Strategy & Leadership, 53(3), 235–255. https://doi.org/10.1108/SL-09-2024-0100

Costa-Climent, R., Haftor, D. M., & Staniewski, M. W. (2024). Intelligent transformation: Navigating the ai revolution in business and technology. In M. T. Del Val Núñez, A. Yela Aránega, & D. Ribeiro-Soriano (Eds.), Artificial Intelligence and Business Transformation: Impact in HR Management, Innovation and Technology Challenges (pp. 19–40). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-58704-7_2

Crimmins, T. B., Tricia. (n.d.). ChatGPT produces the same amount of CO2 emissions as 260 flights from New York City to London each month, study finds. Fortune. Retrieved 29 April 2025, from https://fortune.com/2025/01/21/chatgpt-carbon-dioxide-emissions-study/

Deranty, J.-P., & Corbin, T. (2024). Artificial intelligence and work: A critical review of recent research from the social sciences. AI & SOCIETY, 39(2), 675–691. https://doi.org/10.1007/s00146-022-01496-x

Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ). (2026.). Invisible workers, visible harms: Perils and precarities of AI labour. Retrieved 6 May 2026, from https://www.bmz-digital.global/wp-content/uploads/2026/02/GIZ_2026_InvisibleWorkersVisibleHarms-1.pdf

Diaz, K. (2026, March 26). How AI is changing the nature of entry level work. World Economic Forum. https://www.weforum.org/stories/2026/03/how-ai-is-changing-the-nature-of-entry-level-work/

Hamilton, D. D. (2025, August 27). Why AI adoption fails: The human problem leaders overlook. Forbes. https://www.forbes.com/sites/dianehamilton/2025/08/27/why-ai-adoption-fails-the-human-problem-leaders-overlook/

Hayward, E. (2025, August 22). Why most AI products fail: Key findings from MIT's 2025 AI Report [Mind the product]. Mind the Product. https://www.mindtheproduct.com/why-most-ai-products-fail-key-findings-from-mits-2025-ai-report/

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References

https://smartly.ai/blog/the-carbon-footprint-of-chatgpt-how-much-co2-does-a-query-generate

https://www.mckinsey.com/industries/consumer-packaged-goods/our-insights/consumers-care-about-sustainability-and-back-it-up-with-their-wallets

https://www.timedoctor.com/blog/can-microsoft-teams-monitor-employees/

K.W, J. (2024, December 9). Carbon footprint of AI tools. KnownHost. https://www.knownhost.com/blog/carbon-footprint-of-ai-tools/

Klenert, D., Fernández-Macías, E., & Antón, J.-I. (2023). Do robots really destroy jobs? Evidence from Europe. Economic and Industrial Democracy, 44(1), 280–316. https://doi.org/10.1177/0143831X211068891

Liu, J., Chen, K., & Lyu, W. (2024). Embracing artificial intelligence in the labour market: The case of statistics. Humanities and Social Sciences Communications, 11(1), 1112. https://doi.org/10.1057/s41599-024-03557-6

Matescu, A., & Nguyen, A. (2019). Explainer: Algorithmic management in the workplace. Data & Society. https://datasociety.net/wp-content/uploads/2019/02/DS_Algorithmic_Management_Explainer.pdf

Nguyen, Q. P., & Vo, D. H. (2022). Artificial intelligence and unemployment: An international evidence. Structural Change and Economic Dynamics, 63, 40–55. https://doi.org/10.1016/j.strueco.2022.09.003

Patterson, D. (2022.). The carbon footprint of machine learning training will plateau, then shrink. Retrieved 29 April 2025, from https://www.authorea.com/doi/full/10.36227/techrxiv.19139645.v1?commit=fa97e4ce012d0062185dbcae7fd1dfe207fb69d7

Rohde, F., Wagner, J., Meyer, A., Reinhard, P., Voss, M., Petschow, U., & Mollen, A. (2024). Broadening the perspective for sustainable artificial intelligence: Sustainability criteria and indicators for Artificial Intelligence systems. Current Opinion in Environmental Sustainability, 66, 101411. https://doi.org/10.1016/j.cosust.2023.101411

Simon, L. K. (2025). Is AI responsible for the rise in entry-level unemployment? | Revelio Labs. Revelio labs. https://www.reveliolabs.com/news/macro/is-ai-responsible-for-the-rise-in-entry-level-unemployment

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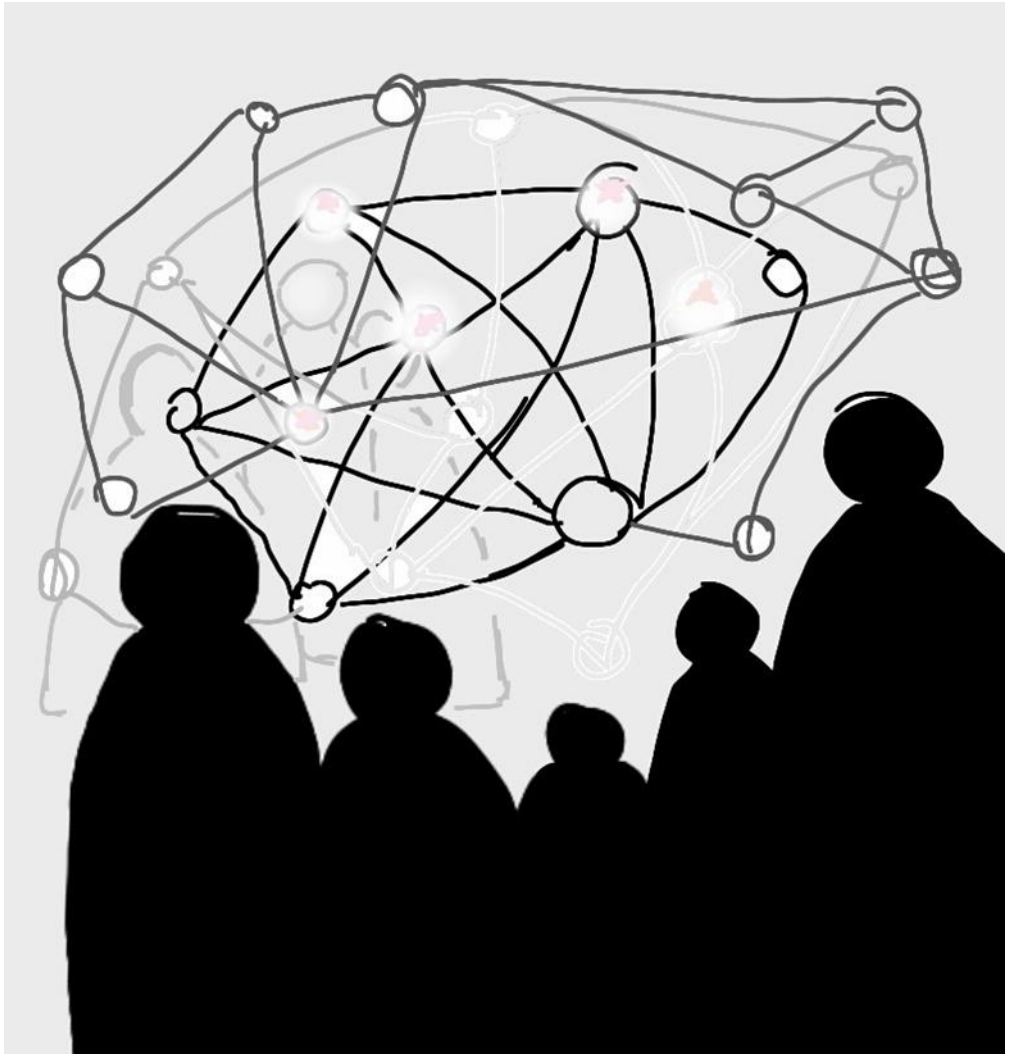
AI 507: Artificial Intelligence and Society

#13 AI & health

Associate Prof. Dr. Lena Frischlich

Flashlight



Flashlight

AI & the finance sector

Risk assessment

Model evaluation

Changing hierarchies of risk evaluation

Traditional modelling versus AI

RAG-set-ups & Jailbreaking

AI & leadership

Leadership skills

AI & business innovation

AI & the job market

What is work

AI as part of the capitalist vision

AI & unemployment

Shifting workforce for beginners

New work markets

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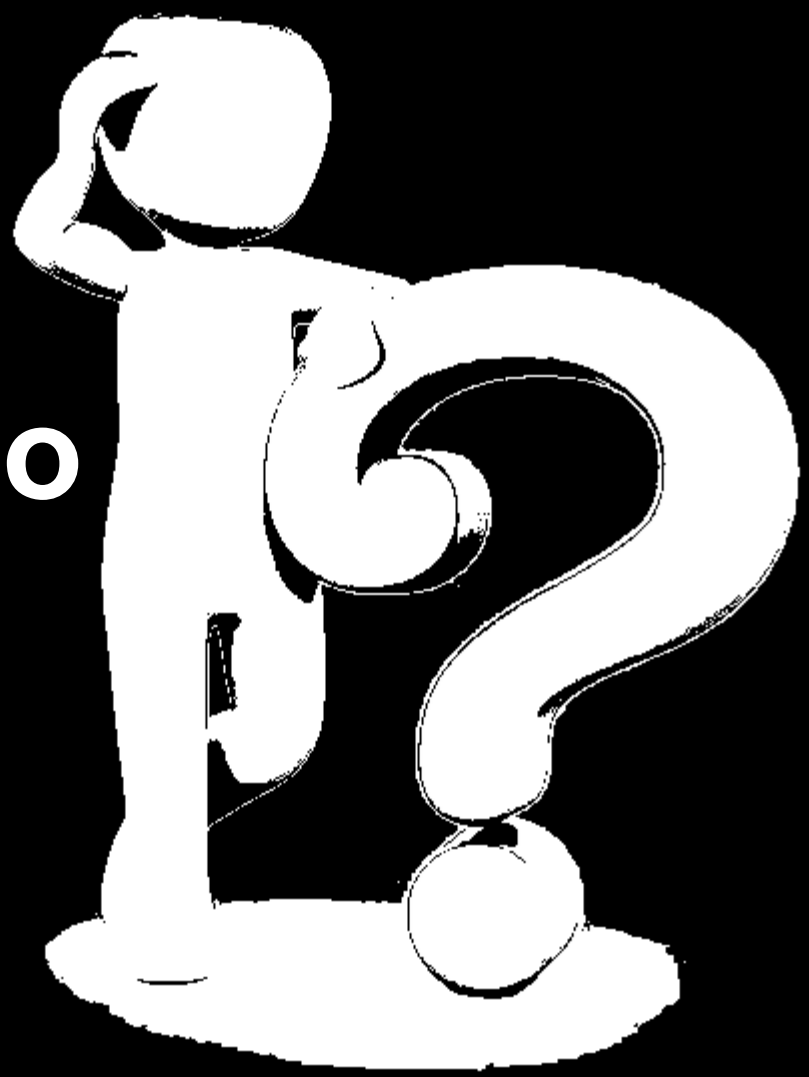
Do you have questions to last week?



Then let’s get started!

After today’s lecture, you will hopefully have…

- ❑ … a better understanding of psychological well-being
- ❑ … insights into the relationship between AI and wellbeing
- ❑ …discussed about AI’s risks for mental health
- ❑ … learned how AI can (not) revolutionize psychotherapy and medicine

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What does “health” mean for you?

Health is a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity (WHO, 1948)

But… (Aaronowsky, 1979; Lange, 2020)

- We all are sick sometimes, most of us will experience terrible events at some point in our life and chances are high that we will be disabled at some point too
- Often focus on factors that make people sick, the *pathogenesis*
- Trauma, stress, pathogens– all of that matters (#lifestyle coach)
- **BUT: A substantial share of people cope with extreme life event, bacteria, viruses and *do well***
- Inspired by research with women who survived the concentration camps in WWI, Aaron Antonovsky started to ask **what is it that makes people healthy**?
- Field of **Salutogenese** (“Saluto” = lat. health)
- He argues that the central force in there is the sense of **coherence**

1 mio.

Danes lived with a chronic condition in 2019

17%

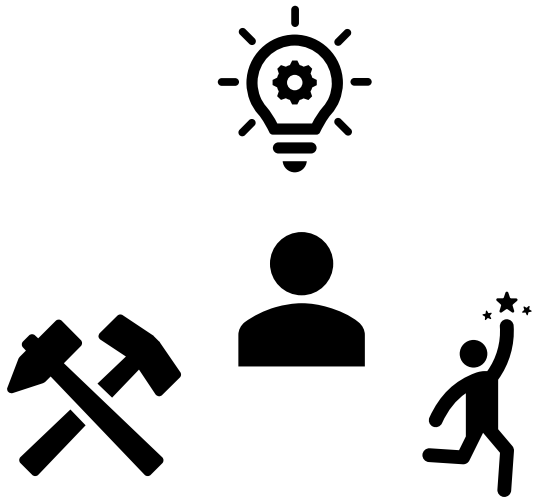
Aaron Antonovsky via https://www.aerzteblatt.de/archiv/aaron-antonovsky-vater-der-salutogenese-ef5462ba-d386-44aa-9bd6-1377ebd00f91

Coherence (Aaronowsky, 1979; Bauer et al., 2020)

“a global orientation that expresses the extent to which one has a

- pervasive, enduring though dynamic feeling of confidence that
  - one’s internal and external environments are predictable (*comprehensibility*)
  - that there is a high probability that things will work out as well as can reasonably be expected (*manageability*)
  - that one matters (*meaningfulness*)

  Can work on the individual, organizational and national level (but attention - ingroup coherence/ meaning can foster outgroup discrimination!)
- Focus lies on well-being rather than “health”

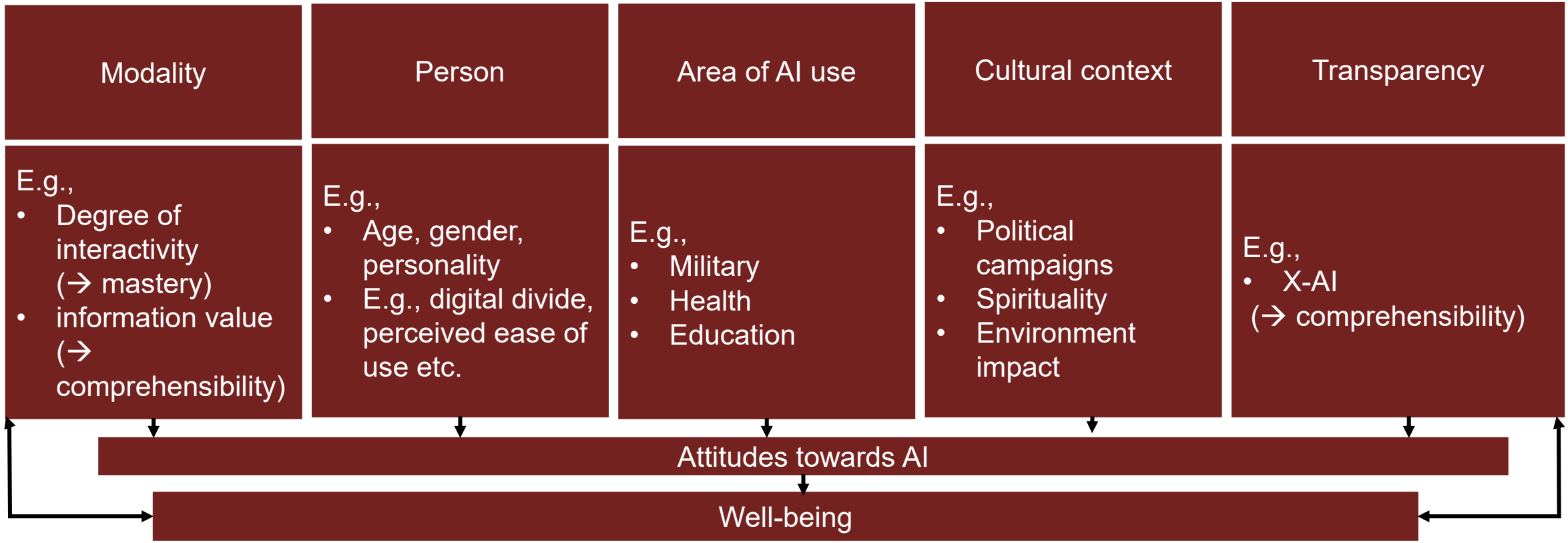


Well-being? (Diener, 1984; Ryan & Deci, 2000; Oishi & Westgate, 2021; Ryff & Keyes, 1995)

- Different conceptualisations over time
- Has been discussed and measured amongst others as:
  - *Life Satisfaction*
  - *Hedonic:* an overall positive balance between negative and positive affect
  - *Eudaimonic:* a meaningful life
  - *Psychologically rich*: an interesting and diverse life
- Satisfaction of intrinsic needs for mastery, autonomy, and affiliation (manageability, comprehensibility, meaningfulness)
- Nowadays, often multidimensional: self-acceptance, purpose in life, control over the environment (mastery); positive relations; personal growth; autonomy

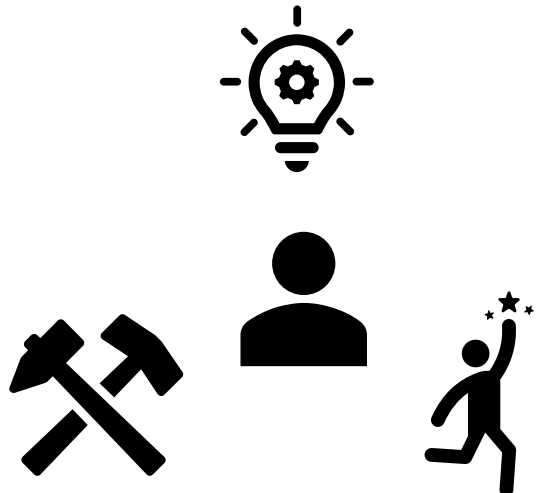
How can AI impact well-being? (Montag et al., 2024)

- Impact framework to understand well-being in the age of AI
- Five “main” dimensions on which AI can affect well-being plus reinforcing effects



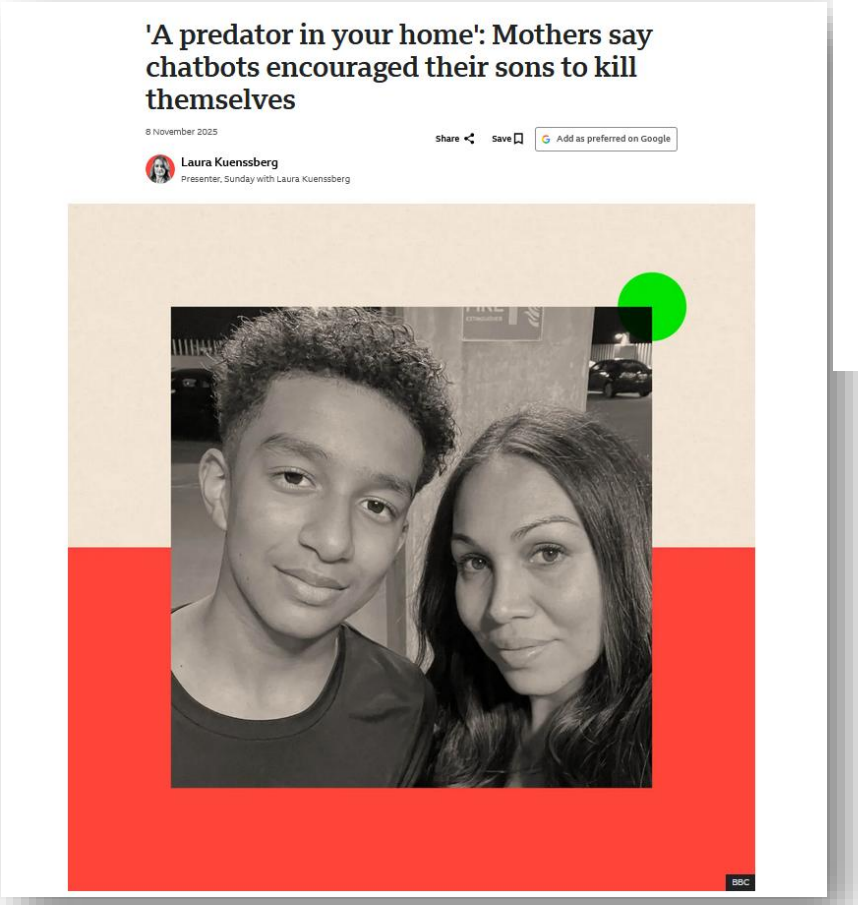
What do you think – how can AI foster well-being?

Modality	Person	Area of AI use	Cultural context	Transparency



Negative consequences on well-being: despair, delusion, death

Cases of AI-psychosis make headlines (Jamali, 2025)


'A predator in your home': Mothers say chatbots encouraged their sons to kill themselves

8 November 2025

Share Save Add as preferred on Google

Laura Kuenssberg

Presenter, Sunday with Laura Kuenssberg

https://www.bbc.com/news/articles/ce3xgwyywe4o

OpenAI faces criminal probe over role of ChatGPT in shooting

21 April 2026

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Kali Hays

Technology reporter

OpenAI is facing a criminal investigation in the US over whether its ChatGPT technology played a part in the murder of two people during a mass shooting at Florida State University last year.

https://www.bbc.com/news/articles/c62j4ldp2jqo

Open AI: 0.07% of weekly users show signs of delusion, 0.15% voice suicidal thoughts = RARE

BUT

56k respectively **120k**

Musk's AI told me people were coming to kill me. I grabbed a hammer and prepared for war

3 May 2026

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Stephanie Hegarty

Population correspondent, BBC World Service

Adam's life was turned upside down by his conversations with the Grok AI

https://www.bbc.com/news/articles/c242pzr1zp2o

Media shaped delusions are not new

(Carlbring & Andersson, 2025; Niederkrotenthaler et al., 2007)

Reading from Goethe's Werther
By Wilhelm Amberg - mnp.art.pl, Public Domain, https://commons.wikimedia.org/w/index.php?curid=131435326

Clinical principle	LLM
Do not collude with delusions	Validates the user
Do not enable suicidal ideation	Provides information asked for
Do not reinforce hallucination	Treats human content as verdict
Equality w. patient	Bias towards certain diagnoses
Therapeutic alliance: support & confrontation	Defaults to agreement

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How does that look like? (Moore et al., 2026 PREPRINT)

- Analysis of 19 chatlogs donated by users who reported “chatbot delusion”
- recruited from a self-help group and an NGO for AI-inflicted harm
- Coding of 391k messages (by both the participants and gemini-3)
- “Validated” against human coders BUT only 10 cases sampled and low reliability values (Cohens’ K = .56 – you would want to have at least >.70)
- Suicide messages were manually validated and only the manual values were reported

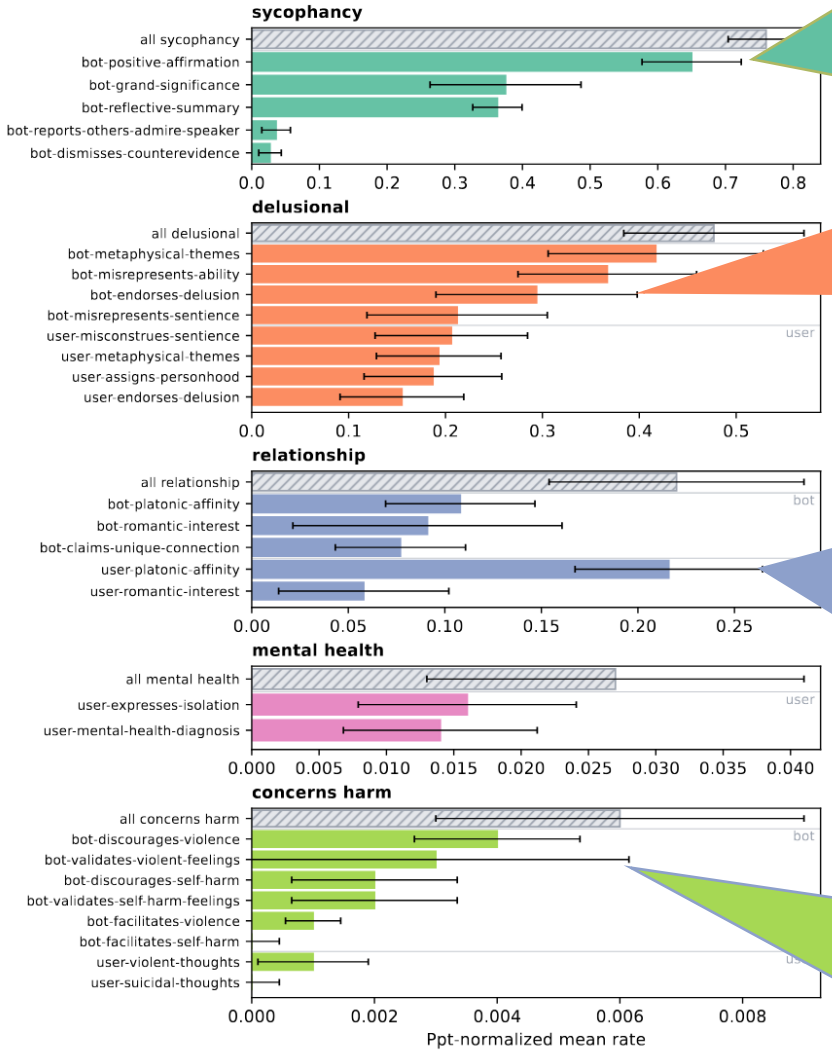

Figure 2: Prevalence of code categories.
Chatbots display sycophancy in more than 70% of their messages, and more than 45% of all (user and chatbot)
signs of delusions. For category descriptions, see §§4.2. See Fig. 8 for these data split by chatbot. Counts for each
Appendix Table 8. Participant-normalized mean annotation rates with 95% confidence intervals on the mean acro

EXPLICIT endorsement/ praise of the user’s ideas, efforts, or feelings

EXPLICITLY endorses/ demonstrates belief in physically impossible ideas

EXPLICITLY acknowledges / refers to a shared PLATONIC relationship

methods, instructions, guidance to facilitate or suggest PHYSICAL violence

Risks are unequally distributed (Buck & Maheux, 2026 PREPRINT)

- Cross-sectional survey *N* = 1004 US-Americans 18-25 years (via Prolific)
- Clinical survey to estimate psychosis risks: High-risk versus low risk
- Those with high risks where more likely to
  - Say that they initiated LLM use 5 or more times per day and on a daily basis
  - Had used a chatbot today (vs. yesterday of earlier)
  - Considered their LLM as a companion or therapist
- Psychosis risk was twice as strongly associated with LLM use for emotional support and dating/sexuality than with learning

➢ **The most vulnerable ones that are at the highest risk**

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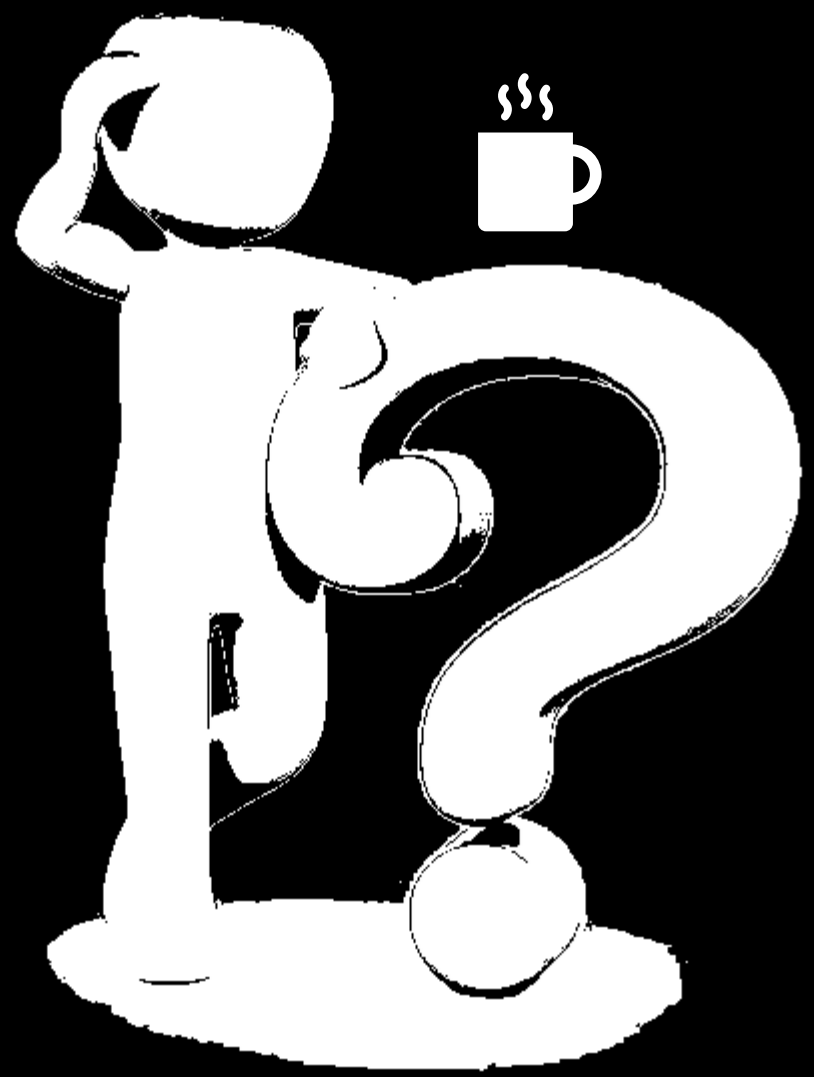
Time for a break



AI and psychotherapy

AI is also of increasing interest for clinical psychology (esundhed.dk, 2026)

- Waiting times are long
  - For children and adolescents, it can take three or more months to get a treatment for depression here in Southern Denmark
  - AI could help to bridge this gap

Diagnosis

Treatment recommendation

Therapeutic exchange

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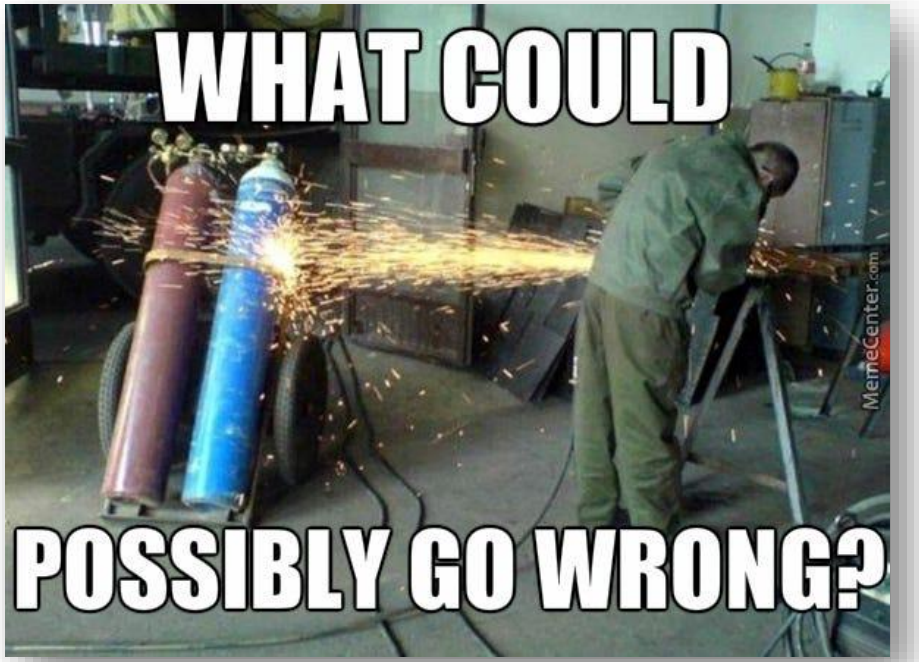
Diagnosis (Park et al., 2024)

- CNNs are nowadays able to predict whether someone has a depression (versus e.g., anxiety) with 98% accuracy – best from smartphone sensor data
- Generally best: multimodal data
- When they are trained with EEG data, they can predict depression nearly perfectly
- Based on social media data, the accuracy is still around 88%



Treatment recommendation (Petersen et al., 2023)

- Traditionally: Fairness = fairness of the *classification,* aka statistical accuracy in different classes (and groups)
- Especially relevant for consequential fields (and especially life-and-death ones),
  - Fairness should not only be focused on the classification alone
  - The *outcome* of the classification should also be considered (e.g., treatment for depression)
  - Are there fair decisions made based on the outcome of the classification?
- Needs to account for the baseline risk (positive versus negative outcome) in each group as baseline risks might be outside of the “typical” range –thus leading to unfair treatment
- Especially for smaller groups: Estimates are less stable, and metrics must be stable across sample sizes to be fair (which they are not)

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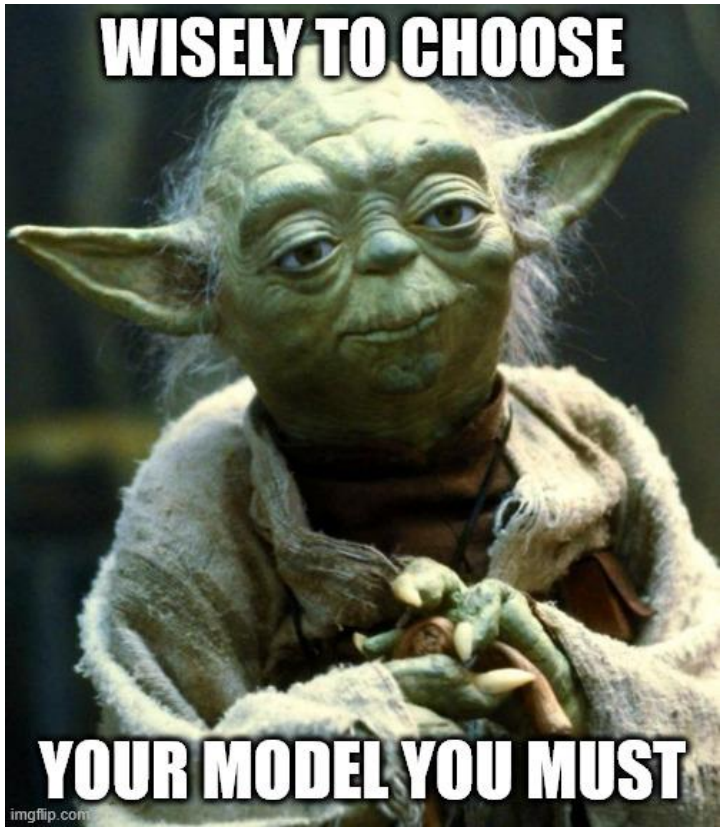
Example: Depression in Denmark

(Petersen et al., 2023; Thielen et al. 2007)

- Danish registry data 2000 – 2018 (> 15 years)
- Everyone diagnosed with depression and an equally large control sample
- ~ 240,000
- Trained a model to predict depression risk
- Overall AUROC >.70 but varying between .60 and .75 in different groups
- Subjects with groups that have better discrimination values are selected earlier → and receive earlier treatment
- For example, men and women are equally likely to suffer from depression, but women are more likely to receive medical help
- Bias-aware calibration can help to mitigate this risk



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Therapy (Beg et al., 2924; Li et al., 2023)

Enhancement not Replacement

- Several attempts to use computer-assisted digital therapy show promising results in a reduction of depression and anxiety
- However: Student samples are overrepresented, and there is still a need for more high-quality samples
- Many studies lack active control groups, focus on rather limited time frames
- big pre-registered cohort experiments are missing

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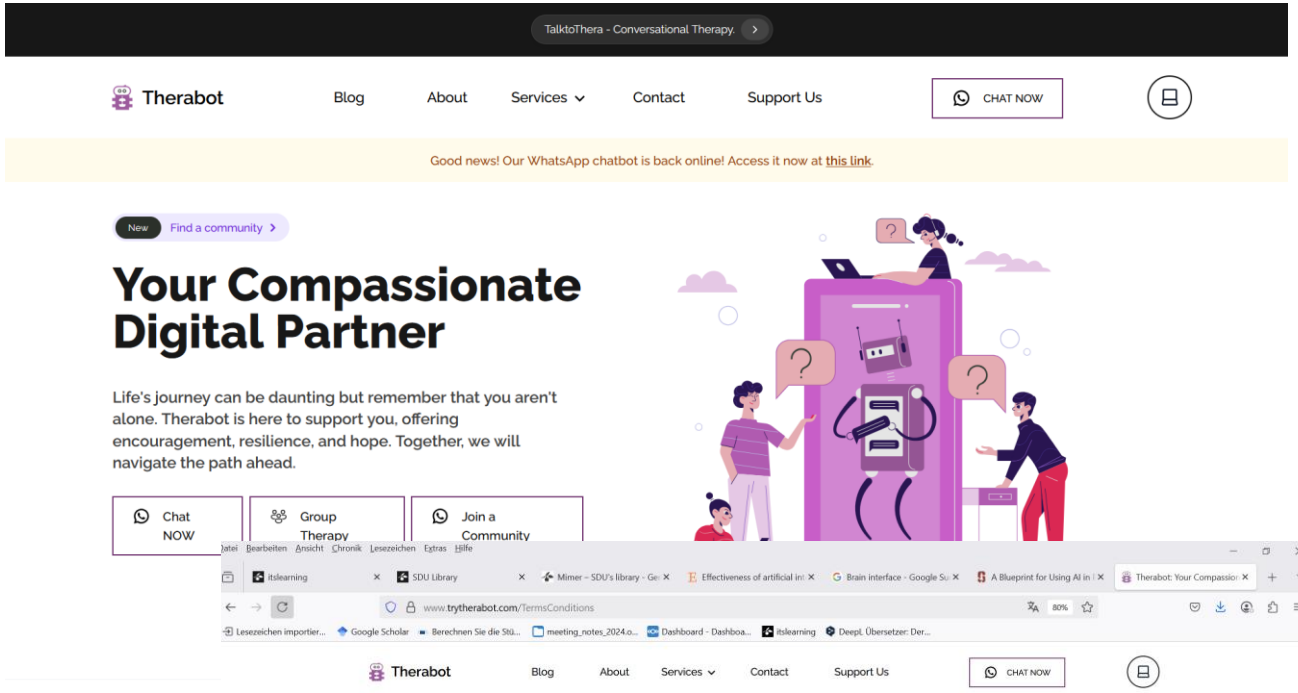
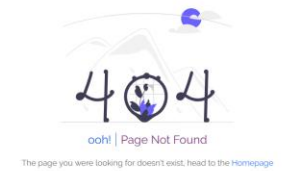
Can it be made better?

- Therabot trained by researchers at Dartmouth colleague (took them five years btw.)
- Tested with only (!) 210 patients (half of them with major depressions, general anxiety or high risk for eating disorders)
- The test lasted only 4 weeks, but it could show reductions of self-reported symptoms
- Reason enough to make it publicly available?



https://www.trytherabot.com/TermsConditions



AI in medicine

Hopes are high

Succes med kunstig intelligens på hospital i Køge skal bredes ud til hele landet, siger Løhde til Berlingske.

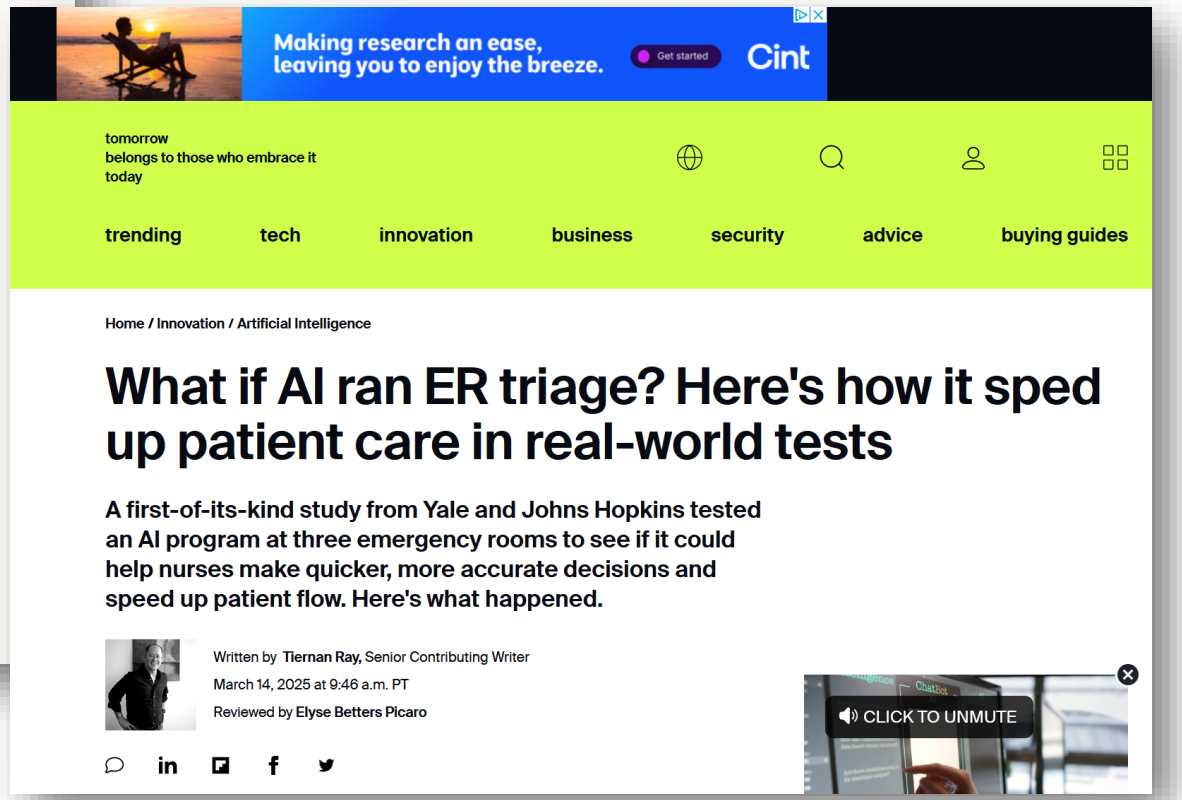
Sophie Løhde: Kunstig intelligens skal løse personalemangel på hospitaler

Kunstig intelligens skal fremover være med til at afhjælpe manglen på sundhedspersonale., mener sundhedsminister Sophie Løhde (V). Foto: Anders Holst Pedersen

LYT TIL ARTIKLEN GEM ARTIKLEN

DANMARK
14. AUGUST 2023, 23.41

https://politiken.dk/danmark/art9476964/Kunstig-intelligens-skal-I%C3%B8se-personalemangel-p%C3%A5-hospitaler



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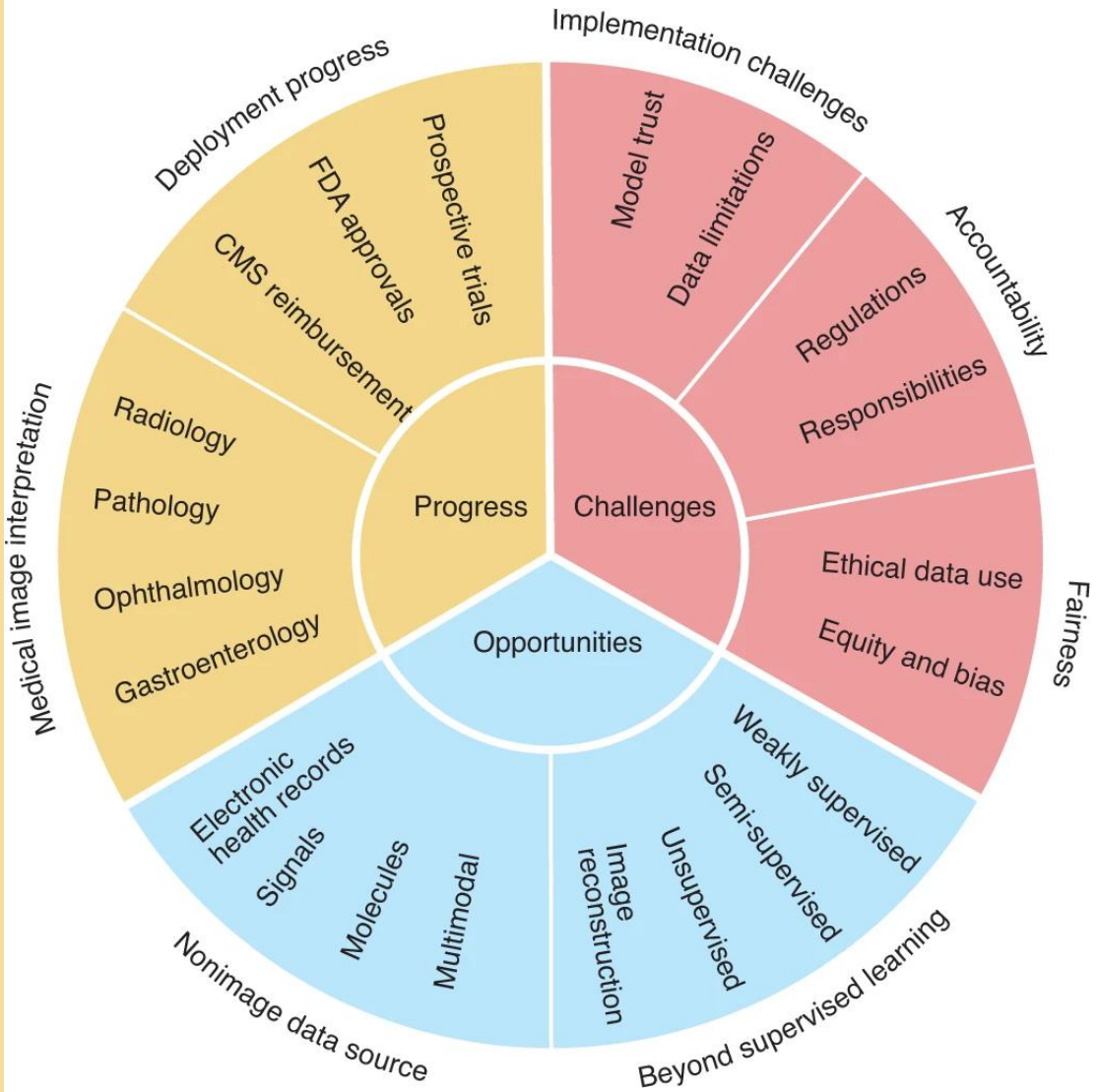
AI in medicine: an overview (Rajpurkar et al., 2022; Yoon et al., 2023)

Big steps forwards in medical image analytics

- Radiology
- Pathology
- Ophthalmology
- Gastroenterology
- E.g., mammography: AI is even slightly better without humans (Δ_{AUC} = .05 -.11)
- Remember: All perception is construction (→#2 Perception)



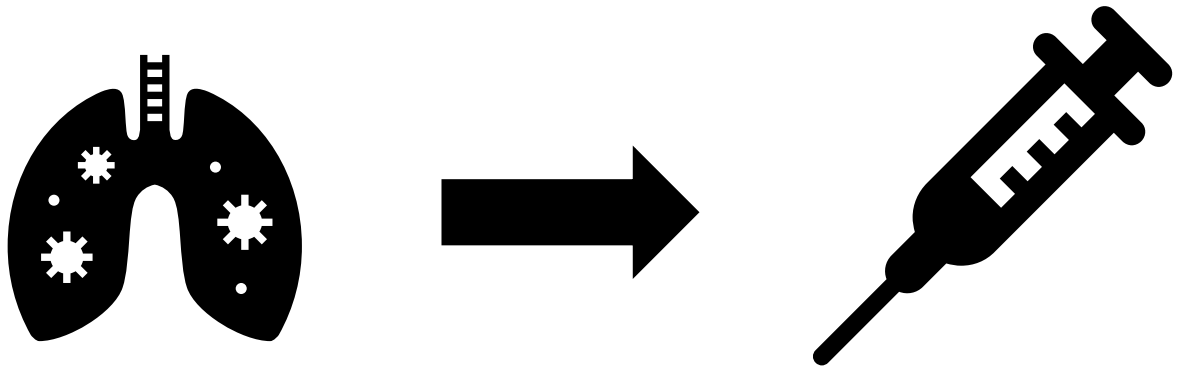
Big steps forwards in the deployment process

- Prospective randomized control trials
- Approvals by official institutions (e.g., FDA)
- Reimbursement by insurance

For example: the promise of personalisation

(Nationalt Genom Centre, *n.d.*)

From



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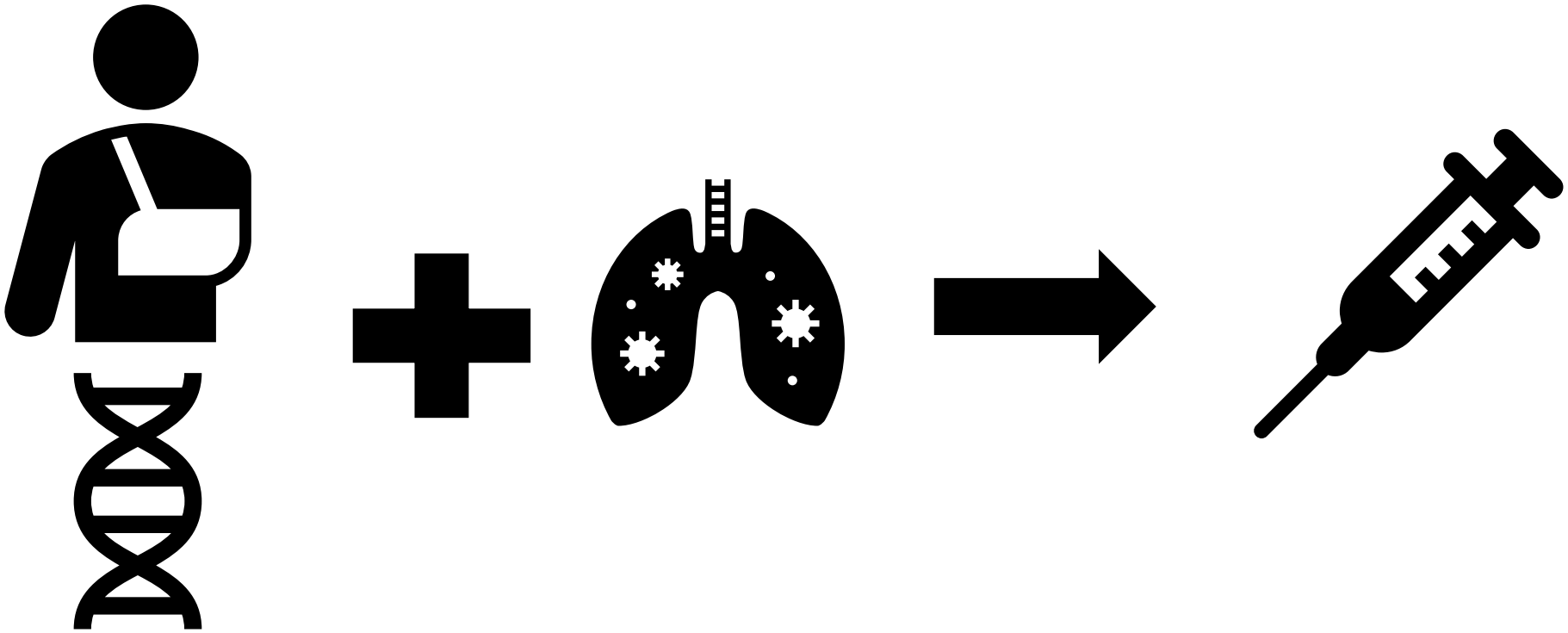

AI can support personalised medicine

(Ghebrehiwet et al., 2024)

- BUT: highly sensitive and personalized data
- One approach: simulated data – a study based on 7000 real data showed that cGANs can be used to simulate patient data with specific characteristics and can recapitulate clinical scoring and results from clinical trials
- Basic idea: Imputing missing values in a data set using cGANs and then use the imputed data
- Another approach: Anonymize patient data using GANs – here the idea is to mimick the original data set closely while ensuring that no single combination of values can identify a specific person
- AI can furthermore improve prediction for individual patients – for example, through predicting better how they will respond to individual drugs

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AI in medicine: an overview

(Ghebrehiwet et al., 2024; Rajpurkar et al., 2022; Yoon et al., 2023)

Big challenges

- Restricted generalisability
- Lack of external evaluation

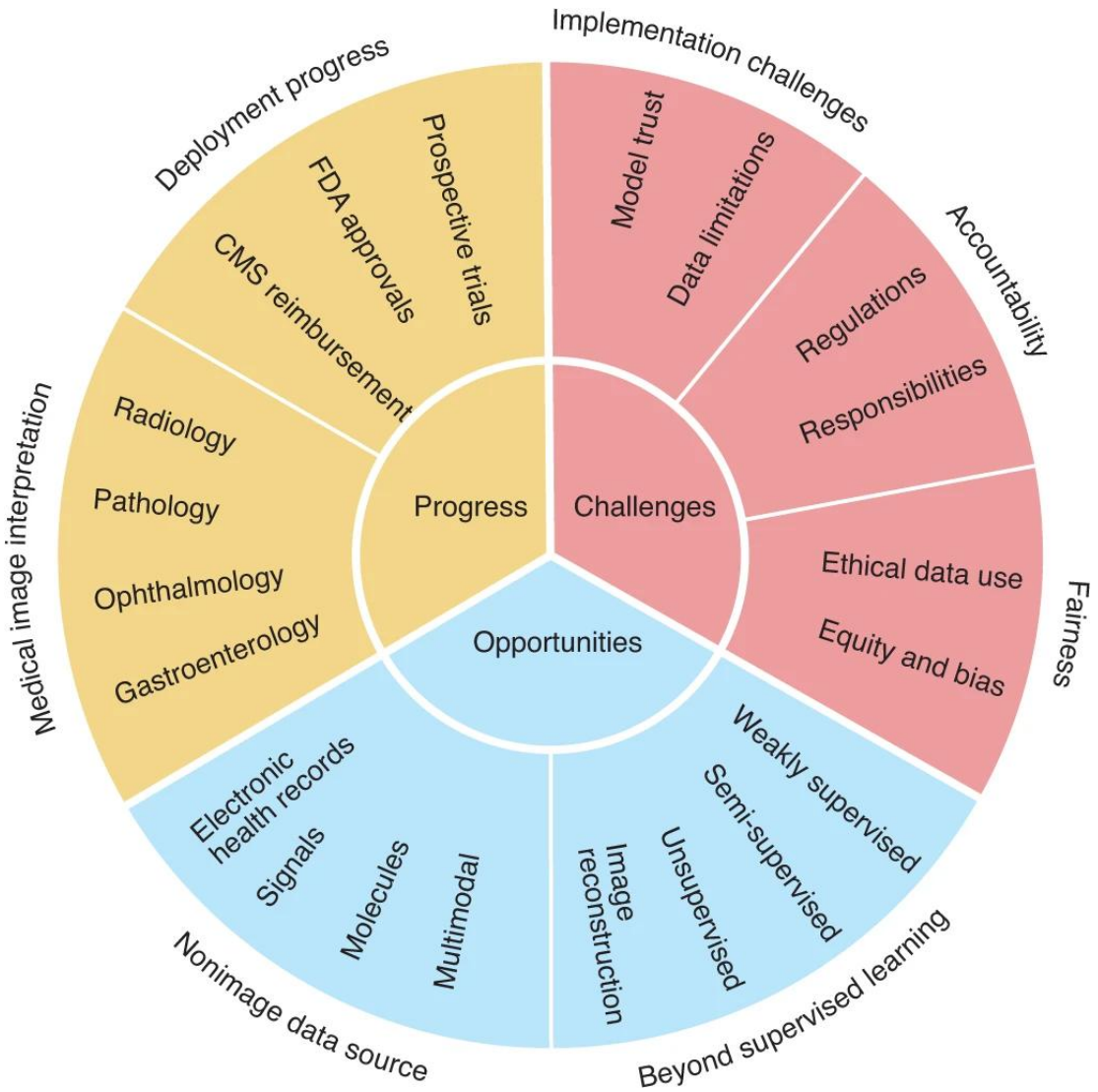
Quality & Fairness

- Bias (e.g., interpreting a ruler as a cancer cue)
- Ethical use

Accountability

- Regulations
- Responsibilities

Whom are you going to sue, if your beloved one dies?



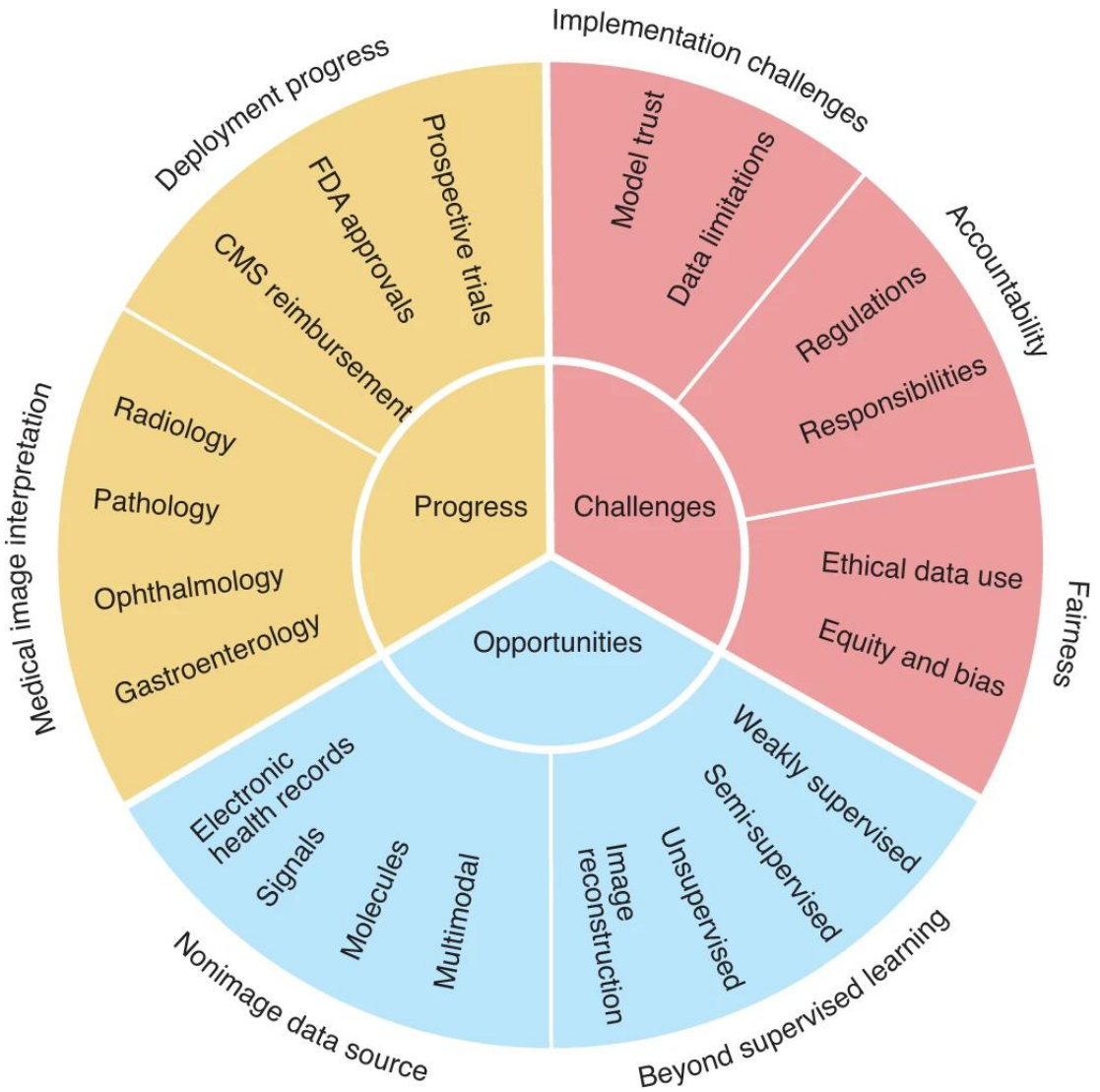
Big challenges

- Trust in the models
  - Hallucination
  - Lack of accuracy
  - False positives
- Limitations of available data (e.g., often trained on data by one provider and on highly processed samples of tissue) → not necessary how it looks in real life

AI in medicine: an overview (Rajpurkar et al., 2022; Yoon et al., 2023)

Big opportunities Beyond supervised learning

- Weakly supervised learning
- Semi-supervised human in the loop approaches
- Unsupervised
- Deep-learning



Non-image data

- Electronic health records
- Natural language processing
- Signals (e.g., EEG)
- Molecules
- Multimodal data
- Genomics → e.g., informing CRISPR editing

What are your questions?

Then I hope by now

You have…

- ❑ … a better understanding of psychological well-being
- ❑ … insights into the relationship between AI and wellbeing
- ❑ …discussed about AI's risks for mental health
- ❑ … learned how AI can (not) revolutionize psychotherapy and medicine

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AI 507: Artificial Intelligence and Society

#Next week: Entertainment & games with Modl.AI

Associate Prof. Dr. Lena Frischlich

References

Aaronowsky, A. (1979). Health, stress and coping. JosseyBass Publishers.

Bauer, G. F., Roy, M., Bakibinga, P., Contu, P., Downe, S., Eriksson, M., Espnes, G. A., Jensen, B. B., Juvinya Canal, D., Lindström, B., Mana, A., Mittelmark, M. B., Morgan, A. R., Pelikan, J. M., Saboga-Nunes, L., Sagy, S., Shorey, S., Vaandrager, L., & Vinje, H. F. (2020). Future directions for the concept of salutogenesis: A position article. Health Promotion International, 35(2), 187–195. https://doi.org/10.1093/heapro/daz057

Beg, M. J., Verma, M., M., V. C. K. M., & Verma, M. K. (2024). Artificial Intelligence for Psychotherapy: A Review of the Current State and Future Directions. Indian Journal of Psychological Medicine, 02537176241260819. https://doi.org/10.1177/02537176241260819

Buck, B., & Maheux, A. J. (2026). Psychosis risk and generative artificial intelligence use frequency, motivations, and delusion-like experiences: Cross-sectional survey study. Journal of Medical Internet Research, 28(1), e85038. https://doi.org/10.2196/85038

Carlbring, P., & Andersson, G. (2025). Commentary: AI psychosis is not a new threat: Lessons from media-induced delusions. Internet Interventions, 42, 100882. https://doi.org/10.1016/j.invent.2025.100882

Diener, E. (1984). Subjective well-being. Psychological Bulletin, 95(3), 542–575.

Ghebrehiwet, I., Zaki, N., Damseh, R., & Mohamad, M. S. (2024). Revolutionizing personalized medicine with generative AI: A systematic review. Artificial Intelligence Review, 57(5), 128. https://doi.org/10.1007/s10462-024-10768-5

Hays, K. (2026, April 21). OpenAI faces criminal probe over role of ChatGPT in shooting. BBC. https://www.bbc.com/news/articles/c62j4ldp2jqo

Hegarty, S. (2026, May 3). Musk's AI told me people were coming to kill me. I grabbed a hammer and prepared for war. BBC. https://www.bbc.com/news/articles/c242pzr1zp2o

Heinz, M. V., Mackin, D. M., Trudeau, B. M., Bhattacharya, S., Wang, Y., Banta, H. A., Jewett, A. D., Salzhauer, A. J., Griffin, T. Z., & Jacobson, N. C. (2025). Randomized trial of a generative AI chatbot for mental health treatment. NEJM AI, 2(4), AIoa2400802. https://doi.org/10.1056/AIoa2400802

https://politiken.dk/danmark/art9476964/Kunstig-intelligens-skal-l%C3%B8se-personalemangel-p%C3%A5-hospitaler

https://venturebeat.com/business/when-ai-flags-the-ruler-not-the-tumor-and-other-arguments-for-abolishing-the-black-box-vb-live/

https://venturebeat.com/business/when-ai-flags-the-ruler-not-the-tumor-and-other-arguments-for-abolishing-the-black-box-vb-live/

https://www.eng.ngc.dk/patients-and-personalised-medicine/what-is-personalised-medicine

https://www.esundhed.dk/Emner/Patienter-og-sygehuse/MitSygehusvalg?treatmentId=46&organizationGroupId=1&regionId=1,2,3,,5&postalFrom=5000

https://www.who.int/about/governance/constitution

https://www.zdnet.com/article/what-if-ai-ran-er-triage-heres-how-it-sped-up-patient-care-in-real-world-tests

Jamali, L. (2025, October 28). ChatGPT shares data on how many users exhibit psychosis or suicidal thoughts. BBC. https://www.bbc.com/news/articles/c5yd90g0q43o

Kuenssberg, L. (2025, November 8). A predator in your home': Mothers say chatbots encouraged their sons to kill themselves. BBC. https://www.bbc.com/news/articles/ce3xgwyywe4o

Lange, M. (2020). Antal borgere med kroniske sygdomme. Sundhedsdata Styrelsen. https://cdn1.gopublic.dk/sundhedsdatastyrelsen/media/16331/Antal%20borgere%20med%20kroniske%20sygdomme%20_%20februar%202020%20(PDF).pdf

Li, H., Zhang, R., Lee, Y.-C., Kraut, R. E., & Mohr, D. C. (2023). Systematic review and meta-analysis of AI-based conversational agents for promoting mental health and well-being. Npj Digital Medicine, 6(1), 1–14. https://doi.org/10.1038/s41746-023-00979-5

Montag, C., Nakov, P., & Ali, R. (2024). Considering the IMPACT framework to understand the AI-well-being-complex from an interdisciplinary perspective. Telematics and Informatics Reports, 13, 100112. https://doi.org/10.1016/j.teler.2023.100112

References

Moore, J., Mehta, A., Agnew, W., Anthis, J. R., Louie, R., Mai, Y., Yin, P., Cheng, M., Paech, S. J., Klyman, K., Chancellor, S., Lin, E., Haber, N., & Ong, D. C. (2026, March 17). Characterizing delusional spirals through human-LLM chat logs. arXiv.Org. https://arxiv.org/abs/2603.16567v1

Niederkrotenthaler, T., Herberth, A., & Sonneck, G. (2007). The 'Werther-effect': Legend or reality? Neuropsychiatrie: Klinik, Diagnostik, Therapie Und Rehabilitation: Organ Der Gesellschaft Osterreichischer Nervenarzte Und Psychiater, 21(4), 284–290.

Oishi, S., & Westgate, E. C. (2021). A psychologically rich life: Beyond happiness and meaning. Psychological Review, advanced online publication, 1–22. https://doi.org/10.1037/rev0000317

Park, Y., Park, S., & Lee, M. (2024). Effectiveness of artificial intelligence in detecting and managing depressive disorders: Systematic review. Journal of Affective Disorders, 361, 445–456. https://doi.org/10.1016/j.jad.2024.06.035

Petersen, E., Ganz, M., Holm, S., & Feragen, A. (2023). On (assessing) the fairness of risk score models. 2023 ACM Conference on Fairness Accountability and Transparency, 817–829. https://doi.org/10.1145/3593013.3594045

Rajpurkar, P., Chen, E., Banerjee, O., & Topol, E. J. (2022). AI in health and medicine. Nature Medicine, 28(1), 31–38. https://doi.org/10.1038/s41591-021-01614-0

Riddle, K. (2025, April 7). The (artificial intelligence) therapist can see you now. NPR. https://www.npr.org/sections/shots-health-news/2025/04/07/nx-s1-5351312/artificial-intelligence-mental-health-therapy

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. American Psychologist, 55, 68–78. https://doi.org/10/bgvcpf

Ryff, C. D., & Keyes, C. L. M. (1995). The structure of psychological well-being revisited. Journal of Personality and Social Psychology, 69(4), 719–727.

Taylor, R. A., Chmura, C., Hinson, J., Steinhart, B., Sangal, R., Venkatesh, A. K., Xu, H., Cohen, I., Faustino, I. V., & Levin, S. (2025). Impact of artificial intelligence–based triage decision support on emergency department care. NEJM AI, 2(3), AIoa2400296. https://doi.org/10.1056/AIoa2400296)

Thielen, K., Nygaard, E., & Diderichsen, F. (2007). Gender differences in depression - a matter of measure: 15th EUPHA conference, The future of public health in United Europe. European Journal of Public Health, supplement 2, 151.

Wang, L., Wang, C., Li, C., Murai, T., Bai, Y., Song, Z., Zhang, S., Zhang, Q., Huang, Y., Bi, X., & Jiang, J. (2025). AI-assisted multi-modal information for the screening of depression: A systematic review and meta-analysis. Npj Digital Medicine, 8(1), 523. https://doi.org/10.1038/s41746-025-01933-3

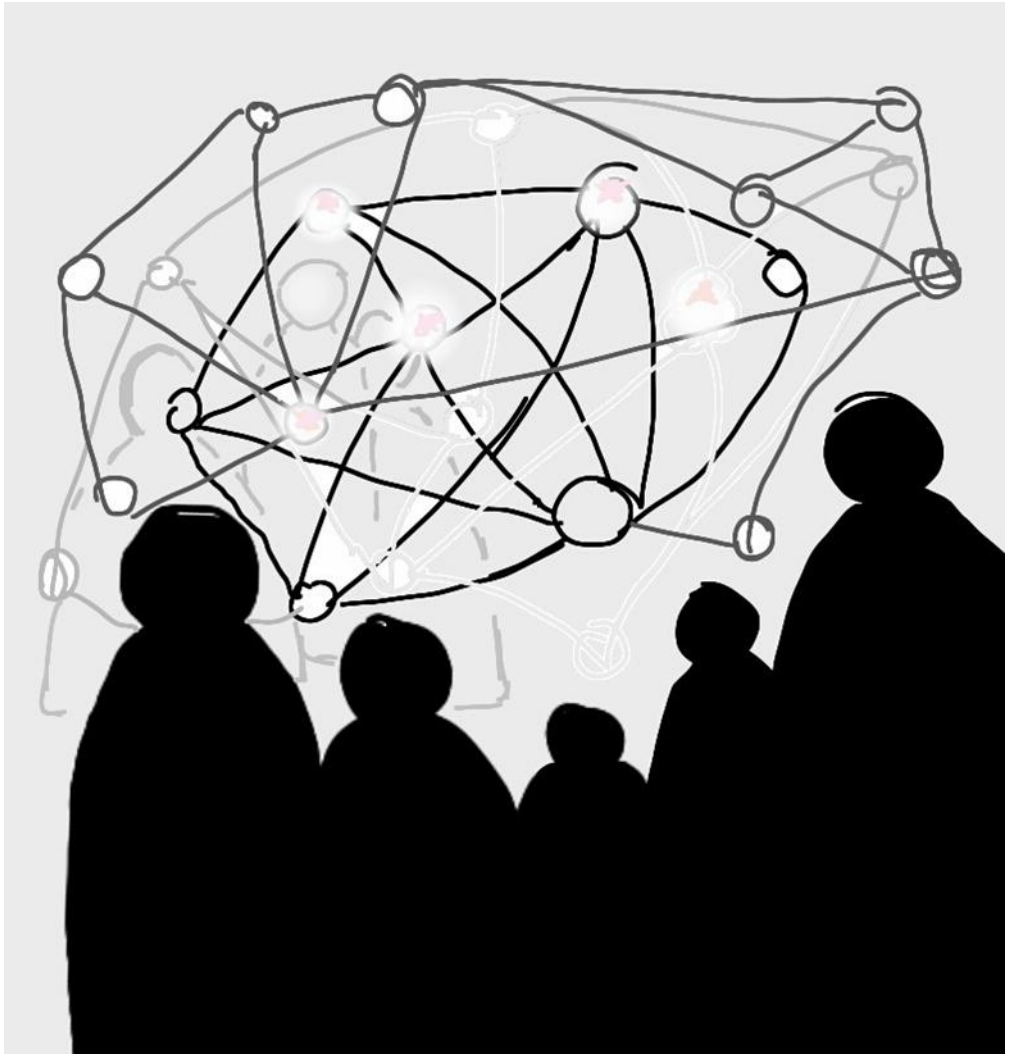
Yoon, J. H., Strand, F., Baltzer, P. A. T., Conant, E. F., Gilbert, F. J., Lehman, C. D., Morris, E. A., Mullen, L. A., Nishikawa, R. M., Sharma, N., Vejborg, I., Moy, L., & Mann, R. M. (2023). Standalone AI for Breast Cancer Detection at Screening Digital Mammography and Digital Breast Tomosynthesis: A Systematic Review and Meta-Analysis. Radiology, 307(5), e222639. https://doi.org/10.1148/radiol.222639

AI 507: Artificial Intelligence and Society

#14 EntertAInment

Associate Prof. Dr. Lena Frischlich

Flashlight



Flashlight

Health & wellbeing

Definition of Health

Salutogenese

Coherence

Well-being

- Life satisfaction
- Satisfaction of intrinsic needs (mastery, autonomy & affiliation)
- Hedonic
- Eudaimonic
- Psychologically rich

AI for wellbeing

AI negative consequences for wellbeing

Delusion Despair Deaths

Small share – but still many people

Sycophancy – validation - violence

AI and psychotherapy

Faster diagnosis (e.g., of depression/ anxiety)

Reduced waiting times

Certified online apps

Therapeutic support of *specialised* models

But: Fairness in different subgroups matters

AI and medicin

Great potentials

- Personalized medicine
- Image interpretation
- Multimodal data
- Approval by central organisations & insurance companies

Great challenges

- Personalized medicine
- Data access
- Accountability

Great opportunities for further development

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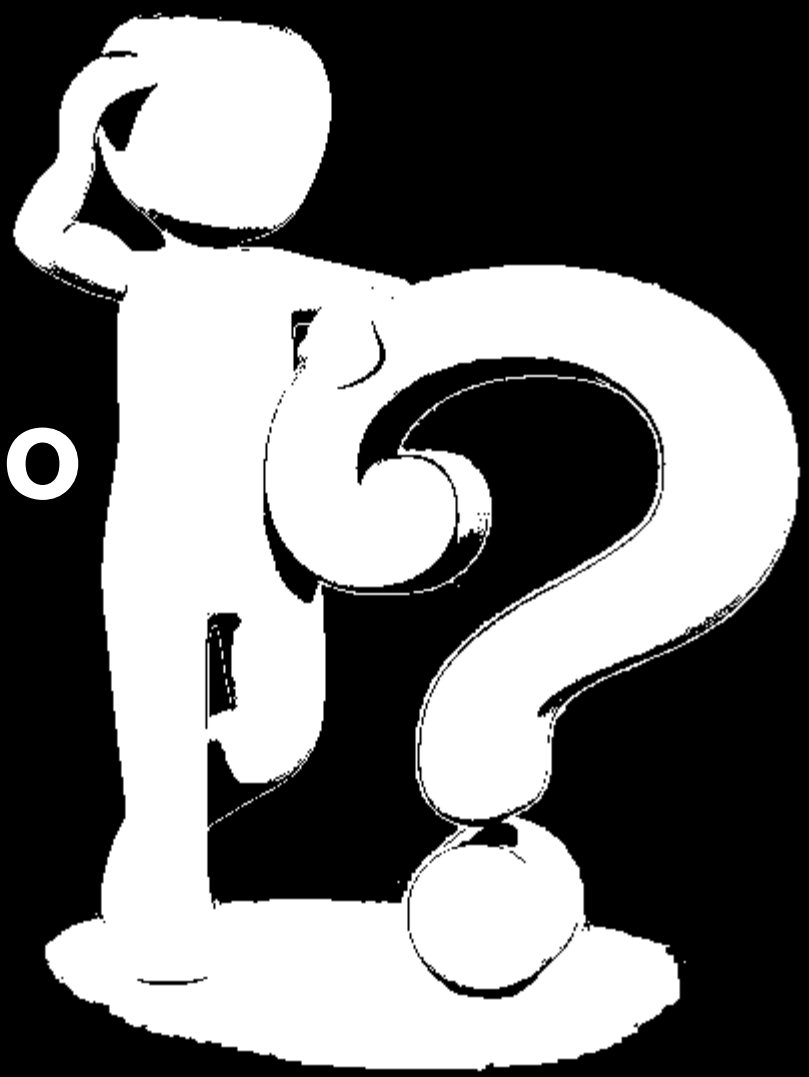
Do you have questions to last week?



Then let’s get started!

After today’s lecture, you will hopefully have…

- ❑ … a better understanding of why people play
- ❑ … learned how people perceive virtual worlds and fiction
- ❑ … discussed how virtual realities affect our empathy
- ❑ … gotten first hand experiences into how one of Copenhagens’ best ranked start-ups https://modl.ai/ uses AI to make games ready for their future fans

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Why do we talk about games?

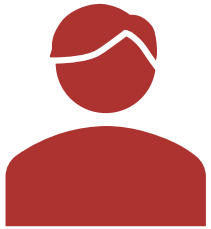
Games are the worlds largest entertainment industry

(Accenture, 2021; https://gamesdenmark.dk/industry-in-numbers; Danske Statistik, 2018)

$300 Billions

Estimated value of the industry – more than film & music combined

50%



52%

3,6 mio. DKK

Revenue 2022

https://www.deviantart.com/miloscat/art/Butterflies-986651824

Between 2017- 2022: 74% more full-time employees in the gaming industry

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Why do people play?

Uses & Gratifications

(Katz, Blumler & Gurevitch, 1973)

Key question: Why do people turn to media?

Basic assumptions

1. People have social and psychological needs
2. They have expectations of mass media (and other sources) regarding the satisfaction of these needs
3. This leads to different usage patterns
4. These patterns of use lead to the satisfaction of needs (gratification) – but can also have unintended consequences

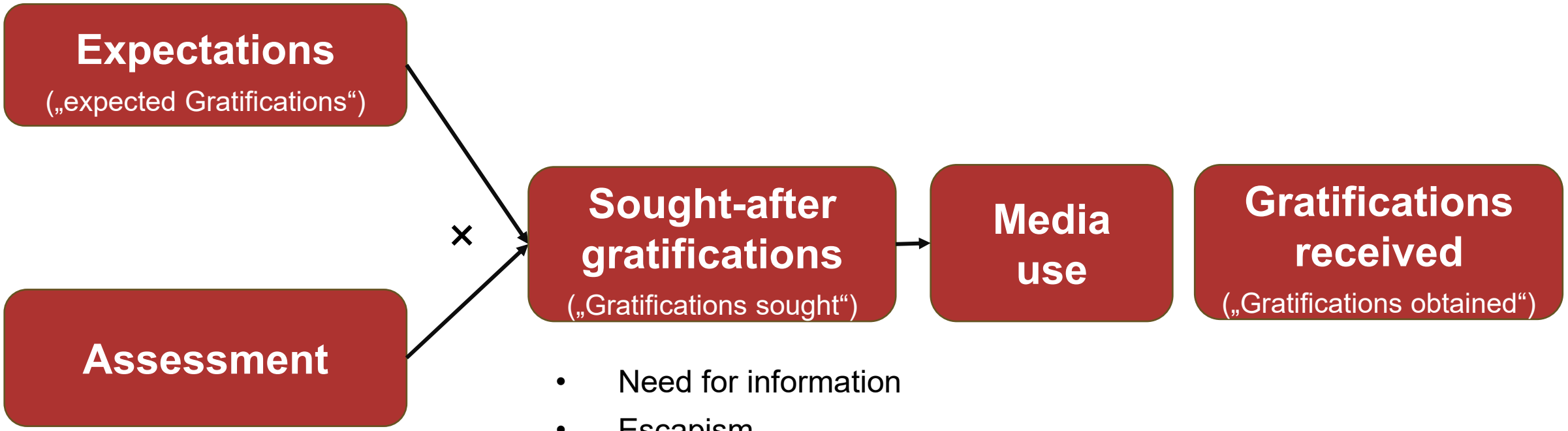
Central: Focus on the user – not the normative evaluation of the use

Elihu Katz via /www.asc.upenn.edu

Jay Blumler via University of Meryland

Michael Gurevitch via http://teoriausosygratificaciones.blogspot.com/2010/05/los-teoricos.html

Or, as a process (Katz, Blumler & Gurevitch, 1973)



- Need for information
- Escapism
- Relaxation
- Social connectedness...

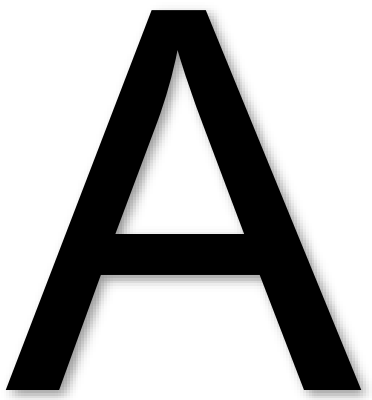
Gaming specific motives (Chea et al., 2022; Sailer et al., 2017; Tang et al., 2020; Yee, 2006)

N = 3000 Massive Multi-Player Online Role Game (MMPoRG) Player

Factor analysis approach

Three key factors

- **A**chievements
- **S**ocial affairs,
- **I**mmersion



- **Gamification** = application of elements from computer games in other contexts
- In particular, features such as badges, leaderboards, and performance graphics satisfy needs for mastery
- Central for e-sports
- Game addiction is more associated with escapist and social motives – goal orientation is a rather „healthy play“

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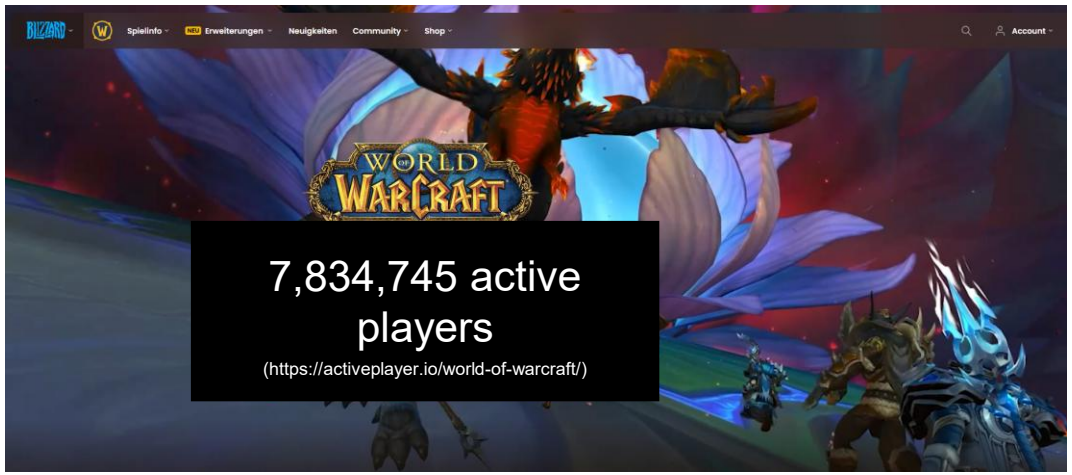
2) Social Interactions/ connectedness

(Bowman, Rieger & Lin, 2022; Domahidi, Festl & Quandt, 2014; Kowert, Domahidi, & Quandt, 2014)

- Gaming is (also) a place for friendships
- Especially those who play a lot and have a strong social motive to play are more likely to make lasting connections online
- Gaming can even help emotionally insecure people to form social relationships
- Can contribute to psychological well-being
- Social experiences are part of the attraction of many games (and game adjacent platforms such as discord, twitch or steam)



https://worldofwarcraft.blizzard.com/de-de/

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3) Identification (Bessiére et al. 2007; Chea et al., 2022; Guadagno et al., 2011; Qi et al., 2023)

Avatars as a means of self-exploration

- Often represent the ideal self ("how you would like to be") rather than the real self ("how you are")
- Opportunity to try out roles (e.g., in violent or gendered games)
- Doesn’t mean that avatars necessarily overcome stereotypes…
- Players identify with their avatars and can experience them as part of their self-concept, up to a more automated processing of the avatar compared to other familiar content

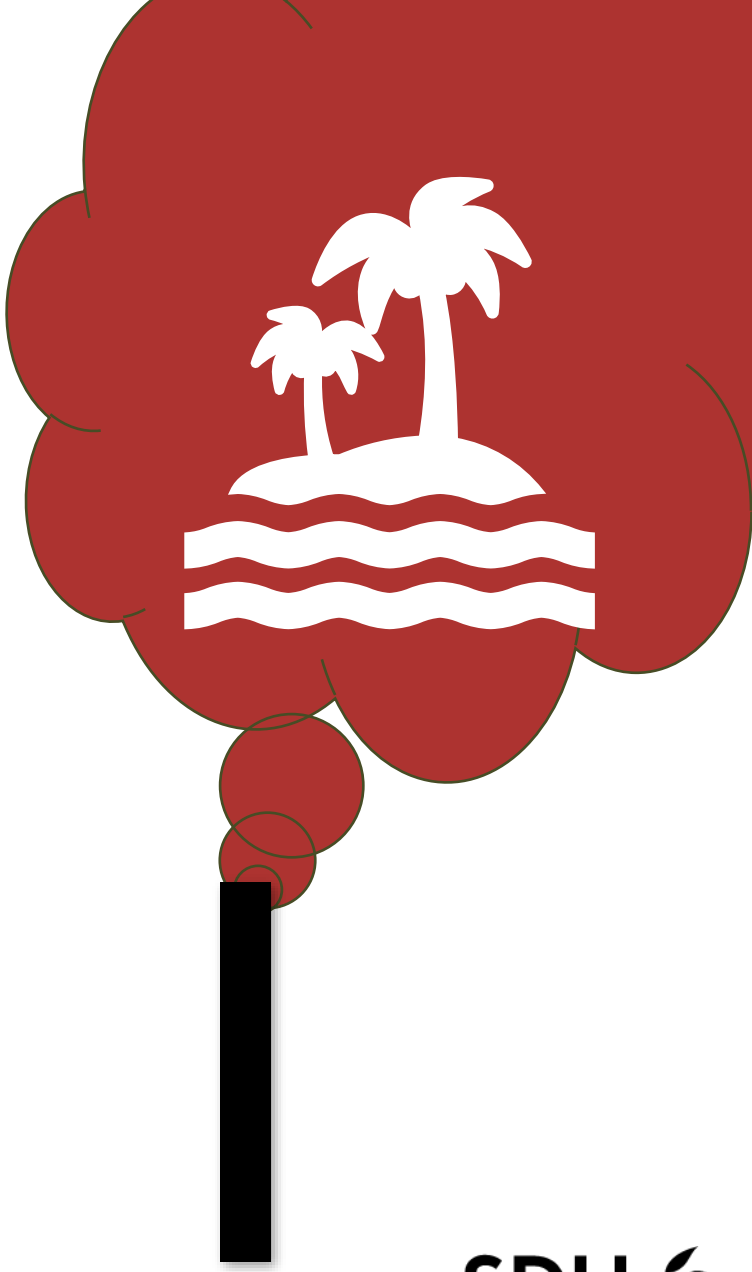
S

4) Immersion and flow (Csikszentmihalyi, 2014; Michailidis et al., 2018; Therrien, 2023)

- **Immersion** = immersion in the virtual environment, credibility, liveliness, sensuality of experience – an imperfect, gradual state
- **Presence** = The feeling of being "there“, of "getting lost" in the digital environment; the feeling "that this is real “
- **Flow** = Holistic state, total involvement
  - Action after action results according to an internal logic
  - We merge with our environment, stimulus and reaction, past, present and future become one

(5) Escapism (Hussain et al., 2021)

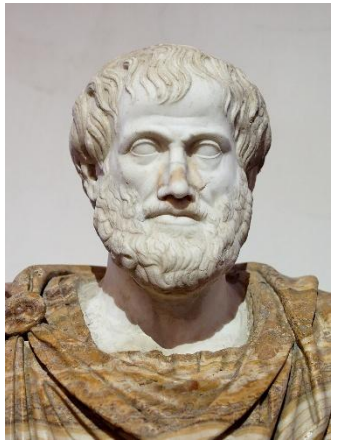
- Immersion to distract yourself from everyday problems
- Negative correlates: depression, social phobias, loneliness, etc.
- Positive correlates: e.g. positive mood, wishes, joy
- But meta-analysis of 27 studies (*N* = 28.893) showed that the association with negative aspects is more pronounced, including addictive behavior (same we also saw in problematic AI use last week)
- Effect more pronounced in Western (i.e., compared to Eastern) cultures
- More cultural comparative research needed!



Content factors: Entertainment processes

(Bartsch et al. 2016; Oliver et al. 2018; Oliver & Raney, 2011; Tamborini et al. 2011; Vorderer & Reinecke, 2015; Wirth et al. 2012; Zillmann, 1988)

- *Hedonic* entertainment (e.g., via comedies)
  - Characterised by feelings of fun, pleasure, or suspense
  - *Central for mood management*
- a more complex, *eudaimonic* entertainment (e.g., via dramatic movies, meaningful, value-laden content)
  - characterized by mixed and often self-transcending emotions such as nostalgia, meaningfulness or appreciation
  - “thought-provoking” entertainment experiences
  - can inspire self-transcending experiences and motivate pro-social behavior



Aristoteles; Lysipp - Jastrow (2006), via https://commons.wikimedia.org/w/index.php?curid=1359807

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Content factors: Narrative processing

(Cohen et al., 2015; de Graaf et al., 2012; Green & Brock, 2000)

1. **Transportation**
   - the feeling of getting lost in the world of a narrative, immersing yourself and leaving the real world behind (→ pretty similar to immersion/ presence)
   - integrated fusion of attention, images, and emotions
2. **Identification**
   - Changing the self-concept by adopting perceived characteristics of a media person
   - Understanding, taking perspective and empathy

➢ VR enhances the transportation and identification through *being* and *acting* in *there* (aka presence)

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Questions

Augmented & Virtual Reality

Augmented Reality (Azuma, 1997; Devarigi et al., 2022; LaViola et al., 2017)

- Approaches to extend (= augmentation) the user's view of a real environment through synthetic or simulated information
- Augmented reality systems typically combine the real and the virtual, are interactive in real time, and are captured in 3D
- Examples
- 1) Augmented reality technology based on mobile phones/tablets, or
- 2) Applications with optically transparent or video-transparent head-mounted displays
- E.g., Meta’s Rayban collaboration

https://www.smk.dk/en/article/titanernes-fald-med-augmented-reality/

Virtual Reality (Biocca, 1992; Fox et al., 2009)

Interactive Media Systems

- Record users' movements
- Responsively create a rich sensory environment
- which is intended to replace the cues from the physical environment

- a wider range of sensory impressions (e.g. visual representations, movement, depth, sound, spatiality, balance)
- Naturally depicted interaction possibilities through head rotation, gestures and body movements
- Higher level of interactivity
- Therefore, more immersive

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VR as an empathy machine? (Herrera et al., 2018)

Suggested in a TED Talk by filmmaker Chris Milk, in which he argues that VR can be the ultimate "empathy machine“

But is that really the case??

Experiment 1

- a) Idea of what it would be like to be homeless vs. b) VR, in which you are homeless in the 1st person perspective
- *N = 113 participated in follow-up surveys for 8 weeks*
- More empathy in VR short term, but not in the long run
- No differences in attitudes but more signatures on a petition 8 weeks later

Experiment 2 (N = 452) showed that VR worked better than a less immersive medium (desktop) and replicated the results

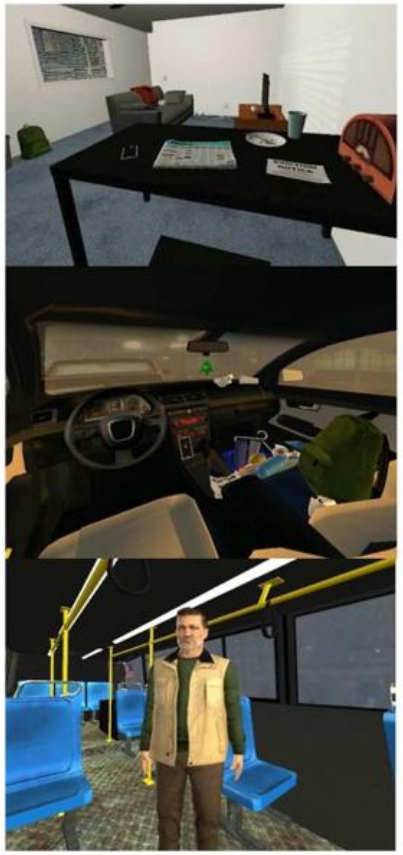


Fig 2. Scene progression for VR perspective-taking task. Top: Participant's point of view in the apartment scene. Middle: Participant's point of view in the car scene. Bottom: Participant's point of view in the bus scene.

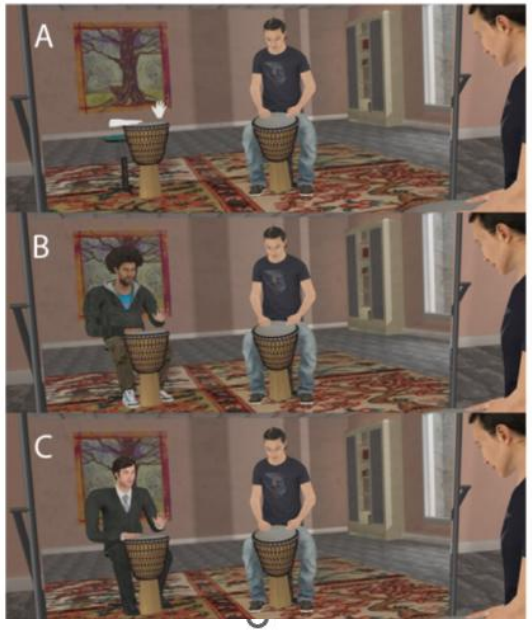
Embodiement & the Proteus Effect

(Bem, 1972; Banakou et al., 2016; Fox et al., 2013, Hoter & Nagar, 2023; Kilteni et al., 2013; Pena et al. 2009; Ratan et al. 2020; Yee & Bailenson, 2007, Yee et al., 2009)

- Similar idea to the SIDE-Model (→ Recall #4)
- But: Here avatars serve as the “salient” identity
- Remember: Our self-concept is based on what we do (working self-concept → Recall #4)
- When users are "embodied", they accept the virtual body as their own and behave accordingly
- Users align their attitudes and behavior with the characteristics they attribute to the avatar= Proteus-Effect

Activates stereotypes

- Better drum performance with Jamaican avatars
- More self-objectification as women (vs. men)



Reduce stereotypes

- Less stereotypes after playing a Black avatar
- Less prejudice and more engagement for disabled people

Time for a break

Then I hope you now have

- ❑ … a better understanding of why people play
- ❑ … learned how people perceive virtual worlds and fiction
- ❑ … discussed how virtual realities affect our empathy
- ❑ … gotten first hand experiences into how one of Copenhagens' best ranked start-ups https://modl.ai/ uses AI to make games ready for their future fans

AI 507: Artificial Intelligence and Society

Wrap up – all things multiple choice exam

Associate Prof. Dr. Lena Frischlich

References

Accenture. (2021, April 29). *Global Gaming Industry Value Now Exceeds $300 Billion, New Accenture Report Finds*. https://newsroom.accenture.com/news/global-gaming-industry-value-now-exceeds-300-billion-new-accenture-report-finds.htm

Azuma, R. T. (1997). A survey of augmented reality. *Presence: Teleoperators and Virtual Environments*, *6*(4), 355–385. https://doi.org/10.1162/pres.1997.6.4.355

Banakou, D., Hanumanthu, P. D., & Slater, M. (2016). Virtual Embodiment of White People in a Black Virtual Body Leads to a Sustained Reduction in Their Implicit Racial Bias. *Frontiers in Human Neuroscience*, *10*. https://www.frontiersin.org/articles/10.3389/fnhum.2016.00601

Bartsch, A., Oliver, M.-B., Nitsch, C., & Scherr, S. (2016). Inspired by the Paralympics: Effects of empathy on audience interest in para-sports and on the destigmatization of persons with disabilities. *Communication Research*. https://doi.org/10.1177/0093650215626984

Bem, D. J. (1972). Self-perception theory. In L. Berkowitz (Hrsg.), *Advances in Experimental Social Psychology* (6. Aufl., S. 2–57). Academic Press Inc. http://books.google.com/books?hl=de&lr=&id=INH3hIDHb-MC&pgis=1

Bessière, K., Seay, F., & Kiesler, S. (2007). The ideal elf: Identity exploration in World of Warcraft. *Cyberpsychology & behavior : the impact of the Internet, multimedia and virtual reality on behavior and society*, *10*(4), 530–535. https://doi.org/10.1089/cpb.2007.9994

Biocca, F. (1992). Virtual reality technology: A tutorial. *Journal of Communication*, *42*(4), 23–72. https://doi.org/10.1111/j.1460-2466.1992.tb00811.x

Bowman, N. D., Rieger, D., & Tammy Lin, J.-H. (2022). Social video gaming and well-being. *Current Opinion in Psychology*, *45*, 101316. https://doi.org/10.1016/j.copsyc.2022.101316

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References

- Cheah, I., Shimul, A. S., & Phau, I. (2022). Motivations of playing digital games: A review and research agenda. *Psychology & Marketing*, *39*(5), 937–950. https://doi.org/10.1002/mar.21631
- Cohen, J., Tal-Or, N., & Mazor-Tregerman, M. (2015). The tempering effect of transportation: Exploring the effects of transportation and identification during exposure to controversial two-sided narratives. *Journal of Communication*, *65*(2), 237–258. https://doi.org/10/f68sm2
- Csikszentmihalyi, M. (2014). *Flow and the foundations of positive psychology*. Springer.
- de Graaf, A., Hoeken, H., Sanders, J., & Beentjes, J. W. J. (2012). Identification as a mechanism of narrative persuasion. *Communication Research*, *39*(6), 802–823. https://doi.org/10/cnnskn
- Devagiri, J. S., Paheding, S., Niyaz, Q., Yang, X., & Smith, S. (2022). Augmented reality and artificial intelligence in industry: Trends, tools, and future challenges. *Expert Systems with Applications*, *207*, 118002. https://doi.org/10.1016/j.eswa.2022.118002
- Domahidi, E., Festl, R., & Quandt, T. (2014). To dwell among gamers: Investigating the relationship between social online game use and gaming-related friendships. *Computers in Human Behavior*, *35*, 107–115. https://doi.org/10.1016/j.chb.2014.02.023
- Fox, J., Arena, D., & Bailenson, J. N. (2009). Virtual reality. *Journal of Media Psychology*. https://econtent.hogrefe.com/doi/10.1027/1864-1105.21.3.95
- Fox, J., Bailenson, J. N., & Tricase, L. (2013). The embodiment of sexualized virtual selves: The Proteus effect and experiences of self-objectification via avatars. *Computers in Human Behavior*, *29*(3), 930–938. https://doi.org/10.1016/j.chb.2012.12.027

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References

Green, M. C., & Brock, T. C. (2000). The role of transportation in the persuasiveness of public narratives. *Journal of Personality and Social Psychology*, *79*(5), 701–721. https://doi.org/10/bkr5dt

Guadagno, R. E., Muscanell, N. L., Okdie, B. M., Burk, N. M., & Ward, T. B. (2011). Even in virtual environments women shop and men build: A social role perspective on Second Life. *Computers in Human Behavior*, *27*(1), 304–308. https://doi.org/10.1016/j.chb.2010.08.008

Herrera, F., Bailenson, J., Weisz, E., Ogle, E., & Zaki, J. (2018). Building long-term empathy: A large-scale comparison of traditional and virtual reality perspective-taking. *PLOS ONE*, *13*(10), e0204494. https://doi.org/10.1371/journal.pone.0204494

Hoter, E., & Nagar, I. (2023). The effects of a wheelchair simulation in a virtual world. *Virtual Reality*, *27*(1), 407–419. https://doi.org/10.1007/s10055-022-00625-7

https://activeplayer.io/world-of-warcraft/)

https://gamesdenmark.dk/industry-in-numbers

https://www.bbc.com/news/articles/cj37z8357e5o

https://www.dst.dk/da/Statistik/nyheder-analyser-publ/nyt/NytHtml?cid=31790

Hussain, U., Jabarkhail, S., Cunningham, G. B., & Madsen, J. A. (2021). The dual nature of escapism in video gaming: A meta-analytic approach. *Computers in Human Behavior Reports*, *3*, 100081. https://doi.org/10.1016/j.chbr.2021.100081

Katz, E., Blumler, J. G., & Gurevitch, M. (1973). Uses and gratifications research. *Public Opinion Quarterly*, *37*, 509–523. https://www.jstor.org/stable/2747854

Kilteni, K., Bergstrom, I., & Slater, M. (2013). Drumming in immersive virtual reality: The body shapes the way we play. *IEEE Transactions on Visualization and Computer Graphics*, *19*(4), 597–605. https://doi.org/10.1109/TVCG.2013.29

Kowert, R., Domahidi, E., & Quandt, T. (2014). The relationship between online video game involvement and gaming-related friendships among emotionally sensitive individuals. *Cyberpsychology, Behavior, and Social Networking*, *17*(7), 447–453. https://doi.org/10.1089/cyber.2013.0656

LaViola, J. J., Williamson, B. M., Sottilare, R., & Garrity, P. (2017). Analyzing slam algorithm performance for tracking in augmented reality systems. *Interservice/Industry Training, Simulation, and Education Conference (I/ITSEC) 2017*.

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References

Michailidis, L., Balaguer-Ballester, E., & He, X. (2018). Flow and Immersion in Video Games: The Aftermath of a Conceptual Challenge. *Frontiers in Psychology*, *9*. https://www.frontiersin.org/articles/10.3389/fpsyg.2018.01682

Oliver, M. B., Raney, A. A., Slater, M. D., Appel, M., Hartmann, T., Bartsch, A., Schneider, F. M., Janicke, S. H., Krämer, N., Mares, M.-L., Vorderer, P., Rieger, D., Dale, K. R., & Das, E. (2018). Self-transcendent media experiences: Taking meaningful media to a higher level. *Journal of Communication*, 380–389. https://doi.org/doi:10.1093/joc/jqx020

Oliver, M.-B., & Raney, A. A. (2011). Entertainment as pleasurable and meaningful: Identifying hedonic and eudaimonic motivations for entertainment consumption. *Journal of Communication*, *61*(5), 984–1004. https://doi.org/10/dh6pws

Oliver, M.-B., Dillard, J. P., Bae, K., & Tamul, D. J. (2012). The effect of narrative news format on empathy for stigmatized groups. *Journalism & Mass Communication Quarterly*, *89*, 205–224. https://doi.org/10/gf3gpd

Palmgreen, P. (1984). Uses and gratifications: A theoretical perspective. *Annals of the International Communication Association*, *8*(1), 20–55. https://doi.org/10.1080/23808985.1984.11678570

Pena, J., Hancock, J. T., & Merola, N. (2009). The priming effects of avatars in virtual settings. *Communication Research*, *36*(6), 838–856. https://doi.org/10.1177/0093650209346802

Qi, Y., Zou, Y., Peng, K., & Wang, F. (2023). The avatar-prioritization effect among online gamers: A perspective from self–avatar identity relevance. *Journal of Applied Research in Memory and Cognition*. https://doi.org/10.1037/mac0000096

Ratan, R., Beyea, D., Li, B. J., & Graciano, L. (2020). Avatar characteristics induce users' behavioral conformity with small-to-medium effect sizes: A meta-analysis of the proteus effect. *Media Psychology*, *23*(5), 651–675. https://doi.org/10/ggp6r9

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References

Sailer, M., Hense, J. U., Mayr, S. K., & Mandl, H. (2017). How gamification motivates: An experimental study of the effects of specific game design elements on psychological need satisfaction. *Computers in Human Behavior*, *69*, 371–380. https://doi.org/10.1016/j.chb.2016.12.033

Tamborini, R., Grizzard, M., David Bowman, N., Reinecke, L., Lewis, R. J., & Eden, A. (2011). Media Enjoyment as Need Satisfaction: The Contribution of Hedonic and Nonhedonic Needs. *Journal of Communication*, *61*(6), 1025–1042. https://doi.org/10.1111/j.1460-2466.2011.01593.x

Tang, W. Y., Reer, F., & Quandt, T. (2020). The interplay of gaming disorder, gaming motivations, and the dark triad. *Journal of Behavioral Addictions*, *9*(2), 491–496. https://doi.org/10.1556/2006.2020.00013

Therrien, C. (2023). Immersion. In M. J. P. Wolf & B. Perron (Hrsg.), *The Routledge Companion to Video Game Studies* (S. 564–571). Routledge.

Vorderer, P., & Reinecke, L. (2015). From mood to meaning: The changing model of the user in entertainment research. *Communication Theory*, *25*(4), 447–453. https://doi.org/10.1111/comt.12082

Wetterberg, N. (2025). *The impact of digital games*. Games Denmark.

Yee, N. (2006). Motivations for Play in Online Games. *CyberPsychology & Behavior*, *9*(6), 772–775. https://doi.org/10.1089/cpb.2006.9.772

Yee, N., & Bailenson, J. (2007). The proteus effect: The effect of transformed self-representation on behavior. *Human Communication Research*, *33*(3), 1–38. https://doi.org/10.1111/j.1468-2958.2007.00299.x

Yee, N., & Bailenson, J. (2007). The proteus effect: The effect of transformed self-representation on behavior. *Human Communication Research*, *33*(3), 1–38. https://doi.org/10.1111/j.1468-2958.2007.00299.x

Yee, N., Bailenson, J. N., & Ducheneaut, N. (2009). The Proteus Effect. *Communication Research*, *36*(2), 285–312. https://doi.org/10.1177/0093650208330254

Zillmann, D. (1988). Mood management through communication choices. *American Behavioral Scientist*, *31*(3), 327–340. https://doi.org/10/bfq9jj

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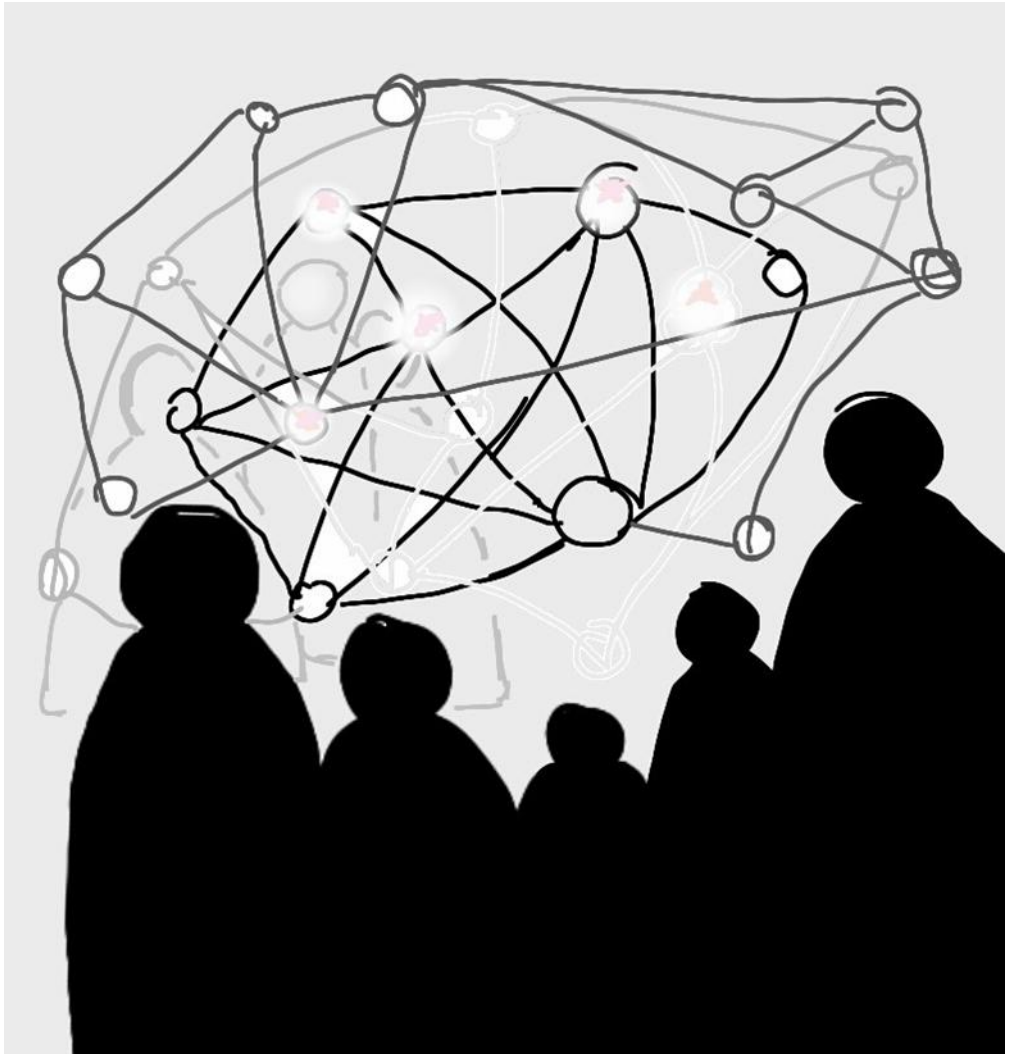
AI 507: Artificial Intelligence and Society

#15 Wrap-up

Associate Prof. Dr. Lena Frischlich

For the last time: flashlight



Flashlight

Computer games

Gaming industry

Uses & Gratifications

Gaming specific motives

Archievement

Gamification

Social affairs

Connectedness

Identification

Immersion

Flow

Escapism

Augmented & Virtual reality

What is what

VR as an empathy machine?

The Proteus effect

Entertainment processes

Hedonic entertainment

Eudaimonic entertainment

Identification

Transportation

Agentic AI for Games development

The role of testing game features in games development

What AI can(not) scale up

From research to successful start-up

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Any last questions on that?

Then let’s get started!

Todays lecture is a wrap-up, so we will

- ❑ … recap what we discussed this semester
- ❑ … clarify open questions regarding the exam

To this end, we will have discussion units in which you will be assigned a role

Draw from all that you learned to fill out your role – also an excellent way to ensure that larger networks form in your brain

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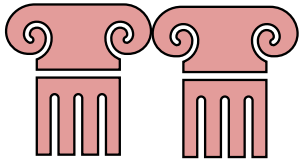
We have come a long way…

Foundations



Intelligence → Learning → Social Cognition → Communication → Innovations

Emil Bakkensen Johansen

Postdoc

emilbj@sam.sdu.dk

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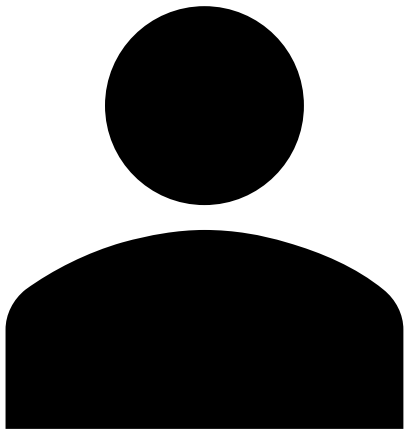
Artificial Intelligence Consultant for NIIT (remote)

Details zum Jobangebot

About NIIT

NIIT provides comprehensive and scalable managed training services for learning administration that maps directly to our customer needs. Training is our business and passion. We are one of the largest training outsourcing providers in the world. NIIT delivers and administers 75,000 days of training each year, in 18 languages. We have designed and developed over 50,000 days of classroom training and 20,000 hours of web-based training localized to more than 25 languages. NIIT is a global service provider. We have a presence in more than 40 countries.



Expected Tech Qualification (or should be able to pick these up quickly. These skills are a must have)
The instructor will debug participant environments live in front of the cohort. Fluency, not awareness (Python, Linux, LLM APIs, Git…)

Teaching and Content Delivery
translate technically dense content around agentic AI into clear, accessible explanations for a diversely skilled audience spanning advanced engineers, intermediate practitioners, and career transitioning participants

What are core differences between human and artificial intelligence your clients have to be aware of?

How do your structure the learning experience in a human-brain optimized manner – and how should your clients use AI?

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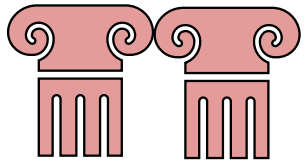
We have come a long way

Foundations



Intelligence

Learning

Social Cognition

Communication

Innovations

Robots

Recommender System

Chatbots

Smart things

Digital platforms

Human-Technology Interactions

Wissenschaftliche Mitarbeiterin

Dr. Jessica Szczuka

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ggf. Assoziation:	**Research Center Trustworthy Data Science and Security / INTITEC: Intimacy with and through Technology**
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Hannah Decker

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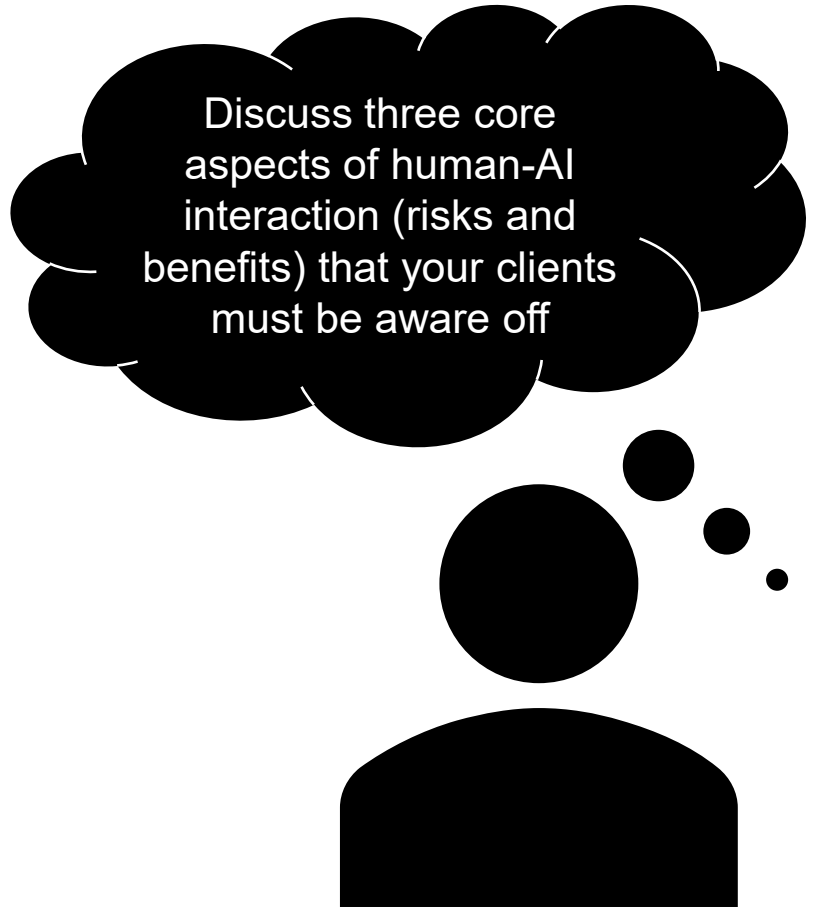
AI Automation Consultant at Vertica (Aarhus)

Details zum Jobangebot

Vil du hjælpe virksomheder med at omsætte AI og automation til konkret forretningsværdi? Trives du i rollen som konsulent, hvor du er tæt på kunden, forstår deres forretning, bygger løsninger og hjælper organisationer med at lykkes med forandring og adoption? Så er det måske dig, vi leder efter hos Vertica.

Som AI Automation Consultant bliver du en central del af Verticas automations- og AI-indsats. Du arbejder tæt sammen med kunderne og guider dem hele vejen fra inspiration og procesanalyse til implementering. Rollen kræver en kombination af stærke konsulentevner til at forstå kundernes virkelighed, og konkret teknisk kunnen til at omsætte idéer til fungerende automationsløsninger og flows.

Din nye hverdag:
Din nye hverdag består af kundevendt konsulentarbejde. Her har du en bærende rolle i at drive hele processen omkring AI- og automationsløsninger, hvor du vil både være udførende specialist, rådgiver og koordinator, afhængigt af opgavens karakter og kundens modenhed.



- You have a solid understanding of AI and LLM technology. You know what AI can do—and, just as importantly, what it cannot do.
- You understand that success with AI and automation isn’t just about technology—it’s largely about people, workflows, and ownership..

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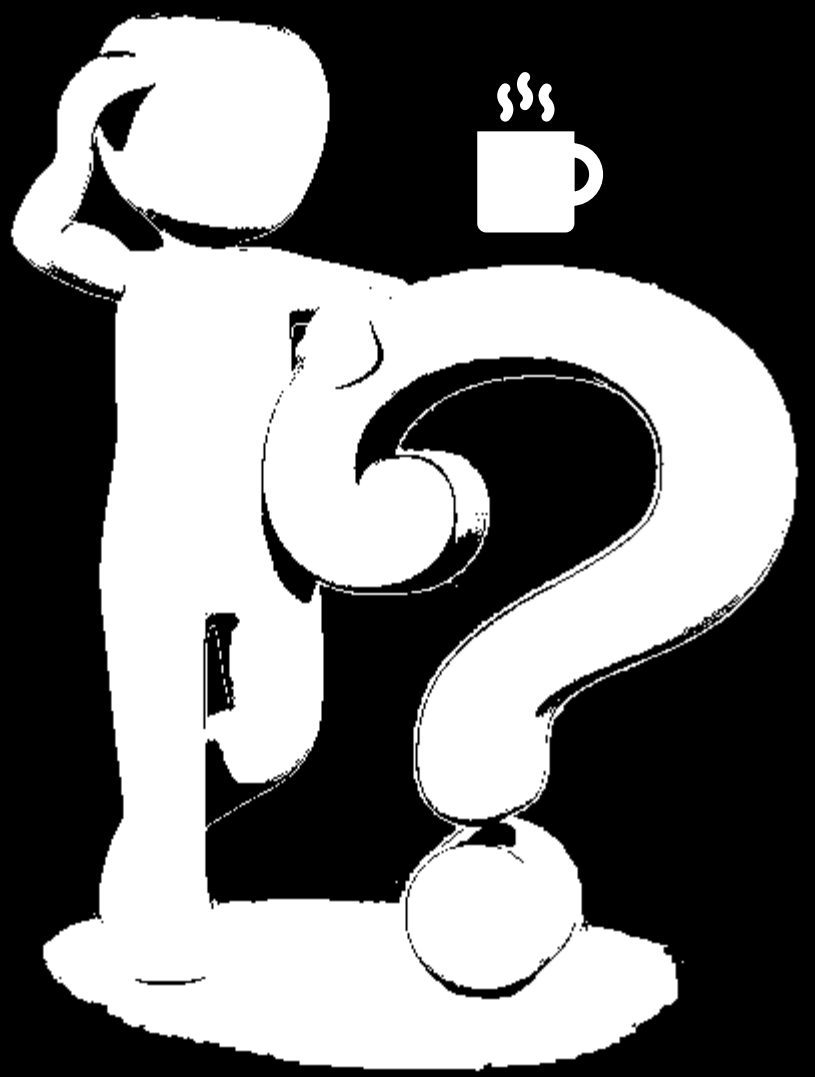
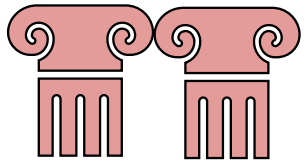
Time for a break



We have come a long way

Foundations



Intelligence

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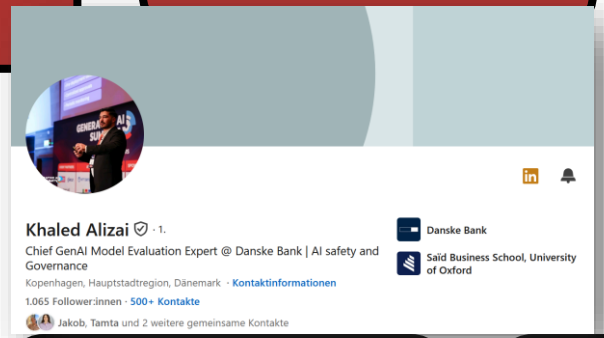
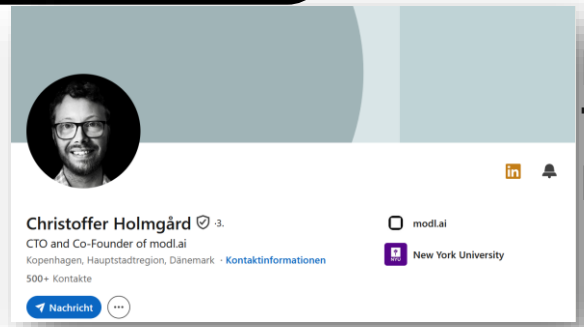
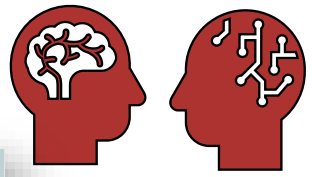
Smart things

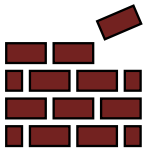
Digital platforms







Human-Technology Interactions



Application

Political Communication

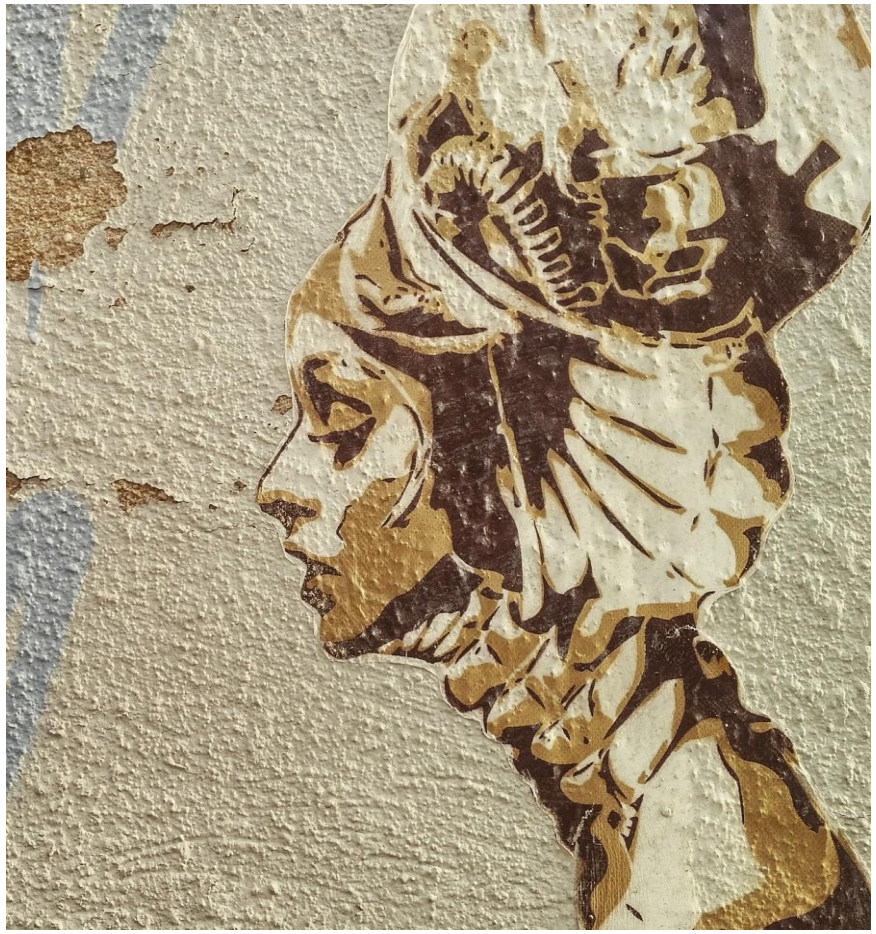
Organisations & Institutions

Health & Wellbeing

Entertainment

Exam

What are the core potentials and risks of AI across these industries?



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Then, you’re ready now

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AI 507: Artificial Intelligence and Society

Tips & tricks for preparing for the exam

Associate Prof. Dr. Lena Frischlich

First of all: What are you preparing for?

Heres' what the course description says about the objectives of the lecture

Expected learning outcome

The course provides students with a theoretical and methodological foundation informed by the social sciences, equipping them to assess both the opportunities and risks associated with the use of AI. Students can take this knowledge into consideration in the development of artificially intelligent systems.

Knowledge: The lecture element of the course will provide students with basic knowledge about central social scientific theories and models (predominantly from psychology and communication) to evaluate the opportunities and risks of the use of AI in different areas including automation, the internet of things, the media, social media and virtual games. The research seminar element of the course will provide students with initial insights into the methodological foundations of the social sciences.

Skills: The course will enable students to master various instrumental techniques: These include (a) critically evaluating scientific evidence about AI and society; and (b) working collaboratively in small research groups to (c) employ AI to answer a currently open research question related to AI in social contexts and (d) presenting their insights following rigorous scientific principles. This hands-on approach ensures that students not only understand theoretical concepts but can also apply them in practical scenarios.

Competences: By the end of the course, students will be capable of forming an empirically grounded opinion about the consequences of AI for individuals, social groups, and society at large. Through the gained knowledge and the obtained skills, they will be able to evaluate social scientific evidence related to AI and integrate their insights into the development of intelligent systems and communicate them to others.

Content

The course has two central elements: A lecture "Introduction into social scientific perspectives on AI" and a research seminar: "Applications of AI in society". The lecture will introduce students to key theoretical and empirical contributions from the social sciences (particularly psychology and communication) relevant to the creation, implementation, and effects of AI in different areas such as automation, the internet of things, the media, social media and virtual games.

The research seminar will introduce students to methodological principles of the social sciences before they work together in small groups to answer an open research question regarding AI and society and present their results following rigorous scientific principles. The question will be agreed upon with the teacher, allowing the class to adapt to current innovations and development in AI and society.

This gives you an initial idea of what the exam will test (as we try to align exams and objectives)

The exam will likely test your knowledge about social scientific theories and models and ask you to evaluate the risks and opportunities of AI in different areas based on evidence

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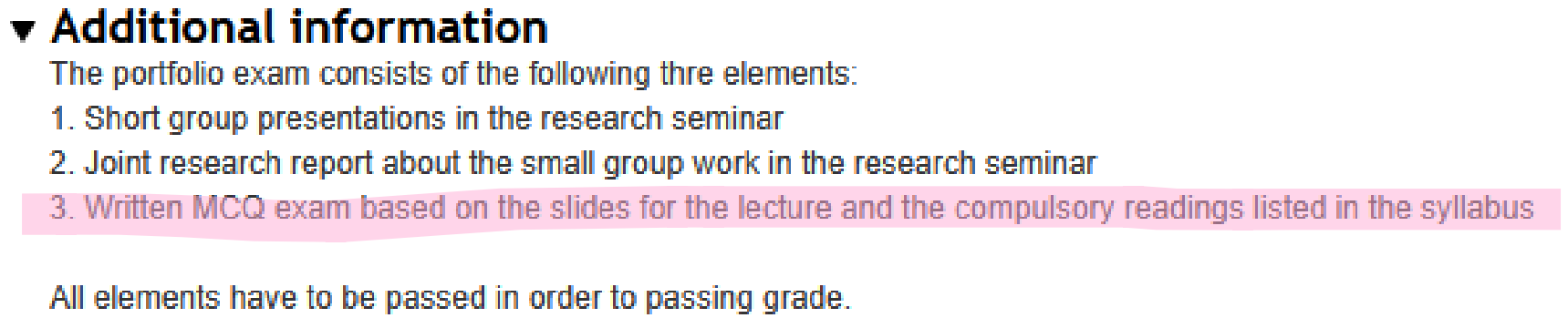
First of all: What are you preparing for?

Let’s see how that looks concretely:



Additional information

The portfolio exam consists of the following thre elements:

1. Short group presentations in the research seminar
2. Joint research report about the small group work in the research seminar
3. Written MCQ exam based on the slides for the lecture and the compulsory readings listed in the syllabus

All elements have to be passed in order to passing grade.

OK, now you also know which format you can expect: It will be a multiple-choice exam, and it will ask for the content of the lecture slides (which you can find in itslearning) and the compulsory readings (not the podcasts!)

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Now: How to prepare?

- First: Get everything in place: Download the slides and texts (which you have hopefully read over the course of the semester already)
- Now: How to best read them? Use the tips for reading literature discussed in the seminar to approach the texts in the lecture and – most importantly – formulate questions about the texts and slides when working with them
- **Remember:** The exam will likely ask you for “social scientific theories and models” and “to evaluate the risks and opportunities of AI in different areas” – try to focus on these: What theories/ models are discussed? What did you learn about the risks and opportunities of AI? How can you apply the theories/ models to advance your understanding of AI?
- Mass learning is very rarely effective – elaboration by posing such questions is much more stable (and more fun;-)). If you are short in time: Consider sending yourself podcast voice messages with the questions you have about the lecture and think about them while taking a walk/ biking to SDU

etc.

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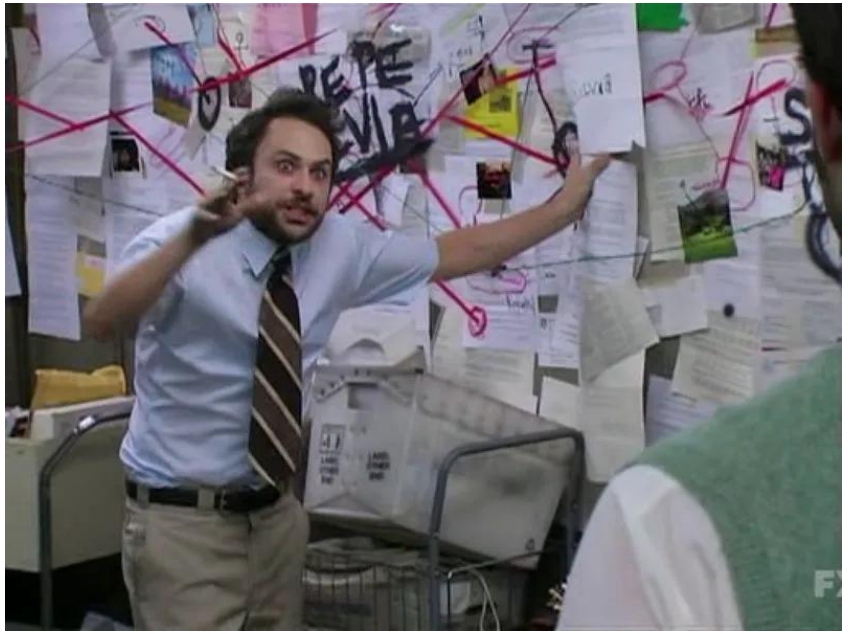
Now: How to prepare?

Remember how we learn

- E.g., embedding knowledge in existing schemata: How does the new information add to/ change what you already knew?
- Through temporal contiguity (i.e., thinking things together that belong together)
- Mind maps are a great way to structure information and identify links – also consider developing a personal colour scheme to structure texts (e.g., general arguments yellow, definitions in blue, names in green, methods in orange) – once you have that stick to it
- Mere reading is <u>not</u> learning – writing things down/ visualising them/ applying



Now: How to prepare?

- Start early – do a bit every week
- Create your own questions for each of the topics and answer them
  You can ask Chatgpt/ LLM xyz to help you find good questions.
- Attention: The AI is crazily focused on historical facts and dates –it's much better first to identify the relevant elements yourself to prompt it precisely
- If the AI answers the questions for you, Chatgpt has learned something and not you – don't trust that reading a generated summary helps you learn anything (use the AI to prompt your *academic* intelligence and not vice versa ;-)
- Sleep and drink enough, go for a walk and take breaks while learning
- Do not panic – it doesn't help you, and if you prepare yourself well, you are 100% able to pass this. I believe in you – you should do the same ☺

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The actual exam

- Multiple-choice, 60 minutes via exam web
- Every question has one correct solution and four alternatives to choose from (guessing likelihood = ¼) – with 30 to max 40 questions
- Spoiler: mere guessing is statistically not enough to pass
- No questions for historical dates, names of individual persons (but remember you are here to learn about theories, concepts, and models)
- Some questions will require application: Take what you have learned and apply it to an AI context

Grade	Description	%	Decision
12	For an excellent performance, displaying a high command of all aspects of the relevant material, with no or only a very few minor weaknesses.	90-100	Surpasses the expectations
10	For a very good performance displaying a high level of command of most aspects of the relevant material, with no or minor weaknesses.	80-89	Passed
7	For a good performance displaying a good command of the relevant material but also some weaknesses.	70-79	
4	For a fair performance displaying some command of the relevant material but also some major weaknesses.	60-69	
02	For a performance meeting only the minimum requirements for acceptance	50-59	
00	For a performance which does not meet the minimum requirements for acceptance	40-49	Not passed
-3	For a performance which is unacceptable in all aspects	0-39	

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AI 507: Artificial Intelligence and Society

And that’s it - let me know, if you have further questions!

Associate Prof. Dr. Lena Frischlich